Panasonic

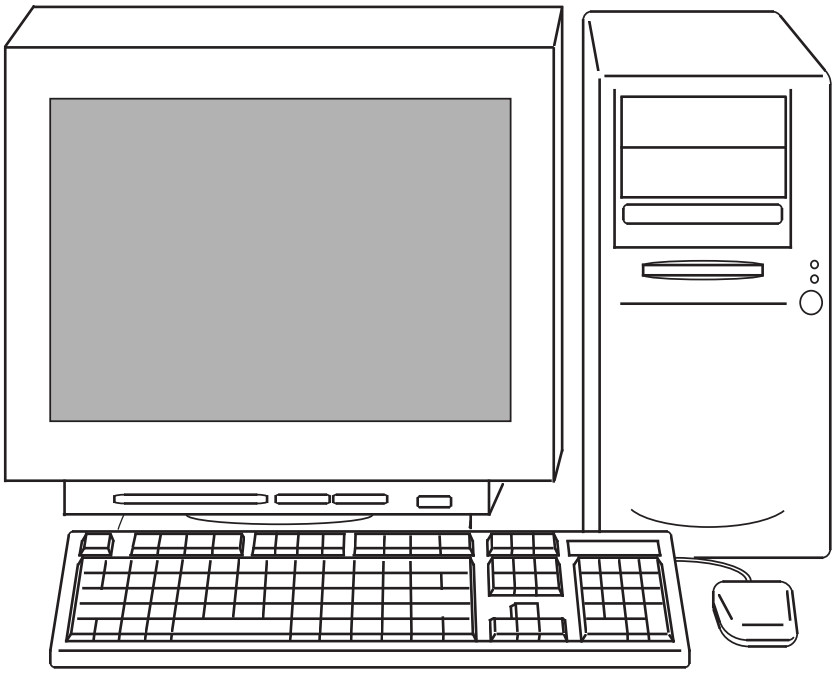
Operating Instructions

Base

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A442E21 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

For Maintaining Product Security

To maintain the security of this product, conduct the following efforts.

User ID and Password Management

- Do not continue using the initial password
  Be sure to change your initial password after first log-in.
- Use a high-strength (resistant to guessing) password
  Avoid easy-to-guess numbers or character string (birth date, nickname, phone number, etc.), words listed in a dictionary or frequently used phrases, and make your password as long as possible.
  For example, the following helps increase the strength of your password:
  - ✓ A combination of uppercase and lowercase letters, numbers and symbols
  - ✓ Long password (the one with at least 8 characters is recommended)
- Properly store and use the password
  Prevent any Web browser from remembering or saving your password.
- Do not tell anyone your password
  Refrain from sharing your user ID with your colleague, or using the user ID of your colleague, even if temporarily.
  It is necessary for the system administrator to draw up guidelines for setting high-strength password, and to eliminate security holes so that password-protected files will not be cracked.
- Avoid repeated use of the password
  In logging in to multiple systems, avoid repeatedly setting, in other words using, the same password.
  Otherwise, information leakage from one of the systems can incur the risk of allowing unauthorized use of the leaked password for purposes such as to gain unauthorized access to the other systems.
- Disclaimer for any password under your management
  We take no responsibility for your password management.

System Configuration Information Management

To prevent the leakage of system configuration information including the one on optional setting parameters, conduct the following efforts.

- Before saving the electronic file containing configuration information, be sure to encrypt and protect it with a password that only the system administrator knows.
- Try saving the electronic file containing system configuration information in an access-limited server.
- In principle, avoid saving the electronic file containing system configuration information in portable external medium such as USB. In cases where you have no choice but to save it in an external medium, be sure that you encrypt the medium before saving it, and completely delete it from the medium upon completion of work.

Connection Environment for Devices Comprising the System

Use the server on which this product is to be installed and the devices (machines, E-mail server, etc.) with which this product is to be communicated in an environment (or firewall-protected environment) that is cut off from the Internet.

= MEMO =

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1. OVERVIEW

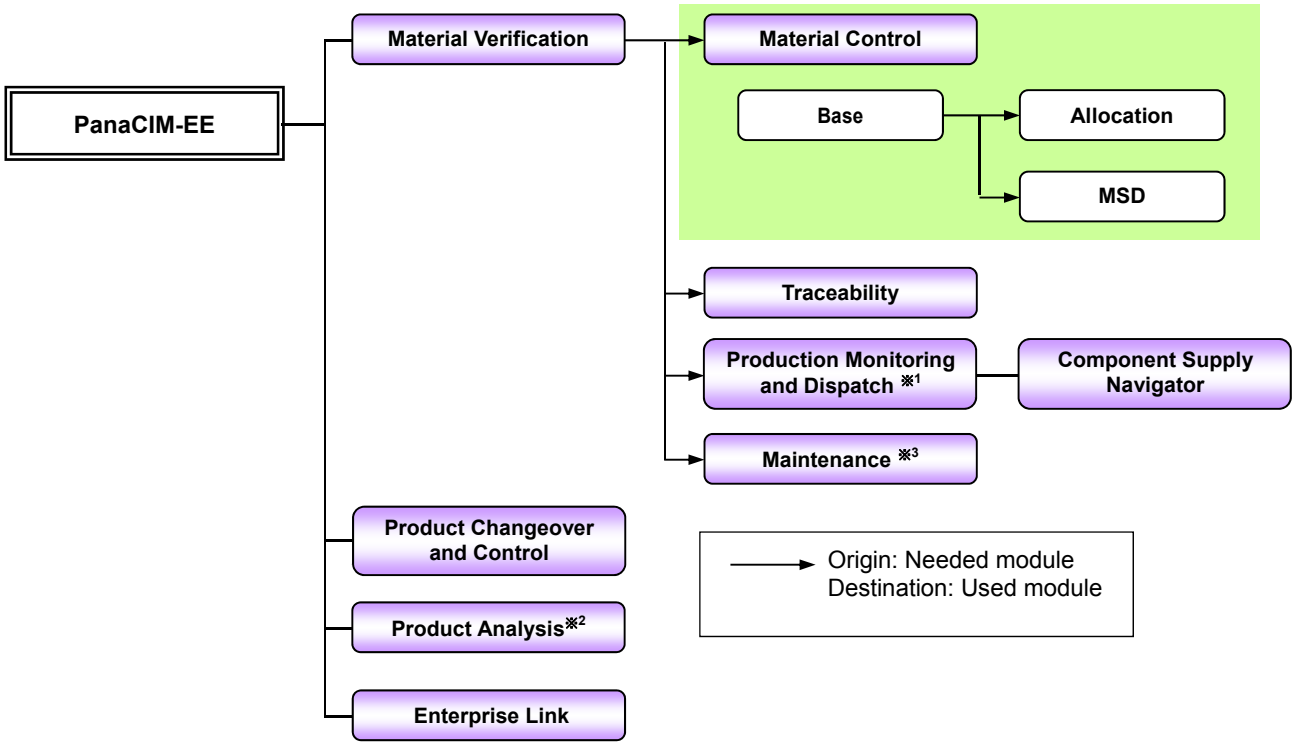
This chapter explains the Overview of PanaCIM-EE.

1.1 PanaCIM-EE Overview

PanaCIM-EE (PanaCIM Enterprise Edition: hereinafter called PanaCIM) is a mount system configured with nine modules[*1] to provide our customers full MES function support that improves OEE (Overall Equipment Efficiency) and aids customer's mount process to have visible productivity, efficiency and quality.

*1: Module means a group of software application functions that is provided per main operation or category in mount process such as Material Verification.

■ PanaCIM Configured modules



*1: Product Analysis needed to show machine's operation information.
*2: One of Material Verification or Traceability or Product Analysis module needed as original data needed.
*3: The license for the operation monitor may be required as necessary.

The following will be achieved by using PanaCIM system.

1. Less bad or discarded or reprocessed PCB
2. Maximum machine usage efficiency
3. Better throughput

To achieve them, PanaCIM has the following functions.

- Program data management
- Real time monitoring
- Production performance, operability history/real time chart generation
- Notification when control limit defined by user exceeded
- Feeder preparation, pat setup, part exhaust verification
- Maintenance history record of mount equipment
- Remaining part count warning
- Part expiration date control
- Moisture control part support (MSD)
- Traceability from parts to PCB/PCB to parts
- Auto changeover *(CM machine only)*
- Parts and feeder material management
- Pat setup and reservation for more different material use
- Data communication between PanaCIM and 3rd-party system
- Web base system that can easily access to application and data

Chapter 2 and later explain PanaCIM basic settings and how to operate it. The following sections are explained for each step using diagrams.

- Initialization
- Creating Line
- Operating Setting
- Troubleshooting

1.2 PanaCIM-EE Product Module

1.2.1 PanaCIM-EE/Material Verification

PanaCIM-EE/Material Verification is an application to verify parts to prevent wrong setup at production setup, changeover, splicing and part exhaust in mount process. Material Verification module is a package consists of software and hardware and allows checking if correct parts are set on the correct supply unit.
This module also equips with “No server package” for customers who do not need the function to control material verification for several lines or to link with the material control module. In “No server package”, you do not require the PanaCIM-EE server. (We have a separate instruction manual for “No server package”).

1.2.2 PanaCIM-EE/Product Analysis

PanaCIM-EE/Operation analysis is a web application that collects operation information of equipment (line, machine) and shows its analysis, statistics and real time to improve productivity of mount line (Lower recognition/pickup error rate, less equipment trouble stop time). Also, creating charts helps understanding trend of production and handling problems on site in advance.

1.2.3 PanaCIM-EE/Traceability

PanaCIM-EE/Traceability has three levels, 1) PCB level, 2) Part level, 3) Mount point level, and you can select a level according to your needs. PCB scanner installed on conveyor reads PCB model ID and serial number and links part and PCB serial numbers in coordination with Material Verification module.

1.2.4 PanaCIM-EE/Material Control

PanaCIM-EE/Materials Management manages parts and feeders at production machine or in support area. It estimates remaining number of each part package and notifies the entire site if the number is getting close to the part exhaust threshold value. Moisture control function monitors an expire data of MSDs on site and notifies if the date is getting close. Material allocation reserves parts for production order to use parts more efficiently on site. Also, it creates missing parts list to check out as less as material from outside storage.

1.2.5 PanaCIM-EE/Product Changeover and Control

PanaCIM-EE/Auto changeover changes product automatically on equipment. After it reads a barcode and detects program change, all machines downloads the new program and starts production automatically.

1.2.6 PanaCIM-EE/Maintenance

PanaCIM-EE manages profiles of assets (equipment, head feeder, etc.) on the mounting floor and records a maintenance history of each asset. By using this feature, you can save your time taken for maintenance planning and management task.

1.2.7 PanaCIM-EE/Production Monitoring & Dispatch

PanaCIM-EE/Operation monitor provide a system where operation status is monitored on real time basis and notification to operators are integrated. It identifies events that may affect production from real time data and notifies operators. It promotes visible factory and maximum communication for the entire production floor to be more visible.

1.2.8 PanaCIM-EE/ Operation Monitor: Component Supply Navigator Option

Operation monitor: Component supply navigator option optimizes the supply sequence in order to choose a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line and contribute to manpower saving.

1.2.9 PanaCIM-EE/Enterprise Link

PanaCIM-EE/External Interface outputs standard format data of XML, socket, database to non-Panasonic system. It also supports custom formats.

1.3 PanaCIM-EE Configuration

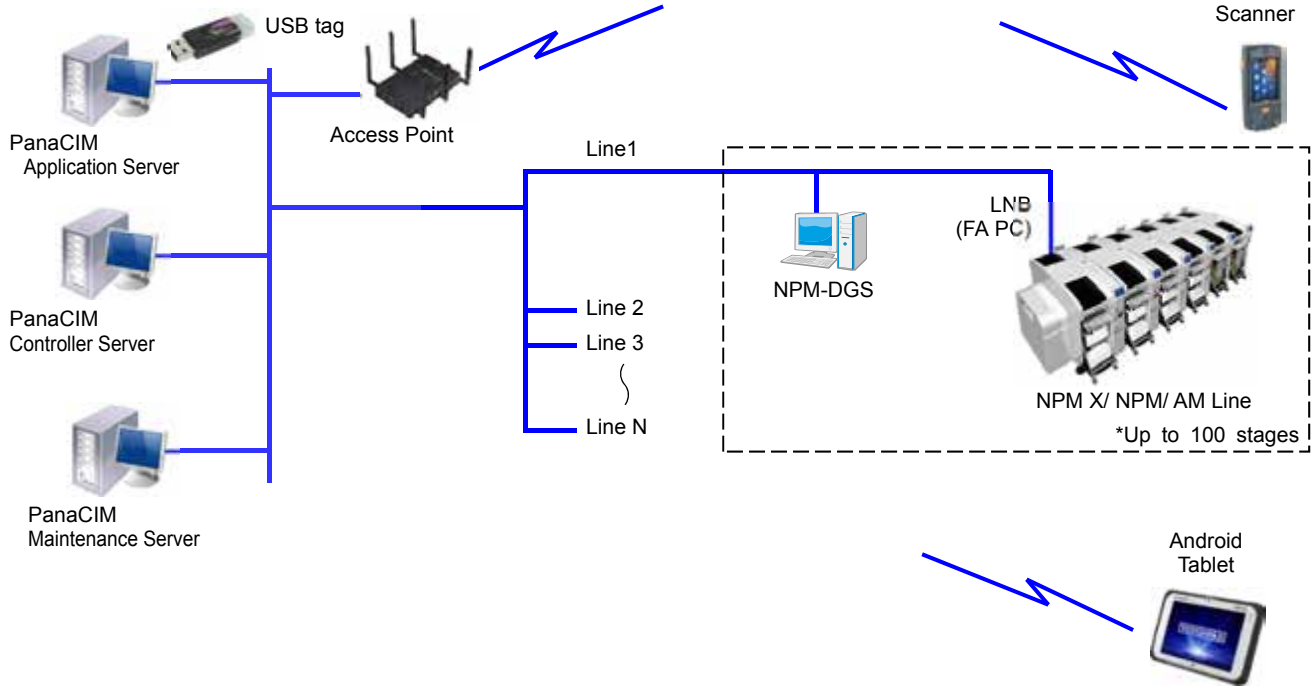
PanaCIM operates in distributed form to balance process loads between multiple work stations. This allows system extension and protects processes between production floor and PanaCIM system.

The table below shows basic PanaCIM design configuration. Its general configuration has the following elements.

Element	OS, Database	Role
	Restrictions	
PanaCIM application server *1	Windows® Server 2008 R2 (32-bit/64bit) *2 Windows® Server 2012 R2 (64-bit) *2 Microsoft® SQL Server 2008 R2 Standard/Enterprise Edition (32-bit/64-bit) *2 Microsoft® SQL Server 2012 Standard/Enterprise Edition (32-bit/64-bit) *2 Microsoft® SQL Server 2014 Standard/Enterprise Edition (32-bit/64-bit) ※2	・Application host ・Provide Web portal (Web server) ・License control ・Data management (Database) ・Minimum configuration RAID 1 recommended
	・Internet Explorer 11 is required. * Use IE8 Compatibility View Settings.	
PanaCIM controller server *1	Red Hat Enterprise Linux 5.4/5.7/6.5 Server (32-bit)	・Server which controls line configuration, settings and all PanaCIM application data/events ・Minimum configuration RAID 1 recommended
	・Max. 100 stages can be managed only by PanaCIM controller server	
PanaCIM Maintenance Server *1	Windows® Server 2012 R2 (64-bit) *2 Microsoft® SQL Server 2012 Standard/Enterprise Edition (32-bit/64-bit) *2	・Manage asset and maintenance performance data (database) ・Provide the maintenance information screen (Web server) ・Minimum configuration RAID 0 recommended
	・It is required when the maintenance module is purchased ・Internet Explorer 11 is required. * Use IE8 Compatibility View Settings.	
PT interface	Windows® 7 Professional (32-bit/64-bit) Windows® 8 Professional (32-bit/64-bit) Windows® 8.1 Professional (64-bit) Windows® 10 Professional (32-bit/64-bit)	・Data communication with PT200
	・One needed for each PT200	
PanaCIM operation monitor Client / Web client (PC for browsing report)	Windows® 7 Professional (32-bit/64-bit) Windows® 8.1 Professional (64-bit) Windows® 10 Professional (32-bit/64-bit)	・Access to the PanaCIM application server and show graph/report. ・Access to the PanaCIM maintenance server and show graph/report
	・No need to purchase new. You can use your computer that can access to PanaCIM application server (Speed such as for response depends on your network environment.) ・Internet Explorer 11 is required. * Use IE8 Compatibility View Settings. *For Windows® 10, IE can be opened from “Start Menu” → “Windows Accessory.”	

LMNC *1	—	・Communicate with fixed barcode readers ・Verify PCB barcodes ・Stop PCB loading into mounter
	・Needed for Traceability and Product Changeover and Control (CM/SP series machine only)	
Fixed barcode reader *1	—	・Read barcode at PCB loading
	・Needed for Traceability and Product Changeover and Control	
Access point *1 *3	—	・Wireless communication between the PanaCIM controller server and the wireless scanner or Android tablet
	・Needed for Material Verification, Material Control, Production Monitoring & Dispatch and Maintenance	
Wireless scanner *1 *3	—	・Read barcodes at Material Verification and Material Control ・Display errors at Production Monitoring & Dispatch and events such as part exhaust
	・Needed for Material Verification, Material Control, and Production Monitoring & Dispatch	
Android tablet *1 *3	Android™ 6.0	・Display maintenance inspection items ・Display spare parts information
	・Required for the maintenance module. ・Google Chrome™ browser is required. *1	

*1: You need to prepare this depending on modules (functions) you use.
The number of depends on machine configuration. Contact us for more details.
*2: Windows® Server and Microsoft® SQL server require the license of the server itself and of access to the server. Contact Microsoft® or sales agent and prepare licenses according to the devices you want to use.
*3: Wireless equipment such as an access point, a wireless scanner an Android tablet must comply with radio regulations of the country or region of use. When you relocate your wireless equipment to a different country or region from the country or region of delivery, it must comply with radio regulations of such a country or region before use. So, when you relocate your wireless equipment, consult us.



PanaCIM Network Configuration Diagram

1.4 Operationg Environment

1.4.1 Hardware Requirement

PanaCIM hardware requirements are determined by two factors, the number of equipment and collecting recipes. The number of equipment determines the processing performance needed and collecting recipe data volume determines the Hard disk space required.[*1]

*1: Contact our sales or service for more details on hardware.

① Server and PC

Item		Specifications			
		PanaCIM application server[*1]	**PanaCIM controller server[*1]**	**PT interface · Web client**	**PanaCIM operating monitor client/ Web client**
Main body		IBM PC/AT compatible (Desktop computer strongly recommended)			
	CPU	Intel® Core™2 Duo E6700 equivalent or higher (For V8.6.0.1 or later, Intel® Xeon E3-1220 3.1 GHz equivalent or higher is strongly recommended)		Intel® Core™ i3/ i5/ i7(8 MB cash, 3.9 GHz) equivalent or higher	
	Mother board	IBM complete compatible			
	Serial I/O	IBM complete compatible			
	Graphic board	SXGA or more(Display 65 536 or more colors) Desktop area: 1 024 × 768 dot or more			
	Memory	16 GB or more	4 GB or more	4 GB or more	
	Hard disk	500 GB or more[*1]		250 GB or more	80 GB or more
		NTFS file system	Ext3 file system	NTFS file system	
	Optical drive	DVD drive			
Keyboard		101 Keys US-English Keyboard			
Mouse		OS standard support item			
Monitor		SXGA supported			
Network card		1000 BASE-T × 2			

*1: Regularly move the purge file, etc., to another area.

② Server and PC (When the maintenance module is purchased)

The following lists hardware requirements besides the PanaCIM application server and controller server when the maintenance module is purchased.

Item		Specificaitons		
		PanaCIM maintenance server	Web client	Mobile termial
Main body		IBM PC/AT compatible (Desktop computer strongly recommended)		Android tablet
	CPU	Intel® Xeon® E3 (4 Core) equivelent or higher	Intel® Core™ i3/i5/i7 (8 MB cash, 3.9GHz) equivalent or higher	Intel® Atom™ x3 equivalent or higher Qualcomm® Snapdragon™ S4 equivalent or higher
	Motherboard	IBM complete compatible		—
	Serial I/O	IBM complete compatible		—
	Graphic board	SXGA or more (Display 65 536 or more colors) Desktop area: 1 280 × 1024 dot or more		7-color LCD WXGA or more Display 65 536 or more colors
	Memory	16 GB or more	4 GB or more	2 GB or more
	Hard disk	1.0 TB or more	80 GB or more	32 GB or more
		NTFS file system		—
	Optical driver	DVD drive		—
Keyboard		101 Keys US-English Keyboard		—
Mouse		OS standard support item		Capacitive multi touch
Monitor		SXGA supported		—
Camera		—		Back side: Eight million or more pixels
Network		1000 BASE-T × 2	1000 BASE-T × 1 or more	Compliance with IEEE802.11a (W52/W53/W56)/b/g/n

③ Server (one configuration)

Item		**Specifications**	
		Without maitenanace module (PanaCIM application server + PanaCIM controller server)	With maitenance module
Main body		IBM PC/AT compatible (Desktop computer strongly recommended)	
	CPU	Intel® Xeon® E5 (6 Core) equivalent or higher	Intel® Xeon® E5 (8 Core) equivalent or higher
	Mother board	IBM complete compatible	
	USB port	USB port × 1 or more	
	Serial I/O	IBM complete compatible	
	Graphic board	SXGA or more(Display 65 536 or more colors) Desktop area: 1 024 × 768 dot or more	
	Memory	20 GB or more	36 GB or more
	Hard disk	1.0 TB or more	1.5TB or more
		NTFS File system	NTFS File system
	Optical drive	DVD drive	
Keyboard		101 Keys US-English Keyboard	
Mouse		OS standard support item	
Monitor		SXGA supported	
Network card		1000 BASE-T × 2	

1.4.2 Software Requirement

① Operational environment for PanaCIM server and PC

Item		Lan-gauge	Application				
			PanCIM application server	PanaCIM controller server	PanaCIM controller client	PanaCIM PT interface	PanaCIM operating monitor client ・ Web client
OS							
	Windows® Server 2008 R2 Standard/Enterprise Edition (32-bit /64-bit) Windows® Server 2012 R2 Standard/Enterprise Edition (64-bit)	Japanese English	○	–	–	–	–
	Red Hat Enterprise Linux 5.4/5.7/6.5 Server (32-bit)		–	○	○	–	–
	Windows® 7 Professional (32-bit/64-bit) Windows® 8.1 Professional (64-bit) Windows® 10 Professional (32-bit/64-bit)		–	–	–	○	–
	Windows® 7 Professional (32-bit/64-bit) Windows® 8.1 Professional (64-bit) Windows® 10 Professional (32-bit/64-bit)		–	–	–	–	○
Other software							
	Microsoft® SQL Server 2008 R2 Standard/Enterprise Edition (32-bit/64bit) Microsoft® SQL Server 2012 Standard/Enterprise Edition (32-bit/64-bit) Microsoft® SQL Server 2014 Standard/Enterprise Edition (32-bit/64-bit)	Japanese English	○	–	–	–	–
	Internet Explorer 11 *Use IE8 Compatibility View Settings *For Windows® 10, IE can be opened from “Start Menu” → “Windows Accessory.”		○	–	–	–	○
	Microsoft .NET 3.5 Microsoft .NET 4.0		○	–	–	–	–
	IIS 6.0		○	–	–	–	–
	Microsoft's Message Queuing Service		○	–	–	–	–

○: Applicable software, –: No applicable software

② Operating environment of PanaCIM server and PC (When the maintenance module is purchased)

The following lists software requirements besides the PanaCIM application server, PanaCIM controller server and Web client.

Item		Lan-gauge	Application		
			PanaCIM Maintenance server	PanaCIM Maintenance client	Android tablet
OS					
	Windows Server® 2012 R2 (64bit)	Japanese English	○	–	–
	Windows® 7 Professional (64bit) Windows® 10 Professional (64bit)		–	○	–
	AndroidTM 6.0		–	–	○
Other software					
	Microsoft® SQL Server® 2012 R2 Standard/Enterprise Edition (64bit)	Japanese English	○	–	–
	Internet Explorer 11 *Use IE8 compatible display setting *For Windows® 10, IE can be opened from “Start Menu” → “Windows Accessory.”		○	○	–
	Google Crome™		–	–	○
	Microsoft® .NET 3.5		○	–	–
	IIS 6.0		○	–	–
	Microsoft's Message Queuing Service		○	–	–

○: Applicable software, –: No applicable software

③ Operational environment for server (one configuration)

Item		Lan-gauge	Application	
			Without maintenance module (PanaCIM application server + PanaCIM controller server)	With maitenance module
Virtual software		Japanese English		
	VMware ESXi 6.0		○	

○: Applicable software, –: No applicable software

*In addition to above, you need to install software required for the PanaCIM application server, PanaCIM controller server, and PanaCIM maintenance server

1.4.3 Antivirus Software

Installing antivirus software application software on a computer where our application system operates may cause the following problems.

- File scanning causes high load on PC and real time characteristics are lost.
- Communication problem occurs due to firewall function. etc.

Nonetheless, installing antivirus software is important given recent information security matters and to protect the application system itself from computer virus. Take cautions on the following points when installing computer antivirus software.

■ Cautions in installing antivirus software

- Run entire HDD scanning when the system software is not running.
- Exclude the port our system uses from firewall.
- Disable other functions that may put high load on PC.
- Current requirement specifications assume that other application systems are not working. If problems arise by running antivirus software, a computer with higher specification may help solving the problems.

■ Reference

Antivirus software used in our system software operation test environment.

- TREND MICRO　Virus Baster™　Cooperate Edition

1.4.4 The Cautions about Password

If you set a password that can be assumed easily, a third-party may use it and cause problems with PanaCIM system. And your data may leak to others.
Set a login password for the PC to be used for PanaCIM, so that the password will be at least 8 characters long and include at least one each of alphabetical, numeric and special characters (“,” , “.” , “-“, “_”, “@”, “!”). Also, do not use the default password but set up a new password.

1.4.5 Handling of Hard Disk

Malicious third-party may steal data that remained in a discarded PC and leak your information to outside. When you discard a PCB, make sure you delete data completely.
Also, if a subcontractor connect his/her private PC to PanaCIM PC via network, the network setting information stays in the PC. Do not allow using any private PC, and use PCs that are internally controlled to prevent network information leaking.
If you ask a repair company to repair broken PC, PanaCIM and PC setting information may be changed or deleted unintentionally. Or setting information may leak outside. We recommend RAID configuration as backup of data including password setting discussed above.

1.4.6 The Cautions at the Time of the Notice Use of E-mail

PanaCIM has a function to notify an administrator of error information using e-mail.
There are the following restrictions apply to the use of this function.

① You must set up an e-mail server on your own
② You must set up an e-mail server under the environment where internet access is blocked by firewall (We do not support PanaCIM operation on the PC that is connected to the internet)

1.5 Object Machine and Functional List

Equipment and functions handled by PanaCIM are shown below.

■ Operation Analysis/Material Verification/Traceability Module Target Machine

			Product Analysis				Material Verification				Traceability			
		Function	Produced PCBs	Operation Rate	Wait Stop Time	Placement Rate Count etc	Operator Notification	Reel Tray Part	Solder Mask Squeegee	Machine Interlock	PCB Trace	Part Trace	Placement Trace	Manage trace data
No.	Machine model	Product No.												
1	NPM-VF	NM-EJR9A	○	○	○	○	○	○	-	○	○	○	○	○
2	NPM-DX※3	NM-EJM8D	○	○	○	○	○	○	-	○	○	○	○	○
3	NPM-D3	NM-EJM6D	○	○	○	○	○	○	-	○	○	○	○	○
4	NPM-D2	NM-EJM5D	○	○	○	○	○	○	-	○	○	○	○	○
5	NPM-D	NM-EJM1D	○	○	○	○	○	○	-	○	○	○	○	○
6	NPM-TT2	NM-EJM1E	○	○	○	○	○	○	-	○	○	○	○	○
7	NPM-TT	NM-EJM3D	○	○	○	○	○	○	-	○	○	○	○	○
8	NPM-WX	NM-EJM9D	○	○	○	○	○	○	-	○	○	○	○	○
9	NPM-WXS	NM-EJM2E	○	○	○	○	○	○	-	○	○	○	○	○
10	NPM-W2S	NM-EJM5E	○	○	○	○	○	○	-	○	○	○	○	○
11	NPM-W2	NM-EJM7D	○	○	○	○	○	○	-	○	○	○	○	○
12	NPM-W	NM-EJM2D	○	○	○	○	○	○	-	○	○	○	○	○
13	NPM	NM-EJM9B	○	○	○	○	○	○	-	○	○	○	○	○
14	AM100	NM-EJM4D	○	○	○	○	○	○	-	○	○	○	○	○
15	CM602-L	NM-EJM8A	○	○	○	○	○	○	-	○	○	○	×	○
16	CM232-M	NM-EJM9C	○	○	○	○	○	○	-	○	○	○	×	○
17	CM212-M	NM-EJM6A	○	○	○	○	○	○	-	○	○	○	×	○
18	CM101-D	NM-EJM5B	○	○	○	○	○	○	-	○	○	○	×	○
19	CM402-L	KXF-4Z4C	○	○	○	○	○	○	-	○	○	○	×	○
20	CM402-M	KXF-4Y3C	○	○	○	○	○	○	-	○	○	○	×	○
21	CM401-L	NM-EJM1A	○	○	○	○	○	○	-	○	○	○	×	○
22	CM401-M	KXF-4X3C	○	○	○	○	○	○	-	○	○	○	×	○
23	DT401-F	KXF-E64C	○	○	○	○	○	○	-	○	○	○	×	○
24	DT401-M	KXF-E53C	○	○	○	○	○	○	-	○	○	○	×	○
25	SPV-DC	NM-EJP7A	○	○	○	-	×	-	○	○	○	-	-	○
26	SPV	NM-EJP7A NM-EJP8A	○	○	○	-	×	-	○	○	○	-	-	○
27	SPD	NM-EJP5A	○	○	○	-	×	-	○	○	○	-	-	○
28	SPG	NM-EJP6A	○	○	○	-	×	-	○	○	○	-	-	○

29	SP70※2	NM-EJP3A	×	×	×	-	×	-	○	○	×	-	-	×
			×	○	○	-	×	-	○	×	-			
30	SP60P-MU	KXF-1J3C	×	○	○	-	×	-	○	×	-			
31	SP60P-L	KXF-1G4C	×	○	○	-	×	-	○	×	-			
32	SP80V-MU	KXF-1K3C	×	○	○	-	×	-	○	×	-			
33	SP80V-L	KXF-1H4C	×	○	○	-	×	-	○	×	-			

(○: Supported, ×: Not Supported ,—: Not Applicable)

*1: Contact us for other machines than those above.
*2: The higher of SP70 shows interface support when connected with LNB and the lower shows that when connected with PT200.
*3: NPM-DX is not tray-compatible.

■ Materials Management/Auto Changeover/Operation Monitor Module Target Machine

No.	Machine model	Product No.	Material Control: Remaining Part Count Monitoring	Material Control: Check in/out Parts Location Management	Material Control: Reserve Parts and Create Missing Part List	Material Control: Control MSD Expire Date	Product Changeover and Control: Read PCB Barcode	Product Changeover and Control: Compare Model Code	Product Changeover and Control: PCB Loading Control	Product Changeover and Control: Machine Control	Production Monitoring & Dispatch: Entire Factory Line Monitor	Production Monitoring & Dispatch: Collect and Warn Real Time Events	Production Monitoring & Dispatch: Display Information Changeover	Production Monitoring & Dispatch: Instruction to Operator	Production Monitoring & Dispatch: (Supply Patrol Order) *4 Component Supply Nafivator
1	NPM-VF	NM-EJR9A	○	○	○	○	-*2				○	○	○	○	○
2	NPM-DX	NM-EJM8D	○	○	○	○	-*2				○	○	○	○	○
3	NPM-D3	NM-EJM6D	○	○	○	○	-*2				○	○	○	○	○
4	NPM-D2	NM-EJM5D	○	○	○	○	-*2				○	○	○	○	○
5	NPM-D	NM-EJM1D	○	○	○	○	-*2				○	○	○	○	○
6	NPM-TT2	NM-EJM1E	○	○	○	○	-*2				○	○	○	○	○
7	NPM-TT	NM-EJM3D	○	○	○	○	-*2				○	○	○	○	○
8	NPM-WX	NM-EJM9D	○	○	○	○	-*2				○	○	○	○	○
9	NPM-WXS	NM-EJM2E	○	○	○	○	-*2				○	○	○	○	○
10	NPM-W2S	NM-EJM5E	○	○	○	○	-*2				○	○	○	○	–
11	NPM-W2	NM-EJM7D	○	○	○	○	-*2				○	○	○	○	○
12	NPM-W	NM-EJM2D	○	○	○	○	-*2				○	○	○	○	○
13	NPM	NM-EJM9B	○	○	○	○	-*2				○	○	○	○	○
14	AM100	NM-EJM4D	○	○	○	○	-*2				○	○	○	○	○
15	CM602-L	NM-EJM8A	○	○	○	○	○	○	○	○	○	○	○	○	×
16	CM232-M	NM-EJM9C	○	○	○	○	○	○	○	○	○	○	○	○	×
17	CM212-M	NM-EJM6A	○	○	○	○	○	○	○	○	○	○	○	○	×
18	CM101-D	NM-EJM5B	○	○	○	○	○	○	○	○	○	○	○	○	×
19	CM402-L	KXF-4Z4C	○	○	○	○	○	○	○	○	○	○	○	○	×
20	CM402-M	KXF-4Y3C	○	○	○	○	○	○	○	○	○	○	○	○	×
21	CM401-L	NM-EJM1A	○	○	○	○	○	○	○	○	○	○	○	○	×
22	CM401-M	KXF-4X3C	○	○	○	○	○	○	○	○	○	○	○	○	×
23	DT401-F	KXF-E64C	○	○	○	○	○	○	○	○	○	○	○	○	×
24	DT401-M	KXF-E53C	○	○	○	○	○	○	○	○	○	○	○	○	×
25	SPV-DC	NM-EJP7A	-	○	○	-	-*2				○	○	-*2	○	×
26	SPV	NM-EJP7A NM-EJP8A	-	○	○	-	-*2				○	○	-*2	○	×
27	SPD	NM-EJP5A	-	○	○	-	-*2				○	○	-*2	○	×
28	SPG	NM-EJP6A	-	○	○	-	-*2				○	○	-*2	○	×
29	SP70*3	NM-EJP3A	-	×	×	-	-*2				○	○	-*2	○	×
			-	×	×	-	○	○	○	○	×				
30	SP60P-MU	KXF-1J3C	-	×	×	-	○	○	○	○	×				
31	SP60P-L	KXF-1G4C	-	×	×	-	○	○	○	○	×				
32	SP80V-MU	KXF-1K3C	-	×	×	-	○	○	○	○	×				
33	SP80V-L	KXF-1H4C	-	×	×	-	○	○	○	○	×				

(○: Supported, ×: Not Supported ,—: Not Applicable)

*1: Contact us for other machines than those above.
*2: Automatic model change for NPM X/ NPM/ AM series, SPV, SPD, SPG and SP70 (LNB connection) is supported as an optional of the machine.
*3: The higher of SP70 shows interface support when connected with LNB and the lower shows that when connected with PT200.
*4: For the mixed line of NPM X/ NPM/ AM series with CM series (CM602, CM101, CM232, CM212, CM402/ DT401), please ask us.

- **Enterprise Link Module Target Machine**

  Exportable data are PCB Trace data, Material Verification data, Machine Status data and Produced PCB Count. To output data collected through Production Analysis, target machine model is the same as Product Analysis module.

■ Maintenance module target machine

No.	Machine model	Product No.	Maintenance: Asset profile management	Maintenance: Maitennace hisotry management	Maintenance: Spare parts replacement management	Maintenance: Defect assets misuse prevention	Maintenance: Maitenance infomration consolidated monitor *1
1	NPM-VF	NM-EJR9A	○	○	○	○	○
2	NPM-DX	NM-EJM8D	○	○	○	○	○
3	NPM-D3	NM-EJM6D	○	○	○	○	○
4	NPM-D2	NM-EJM5D	○	○	○	○	○
5	NPM-D	NM-EJM1D	○	○	○	○	○
6	NPM-TT2	NM-EJM1E	○	○	○	○	○
7	NPM-TT	NM-EJM3D	○	○	○	○	○
8	NPM-WX	NM-EJM9D	○	○	○	○	○
9	NPM-WXS	NM-EJM2E	○	○	○	○	○
10	NPM-W2S	NM-EJM5E	-	-	-	-	-
11	NPM-W2	NM-EJM7D	○	○	○	○	○
12	NPM-W	NM-EJM2D	○	○	○	○	○
13	NPM	NM-EJM9B	○	○	○	○	○
14	AM100	NM-EJM4D	○	○	○	○	○
15	CM602-L	NM-EJM8A	×	×	×	×	×
16	CM232-M	NM-EJM9C	×	×	×	×	×
17	CM212-M	NM-EJM6A	×	×	×	×	×
18	CM101-D	NM-EJM5B	×	×	×	×	×
19	CM402-L	KXF-4Z4C	×	×	×	×	×
20	CM402-M	KXF-4Y3C	×	×	×	×	×
21	CM401-L	NM-EJM1A	×	×	×	×	×
22	CM401-M	KXF-4X3C	×	×	×	×	×
23	DT401-F	KXF-E64C	×	×	×	×	×
24	DT401-M	KXF-E53C	×	×	×	×	×
25	SPV-DC	NM-EJP7A	×	×	×	×	×
26	SPV	NM-EJP7A NM-EJP8A	×	×	×	×	×
27	SPD	NM-EJP5A	×	×	×	×	×
28	SPG	NM-EJP6A	×	×	×	×	×
29	SP70	NM-EJP3A	×	×	×	×	×
			×	×	×	×	×
30	SP60P-MU	KXF-1J3C	×	×	×	×	×
31	SP60P-L	KXF-1G4C	×	×	×	×	×
32	SP80V-MU	KXF-1K3C	×	×	×	×	×
33	SP80V-L	KXF-1H4C	×	×	×	×	×

(○: Applicable, ×: Not applicable, -: Non-target)

*1 The operating monitor license is required.

2. IN READING AND ADVANCING BASE EDITION

In reading and advancing introductory section, the required thing is explained in this chapter.

2.1 Overview

Chapter 7.1 provides terminology. However this section explains PanaCIM terminology used often in this manual.

No.	Word	Note
1	Route	Route in PanaCIM means line in data generation system or PanaPRO or PT200. PanaPRO/PT200 calls mount equipment line “line” but PanaCIM calls all processes (routes) including PCB loader/reflow/manual mount “Route”.
2	Zone	A process unit that includes equipment/transfer unit/manual mount area is called Zone. Each zone should have at least one of those units.
3	Mix	A program which has intergrated multiple recipe configuratio information (coordinates, parts, feeder, nozzle, part setup information etc.) is called “Mix”.
4	PT200	Panasonic data generation system for CM/DT/SP series machines.
5	DGS	Panasonic data generation system for NPM series machines*1 and CM/DT series machines.
6	LNB	Recipe manager for NPM series machines*1.
7	iLNB	A controller which controls the production program of the third party’s machines*2.
8	PT interface	Interface application for communication between PanaCIM controller and PT200.
9	CM series machine	Panasonic mount equipment line. See Chapter 1.5 for PanaCIM-EE target equipment.
10	NPM series machine*1	Panasonic mount equipment line. See Chapter 1.5 for PanaCIM-EE target equipment.
11	BM series machine	Panasonic mount equipment line. This is not supported. (Example: BM231)

*1: NPM series machine includes AM100, SPD, SPG, SPV and SP70 machine models.
*2: The third party’s machines include SPI, OVEN and AOI machine models.

3. INITIALIZATION

This chapter explains initial setting of PanaCIM-EE.

3.1 Preparation of PanaCIM-EE

This chapter guides you through the configuration process for the PanaCIM Base System.

Note: Individual Module settings are described in detail in the applicable Module's documentation.

Reference the below table for a brief overview of what Applications will be used to fulfill configuration settings for PanaCIM Base.

Application	Overview
PanaCIM Console	The PanaCIM Console Application is installed on your Windows system. This Console houses various utilities that you will use to perform some of your Base configurations and operations. • Access the General Settings Utility to enter and validate your license string. • Access the Email Notifications Utility to define email notification settings. **Note:** Ensure your Keytag has been inserted into the USB port of your computer.
System Configuration	The System Configuration Application is used to configure the site-dependent aspects of the PanaCIM system.
Route Editor	The Route Configuration Editor Application is used to create, edit, and delete routes, zones, and equipment. The initial creation of all objects; Routes, Zones, Machinery, or Equipment, follows a top-down approach. That is to say that before a Zone can be created, a Route must exist, into which the zone is added. Before a piece of Machinery or Equipment can be created, a Zone must exist, into which the machines and equipment is inserted. **Note:** Route Editor only allows major modifications to be made while PanaCIM is stopped to ensure the integrity of data in the database. Privilege levels are used to protect the configuration from accidental corruption by inappropriate users.
Machine Settings	There are various configurations that need to be made in order for machines to communicate with PanaCIM.
PC Settings	Various PC settings must be configured so that your Traditional Machines have the ability to communicate with PanaCIM Machine Communication Settings.

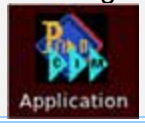
3.1.1 Installation

PanaCIM install setup will be done by Panasonic service engineer.

3.2 Starting of PanaCIM-EE

3.2.1 How to Start

This section explains how to start PanaCIM.

Step	Procedure
1	Start PanaCIM application server (Windows Sever).
2	Login authentication dialog appears. Log in with Administrator account set on Windows Server.
3	Start SQL Server Management Studio and make sure PanaCIM Database is running.
4	Start PanaCIM controller server (Linux).
5	Log in with panadmin/panasupr/panaoper account.  Account kinds • **panadmin**: With PanaCIM administrator privilege • **panasupr**: With PanaCIM supervisor privilege • **panaoper**: With PanaCIM operator privilege • **root**: With Red Hat Linux administrator privilege. **Note:** Default password is the same as account name. To change password, login with root account and go to [System]> [Control]> [Users and Groups] and change password for accounts.
6	PanaCIM service starts on background after login. Wait 1-2 minutes and click the following PanaCIM Application icon to start PanaCIM controller. 

7	Check the valid date of the license. If it is still valid, the following PanaCIM application tool bar appears.  **Note:** Displayed icons vary depending on login account.
8	If clients are installed, start client sequentially.

3.2.2 How to End

This section shows how to terminate PanaCIM. After terminated PanaCIM, you cannot use applications including Material Verification.

Step	Procedure
1	Make sure production line is stopped.
2	Terminate Red Hat Linux system.
3	Terminate Windows Server system.
4	Terminate client(s) if there is any client.

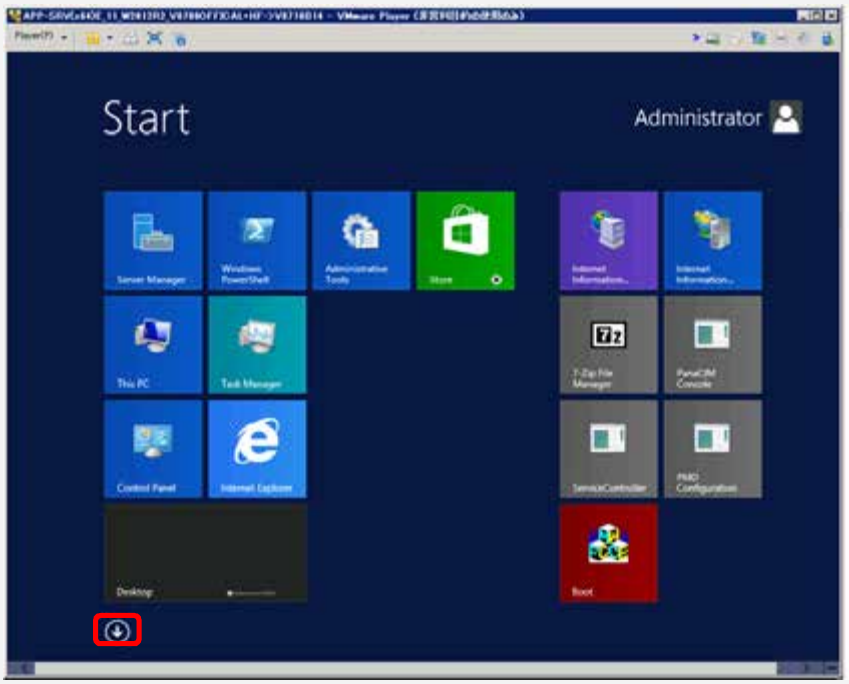
3.2.3 Starting and End of Related System

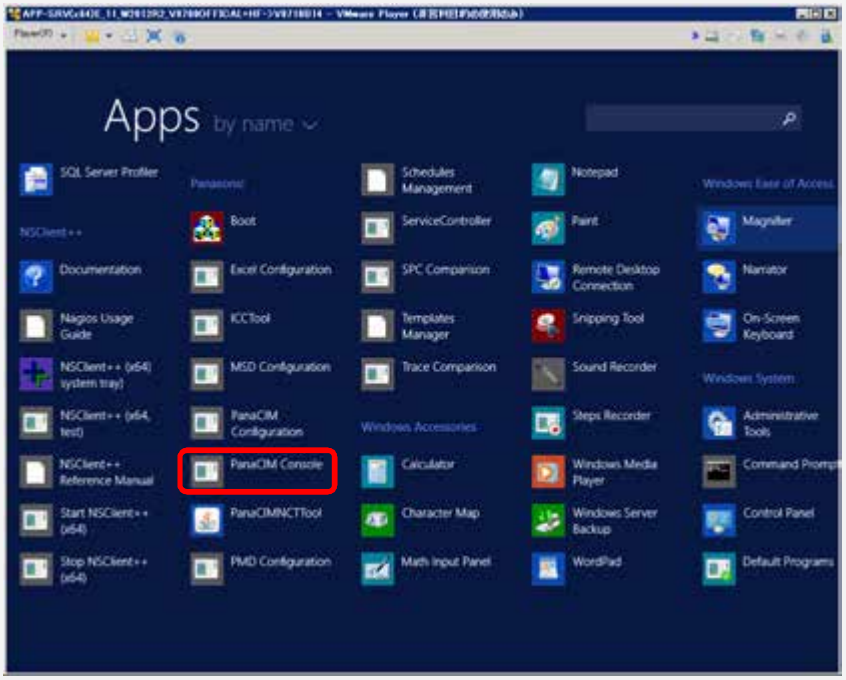
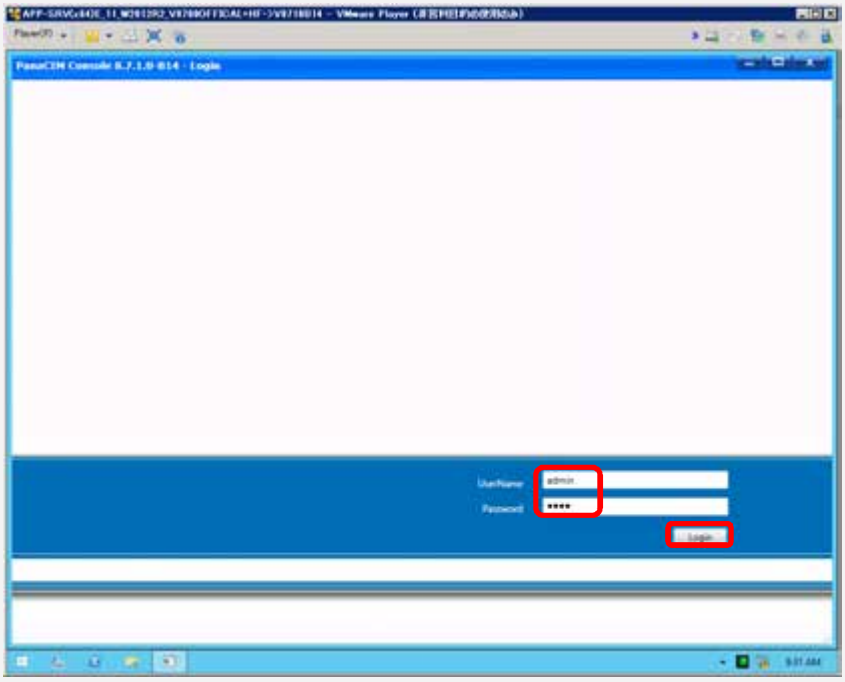
There is no specific order for starting/terminating equipment, PT200, LNB, DGS, PT interface, LMNC and PanaCIM. The systems recognize a system they connect to after they are started.

3.3 Stating of PanaCIM Console and Checking of Version

3.3.1 How to Start

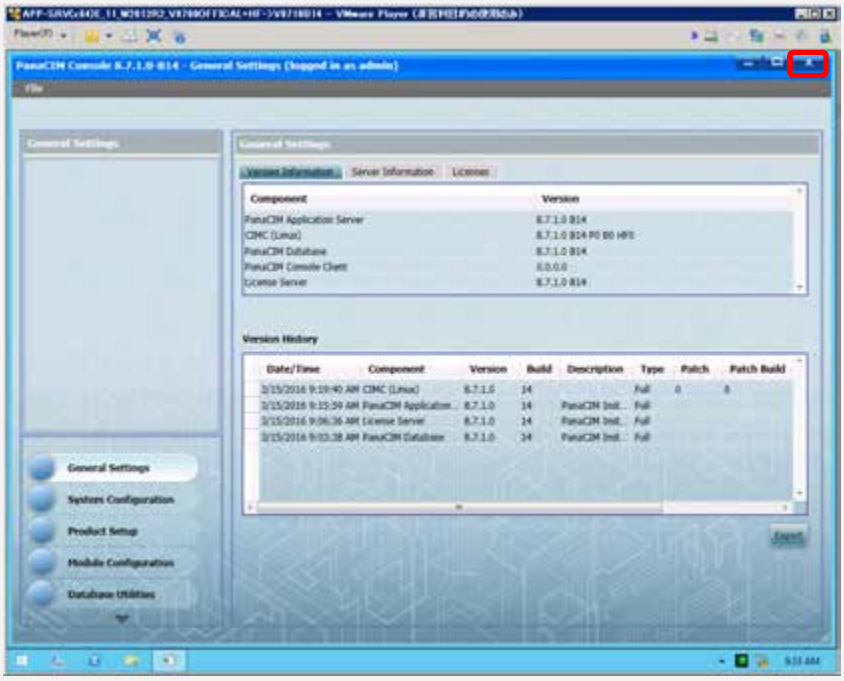
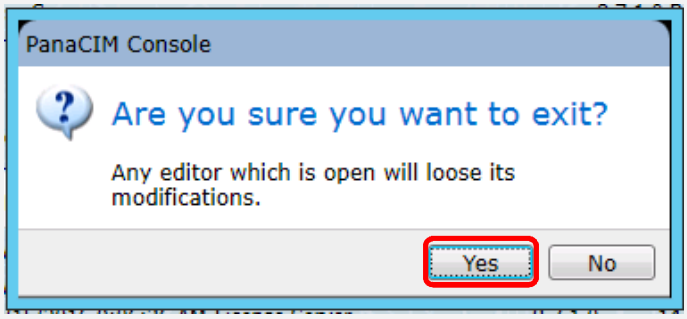
This section shows the procedure for starting the PanaCIM console on the application server.

Step	Procedure
1	Click the shortcut icon for "PanaCIM console" on the application server (Windows Sever). ■When there is the shortcut ・ Double-click the shortcut icon for [PanaCIM Console].  ■When there is not the shortcut icon 1) Click the [Windows Start] icon on the task bar.  2) Click [↓] on the "Start" screen. 

Step	Procedure
	3) Click the [PanaCIM Console] icon on the “App” screen. 
2	The login information entry screen will be displayed where you should enter the following information and then press the [Login] button. ・User: **admin** ・Password: **pass** 
3	The PanaCIM console screen will be displayed.

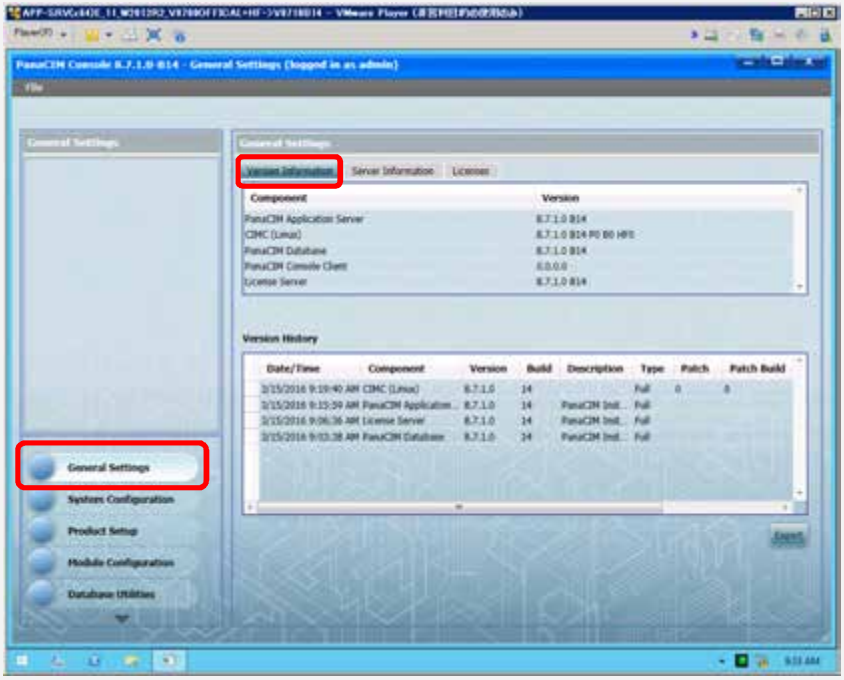
3.3.2 How to End

This section shows how to terminate the PanaCIM console.

Step	Procedure
1	Click the [×] icon on the PanaCIM console screen. 
2	The confirmation pop-up window will be displayed where you should pree the [Yes] button. 

3.3.3 How to Check the Version

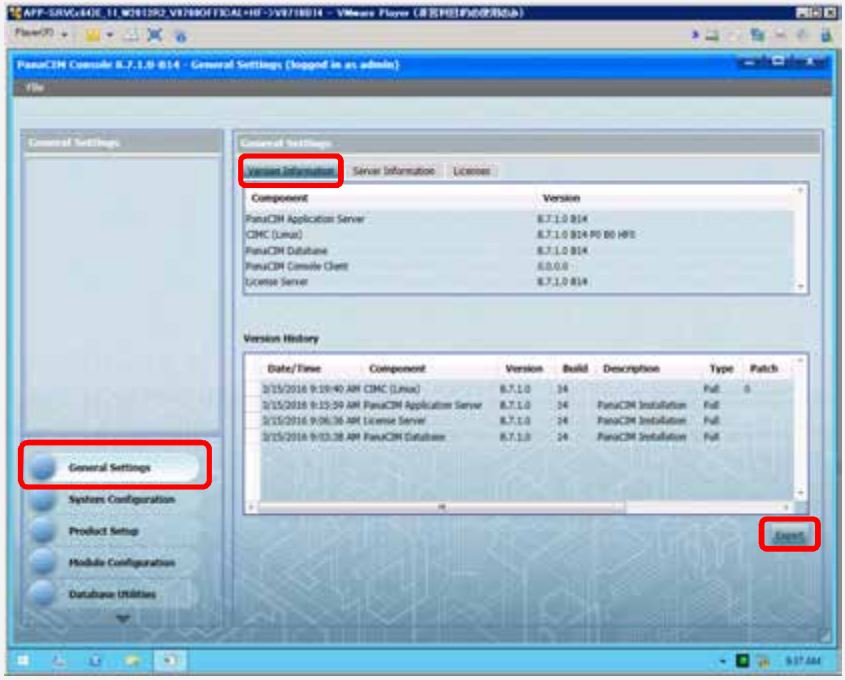
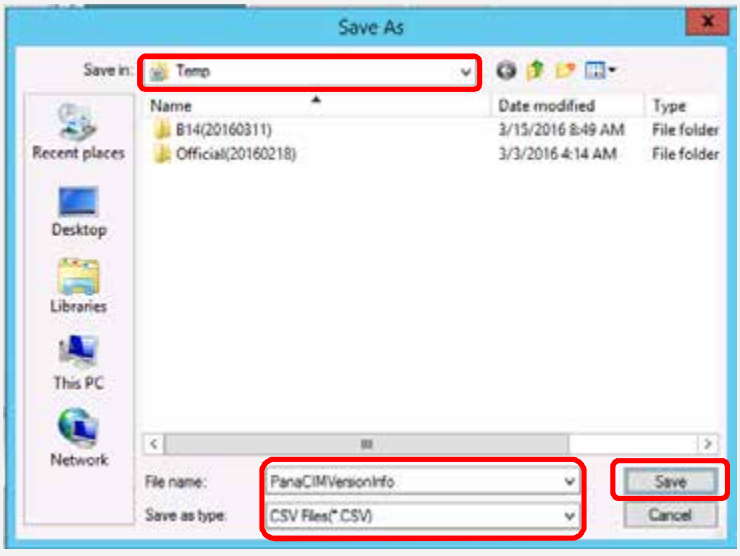
This section shows how to check the PanaCIM version with the PanaCIM console.
***Patch and HotFix are not supported in current phase 1.**

Step	Procedure
1	On the PanaCIM console, select [General Settings] - [Version Information] tab. *This screen is displayed as the initial screen for the PanaCIM console.  ■Current version display The upper list on the screen above shows the information on the current components and their versions. **Component:** Shows the names of the following components. ・PanaCIM application server ・CIMC (Linux) ・PanaCIM database ・License server *Additionally, if any tool or client is installed on the install screen, PanaCIM tools or console clients will also be displayed. **Version:** Shows the version of each of the components above. ■Version history display The lower [Version History] list on the screen above shows the information on installation or upgrades (including application of Patch or HotFix) performed in chronological order. Date/Time: Shows the date and time when installation or upgrade (including application of Patch or HotFix) was performed. Component: Shows the names of the following components ・PanaCIM application server

	・CIMC (Linux) ・PanaCIM database ・License server *If any tool or client has been installed, PanaCIM tools or console clients are also displayed. Version: Shows the PanaCIM version after installation or upgrade (including application of Patch or HotFix). Build: Shows the build number corresponding to the PanaCIM version after installation or upgrade (including application of Patch or HotFix). *This is not shown when the PanaCIM version is OFFICIAL. Description: “PanaCIM Install” is shown. Type: The following three types are shown. Full: Indicates that PanaCIM has been installed/upgraded. Patch: Indicates that Patch has been applied to PanaCIM. HotFix: Indicates that HotFix has been applied to PanaCIM. Patch: Shows the number of the Patch applied to PanaCIM. Patch Build: Shows the Build for the number of the Patch applied to PanaCIM. HotFix: Shows the number of the HotFix applied to PanaCIM. Host Name: Shows the name of the host (server name) on which the component is operating.

3.3.4 How to Output Version Information to Files

This section shows how to output the PanaCIM version information to a CSV file.

Step	Procedure
1	On the [General Settings] - [Version Information] screen for PanaCIM console, press the [Export] button. 
2	The [File Output] pop-up window will be displayed where you should enter the following information and then press the [Save] button. ・Where to save: Select any destination folder. ・File name: Enter the name of the output CSV file. ・File type: **CSV Files (*CSV) [Fixed]** 

■ **Output file**

The following shows a sample output file. File name: PanaCIM “Version Inforomation”.csv

```
Date/Time,Component,Version,Build,Description,Type,Patch,Patch Build,HotFix,Host Name
3/15/2016 9:19:40 AM,CIMC (Linux),8.7.1.0,14,,Full,0,0,0,CIMC-SRV64E
3/15/2016 9:15:59 AM,PanaCIM Application Server,8.7.1.0,14,PanaCIM Installation,Full,,,,APP-SRV64E
3/15/2016 9:06:36 AM,License Server,8.7.1.0,14,PanaCIM Installation,Full,,,,APP-SRV64E
3/15/2016 9:03:38 AM,PanaCIM Database,8.7.1.0,14,PanaCIM Installation,Full,,,,APP-SRV64E
```

3.4 Configuration of System Config

Set the basic configuration of the PanaCIM system via the System Configuration Application.

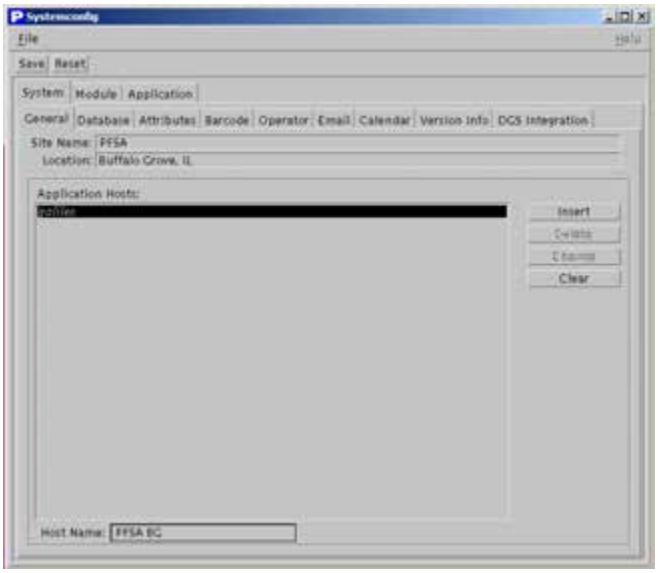


Select **System Cfg** icon on the application toolbar to open the System Configuration Application.



3.4.1 General

Select the **System** tab followed by the **General** tab.



Set the following items on this page.

- Populate **Site Name** and **Location**.
- Define the PanaCIM hosts (controller server and client PCs).

■ Setting the site name and location

The Site Name and Location fields are used for description purposes only. Information need not always be identical with the license information.

Step	Procedure
1	**Site Name:** Enter your company name.
2	**Location:** Enter the geographical location of your company.
3	Click the **Save** button to save changes.

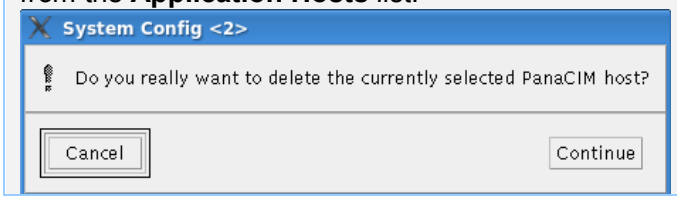
Adding a host

If the CIM Server or CIM Client is not viewable in the Application Hosts list after installation, follow these next steps to add a host:

Step	Procedure
1	Enter the name of the PanaCIM controller server/client to be added in the **Host Name** field.
2	Click the **Insert** button to add the name of the new host to the **Application Hosts** list.
3	Click the **Save** button to save changes, or the **Reset** button to discard changes and revert to original data.
4	Click the **Clear** button to clear the selected host name. Saved host name will not be deleted.

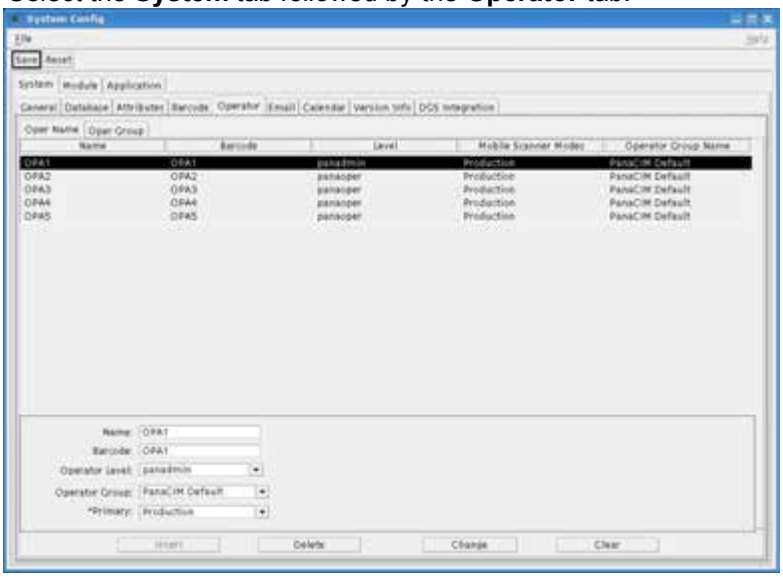
Deleting a host

If a host is no longer being used (i.e. ghost computer), follow these next steps to delete a host:

Step	Procedure
1	Select the **network name** of the host to be deleted. The network name will be displayed in the **Host Name** field. Click the **Delete** button.
2	The following message dialog appears. Click **Continue** to remove the selected host from the **Application Hosts** list. 
3	Click the **Save** button to save changes, or the **Reset** button to discard changes and revert to original data.

3.4.2 Operator

Select the **System** tab followed by the **Operator** tab.

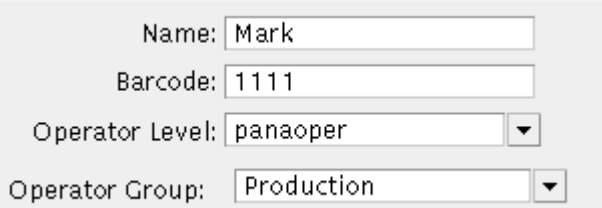


Administrators can configure the System Configuration Application to associate an operator barcode to an operator name. If an association exists, the name will be displayed in reports/popups instead of the barcode.

This page is used to:

- add, change or delete operators.

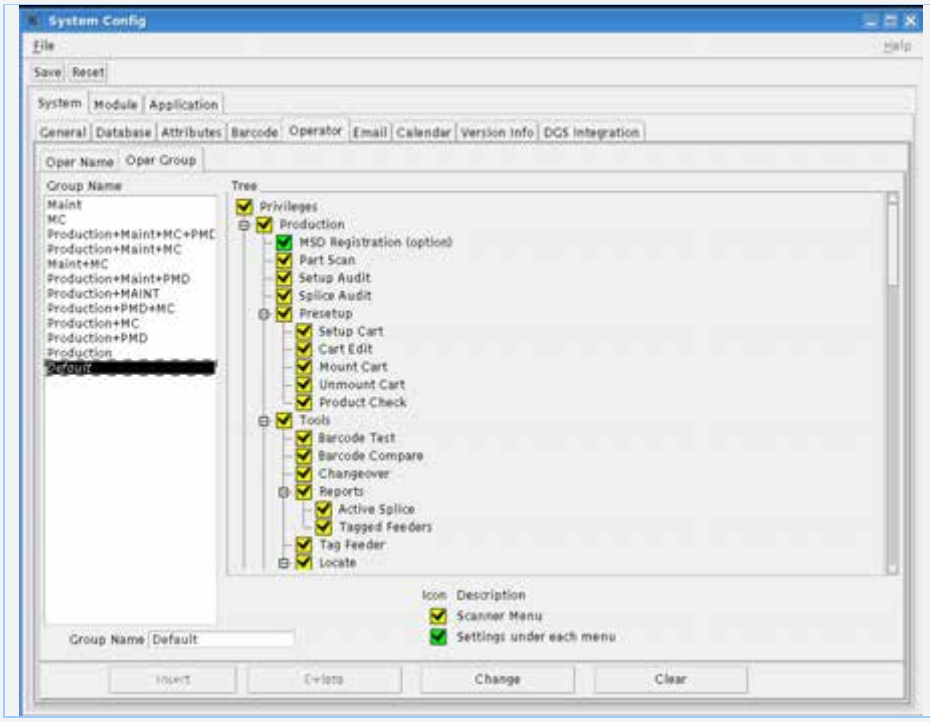
Note: Make sure operator name and barcode do not duplicate. If existing operator name or barcode is entered, a warning message dialog appears and added item will not be reflected.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **System** tab, and then click the **Operator** tab.
3	Click the **Oper Name** tab.
4	Click Clear to rid any unwanted text or selections.
5	a. Enter the **Operator name**. b. Enter the **Operator barcode**. c. Select the **operator level**: • **Panadmin**. Allows users to perform configuration changes and maintenance checks to the PanaCIM system. Users also acquire the same permissions provided to panasupr and panaoper users. • **Panasupr**. Allows users to create products. Users also acquire the same permissions provided to panaoper users. • **Panaoper**. Allows users to transfer products to the Machine Zones as well as display Route Status. d. Select the **Operator** Group.  Refer to section 2.5.1 for description of operator groups.

6	Select Insert after you have satisfied your selections. Oper Name | Oper Group Name / Barcode / Level / Operator Group Name p1a / p1aaa / panaoper / Production p / p / panaoper / Production ad / ad / panasupr / Production 55 / 55 / panaoper / Production 3 / 3 / panaoper / Production+MC 2 / 2 / panaoper / Production+MC 1 / 1 / panaoper / Production Mark / 1111 / panaoper / Production
7	Repeat these steps for each operator. **Note:** To modify an operator, select the Operator Name and modify the text fields. To delete an operator, select the Operator Name and click Delete and **Continue** when prompted.
8	Click the Save button to save changes, or the Reset button to discard changes and revert to original data.

■ Operator Groups

Operator groups are defined using the **Oper Group** tab. This tab allows the user to define the Operator Groups and assign the menus in the mobile scanner that will be visible to operators.

Step	Procedure
1	In **Oper Group** the user has ability to define any number of groups. Each group can have its own unique configuration.
	
2	In order to accommodate existing installations with existing operator's configured the following **default Group Name Templates** are supplied with PanaCIM and have the following attributes: ➢ **Default** - This group name template cannot be deleted and by default have all the functions and their sub functions enabled (checked). This template can be assigned in the previous screen to any operator. It can also be customized which means the user can check or uncheck any of the functions. This template applies to all scanners in the event where Operator IDs are not enabled in PanaCIM. If the Default Template is assigned to an Operator, and the Operator logs into the scanner, only the licensed functions will be visible on the scanner. ➢ **Production** - This template includes all the production functions and sub functions enabled by default. ➢ **Production+PMD** - This template includes the following functions in enabled state: ▪ all the production functions and sub functions ▪ PMD functions ➢ **Production+MC** - This template includes the following functions in enabled state: ▪ all the production functions and sub

functions
- all Material Control functions and sub functions

- **Production+PMD+MC** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - PMD functions
  - All Material Control Functions sub functions

- **Production+Maint** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions

- **Production+Maint+PMD** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - PMD functions

- **Maint+MC** - This template includes the following functions in enabled state:
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions

- **Production+Maint+MC** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions

- **Production+Maint+MC+PMD** - This template includes the following functions in enabled state:
  - all the production functions and sub functions
  - all Maintenance Functions and sub functions
  - all Material Control functions and sub functions
  - PMD functions

- **MaterialControl** - This template includes the following functions in enabled state:
  - all Material Control functions and sub functions

- **Maintenance** - This template includes the following

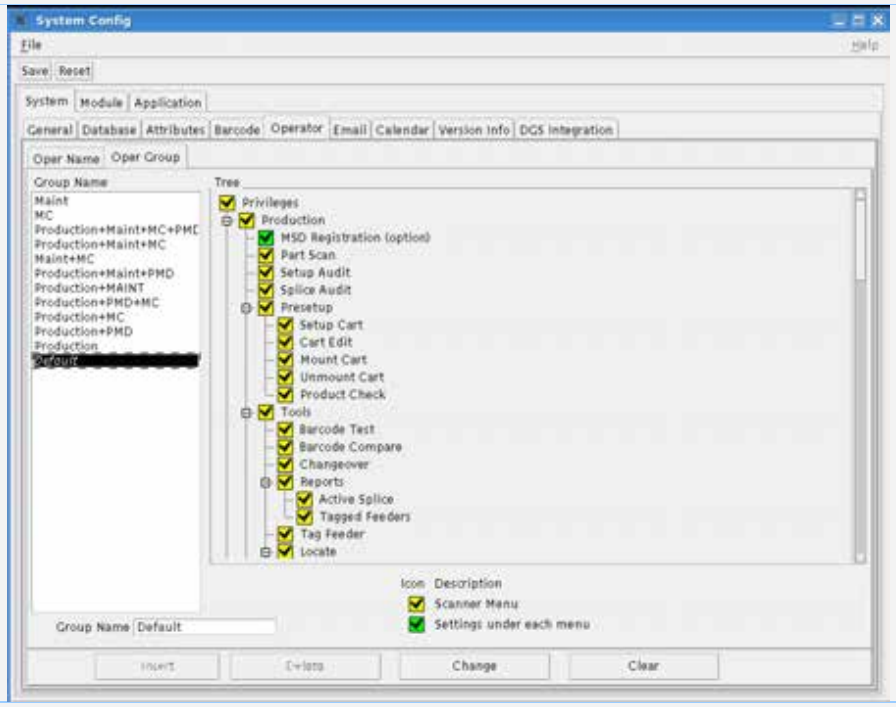
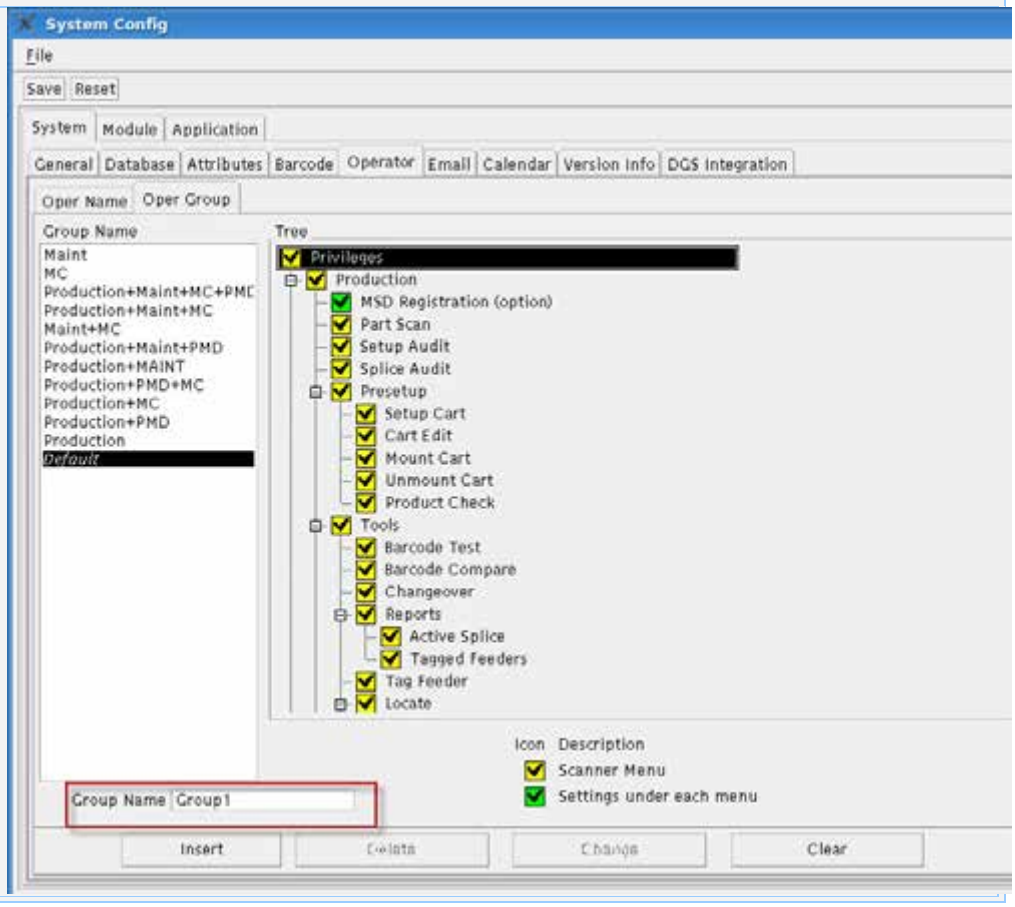
	functions in enabled state: ▪ all Maintenance functions and sub functions

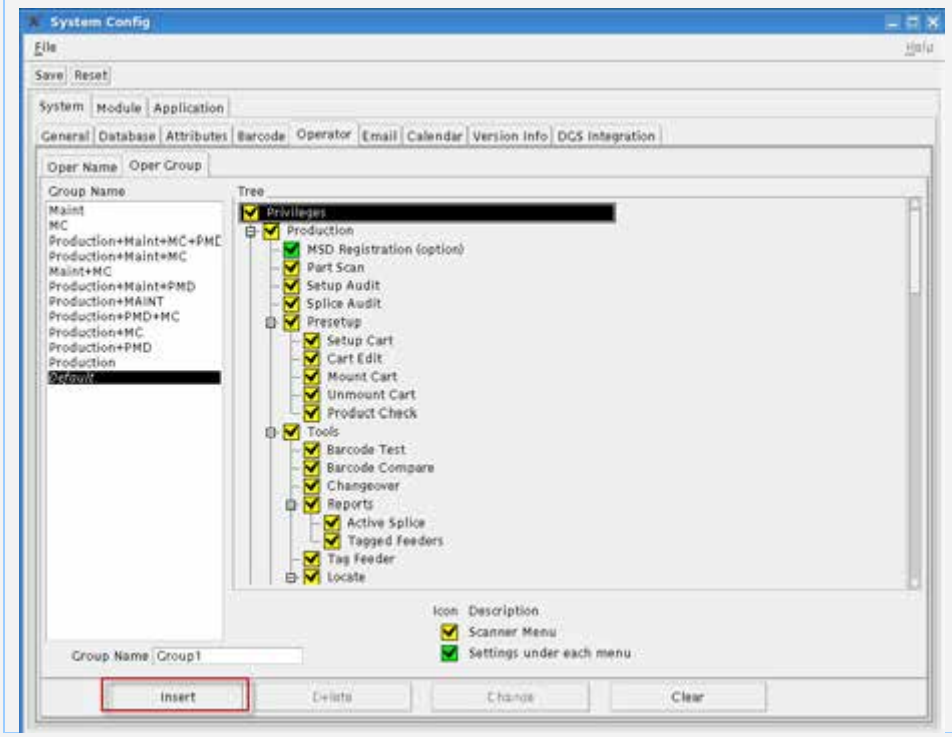
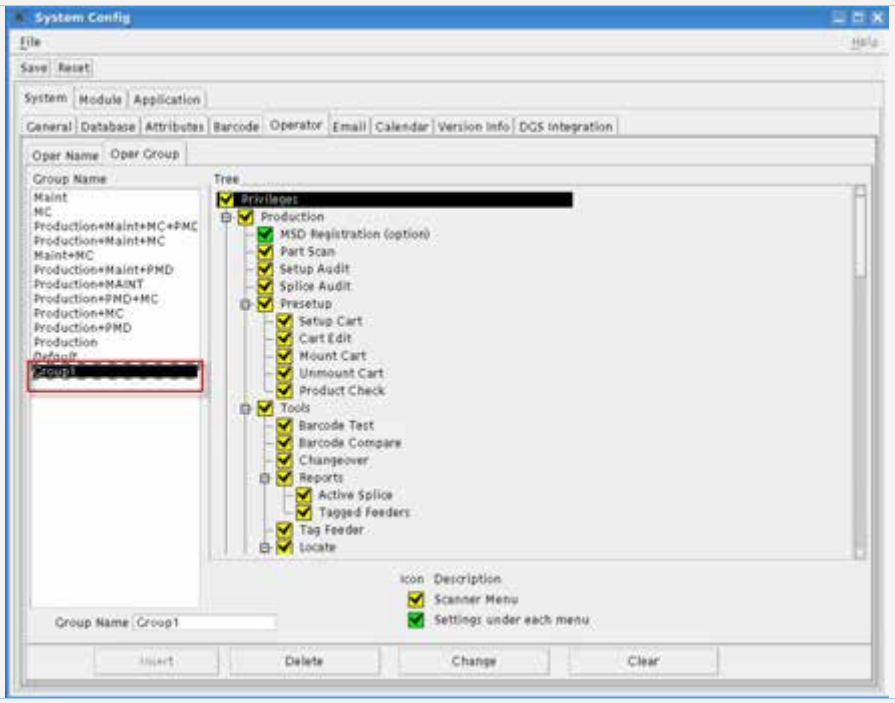
All above Templates can be edited, renamed or even deleted by the user except for the “Default” Template. The Default Template cannot be renamed or deleted. It can however be customized to display any combination of the existing licensed functions.

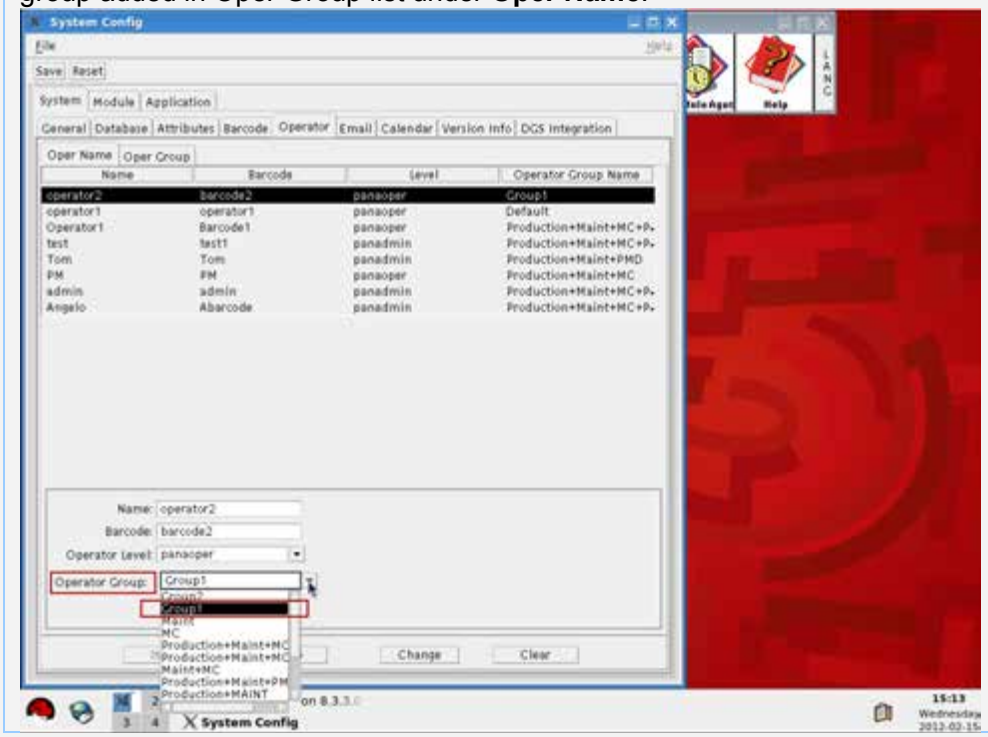
Licensing supersede the above configuration, which means that even if a function is checked in the above configuration, it will not be displayed unless the factory has the corresponding licenses.

■ Adding Operator Group

Following is the procedure to add group:

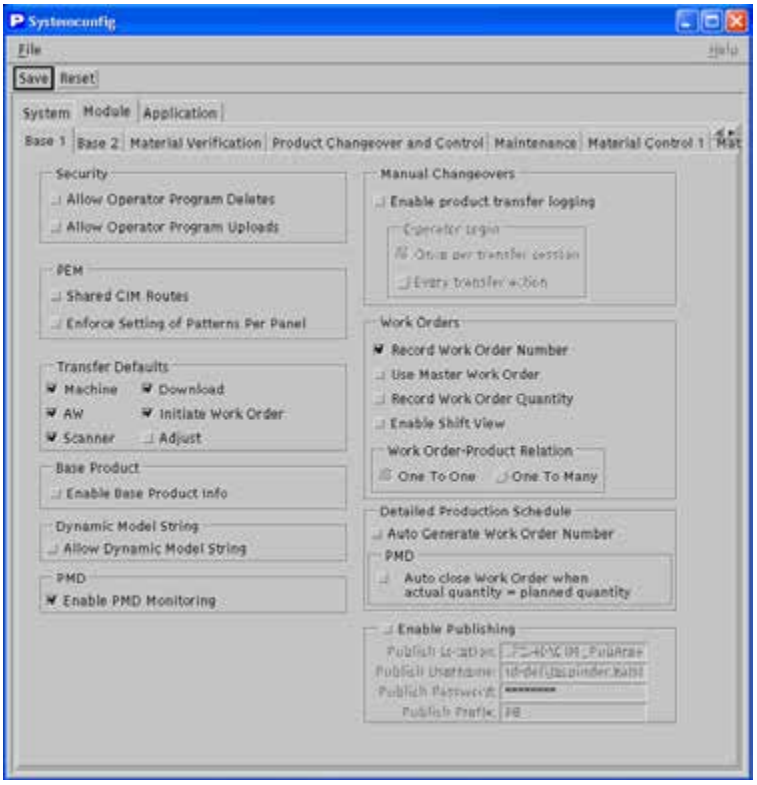
Step	Procedure
1	Enter the name of the operator in System Config > **System** > **Operator** > **Oper Group**.
	
2	Enter the **Group Name**.
	

3	Click **Insert**.
	
4	Group name gets listed with the privileges tree.
	
5	Select the privileges desired to be assign to the group and click Save.

6	Close the System Config screen and reopen the System Config to see the group added in Oper Group list under **Oper Name**. 

3.4.3 Base1

Select the **Module** tab followed by the **Base1** tab.



Set the following items on this page.

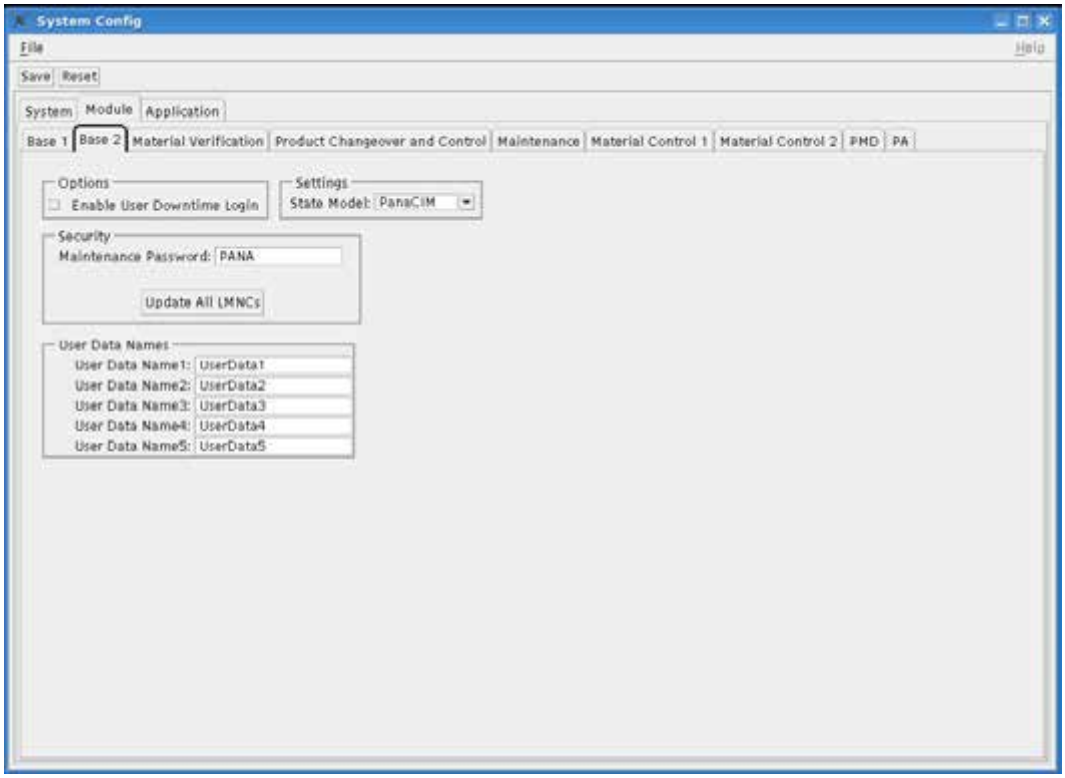
- Select the panaoper security options.
- Configure the program-related data.
- Configure the program transfer defaults.
- Configure a work order.

Step	Procedure
1	Select **Security** options: a. Check **Allow Operator Program Uploads** to grant operators the privilege to upload programs/recipes from the equipment. b. Check **Allow Operator Program Deletes** to grant operators the privilege to delete programs/recipes from the equipment.
2	Select **PEM** options: a. The **Shared CIM Routes** controls how files uploaded by PanaCIM are copied between PanaCIM routes that are mirrored or shared for a single PanaPRO line. If multiple routes are mapped to a single PanaPRO line, this checkbox controls whether or not files uploaded by PanaCIM on one route are also copied to all other shared PanaCIM routes. If this flag is disabled, the uploaded files on each PanaCIM shared/mirrored line can be different. PanaPRO will always have the latest file uploaded from any of the shared lines regardless of this flag configuration. b. Check **Enforce Settings of Patterns Per Panel** to force supervisor or administrator level users to pre-define the number of patterns per panel for each Product.

3	Select **Transfer Defaults**: Set the default values for the program transfer (Product Xfer) discussed in 5.4. • **Machine/AW/Scanner:** Select the equipment type(s) which should be automatically checked **ON** for download in Product Transfer. • **Download/Initiate Work Order/Adjust**: Select the operation(s) which should be automatically checked **ON** in Product Transfer. **Note:** Refer to 5.4 Product Transfer for details.
4	**Base Product - Enable Base Product Info:** Check this option to set the program attributes beforehand regardless of the version or revision. **Note:** Usually uncheck this option.
5	**Dynamic Model String - Allow Dynamic Model String:** Check this option to change the model string when transferring a product program in Product Transfer (Product Xfer). **Note:** Usually uncheck this option.
6	**PMD**
7	Select **Manual Changeovers**: a. Select the **Enable product transfer logging** to force operator login, and log all actions manipulating product data in Product Transfer. b. If you selected the checkbox above, please select either of the following: • Select **Once per transfer session** to force operator login only once during a Product Transfer or Data Transfer session. Operator login will be obtained upon first action against product data. • Select **Every transfer action** to force operator login for every action against product data in Product Transfer.
8	Select **Work Order** information: a. Select the **Record Work Order Number** to force the entry for Work Order Number when a Product is selected on the machine using Product Transfer. b. Select the **Master Work Order** to define master work order for a set of one or more child work orders. c. Select **Record Work Order Quantity** to force the entry for Work Order Quantity when a Product is selected on the machine using Product Transfer. d. Select **One to One Work Order - Product Relation** if a Work Order can contain only one product, or, select **One to Many Work Order - Product Relation** if a Work Order can contain more than one product.
9	**Detailed Production Schedule - Auto Generate Work Order Number:** Prompts operator to enter a work order number when selecting a program on the machine in Product Transfer. • **PMD - Auto close Work Order when actual quantity = planned quantity:** Check this option to close the work order when the actual count equals to the planned count.
10	Select **Enable Publishing** to use the publishing feature. Enter the following information: **Publish Location**, Publish Username, Publish Password, and Publish Prefix.
11	Click the Save button to save changes, or the Reset button to discard changes and revert to original data.

3.4.4 Base2

Select [Module] tab > [Base2] tab.



Set the following on this page.

- Enable login when downtime is set
- Select password for LMNCMaintenance

Step	Procedure
1	**Options - Enable User Downtime Login:** Require operator to login to input equipment stop information for Route Status that will be explained in Chapter 4.8.
2	**Settings - State Model**: Select a status display model used by Route Status. • **PanaCIM**: Selected for CM/DT/NPM series (including AM100) machines • **SEMI**: Selected for IPAC machines (IPAC is not supported.)
3	**Security - Maintenance Password:** Set LMNC password.
4	**Update All LMNCs**: Apply modified password to all LMNCs.
5	**User Data Names:** Input user names at ‘User Data Name1’ ~ ‘User Data Name5’ that are to be displayed at scanner’s user data area.

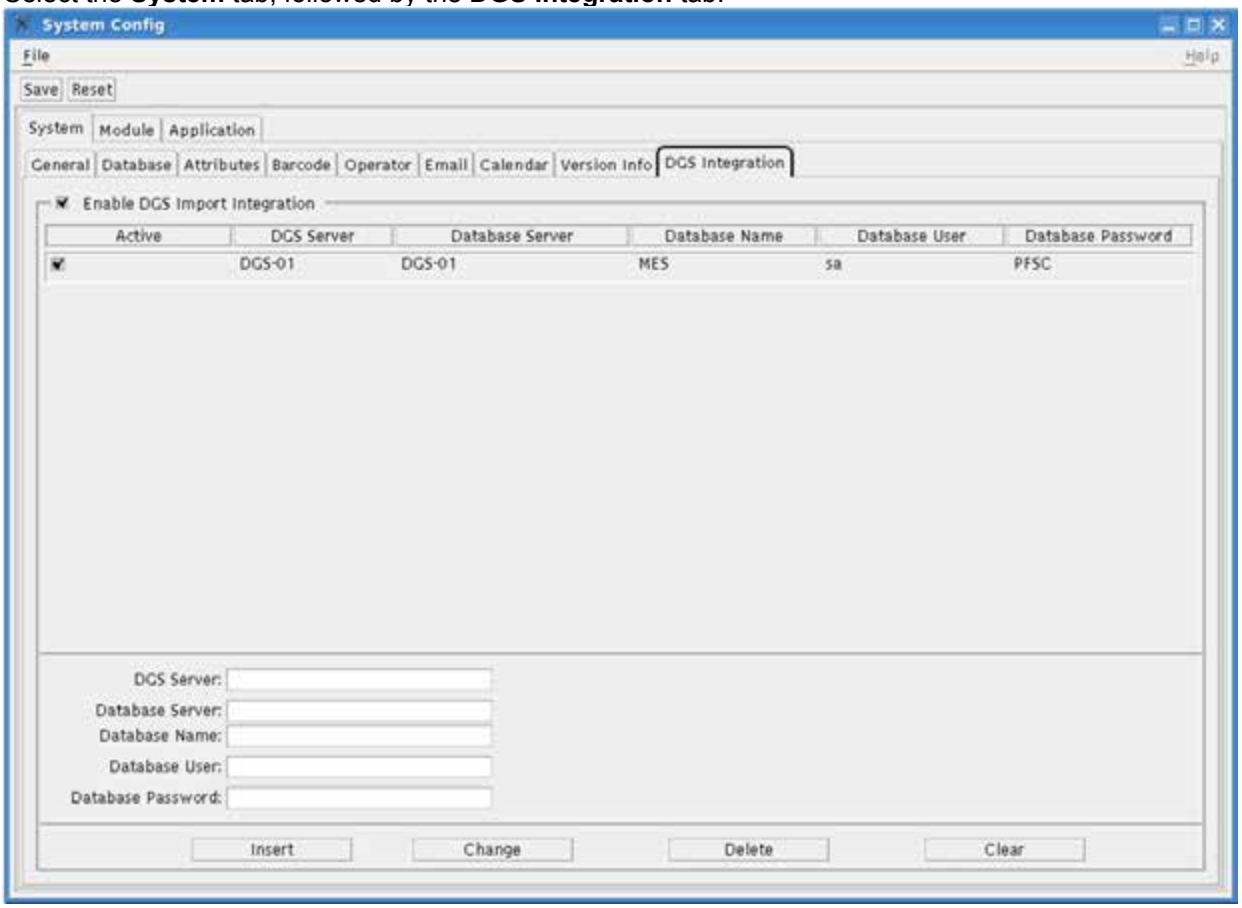
4. CREATING LINE

This chapter explains line construction.

4.1 Setting Production Data Management System

4.1.1 DGS

Select the **System** tab, followed by the **DGS Integration** tab.



This page is used to add, change or delete the DGS Data Generation System controlled by PanaCIM. The Data Generation System, or DGS, creates optimized Product programs that PanaCIM imports for use in its Applications. Adding DGS system related information into the System Configuration Application establishes a direct connection between PanaCIM and the DGS system, thus making this information directly available to PanaCIM.

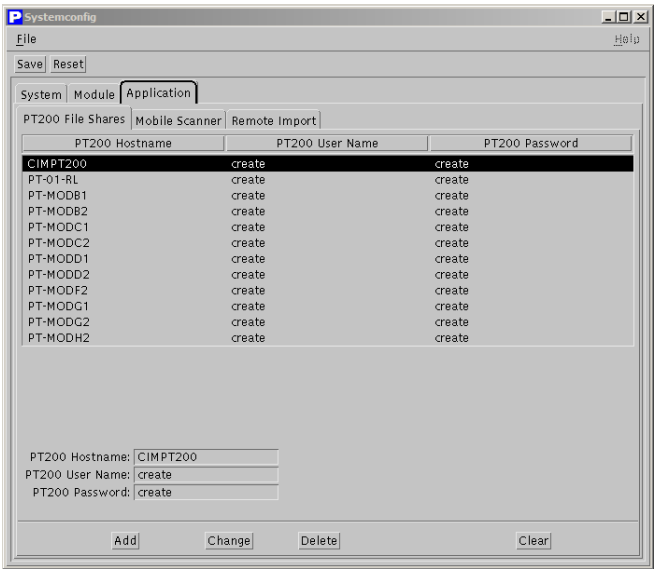
Refer to the action table below to configure DGS Integration.

Step	Procedure
1	**Enable DGS Import integration:** Check this checkbox to enable DGS import. • **DGS Server:** Enter the host name of the DGS server. • **Database Server:** Enter the host name of the SQL Server. • **Database Name:** Enter the database name. • **Database User:** Enter the login account for the administrator of the SQL Server. • **Database Password:** Type the password of the account specified in **Database User.**

4.1.2 PT200

■ PT200 File Shares

Select the **Application** tab, followed by the **PT200 File Shares** tab.

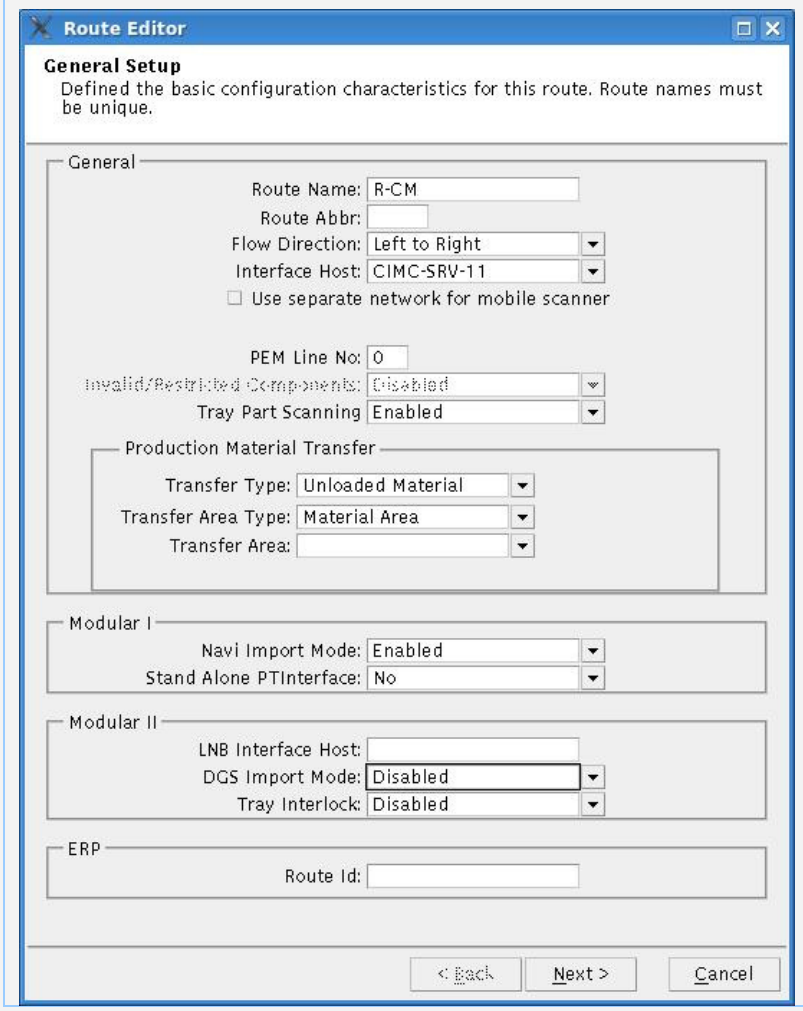


The PT200 Files Shares tab in System Configuration allows you to add, change and delete the PT200 Files shares defined on the system. The information in this page can be selected when adding modules of route editor (CM/DT/NPM (including AM100) series machines) described in 5.3.

Step	Procedure
1	**PT200 Hostname:** Enter the host name of the PT200.
2	**PT200 User Name:** Enter the login account for the administrator of the PT200.
3	**PT200 Password:** Type the password of the account specified in **PT200 User Name**.

■ Navi import

Enabling the Navi Import mode imports the product program and feeder setup information to PanaCIM automatically when the product program is downloaded from PT200 to the CM/DT series machines.

Step	Procedure
1	Select the **Route Editor** icon on the application toolbar of the PanaCIM controller to open the Route Editor Application.
2	Select the route and double-click it.
3	On the **General Setup** screen, set **Navi Import Mode** to **Enabled**. 
4	Click **Next**.
5	Click **Save**.

4.2 Creating Route

4.2.1 Route Setting Editor

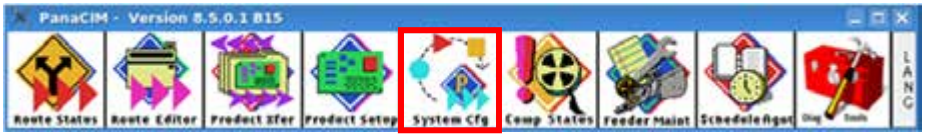


A new route (line) can be configured from Route Editor. After defined routes, machine, equipment transfer unit and production area shown below can also be set.

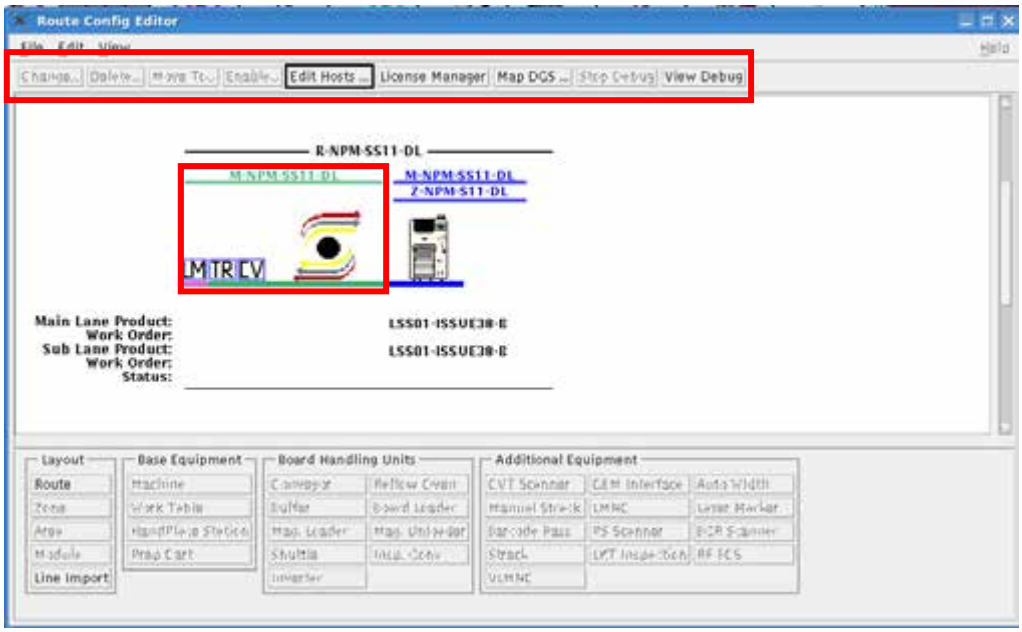
- Machine (mounter, printer, reflow and other equipment)
- Controller (LMNC, Auto adjustment controller and others)
- Material Verification scanner (For CM/DT series machines, BM series, Material Control and others)
- PCB scanner (fixed, handy)
- PCB transfer unit (conveyor, loader and others)
- Communication interface (communication unit and others)
- Production area

Set names and others for the above items on each Editor. When factory's line or machine/network setting was altered, you can create, move, delete, and change entire route or parameters of equipment from Route Editor.

Start <System Cfg> from PanaCIM application tool bar.



Route Setting Editor startup screen is as below.



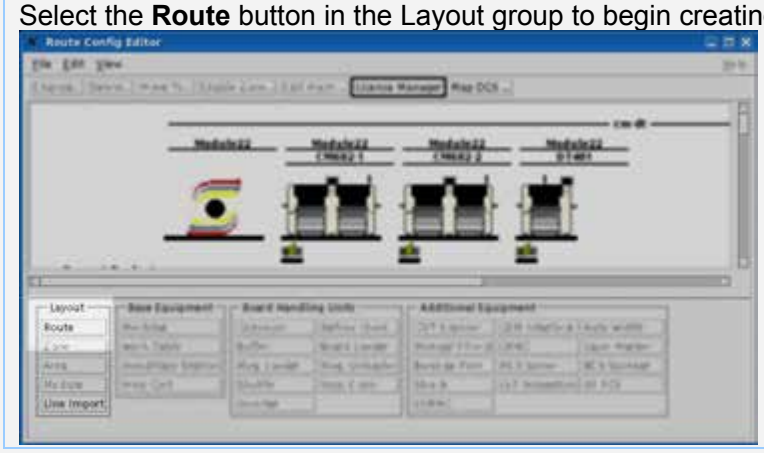
4.3 By Manually

Using the Route Configuration Editor, you can:

- Create, save and change the route.
- Display the route graphically, with icons of machines in the respective zones in the route.

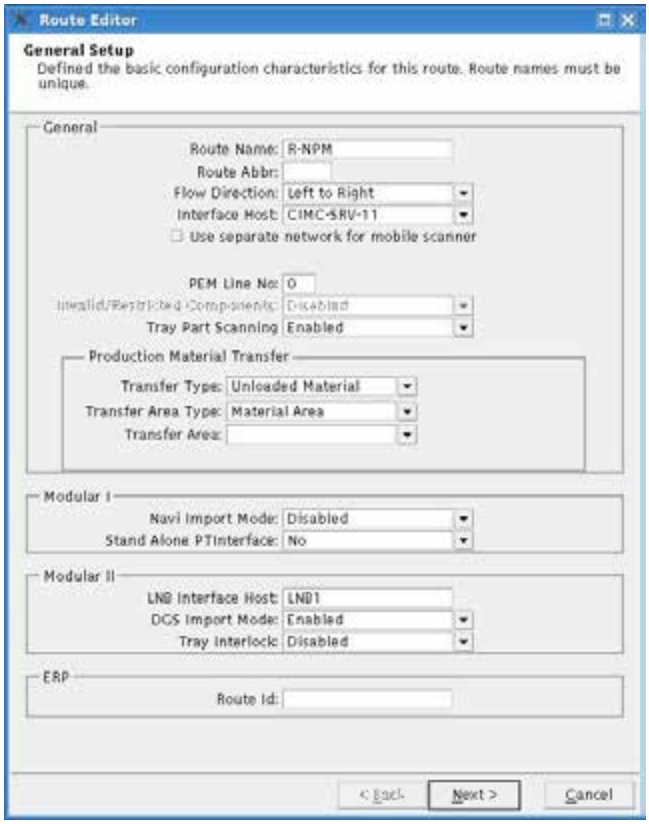
The Route Configuration Editor Application includes:

- Creating, changing and deleting in the route and components contained
- Moving and editing zones in the route
- Moving and editing equipment in the route

Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **Route** button in the Layout group to begin creating your Route. 

■ General Setup

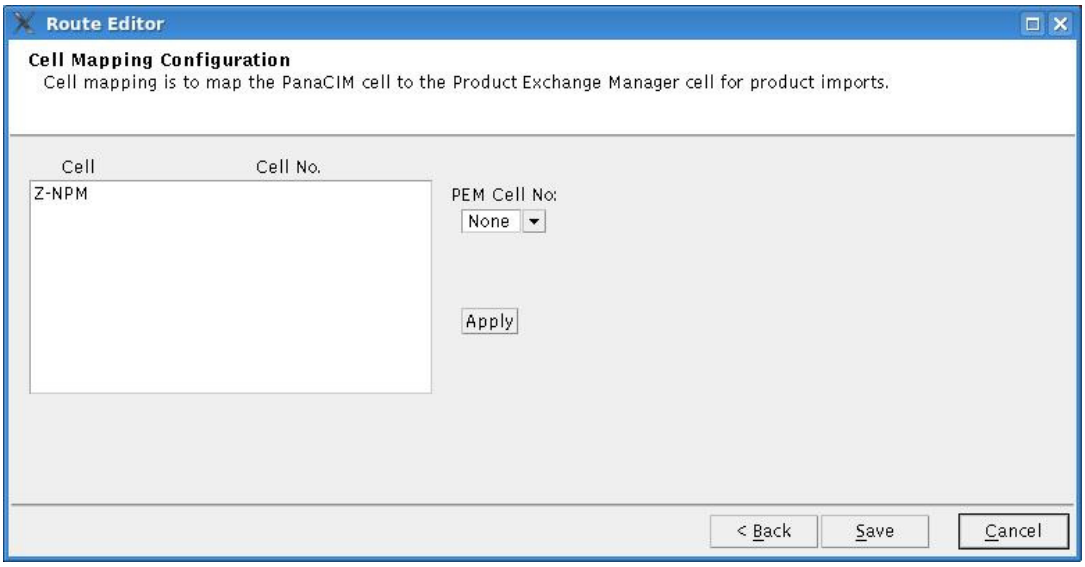
Follow the steps below on the General Setup screen.



Step	Procedure
1	Type in a unique **Route Name**.
2	Enter a **Route Abbreviation**. (Can be omitted)
3	Specify the **Flow Direction** to match the physical Route.
4	Select the host name of the PanaCIM Controller Server from the **Interface Host** drop-down list.
5	Use mobile scanner individual network: Set an IP address (machine LAN) of CIMC. Note) If the checkbox is not marked, ・There is no problem if the same network is used between the base LAN and the machine LAN. ・If the different network is used between them, the scanner may be connected to the base LAN. When this phenomenon occurs, connection with PanaCIM is cut during setup, and the parts list cannot be displayed.
6	Set your **PanaPRO Line Number** if you will be using PEM Import.
7	**Tray Part Scanning: Enabled** is set by default. If **Disabled** is selected, tray parts will be removed from the program and mounted parts have been set as unmounted. Refer to the Material Verification documentation for details.
8	Clicking **Next** creates a route and advances to the Cell Mapping Setup screen. Clicking **Cancel** discards any changes made and closes the editor.

Cell Mapping Setup

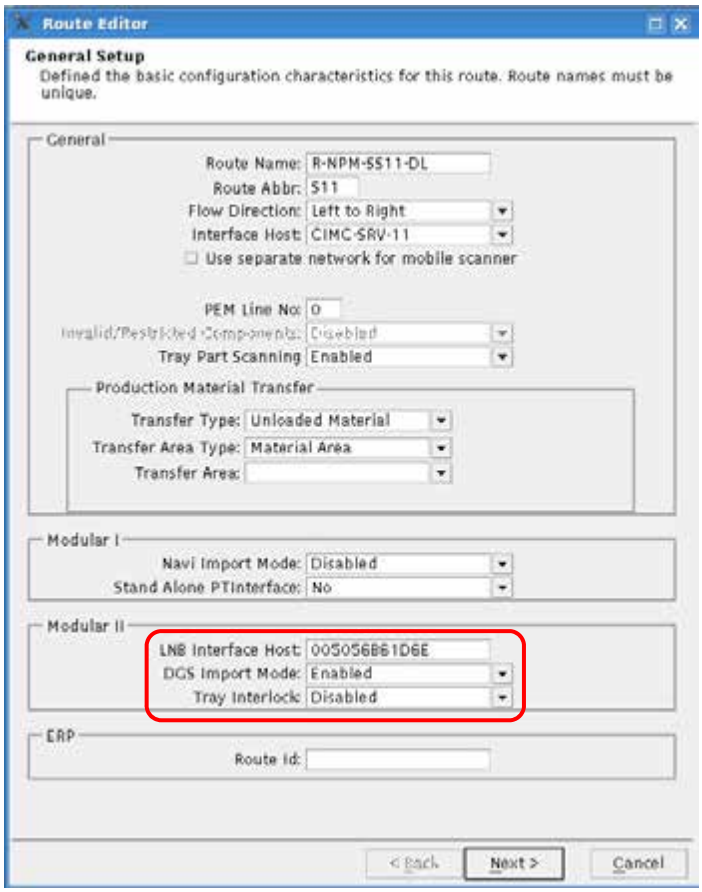
This screen lets you import the product program created on the PanaPRO/IPO to the PanaCIM via the PEM. Set the equipment number of the line specified in **PEM Line No.** for the PanaPRO on the General Setup screen above.



Step	Procedure
1	Assign PanaPRO equipment number to the equipment and click **Apply**.
2	Click **Save** to save the setting, or click **Cancel** to discard any changes made and to close the editor.

4.3.1 Setting the Route - Set LNB

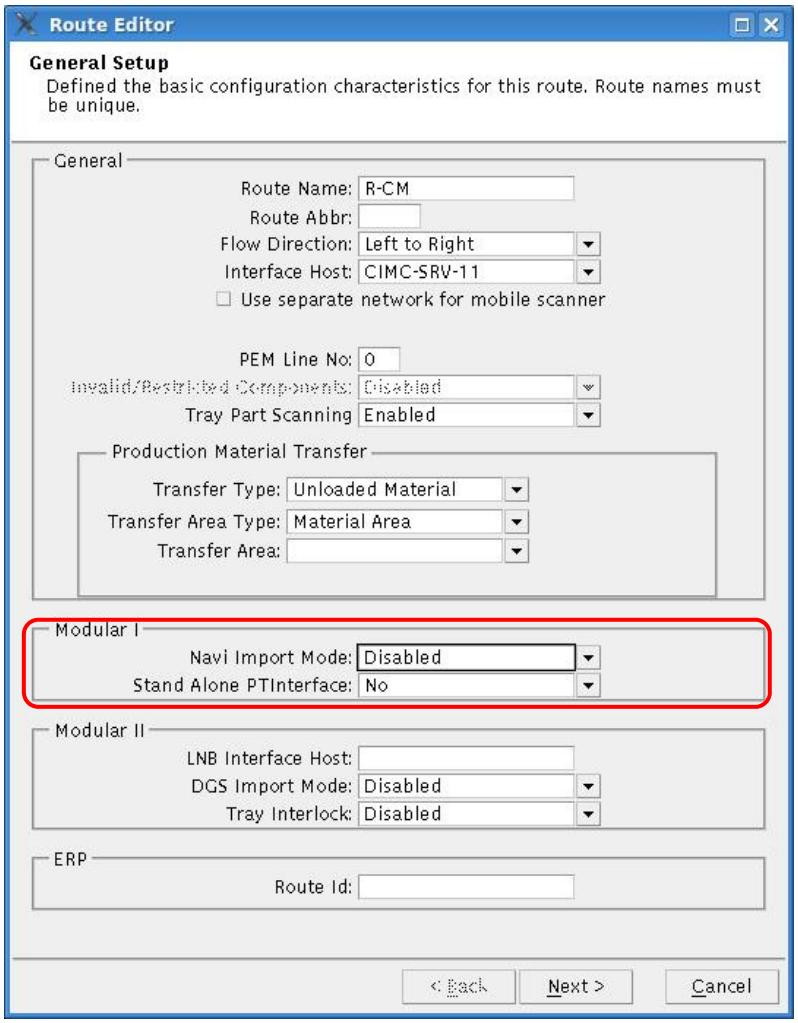
Set the following items on the General Setup screen.



Step	Procedure
1	LNB I/F host: Enter the host name of LNB. (NPM series (including AM100) machines only)
2	DGS import mode: (NPM series (including AM100) machines only) • Enabled: Select this option when DGS is being used. Set up DGS using PanaCIM system configuration application. (For details, refer to “4.1.1 DGS”.) • Disabled: Select this option when DGS is not being used.

4.3.2 Setting the Route - Set PT200

Set the following items on the General Setup screen.

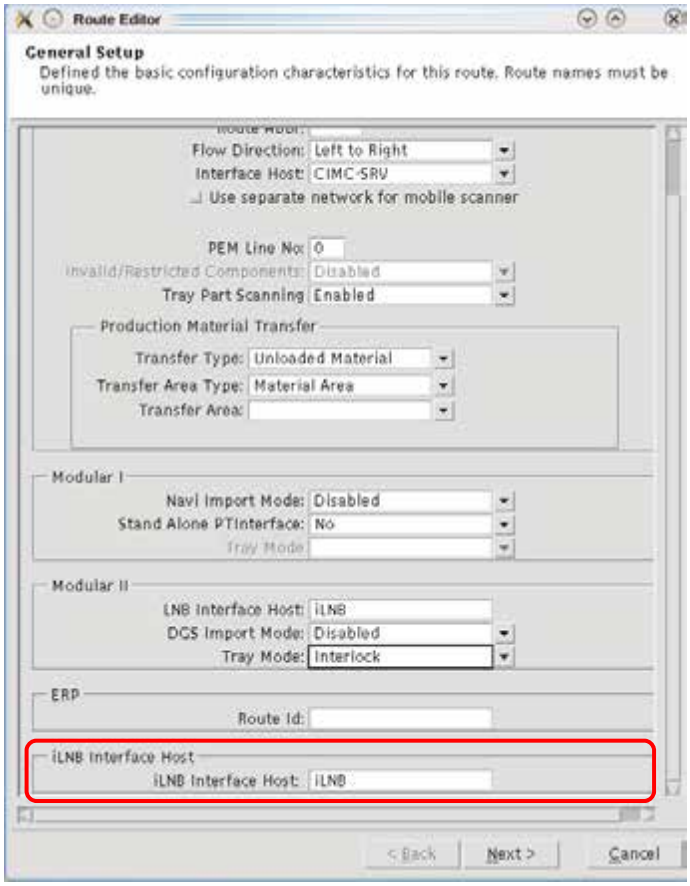


Step	Procedure
1	**Navi Import Mode:** (CM/DT series machines only) • **Enabled:** Select this option to enable PT200 Navi folder set in the controller client that controls this route. The contents of the feeder setup changed on PT200 will be reflected in the product setup of the controller server. (Refer to 5.1 Product Setup for details.) • **Disabled:** Select this option when Navi file is not to be imported.
2	**Stand Alone PT Interface:** • **Yes:** Select this option when CM/DT series machine is used. • **No:** Select this option when the machine other than CM/DT series is used.

4.3.3 Setting the Route - Set iLNB

■ General setup

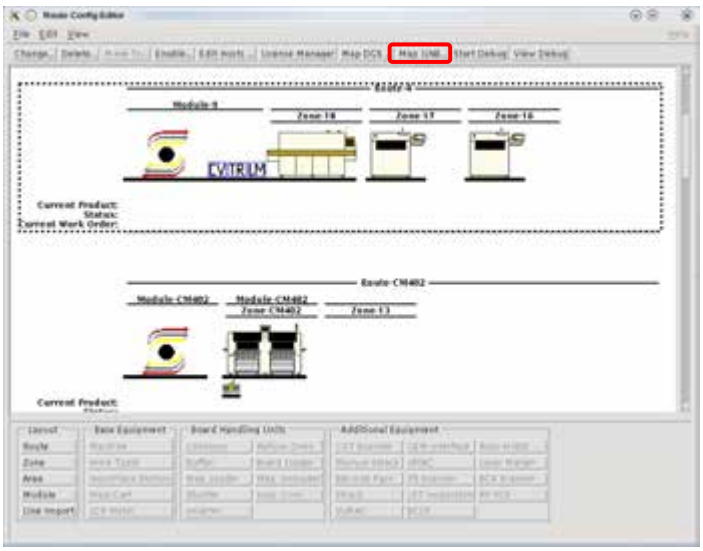
Configure the basic settings for the route.

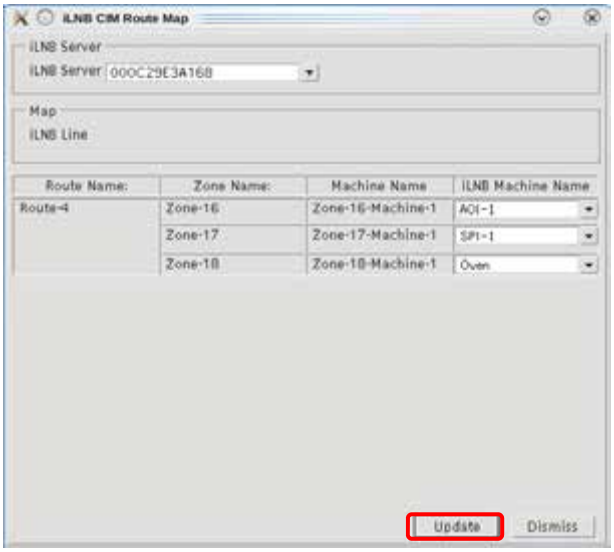


Step	Procedure
1	iLNB Interface Host: Enter an iLNB host name[*1]. (Third party's machine[*2] only) *1: It must be the same host name as LNB. *2: The third party's machine includes SPI, OVEN and AOI machine models.

■ Mapping

In the route using iLNB, mapping is performed by obtaining line configuration information storing in iLNB.





Step	Procedure
1	Select a route on **Route Config Editor**.
2	Click the **Map iLNB** tab.
3	Click the **Update** button.

Note) In Dual lane, if the machine name of Front differs from the one of Rear, the names are shown such as Machine-1/Machine-2, and if the names are the same, they are shown as Machine-1.

4.3.4 Creating a Zone

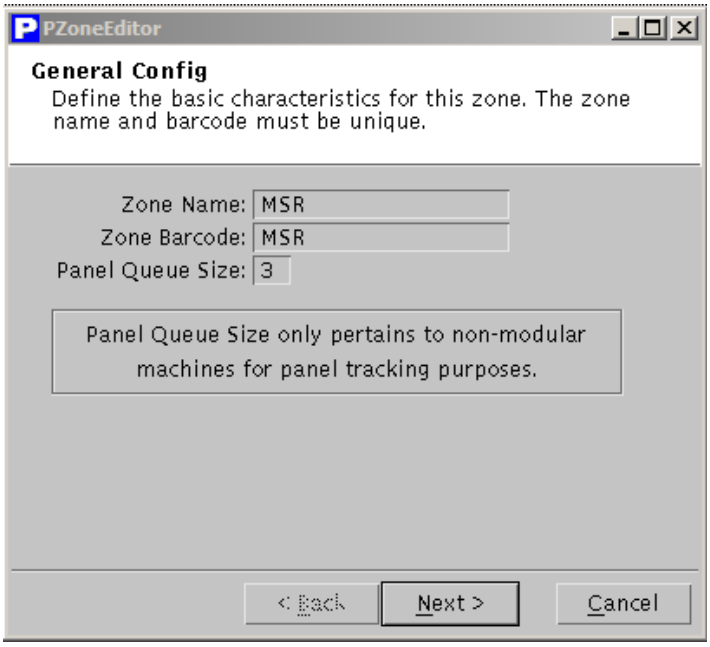
Launch the Zone Editor to create a zone.

Step	Procedure
1	Select a **Route** by clicking it. Upon selection, the Route will be enclosed with dotted line.
2	Click the **Zone** button in the Layout group to invoke the **Zone Editor**.

4.3.5 Setting a Zone

- General Setup

  This screen allows the basic information of the zone to be set.



Step	Procedure
1	Enter a unique **Zone Name**.
2	Enter a unique **Zone Barcode**.
3	Enter a value in **Panel Queue Size**. (BM series machine only) **Panel Queue Size** shows the number of panels that can be held inside the machine. Based on the number of panels to be unloaded and queue size on the machine, it is possible to estimate when the board of a particular serial number is unloaded.
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made.

■ Line Managed Config

This screen is for setting the line control level and control mode of the zone. Displayed Items vary with the model of the machine.

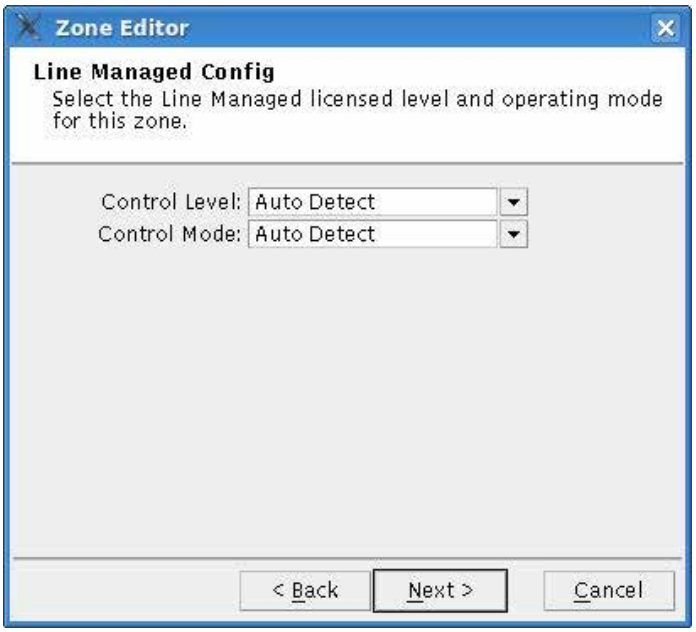
Note: The mode set here may not be used depending on the authenticated license.

Note) You must follow the step below to change to [Placement] in Traceability.

① Disable the target route.
② Change to [Placement] in Traceability.
③ On the [ProductSettup] screen, delete all production data of a target route.
(For the deletion procedure, see “51.5 Deleting Panel/Product Setup”
④ Enable the target route.
⑤ Import all production data used for the changed route from DGS.



CM/DT/NPM series (including AM100) machines



BM series machines

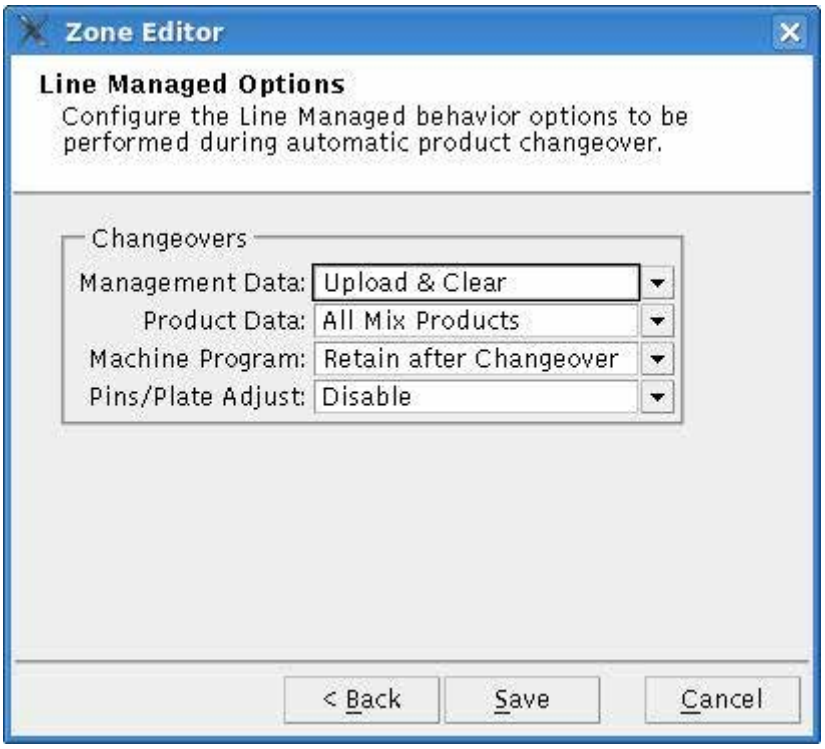
Step	Procedure
1	Select a level of control in the Zone from the **Control Level** drop-down. Set the level that enables operations to the maximum extent in the zone.
2	Select a **Control Mode** for the selected Zone from the drop-down list. The Control Mode determines the Zones operational mode.
3	Select the desired traceability type from the **Traceability** drop-down list. Traceability collects information such as panel model IDs, panel serial numbers, program names and Equipment names.
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to close the editor.

Available options in the respective drop-down lists are explained in the table below. They vary with the model of the machine used.

Drop-down list	Options	
Control Level	Auto Detect	Detects changeovers but does not perform any automated processing.
	Auto Download	Detects changeovers and performs automatic product transfers, selections, and width changes.
	Line Managed	Detects changeovers and performs full automated changeover processing including product transfers, width changes, and machine starting.
Control Mode	The options are the same as in Control Level. Current operation mode is shown.	
Traceability (Refer to [PanaCIM/ Traceability] documenta-tion for details.	None	Traceability is not provided .
	Panel	Panel traceability is provided through the use of fixed barcode readers connected to a controller unit to scan and record the model code and serial number of each panel as it approaches a piece of Equipment.
	Component	Component to panel traceability is provided when used in conjunction with Material Verification. Material Verification records the time and attributes detail for the materials mounted on each piece of Equipment. This information along with the panel traceability data, allows identification of each panel serial number that could have been produced with specific material including: part number, lot/date code, vendor code, initial quantity, and one user-defined attribute.
	Placement	Placement traceability is provided, which includes panel and component to panel traceability. This option enables you to identify the panel and component information from the specified reference (circuit number). (NPM series (including AM100) machines only)

Line Managed Options

This screen is used to set the options for automatic changeover.



Step	Procedure
1	Select how **Management Data** will be handled during a product changeover
2	Select how **Product Data** will be downloaded to the scanner. Available options vary with the machine type.
3	Select how to manage the last downloaded program to the machine. Available options vary with the machine type. This option enables the user to remotely control old program files; some machines have limited disk space, and downloading many different programs quickly fills it up. (BM series machines only)
4	When **Line Managed** is the selected control mode, the **Pins/Plate Adjust** option can be configured. If set to Enabled, you will have the ability to pause before adjusting the rails so the pins and plate may be adjusted for the new product. The pause only occurs if the width of new product panel is less than that of the current product panel. Available options vary with the machine type.
5	Click **Save** to create a zone. Or click **Cancel** to discard any changes made and to close the editor.
6	A newly created Zone will be placed in the Route's last or downstream position. The zone can be moved under the **Move To** page on the Route Configuration Editor.

Available options in the drop-down lists are explained below.

Drop-down list	Option	
Management data	None	Management Data will not be uploaded between changeovers.
	Upload Only (Default)	During a changeover, data will be uploaded for all supported data types.
	Upload Only & Clear (Recommended)	During a changeover, data will be uploaded and machine totals will reset. This option also prevents machine data Overflow.
Product Data	All Mix Products	Downloads all products in the mix to the CVT. Each time a changeover is initiated, a comparison of the mix and product setups for current and target products will be performed. If there are no differences, CVT download will be skipped. If differences are present, CVT receives a full download of the mix and the product mixes. Choose this option if you change setups often.
	Target Product Only	Each time a changeover is initiated, the mix and product setups for the target product will be downloaded. All other products will be skipped. Choose this option if you have more than 32 products in a mix, or if you change setups often.
Machine Program	Retain after Changeover	No product programs will be remotely deleted from the machine unless they are initiated by the user in Product Transfer or Data Transfer.
	Delete after Changeover	Upon a successful product changeover and machine start, the product files associated with the previous product (POS, SET, and BRD) will be remotely deleted. If using the MSET file rather than the SET file on the Assembly machine, the MSET will not be deleted unless the new product does not belong to that MSET. (BM series machines only)
Pins/Plate Adjust	Disable	Choosing this option does not adjust the support pins and plates during automatic changeover.
	Enable	Choosing this option adjusts the support pins and plates during automatic changeover.

4.3.6 Adding Machine to the Zone

To add a piece of Equipment to a Zone, that Zone must first be selected. Select a Zone by clicking it (a dashed line will form around the Zone upon selection). When the Zone is selected some of the options in the Layout, Base Equipment, Board Handling Units, and Additional Equipment button groups will become active.

There are certain limits to the amount of Equipment that can be present in a given Zone. Once these limits have been exceeded, Equipment can no longer be added into the Zone.

Route Editor provides Equipment Editors that are used to create and edit equipment. The table below describes the different types of Equipment Editors available based on the type of Equipment to be created or edited.

Equipment Editors	Purpose
Machine Editor	Used to create and edit machine configuration options.
Ancillary Equipment Editor	Used to create and edit most of the additional Equipment types. Some additional Equipment except for the scanners and LMNC are created and edited via this editor.
Scanner Editor	Used to create and edit RFCVT Scanners for all types of Equipment. • CCS scanner (CM/DT/NPM series (including AM100) series machines) • CVT scanner (BM series machines) • HP scanner (Hand Placement scanner) • Presetup scanner • FM scanner (Feeder Maintenance scanner) • MMS scanner (Material Control scanner)
LMNC Equipment Editor	Used to create and edit LMNC configurations.
BHU Equipment Editor	Used to create and edit Board Handling Units.
Module Editor	Used to create and edit a module for managing as many as six CCS scanners on the modular machines (CM/DT/NPM series (including AM100) machines).

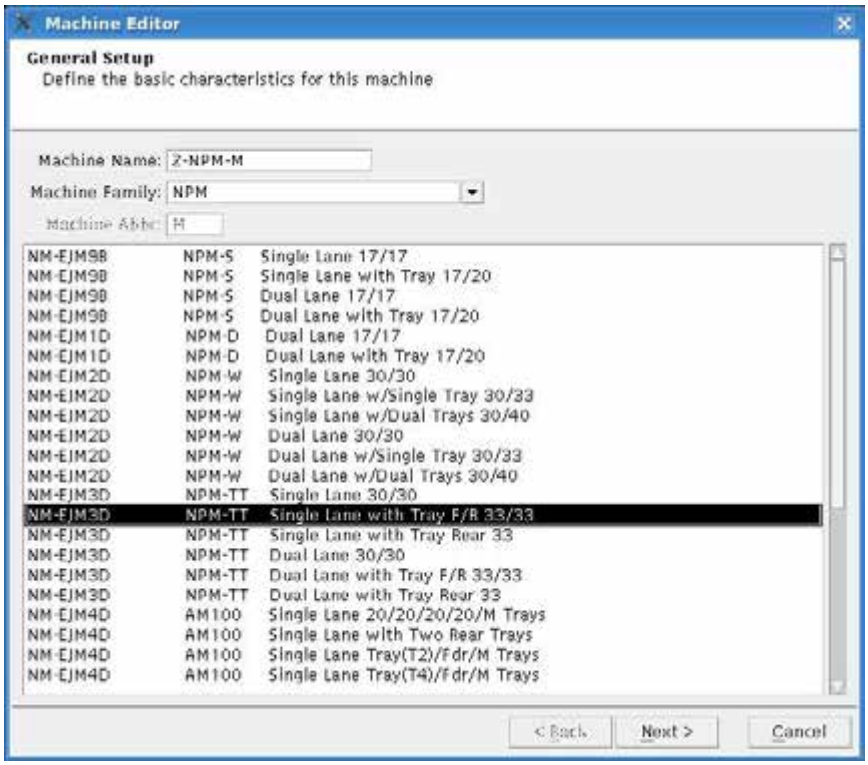
➢ Adding a machine

If no equipment has been set in the production zones, equipment can be added to any zones.

Note: To use the PanaCIM/Product Changeover and Control or Traceability, the equipment required for these options need to be set on the machine at the head of the zone.
Example: To add an LMNC or a scanner to a zone, add a machine to the zone first.

■ General Setup

Select a machine to be added.



Step	Procedure
1	Select a machine group from the **Machine Family** drop-down list. For CM/DT series machines, select **KX**. In case of NPM series (including AM100) machines, select **NPM**.
2	Select the desired machine from the list.
3	Type a unique name in the **Machine Name** field.
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

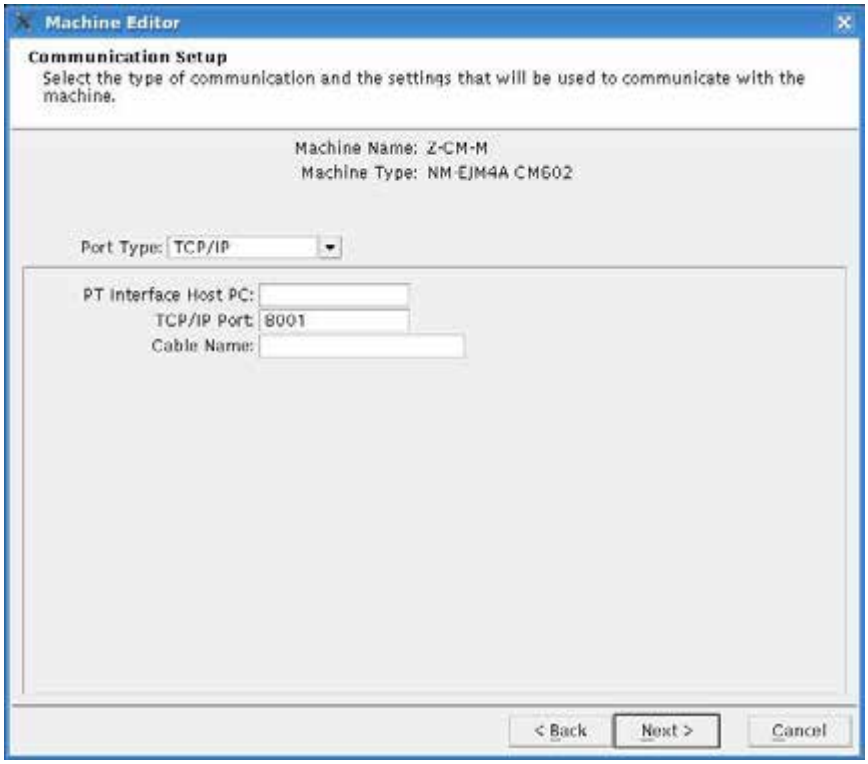
Communication Setup

This page permits the properties to be set for communication between the equipment and communication controller. Since the same set of communication parameters cannot be shared among multiple machines, avoid using the existing combinations of the host name, port and cable.

CM/DT/NPM series (including AM100) machines

Select **TCP/IP** from the Port Type drop-down list and set other items.

Note: The host PC to be set in the **PT Interface Host PC** textbox must be set in advance under **Edit Hosts** on the Route Config Editor.

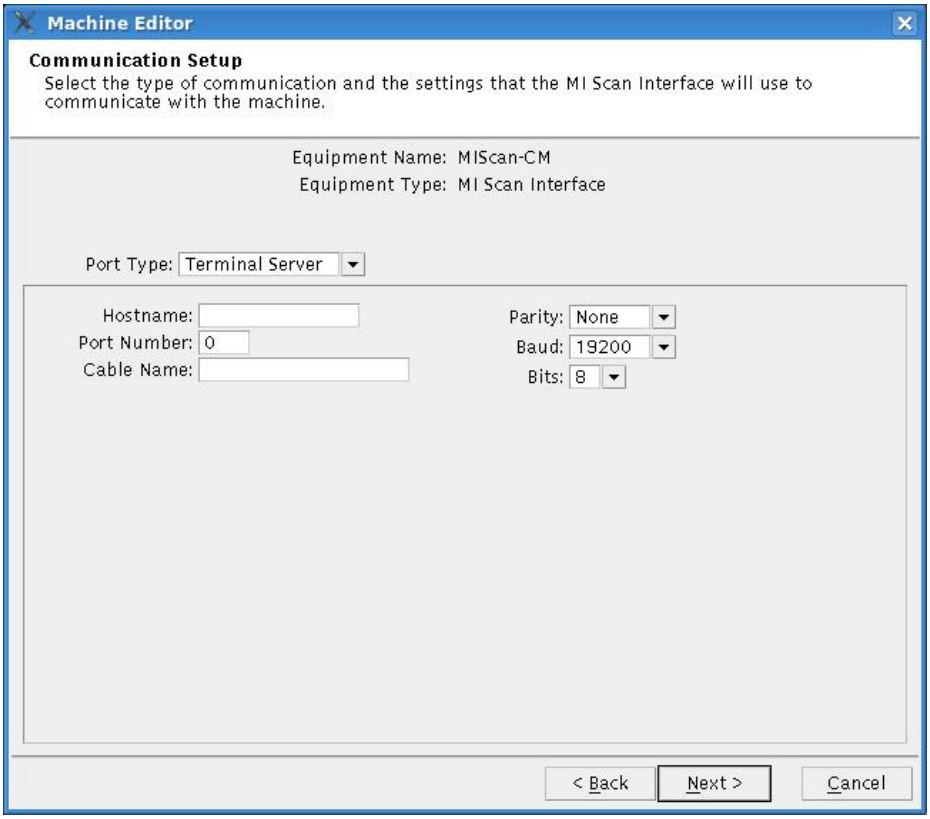


Port	TCP/IP
1	Type the name of computer in which the PT interface has been installed.
2	Type a port number in the **TCP/IP Port** textbox. (Must be an odd number between 8001 and 8099)
3	Type a cable name in the **Cable Name** textbox. (Can be omitted.)
4	Click **Next**.

BM series machines

Select the desired port type from the **Port Type** drop-down list and populate or set other items.

Note: After typing a host name, click the **Edit Hosts** tab along the upper part on the Route Config Editor.to add an IP address.



Port	Terminal Server
1	Type a unique host name in the **Host Name** textbox.
2	Type a unique port number in the **Port Number** textbox.
3	Set the desired option from the **Parity** drop-down list.
4	Set the desired option from the **Baud Rate** drop-down list.
5	Set the desired option from the **Bits** drop-down list.
6	Type a cable name in the **Cable Name** textbox. (Can be omitted)
7	Click **Next**.

Port	Serial
1	Type a device name.
2	Set the desired option from the **Parity** drop-down list.
3	Set the desired option from the **Baud Rate** drop-down list.
4	Set the desired option from the **Bits** drop-down list.
5	Type a cable name in the **Cable Name** textbox. (Can be omitted)
6	Click **Next**.

Port	LMNC Server
1	Type a unique host name in the **Host Name** textbox.
2	Type a unique port number in the **Port Number** textbox.
3	Set the desired option from the **Parity** drop-down list.
4	Set the desired option from the **Baud Rate** drop-down list.
5	Set the desired option from the **Bits** drop-down list.
6	Type a cable name in the **Cable Name** textbox. (Can be omitted)
7	Click **Next**.

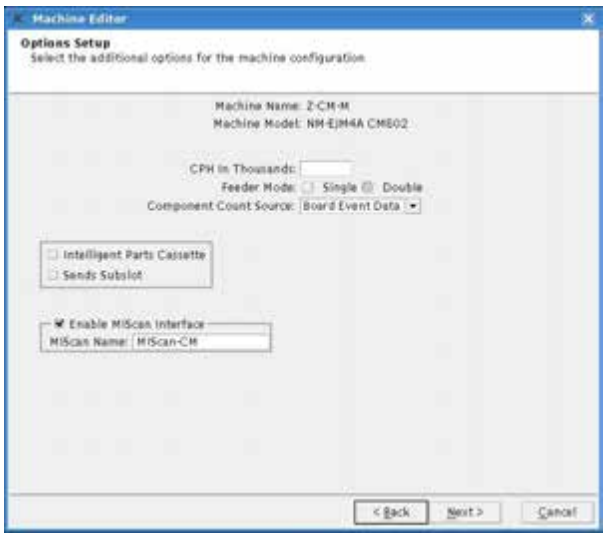
Port	Lantronix TS
1	Type a unique host name in the **Host Name** textbox.
2	Type the desired port number in the **TCP Port** textbox.
3	Type a cable name in the **Cable Name** textbox. (Can be omitted)
4	Click **Next**.

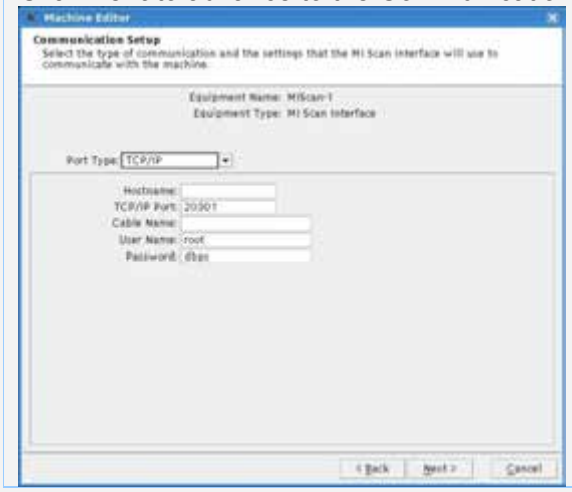
Port	TCP/IP
1	Type a host name of the placement machine in the **Host Name** textbox.
2	Type the desired port number in the **Data Port** textbox.
3	Type the desired port number in the **RTD Port** textbox.
4	Click **Next**.

Options Setup

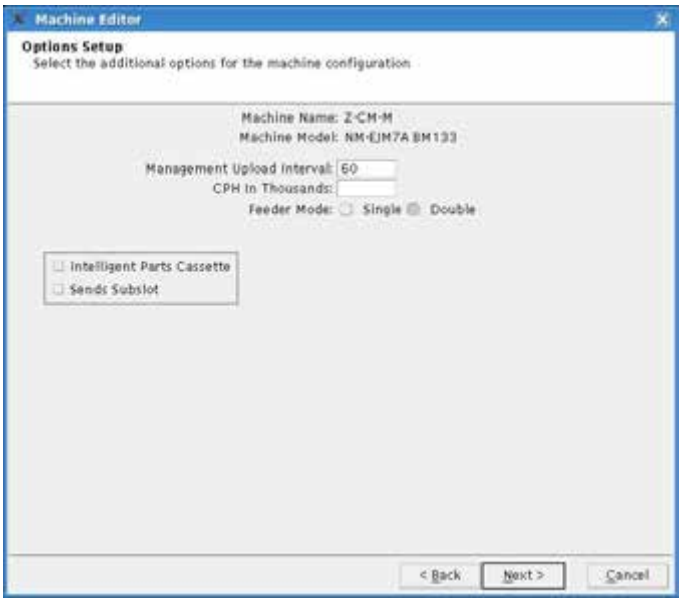
This page is for setting added options of the machine.

CM/DT/NPM series (including AM100) machines



Step	Procedure
1	Select **Double** for the **Feeder Mode**.
2	Uncheck the **Intelligent Parts Cassette** and **Send Subslots** checkboxes.
3	Check the **Enable Communication Unit Interface** checkbox. It is not necessary to change the name in the **Com Unit Name** textbox. **Note:** Communication unit is a name used to collectively describe the TCP/IP to serial converter incorporated in the equipment.
4	Click **Next** to advance to the Communication Setup page. 
5	Select **TCP/IP** from the **Port Type** drop-down list.
6	Type a unique host name in the **Host Name** textbox.
7	Type the TCP/IP port set in the communication unit concerned in the TCP/IP Port textbox.
8	Type a cable name in the **Cable Name** textbox. (Can be omitted)
9	The **User Name** and **Password** textboxes have already been populated. Change these fields only when the login name or route password is changed.
10	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

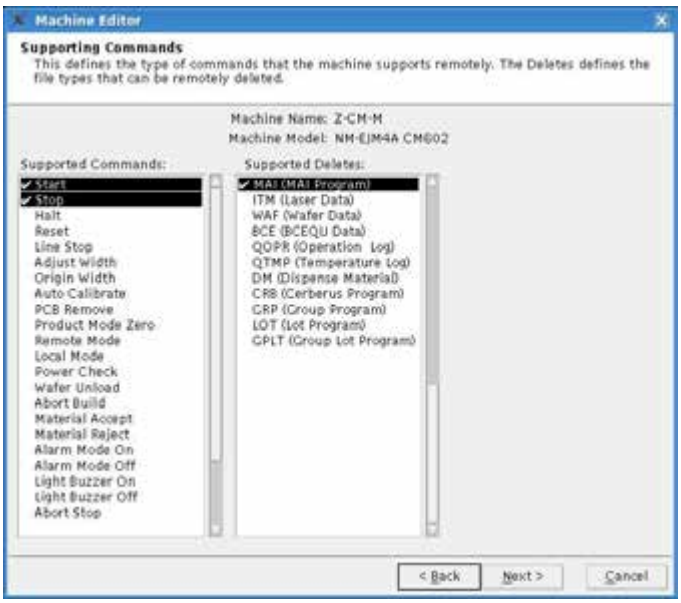
BM series machines



Step	Procedure
1	Type the interval in minutes for uploading data to the host in the **Management Data Upload Interval** textbox.
2	Select the feeder mode between **Single** and **Double**.
3	Determine whether or not to enable the IPC configuration mode. Check the **Intelligent Parts Cassette** checkbox to enable the mode.
4	Check the **Send Subslots** checkbox to include the subslots in the IPC verification.
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

■ Supporting Commands

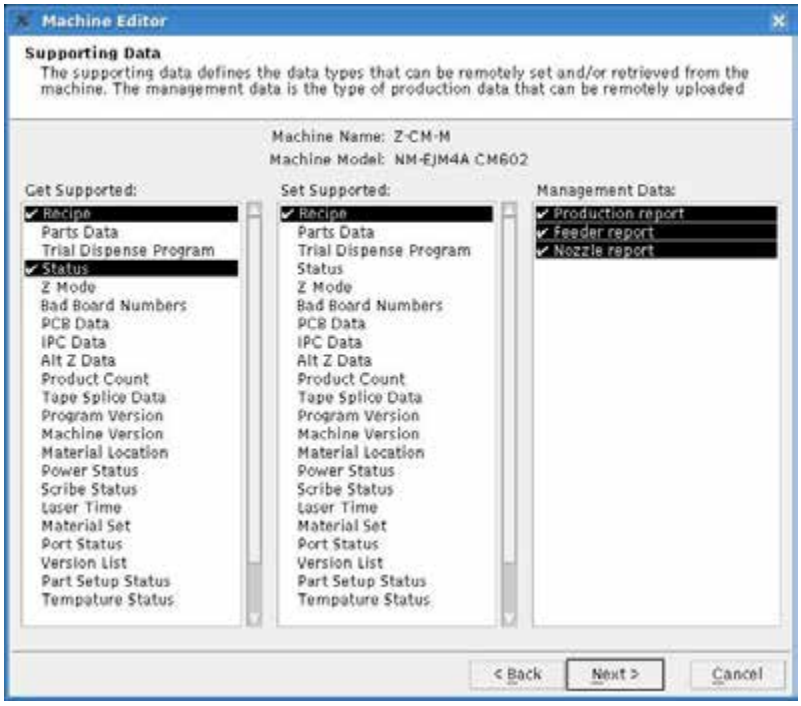
This page allows you to define command properties to specify remote commands that can be operated from the machine.
Select commands to be used on the machine while scrolling through the list. Note that the highlighted items are active and the items with a check mark are the default.



Step	Procedure
1	Select the desired command from the **Supported Commands** list.
2	Select the file to be deleted from the **Supported Deletes** list.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

■ Available Files

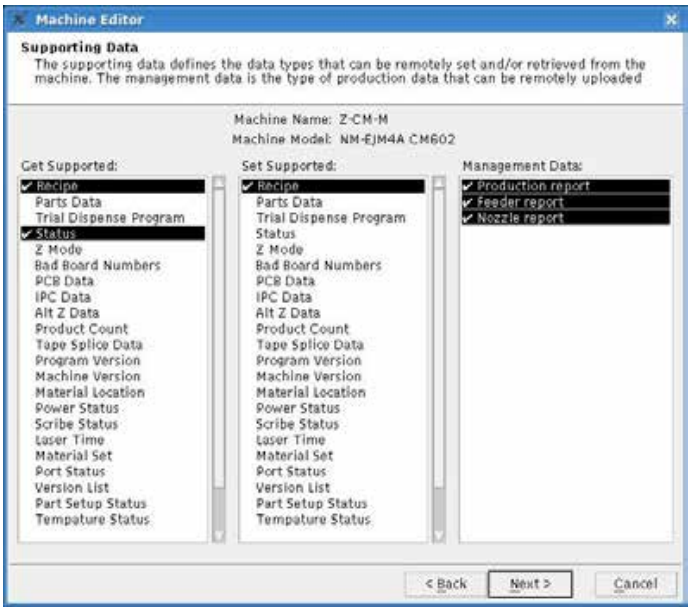
This page permits you to select files that can be remotely handled from the machine with directory, upload and download functions. Note that the highlighted items are active and the items with a check mark are the default.



Step	Procedure
1	Select the desired files from the **Available Directories** list.
2	Select the desired files from the **Set Upload** list.
3	Select the desired files from the **Download Data** list.
4	Click **Next** to continue the setting. Click **Cancel** to cancel the editing.

Available Data

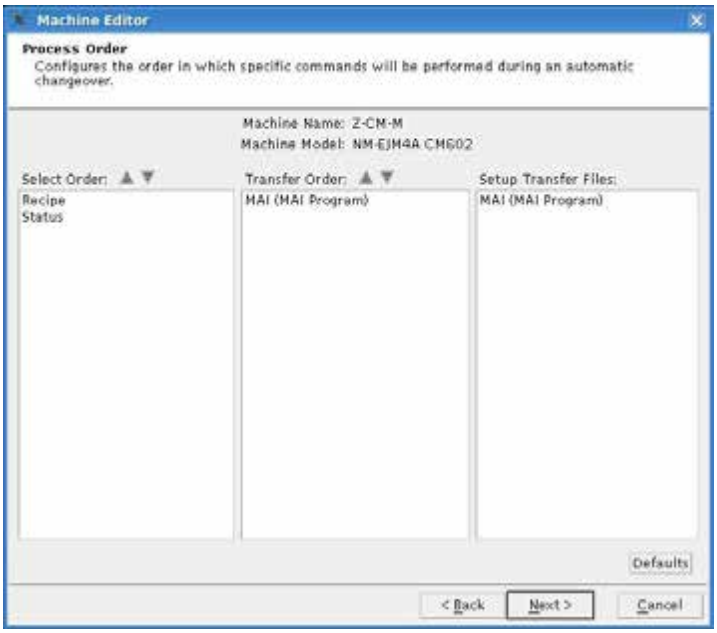
This screen is for specifying the data to be remotely set or retrieved from the machine. Management data refers to the product data that can be uploaded remotely. Note that the highlighted items are active and the items with a check mark are the default.



Step	Procedure
1	Select the desired command from the **Get Supported** list.
2	Select the desired command from the **Set Supported** list.
3	Select the production management data corresponding to the commands from the **Management Data** list.
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

Work Orders

This screen is used to select a program and to specify the order of the files to be downloaded. Set the commands for transferring programs and the order of the files to be downloaded.

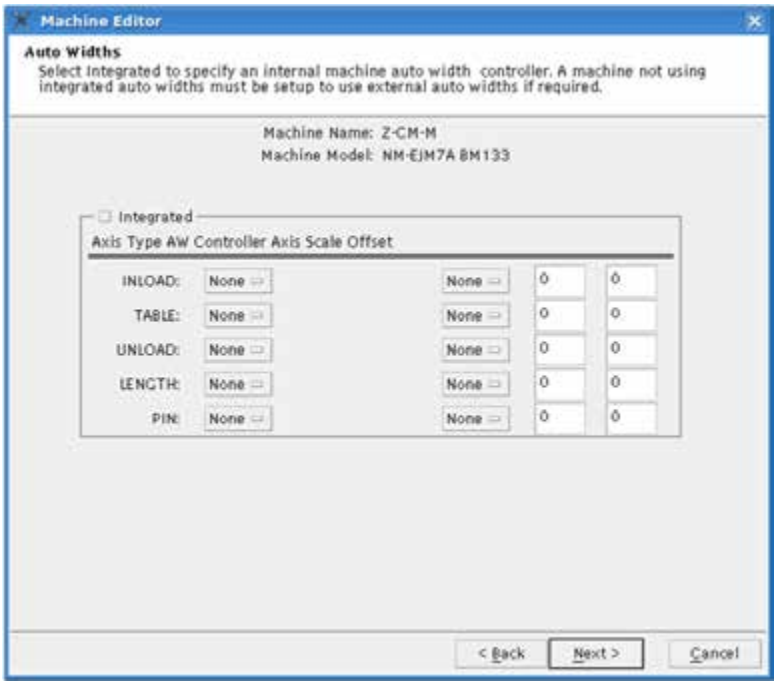


Step	Procedure
1	Select the desired item from the **Select Order** list. Click the Up and Down buttons to change the order.
2	Select the desired item from the **Send Order** list. Click the Up and Down buttons to change the order.
3	Select the desired item from the **Setup Transfer File** list.
4	Clicking the **Default** button resets the lists to the default.
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

■ Auto Width Adjustment

The automatic width adjustment controllers are employed, only on the BM series machines, to adjust the conveyor width according to the product width. Each machine has up to five adjustment axes.
On this page, set the parameters of the automatic width adjustment controller for each axis type.

BM series machines



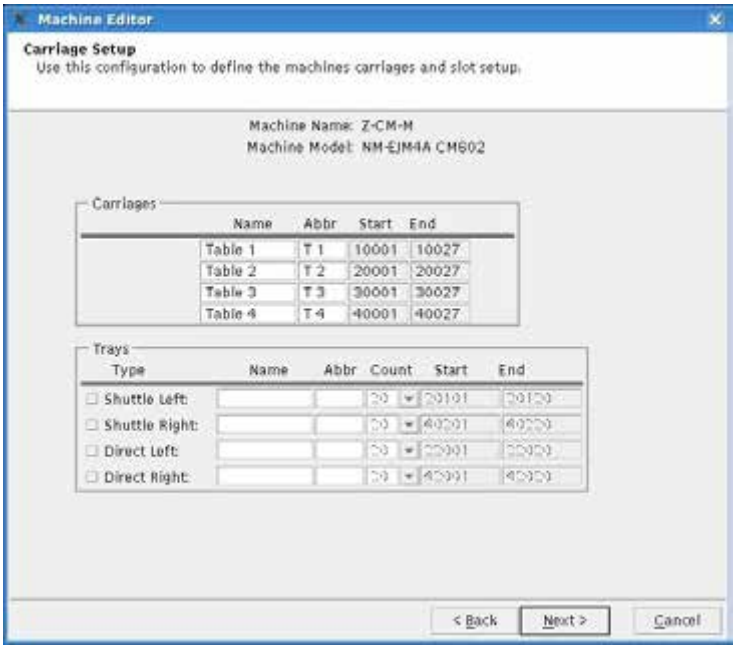
Step	Procedure
1	Set the desired option under **Auto Width Controller** for each axis type. This includes the axes contained in the option menu in the automatic width adjustment controller.
2	Set the number of axes under **Axis** for the selected automatic width adjustment controller. In case of integrated controller, the default number of axes applies, and no other options can be selected.
3	Type the scale in the **Scale** textbox.
4	Type the offset in the **Offset** textbox.
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

Cart Setup

This page enables you to define the information on the carts to be used and the feeder slots on the carts. Each machine can have up to four carts.

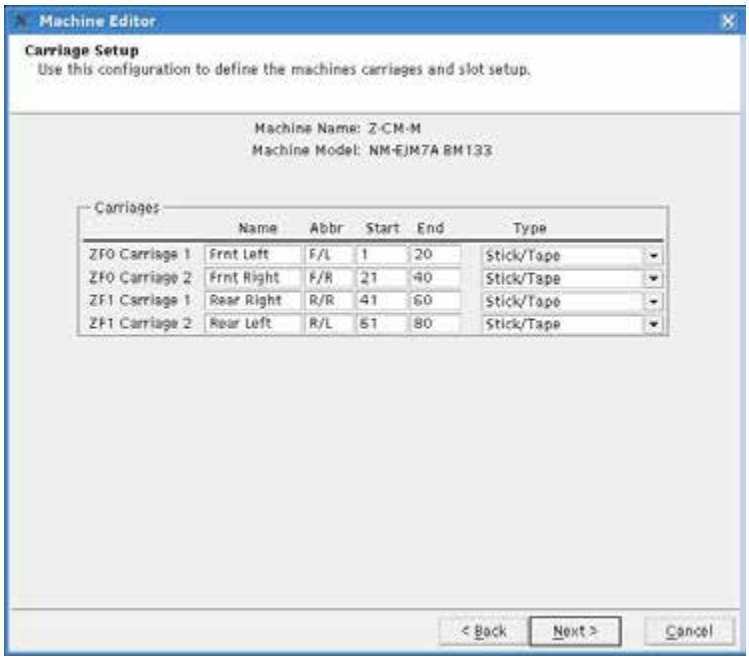
CM/DT/NPM series (including AM100) machines

Start and **Stop** slot numbers designate the mountable feeder locations.



Step	Procedure
1	Type the name and symbol of the cart in the respective textboxes under **Cart** field.
2	When the tray feeders have been installed, type the name, symbol and quantity of the trays to be used in the respective textboxes under the **Tray** field.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

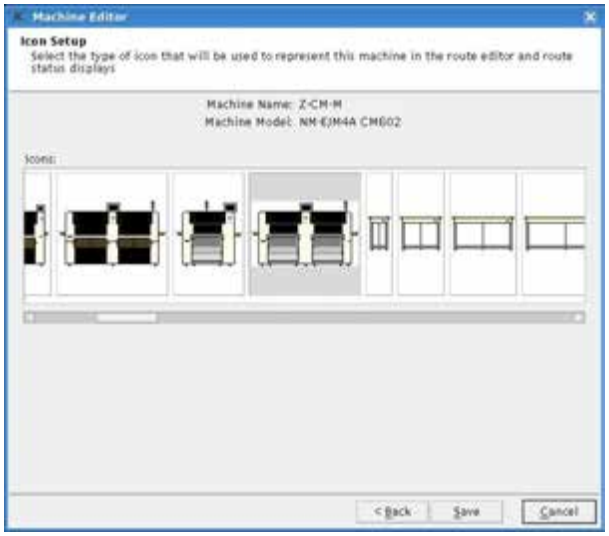
BM series machines



Step	Procedure
1	Type the name and symbol of the cart in the respective textboxes
2	Type the slot number (integer) of the at the start position of the cart in the **Start** textbox.
3	Type the slot number (integer) of the end slot position of the cart in the **End** textbox.
4	Select the type of the cart from the **Type** drop-down list.
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor

■ Icon Setup

This page is used to set the icons of the machines displayed on the Root Editor and Root Status screen. Although the default icons have been assigned to the machines, they can be changed to another by scrolling the list.



Step	Procedure
1	Select the desired icon while scrolling through the list.
2	Click **Save** to change the icon. Click **Cancel** to discard any changes made and to exit the editor.

➢ Adding Units

Using the Ancillary Editor, create and edit standard units as necessary. This editor is the basic board control system without the communication feature.

Board handling units controlled via the Ancillary Editor:

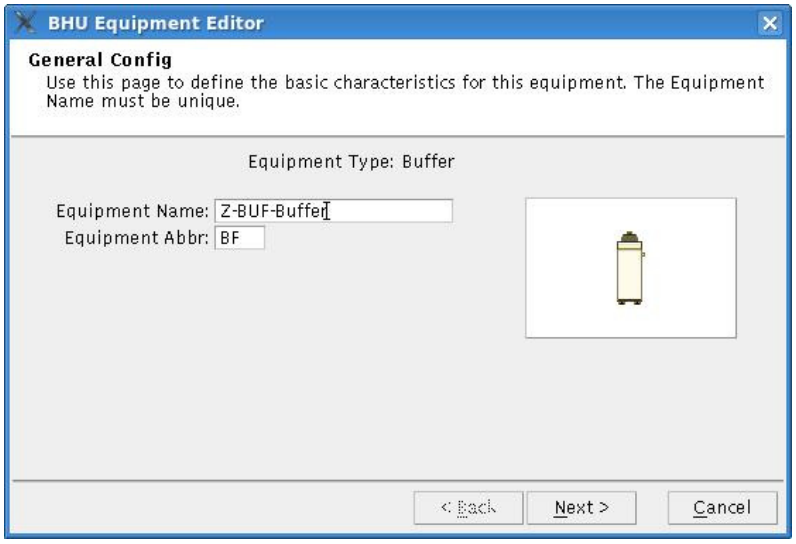
- Conveyor
- Buffer
- Mag. Loader (Loader)
- Shuttle
- Inverter
- Reflow Conv
- Board Loader
- Mag. Unloader (Unloader)
- Insp. Conv
- HP (Hand Place) Station
- Work Table

After selecting a zone, select a unit and launch the Ancillary Editor. The HP (Hand Placement) station and work table are located inside the **Base Equipment** area.

Note: Buttons of the units that cannot be used in the selected zone or area are grayed out.

General Setup

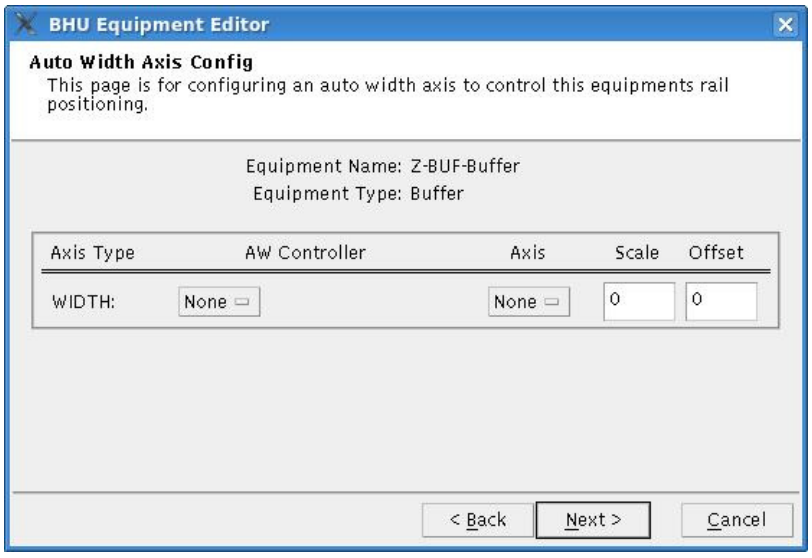
This page allows you to set the name of each unit.



Step	Procedure
1	Type a unique name in the **Equipment Name** field.
2	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

Auto Width Adjustment Setup

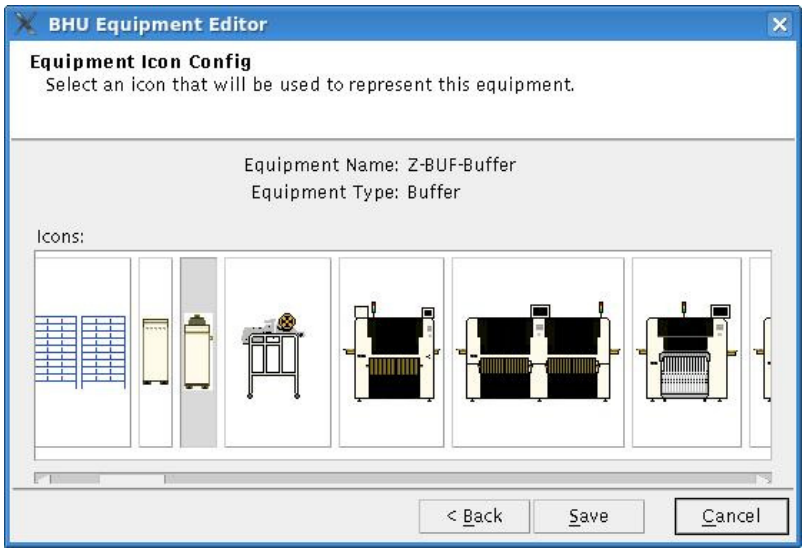
Setting of the board handling units includes the Auto Width Adjustment Axis Setup page (excluding work table, setup cart and HP station).When auto width controller is used to control the unit, set the parameters of the controller on this page.



Step	Procedure
1	Select the desired option under **Auto Width Controller**.
2	Select the desired option under **Axis**.
3	Type the scale and offset in the **Scale** and **Offset** textboxes.
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Icon Setup

This page is used to set the icons of the machines displayed on the Root Editor and Root Status screen. Although the default icons have been assigned to the machines, they can be changed to another by scrolling the list.



Step	Procedure
1	Select the desired icon while scrolling through the list.
2	Click **Save** to change the icon. Click **Cancel** to discard any changes made and to exit the editor.

➢ Adding peripheral units

Using the Ancillary Equipment Editor, add necessary peripheral units in a zone. These units have the communication feature, but will not carry out any production process. Usually, they are dependent on and controlled by the application or host processing.

Peripheral units include:

- PCB scanner
- LMNC / VLMNC
- PS Scanner
- Barcode Pass
- Setup cart
- Gem I/F
- Auto Width (Automatic width adjustment controller)

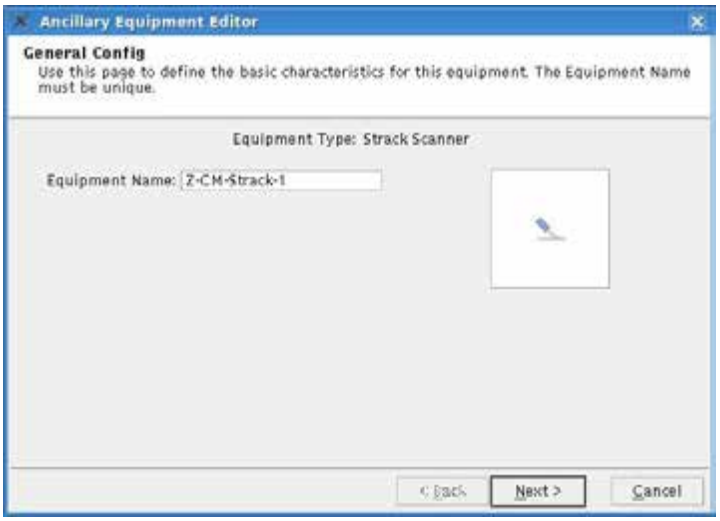
Before launching the Ancillary Equipment Editor, select **Zone**, followed by the peripheral units under the **Additional Equipment** field.

Buttons under the **Additional Equipment** field are available only when the peripheral units can be added to the selected zone and area.

General Setup

This is for making basic setting of the peripheral unit. The items to be set vary with the type of equipment (auto width controller or Gem interface, etc.), but are similar in most cases.

General equipment



Step	Procedure
1	Type a unique name in the **Equipment Name** field. (In case of auto width controller, the **Type** and **Axis** fields are added.)
2	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

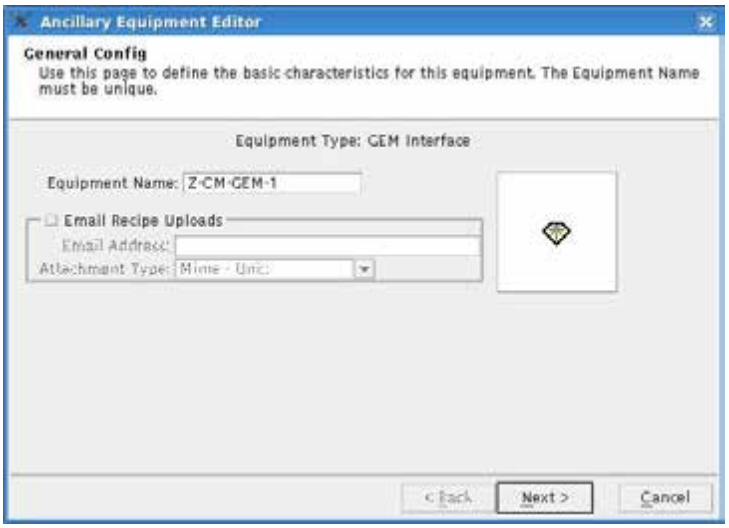
Gem interface

Only the IPAC machines support the Gem interface. It is usually used for remote communication with the machine using a third-party controller. The Gem interface permits communications with a third-party controller via the **SECS-II/GEM**.

The Gem interface is used with the following models:

- IPAC DA
- IPAC CS
- UVS
- IPAC PNP
- IPAC SD

The basic setting of the Gem interface is described below.



Step	Procedure
1	Type a unique name in the **Equipment Name** field.
2	Click **E-mail Product Data Upload** checkbox to receive a notification mail each time a program has been uploaded.
3	Type an e-mail address of the recipient in the **E-mail Address** textbox. If two or more recipient addresses are typed in, delimit them by a comma.
4	From the **Attachment Format** drop-down list, select the format of the e-mail attachment to be received and the system of the sender (Unix, Windows, etc.).
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Communication Setup

This screen is used to set the parameters for communication between the equipment and processing process. Make sure that the same set of communication parameters has not been shared among used in multiple machines via the Route Editor. Avoid using the existing host name, port and device.

Step	Procedure
1	Select the desired port type from the **Port Type** drop-down menu. Set other options that vary with the selected Port Type.
2	Port Type: **TCP/IP** 1. Type a unique host name in the **Host Name** textbox. 2. Type a port number in the **TCP/IP Port** textbox. 3. Type a cable name in the **Cable Name** textbox. (Can be omitted.) 4. Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.
3	Port Type: **TCP/IP - Gem Interface** 1. Select the desired host operation from **Server/Client**. 2. Set the desired option from the **Machine Device ID**. 3. Type an address in the **Serial IP Address** textbox. 4. Type a unique number in the **Server TCP Port** textbox (in the range between 5000 and 5999). 5. Keep the timeout default intact. 6. Click **Save**.
4	Port Type: **MuxLink** 1. The host name is defaulted to **Imnc**. 2. When the port number has been set by default, it cannot be changed. 3. Set the desired option from the **Parity** drop-down list. 4. Set the desired option from the **Baud Rate** drop-down list. 5. Set the desired option from the **Bits** drop-down list. 6. Type a unique cable name in the **Cable Name** textbox. (Can be omitted) 7. Click **Save**.
5	Port Type: **Serial** 1. Type a unique name in the **Device Name** textbox. 2. Set the desired option from the **Parity** drop-down list. 3. Set the desired option from the **Baud Rate** drop-down list. 4. Set the desired option from the **Bits** drop-down list. 5. Type a unique cable name in the **Cable Name** textbox. (Can be omitted) 6. Click **Save**.

➢ Adding RFCVT scanners (BM series machines)

Up to two scanners can be set for each zone of the BM series machines. Once the scanners are set, the **CVT Scanner** button will be disabled.
To launch the editor, select **Zone** and click **CVT Scanner**.

Note: The scanners (mobile scanners) used with the CM/DT/NPM series (including AM100) machines are to be set via the Module Editor. (Refer to “5.3 Adding a Module (CM/DT/NPM series (including AM100) machines only) for details.)

■ Scanner General Setup

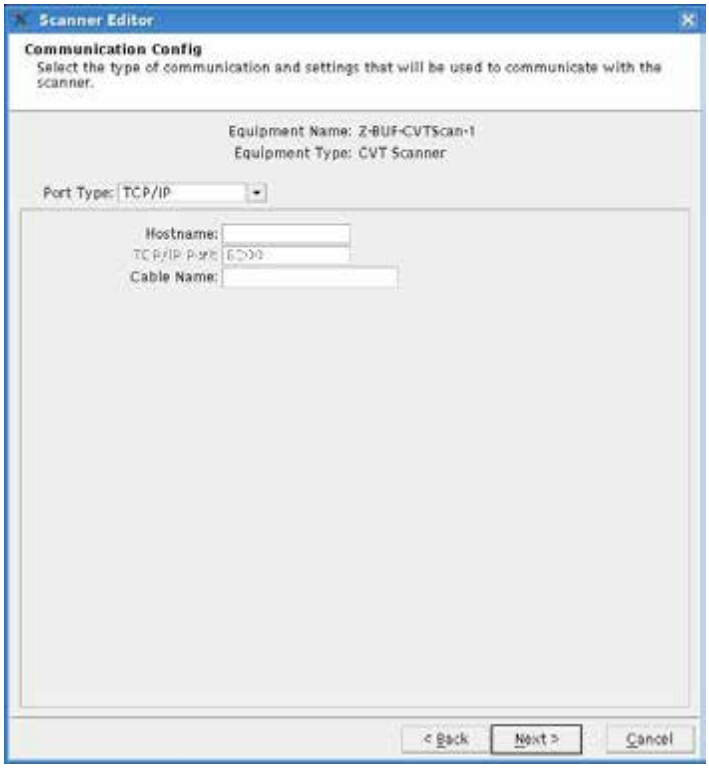
This page enables the basic setting of the scanners to be made.



Step	Procedure
1	Type a unique scanner name in the **Scanner Name** textbox.
2	Check the **Dual Scanners** checkbox when using two scanners in a zone and type scanner names in the **Dual Scanners** textbox.
3	Select the desired mode from the **Operation Mode** drop-down list.
4	Determine whether or not to use the scanner in several zones or in a single zone from the **Move Mode** drop-dist.
5	To use substitute parts in the zone, check the **Use Substitutes** checkbox before clicking **Next**. Select either of the options from the **Check Substitute Parts** drop-down list. • **None:** Scans the substitute parts without a warning message. • **Password:** Prompts the operator to enter the password when substitute parts are used.
6	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Communication Setup

This screen is for setting the parameters for communication between the equipment and scanner. Check the Route Editor to make sure the same set of communication parameters is not shared among multiple units. Make sure the host name, port number and device name for a scanner do not duplicate any existing ones.



The scanner type is determined by the selected port type. For the RF scanner and free zone scanner, select **TCP/IP** from the **Port Type** drop-down list.

Port Type: Serial

Step	Procedure
1	Type a unique name in the **Device Name** textbox.
2	Set the desired option from the **Parity** drop-down list.
3	Set the desired option from the **Baud Rate** drop-down list.
4	Set the desired option from the **Bits** drop-down list.
5	Type a unique cable name in the **Cable Name** textbox. (Can be omitted)
6	Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.

Port Type: Terminal Server

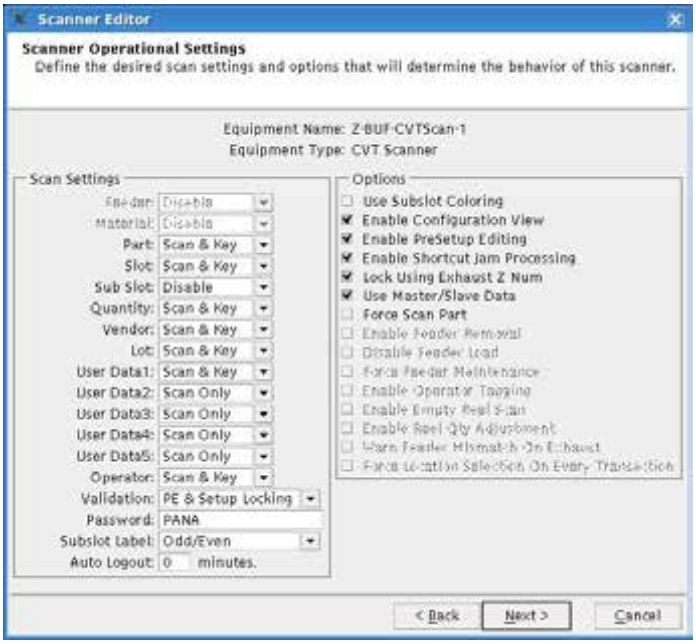
Step	Procedure
1	Type a unique name in the **Host Name** textbox.
2	Type a unique number in the **Port Number** textbox.
3	Set the desired option from the **Parity** drop-down list.
4	Set the desired option from the **Baud Rate** drop-down list.
5	Set the desired option from the **Bits** drop-down list.
6	Type a unique cable name in the **Cable Name** textbox. (Can be omitted)
7	Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.

Port Type: TCP/IP

Step	Procedure
1	Type a unique name in the **Host Name** textbox.
2	The **TCP/IP Port** cannot be edited.
3	Type a unique cable name in the **Cable Name** textbox. (Can be omitted)
4	Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Scanner Operation Setup

This page allows you to set how the scanner is operated.



Step	Procedure
1	Set the parameters under the **Scan Settings** and **Options** fields.
2	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Configure Setup Mode

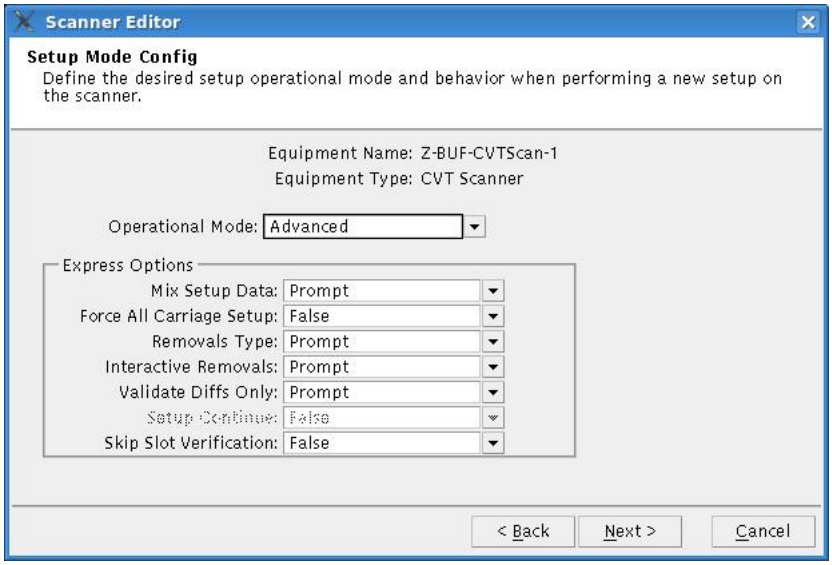
This page is for setting the operation mode for presetup and operations of the scanner.

Available menus in the respective list are explained in the table below.

Drop-down list	Description
Mix Setup Data	Select the scanning method for presetup.
All Carts Force Setup	Select whether or not to scan the entire presetup when entering the scan menu items.
Feeder Removal Method	Select whether or not to remove all products before the next product is scanned.
Interactive Remove	Select whether or not to remove the feeder that differs from the next setup information.
Differences Only	Select whether or not to scan only the new feeders added to the current ones.
Continue Setup	Select whether to continue setup or to start it from the beginning after the current setup is finished.
Omit Slot Verification	Select whether or not to omit scanning of slots.

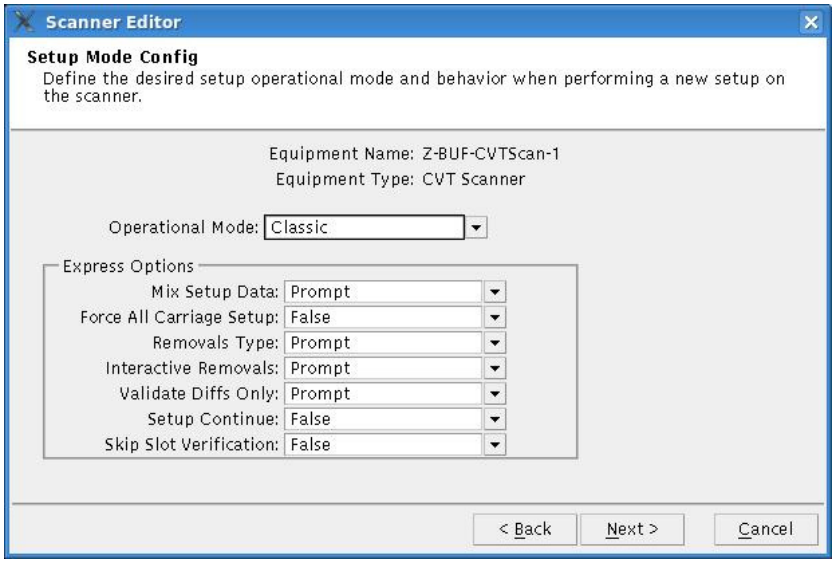
Operation mode - Advanced (Details)

In the Advanced mode, some items under the **Expressions Options** field may be disabled depending on the options selected for other items.



Step	Procedure
1	Select **Advanced** from the **Operation Mode** drop-down list.
2	Set the desired options for items under **Expressions Options**. Some items may be disabled depending on the selection of other items.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

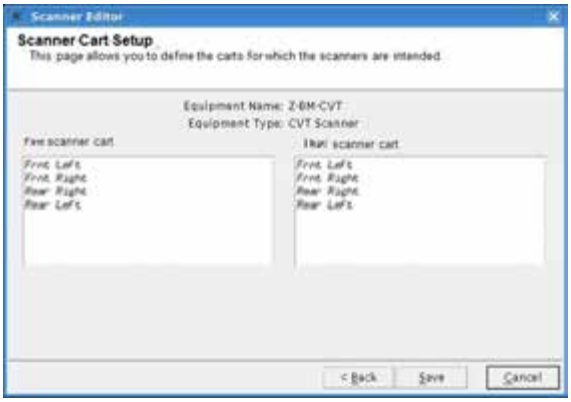
Operation mode - Classic (Normal)



Step	Procedure
1	Select **Classic** from the **Operation Mode** drop-down list.
2	Select the desired options for the items under **Expressions Options**.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

Scanner Cart Setup

This page allows you to define the carts for which the scanners are intended. Dual Scanner Cart First

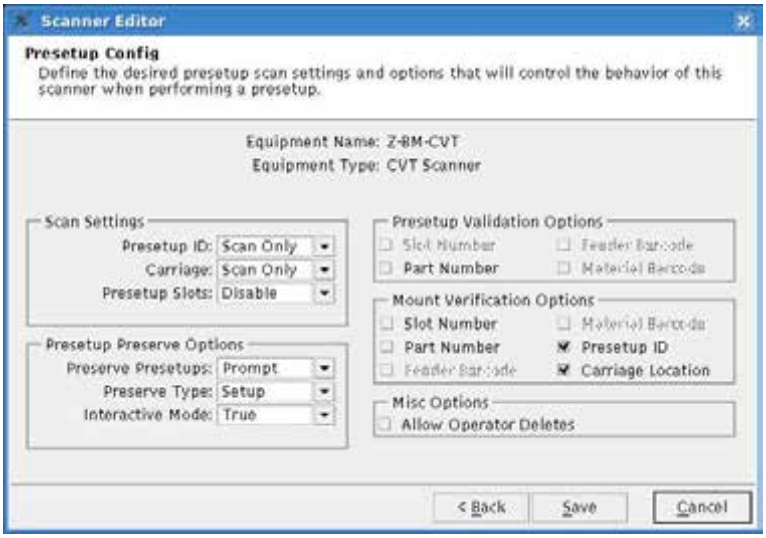


Note: This screen does not appear when the single scanner has been selected.

Step	Procedure
1	Select the carts to which the CVT scanners were assigned.
2	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

Configure Presetup

This page is used to set the conditions for determine how the scanner is activated during presetup.
It also permits the options on the movement of the RF CVT scanners to be set during presetup.



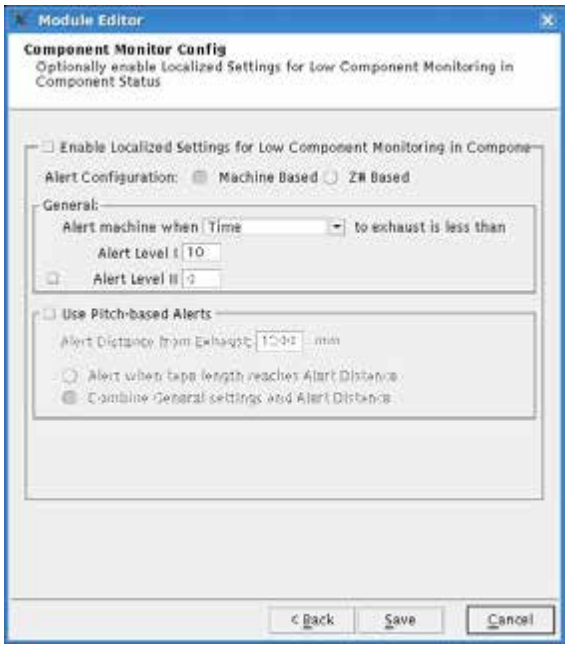
Step	Procedure
1	Set the desired options for the items under **Scan Setting**. **Presetup Slots** may be omitted.
2	Set the desired options for the items under **Save Presetup Options**.
3	Check the desired options under **Verify Presetup Options**.
4	Check the desired options under **Verify Mount Options**.
5	Check the **Enable Removal by Operator** checkbox as necessary.
6	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Part Monitor Setup

The displayed items vary with the part monitoring modes between **Machine Reference** and **Z#** (feeder address number).

Machine Reference

An alert message appears if the remaining component count on the reel is less than the threshold of the alert level 1 or 2 required to produce the preset number of products.



The threshold should be set product by product in the product setup.

Step	Procedure
1	Check the **Enable Local Setting for Parts Remain Monitoring** checkbox.
2	Select **Machine Reference** from **Alert Setting**.
3	From the **Alert Output** drop-down list, select **Panels** (product count) or **Time** (time). If these values are lower than the given values, an alert massage will be sent to the machine.
4	Type a value in the **Alert Level 1** textbox.
5	Check the **Alert Level 2** textbox and type a value in the textbox if necessary. The level 2 is an urgent level. Set a value less than the Level 1.
6	Click **Next**.

Z# Reference



Components are monitored as shown below:

- An alert message appears if the quantity of remaining panels required to complete the production set for the Z number is less than the preset threshold.
- An alert message appears if the remaining component count on the reel is less than the preset threshold.
- Set the thresholds panel by panel in the product setup.

Step	Procedure
1	Check the **Enable Local Setting for Parts Remain Monitoring** checkbox.
2	Select ***Z#*** from **Alert Setting**.
3	From the **Output Alert (as shown right)** drop-down list, select **Estimated Remaining Component Count** or **Panels to Exhaust**.
4	Click **Next**.

➢ Adding the RFCVT scanner (BM series machines)

To make presetup on the BM series machines, a presetup scanner is required in addition to the scanners for production.
Select **Zone** and click **PS Scanner** to launch the Scanner Editor.

Note: In case of the CM/DT/NPM series (including AM100) machines, the mobile scanner can be used as the preset up scanner.

■ General Scanner Setup

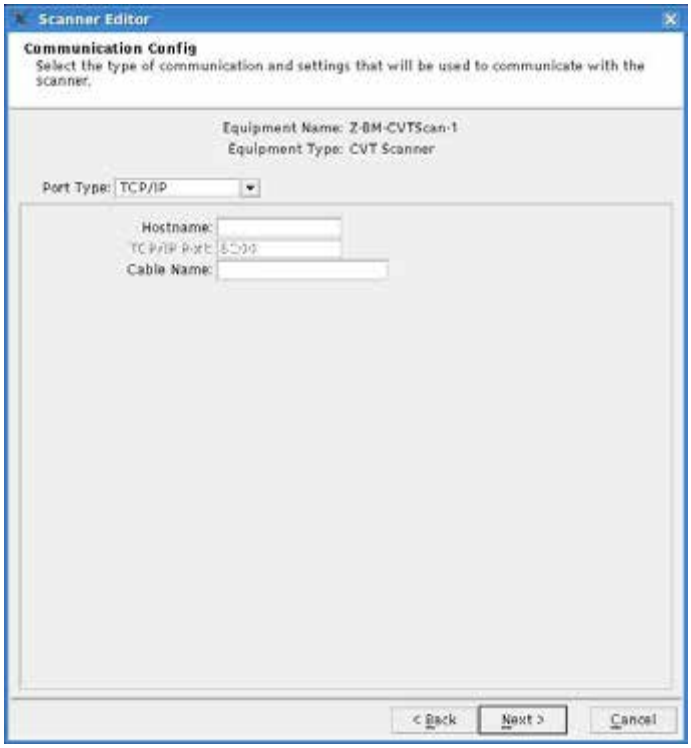
This page enables the basic setting of the scanner to be made.



Step	Procedure
1	Type a unique name in the **Scanner Name** textbox.
2	Specify the method of entering the operator ID from the **Operator** drop-down list.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

Communication Setup

This page is for setting the parameters for communication between the equipment and control process. Check the Route Editor to make sure the same set of communication parameters is not shared among multiple units. Make sure the host name, port number and device name for a scanner do not duplicate any existing ones.



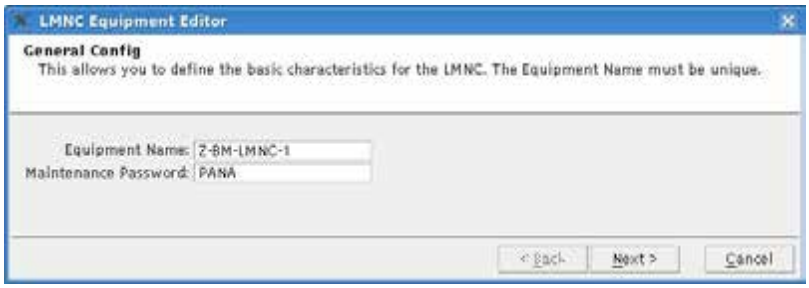
Step	Procedure
1	The **Port Type** is defaulted at **TCP/IP** and it cannot be changed.
2	Type a unique name in the **Host Name** textbox.
3	The **TCP/IP Port** has been set automatically and cannot be changed.
4	Type a unique cable name in the **Cable Name** textbox. (Can be omitted)
5	Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.

➢ Adding LMNC

The LMNC helps to support the production setup, management and traceability.
Only one LMNC can be set in each zone. To add LMNCs to the CM/DT/NPM series (including AM100) machines, create a module first, and add the machine to the modules, then add an LMNC to a zone. Once an LMNC has been set to a zone, the **LMNC** button will be disabled.
Select **Zone** and click **LMNC** to launch the LMNC Machine Editor.

■ General Setup

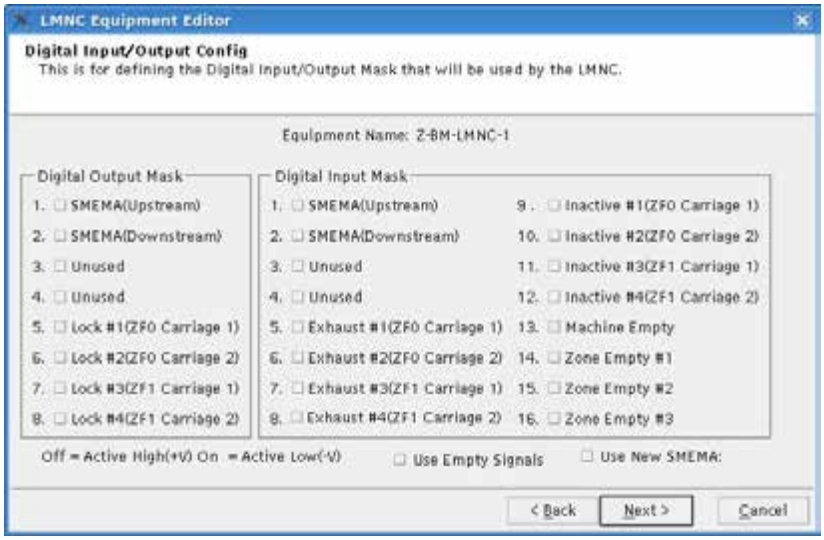
This page is for making the basic setting of the LMNC.



Step	Procedure
1	Type a unique name in the **Machine Name** textbox.
2	Type the password in the **Maintenance Password** textbox. Default: **PASS**
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

■ Digital Input/Output Setup

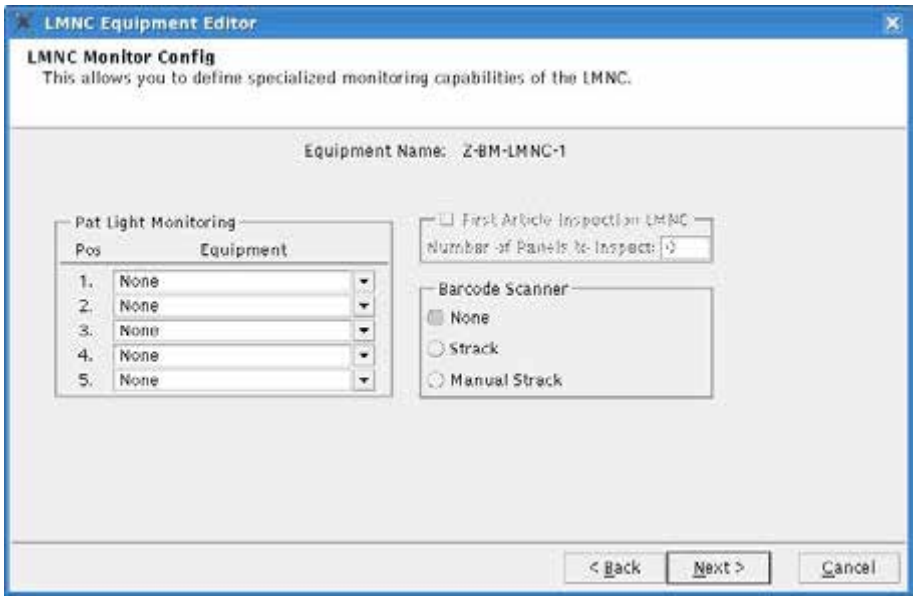
This page is used to set the digital input/output mask used on the LMNC.



Step	Procedure
1	Check the option to be enabled under the **Digital Output Mask** field.
2	Check the option to be enabled under the **Digital Input Mask** field.
3	Check the **Use Empty Signal** checkbox. Check this box when the digital cable is available for detecting the presence of panels in the machine in the zone. (BM series machines only)
4	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

LMNC Monitor Setup

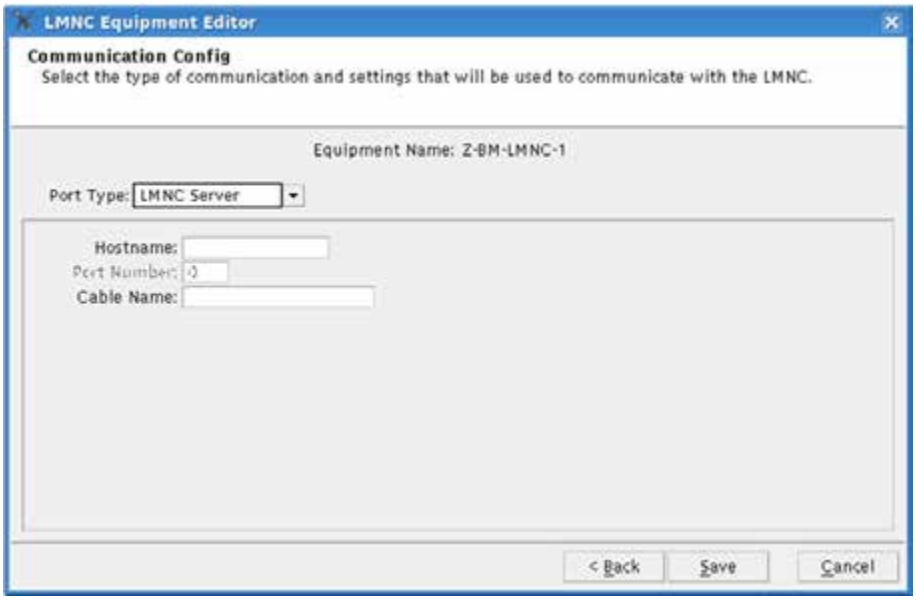
This page allows you to set the monitoring function of the LMNC.



Step	Procedure
1	To add the LMNC for the first products, check the **First Lot LMNC** checkbox and type the number of panels to be inspected during the inspection period.
2	Operators with the PMD license can monitor the status of the third-party machines with the signal tower. Select the machines at the respective locations from the drop-down lists. Pos1: Machines connected to the input ports 16, 17 and 18 Pos2: Machines connected to the input ports 19, 20 and 21 Pos3: Machines connected to the input ports 22, 23 and 24 Pos4: Machines connected to the input ports 25, 26 and 27 Pos5: Machines connected to the input ports 28, 29 and 30
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and to exit the editor.

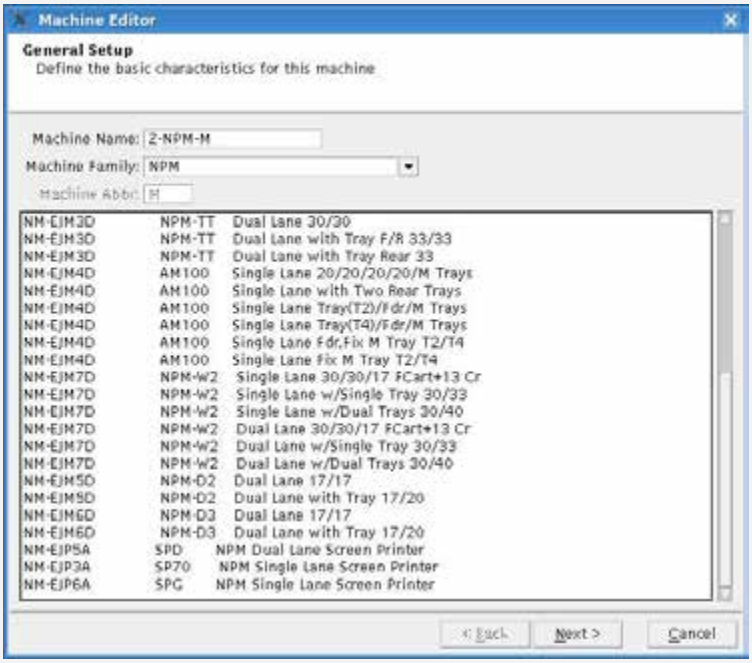
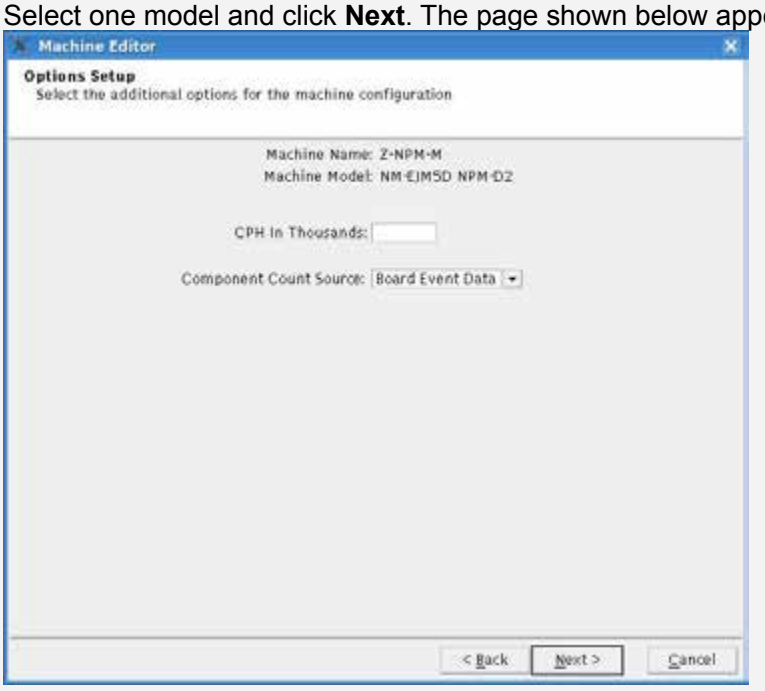
■ Communication Setup

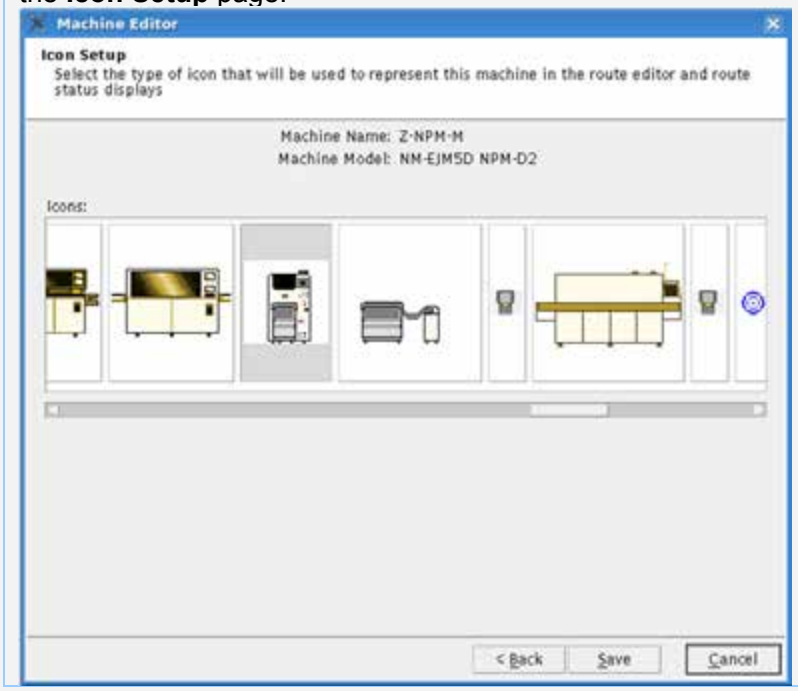
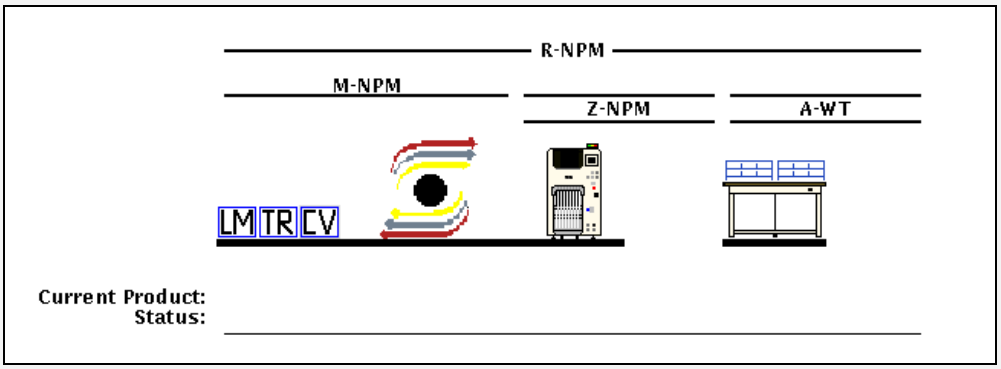
This page is for setting the parameters for communication between the equipment and control process. Check the Route Editor to make sure the same set of communication parameters is not shared among multiple units. Make sure the host name, port number and device name for the LMNC do not duplicate any existing ones.



Step	Procedure
1	From the **Port Type** drop-down list, select **LMNC Server**.
2	Type a unique name in the **Host Name** textbox.
3	Port Number is defaulted at 0 and cannot be changed.
4	Type a unique cable name in the **Cable Name** textbox. (Can be omitted)
5	Click **Save** to save the setting. Click **Cancel** to discard any changes made and to exit the editor.

4.3.7 Setting the Machine - NPM

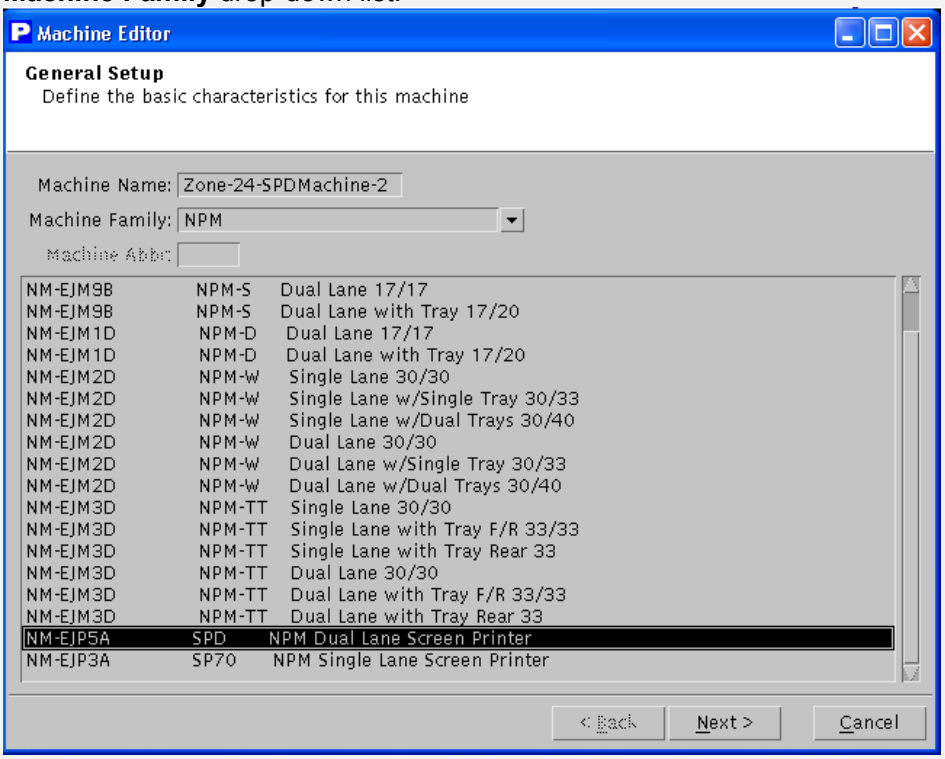
Step	Procedure
1	Select a zone in the route to which NPM machine needs to be added and click the **Machine** button along the bottom of the screen. The **General Setup** page of the **Machine Editor** appears.
2	On the **General Setup** page of the **Machine Editor**, select **NPM** from the **Machine Group** drop-down list. The list of NPM machines appears as shown below. 
3	Select one model and click **Next**. The page shown below appears. 

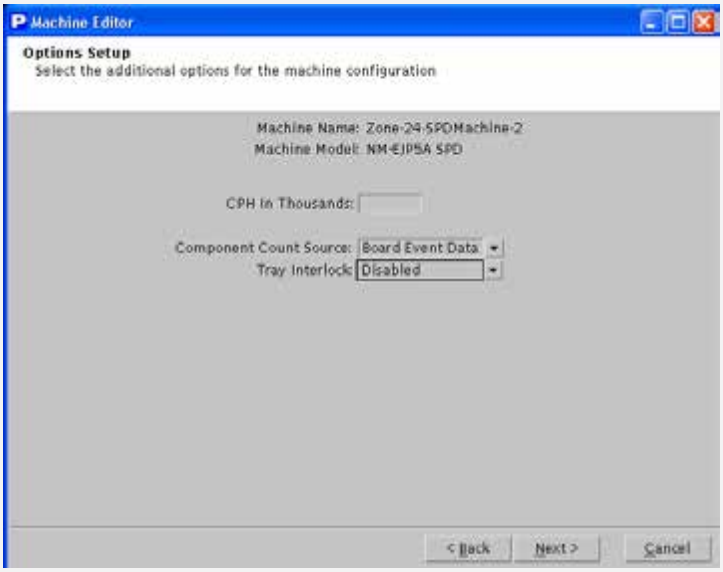
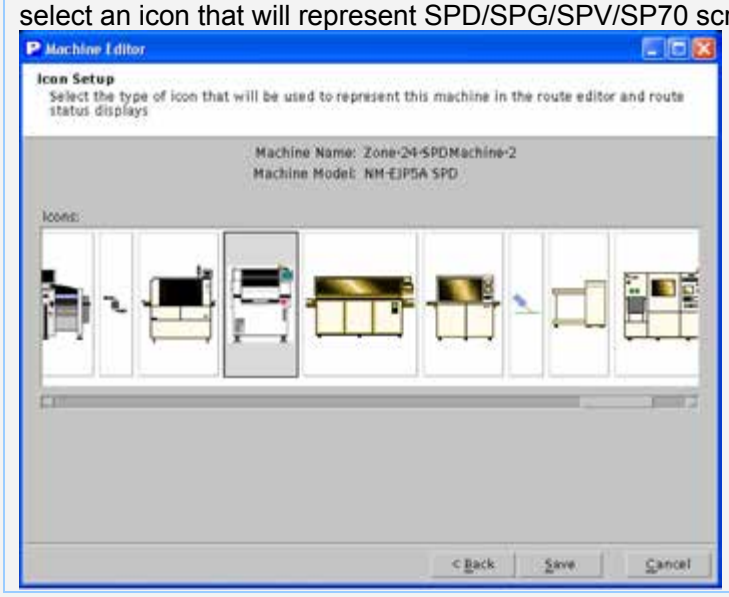
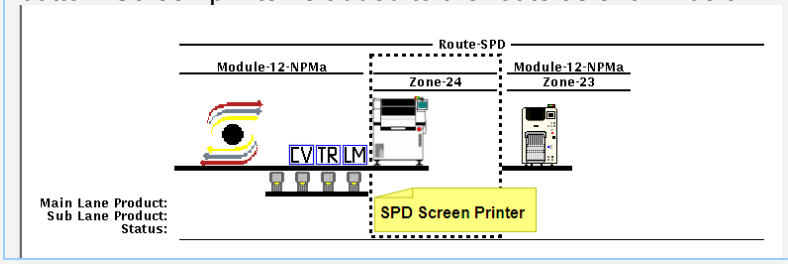
4	Keep the Option Setup page unchanged and click **Next**. The display changes to the **Icon Setup** page. 
5	Click **Save**. The icon of the added machine (NPM series) appears in the zone as shown below. 

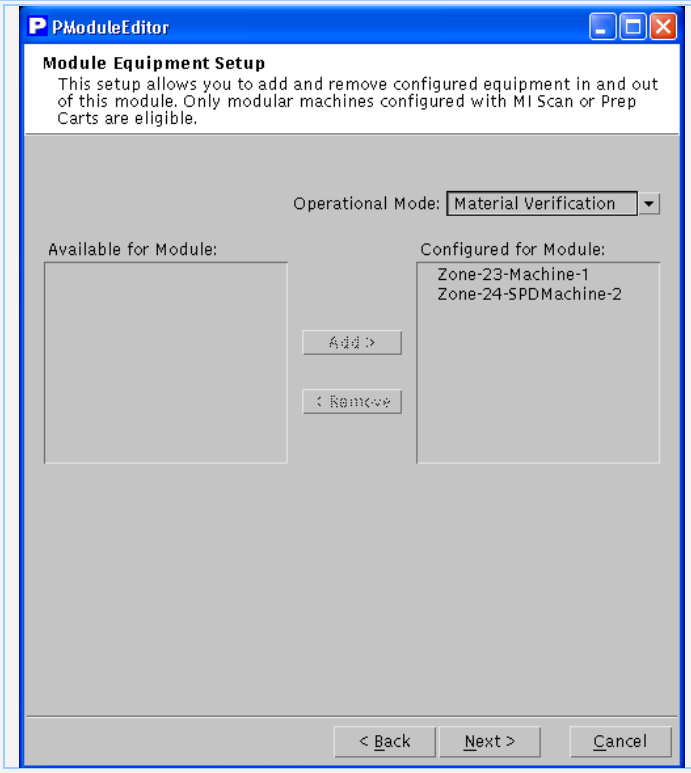
4.3.8 Setting the Machine - SP

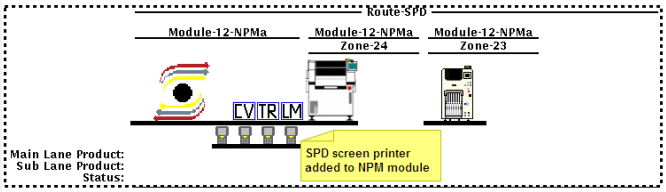
SPD, SPG and SPV are dual lane screen printers that communicate with NPM and AM100 via LNB. The screen printers in PanaCIM are categorized as NPM series machine.
SPD and SPV (Dual specifications, Single specifications with two machines in parallel , 2in1 specifications) are the screen printers for dual lane. SP70, SPG and SPV (Standard specifications, 1to2 specifications) are the screen printers for single lane.
If two different types of products are printed by the screen printer at the same time, SPD or SPV (Dual specifications, Single specifications with two machines in parallel) will be used. Even though SPV (2in1 specifications) is the screen printer for dual lane, you can use it only for screen printing of the same product.
The SP70 screen printer can also be connected to NPM and AM100 machines via LNB.

SPD, SPG, SPV or SP70 machine with new DGS version 7.02 installed must be required to import the products into a route including SPD, SPG or SP70 and NPM through PanaCIM automatic import (Commit to PPD in PPD manager).

Step	Procedure
1	Select a zone in the route to which screen printer needs to be added and click the Machine button within the **Base Equipment** area of the **Route Editor** window. **General Setup** page of the **Machine Editor** appears.
2	Enter the machine name in the **Machine Name** textbox and select **NPM** from **Machine Family** drop-down list.  NOTE) Select NM-EJP3A to add the SP70 screen printer for NPM. Select NM-EJP5A to add the SPD screen printer for NPM. Select NM-EJP6A to add the SPG screen printer for NPM. Select a target model of NM-EJP7A to add the SPV screen printer for NPM. Select KX from the list in the machine group to add the SP70 screen printer for CM.

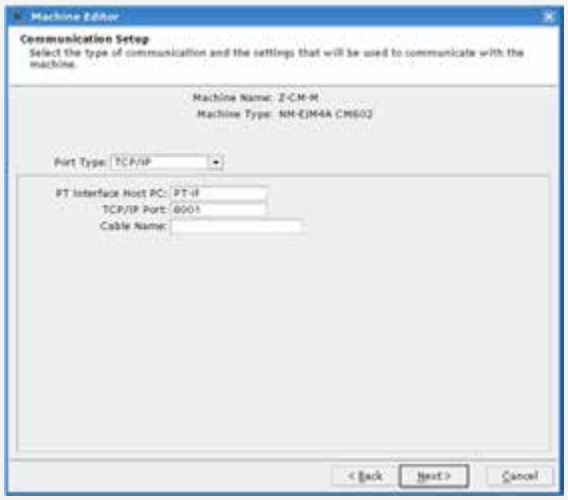
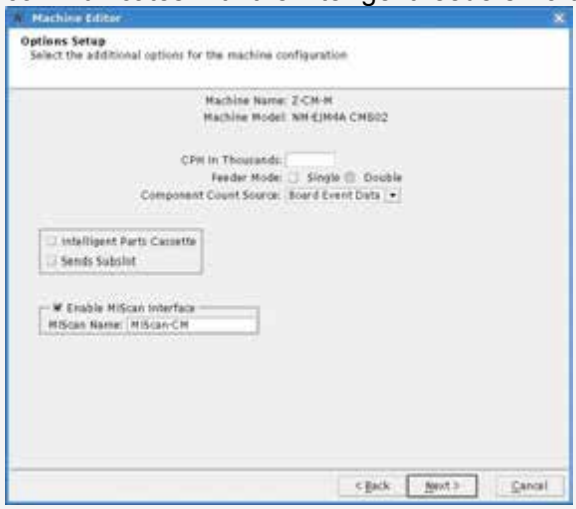
3	Select the name of the screen printer as shown in step 2 and click the **Next** button in the **General Setup** page. Following page appears:  **Note: Tray Interlock option is not available for Screen Printers.**
4	Click the **Next** button in the **Options Setup** page. **Icon Setup** page appears to select an icon that will represent SPD/SPG/SPV/SP70 screen printer in the Route. 
5	Select an icon for the screen printer in the **Icon Setup** page and click the **Save** button. Screen printer is added to the route as shown below: 
6	Double-click the module to open the Module Editor. Settings need to be done in the Module Editor so that the NPM module will control this screen printer machine. Click the **Next** button until **Module Equipment Setup** page of Module Editor appears. Click the **Add** button to add the screen printer to the module. The screen printer now appears underneath **Configured for Module** list. This is shown below:

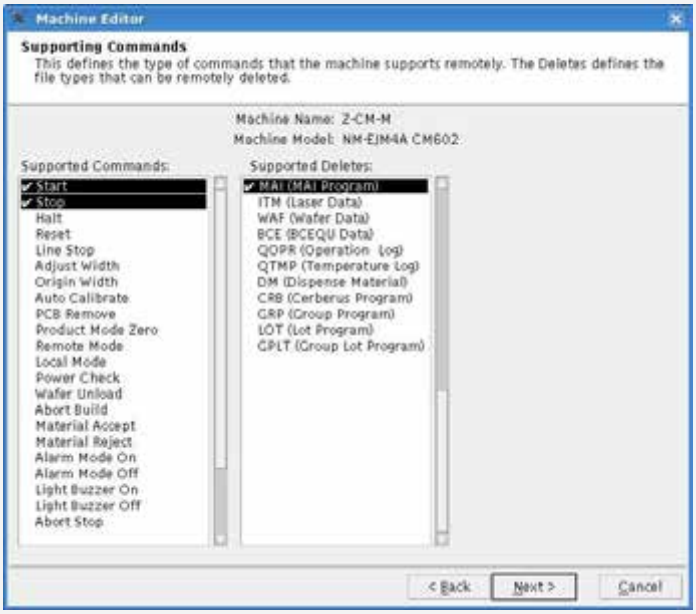


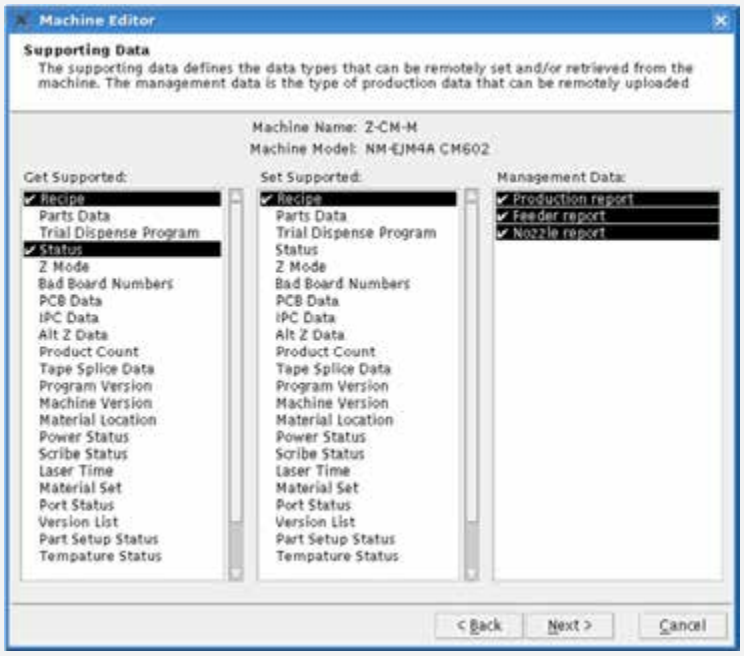
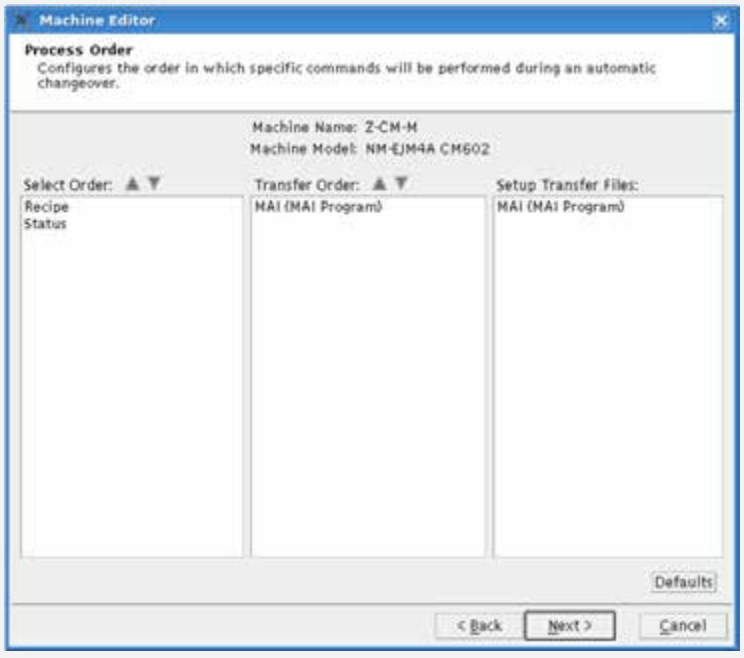
7	Keep on clicking the **Next** button until **Save** button appears in the **Component Monitor Config** page. After clicking the **Save** button, screen printer is now added to the NPM module: 

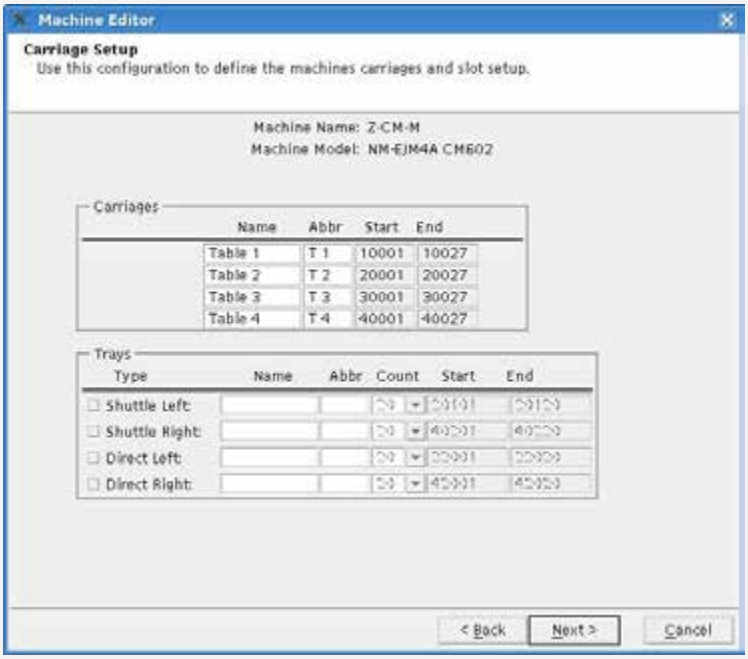
4.3.9 Setting the Machine - CM

This section provides the procedure for setting the CM/DT series machines required to use the PanaCIM/Material Verification.
(It assumes that the production line has been configured beforehand.)

Step	Procedure
1	Double-click **Machine** on the Route Editor screen to open up the Machine Editor screen.
2	Click the **Next** button to advance to the Communication Setup page, and set the following: • **PT Interface Host PC:** Type the host name of the PT interface. • **TCP/IP Port:** Type a port number. **Note:** An odd number between 8001 and 8099 not duplicated with other port numbers must be entered in the **TCP/IP Port** textbox.  Click the **Next** button.
3	Set the options as follows on the Option Setup page. • **Feeder Mode:** Double • **Intelligent Parts Cassette / Send Subslots:** Uncheck the textboxes. • **Enable Communication Unit Interface:** Check the textbox. • **Communication Unit Name:** Automatically populated Communication unit refers to a communication terminal unit known as Digi-One (optional kit for PanaCIM and CM/DT series machines) incorporated in the machine. It communicates with the intelligent feeders via the machine.  Click the **Next** button.

4	Set the options as follows on the Communication Setup page. • **Port Type:** TCP/IP • **Host Name:** Type the host name of the communication unit (Digi-One). Make sure that the typed host name and IP address have been registered using the Host Editor (Refer to “2.2 Edit Hosts” in the Material Verification documentation.). • **TCP/IP Port:** 20301 • **User Name/Password:** root/dbps  Click the **Next** button.
5	The **Supporting Commands** page appears. To verify the tray materials, set the following item. (To verity the reel parts only, this step is not necessary.) • Check **Tray Release** under **Supported Commands**.  Click the **Next** button.

6	Click **Next** until the **Supporting Data** page appears. To verify the tray materials, set the following item to enable tray locking. (To verity the reel parts only, this step is not necessary.) • **Set Supported:** Check **Part Setup Status**.  

7	Set the following items on the **Cart Setup** page. • When the machine is provided with the tray feeders, check the checkboxes of the tray feeders concerned. • Change **Name** and **Symbol** (abbreviation) of the cart if necessary. • Make sure that the quantity of the tray reflects the number of available tray slots. • Make sure that the **Start** and **End** values reflect the actual tray feeder and number of tray slots in use.  Click the **Next** button until the last page is reached.
8	Click the **Save** button.

4.3.10 Create or add Module

Refer to the following:

- 5.3 Starting Production

4.3.11 Setting the Module - LNB

Refer to the following:

- 5.3 Starting Production

4.3.12 Setting the Module - PT200

Refer to the following:

- 5.3 Starting Production

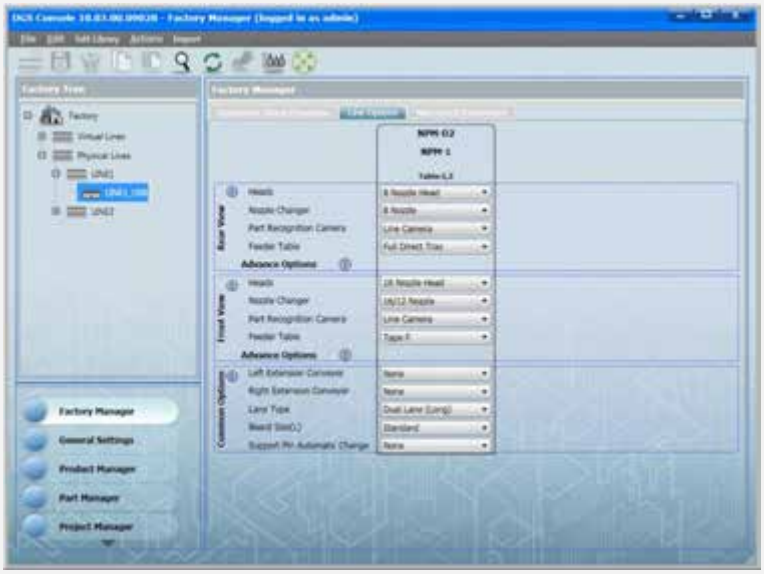
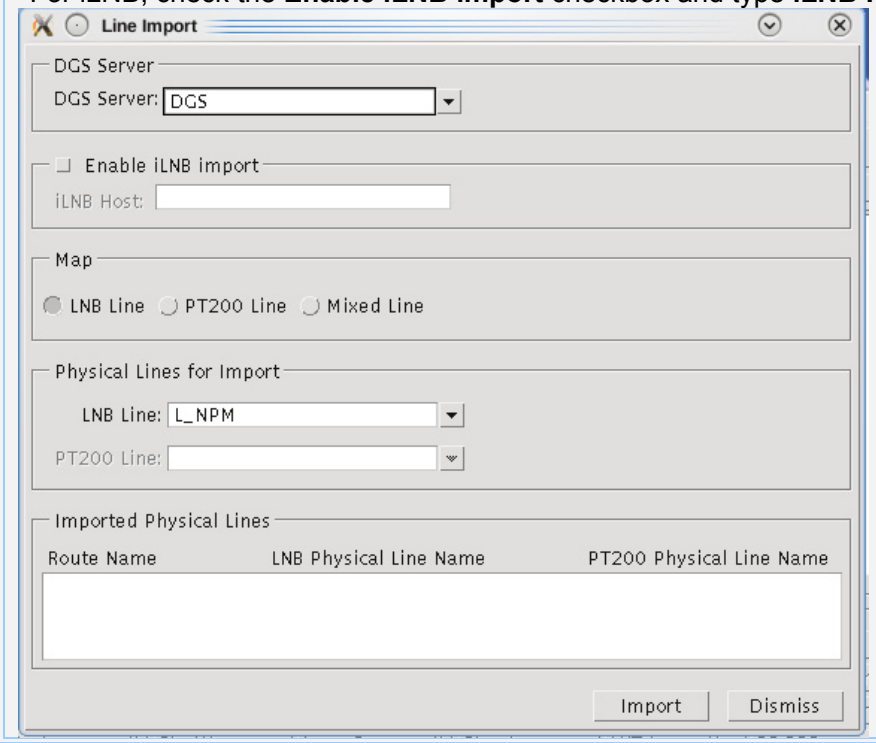
4.3.13 Setting the Module - iLNB

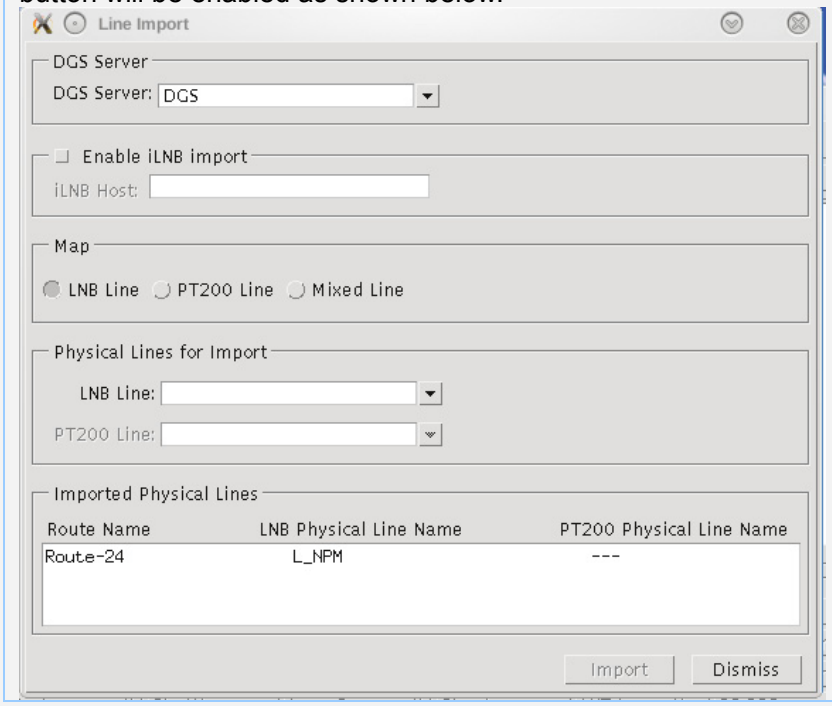
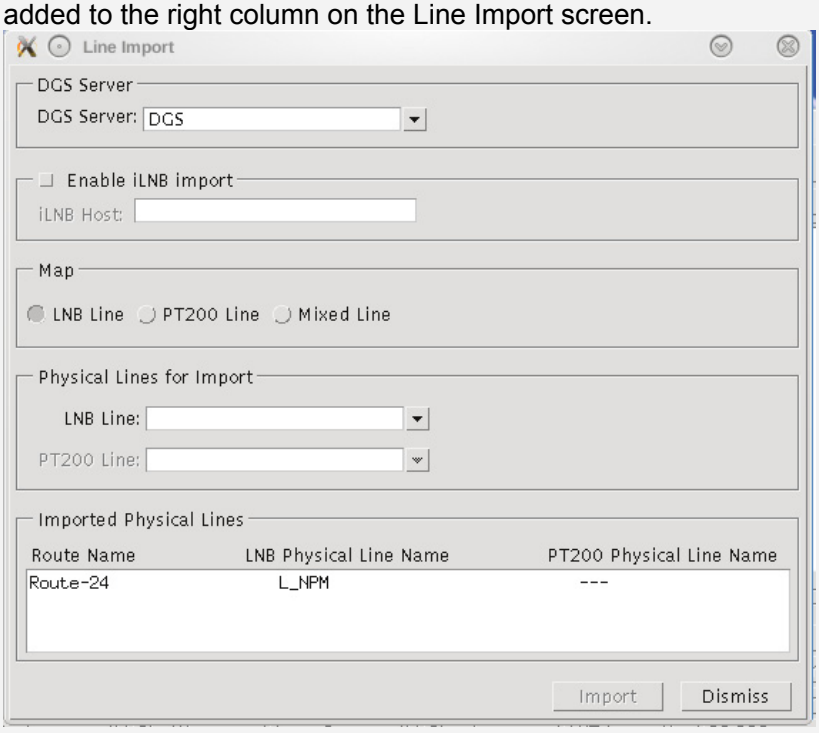
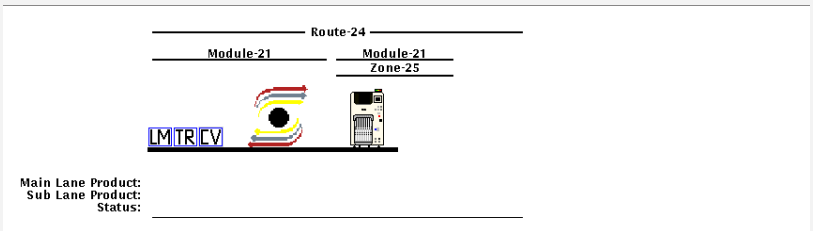
Refer to the following:

- 5.3 Starting Production

4.4 Creating Route Automatically

Follow the steps below to import a line of the machines.

Step	Procedure
1	Make sure that the machines are shown in the actual line on the DGS side. 
2	Click the **Line Import** button under **Layout** field on the Route Config Editor. The Line Import screen appears. *For iLNB, check the **Enable iLNB import** checkbox and type **iLNB Host**. 

3	If the actual line containing the machine to be imported is selected, the **Import** button will be enabled as shown below. 
4	Clicking the **Import** button adds a route consisting of one or more machines of the actual line on the Route Config Editor. After imported, the route name will be added to the right column on the Line Import screen.  

4.4.1 Setting the Route - Set LNB

4.4.2 Setting the Route - Set PT200

4.4.3 Setting a Zone

4.4.4 Setting the Machine - NPM

4.4.5 Setting the Machine - SP

4.4.6 Setting the Machine - CM

4.4.7 Setting the Module - LNB

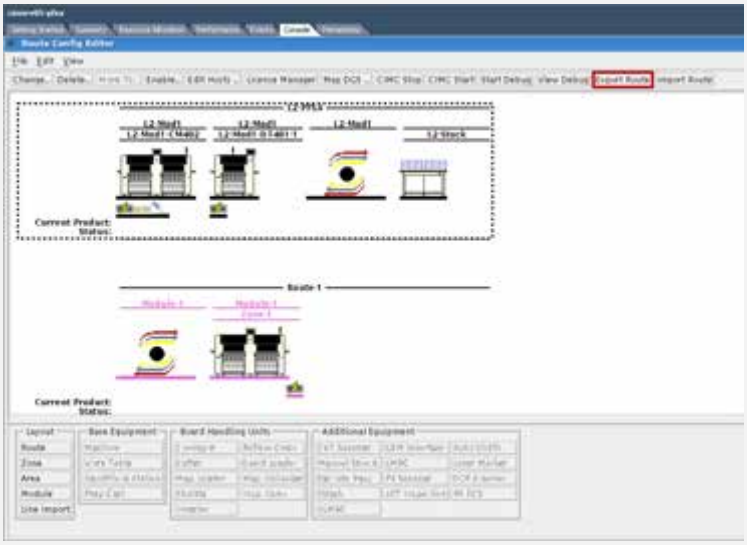
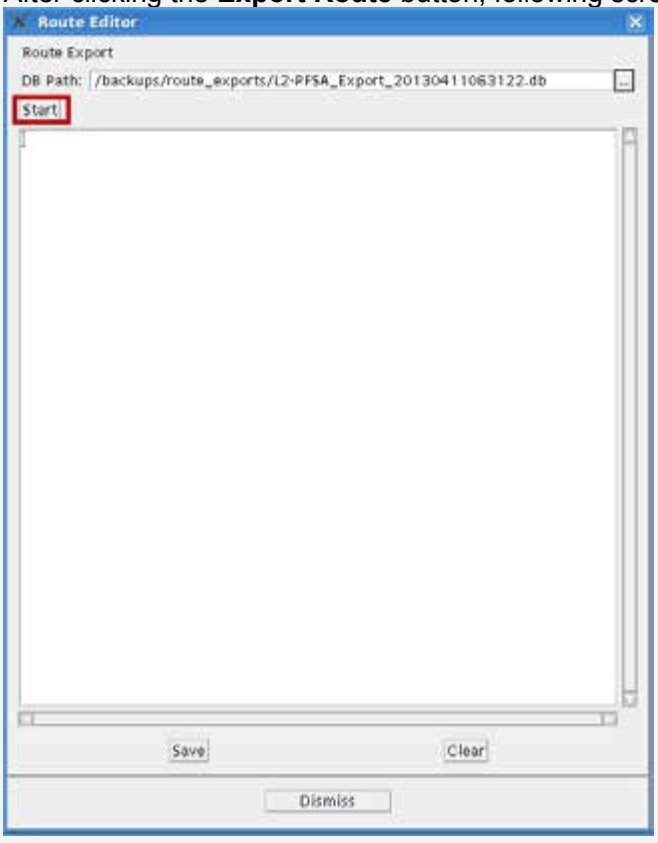
4.4.8 Setting the Module - PT200

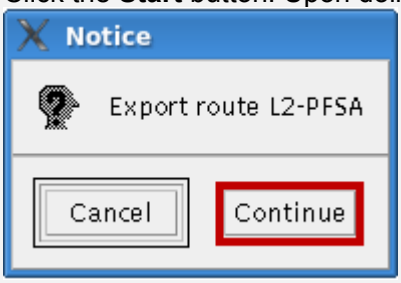
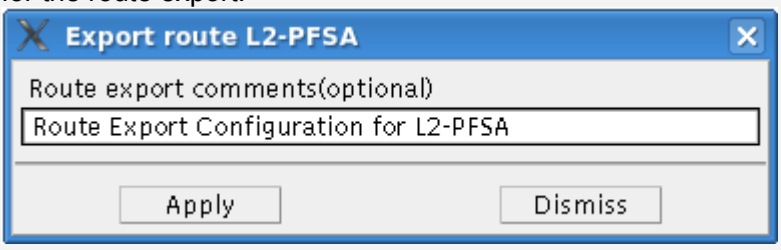
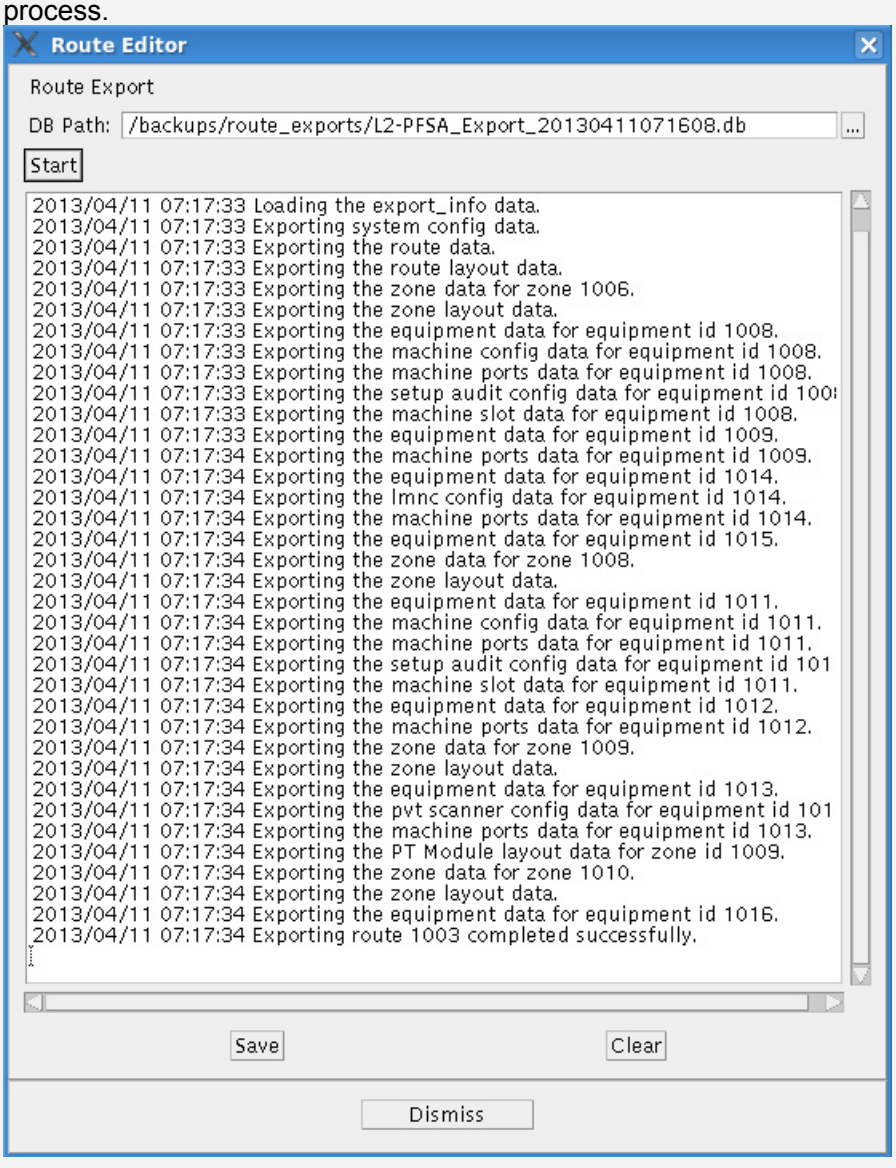
4.5 By Copy

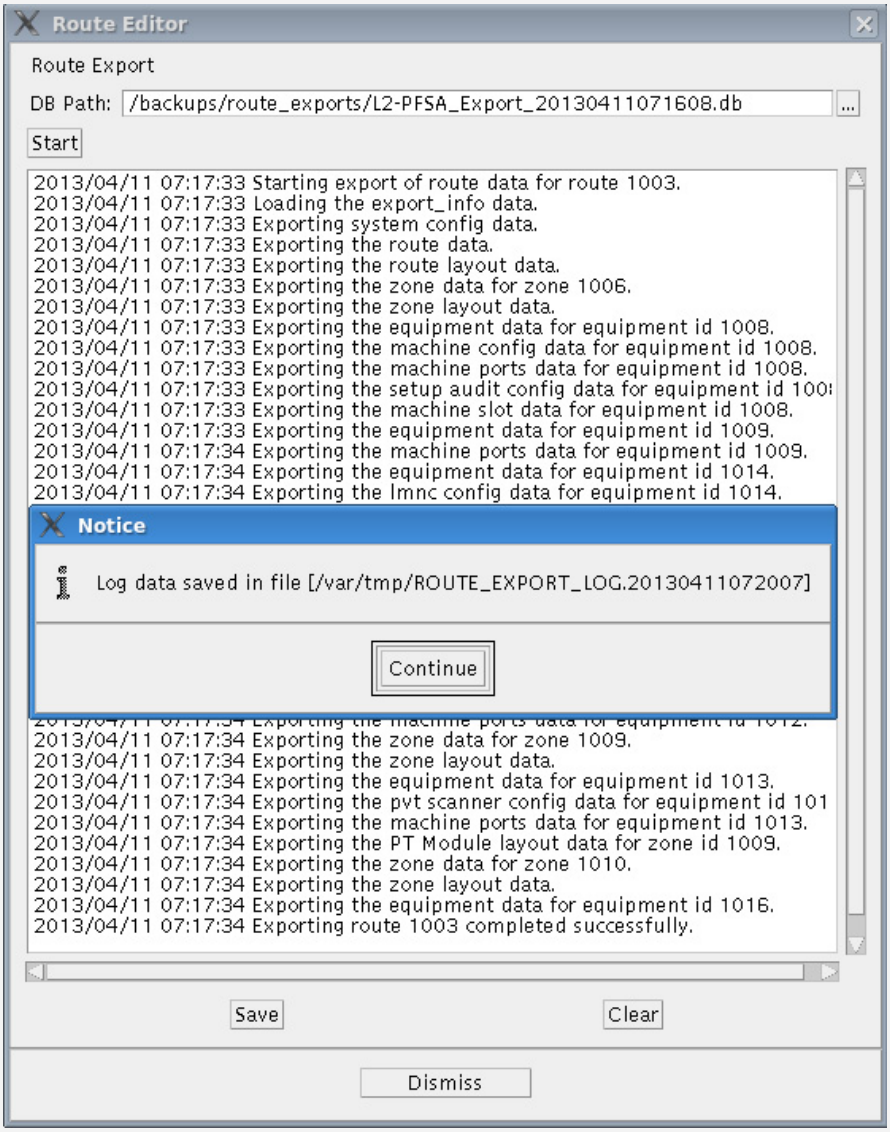
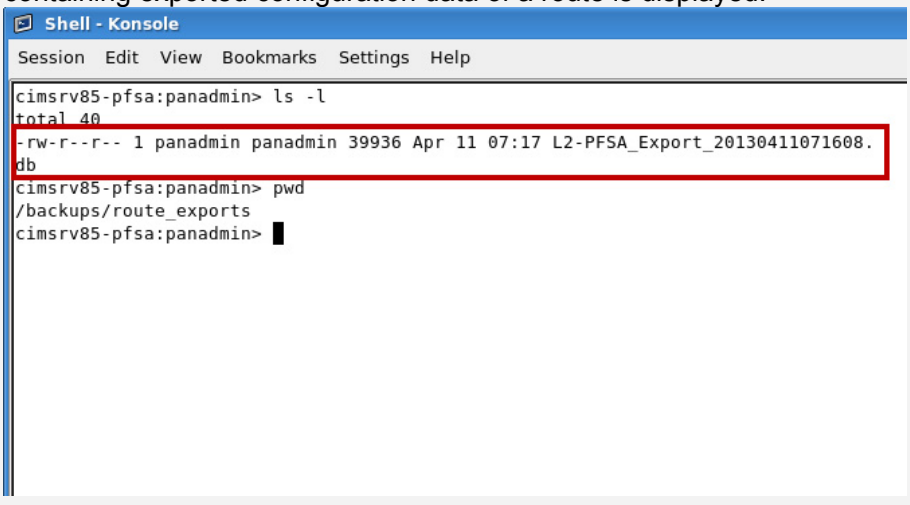
Route import/export utility is added to facilitate exporting of route configuration. This configuration can then be imported back to create another route with this configuration on a same/different CIMC server. Thus essentially this utility implements a route based Cloning system where a route can be exported to a predefined data structure and imported using same structure.

4.5.1 Exporting Route Configuration from CIMC Server

Refer to the action table below listing the steps to export route configuration.

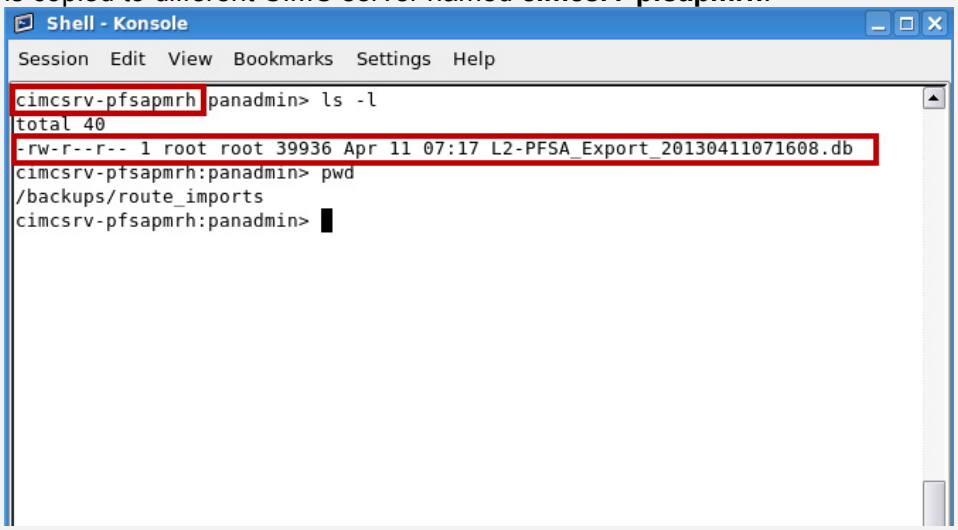
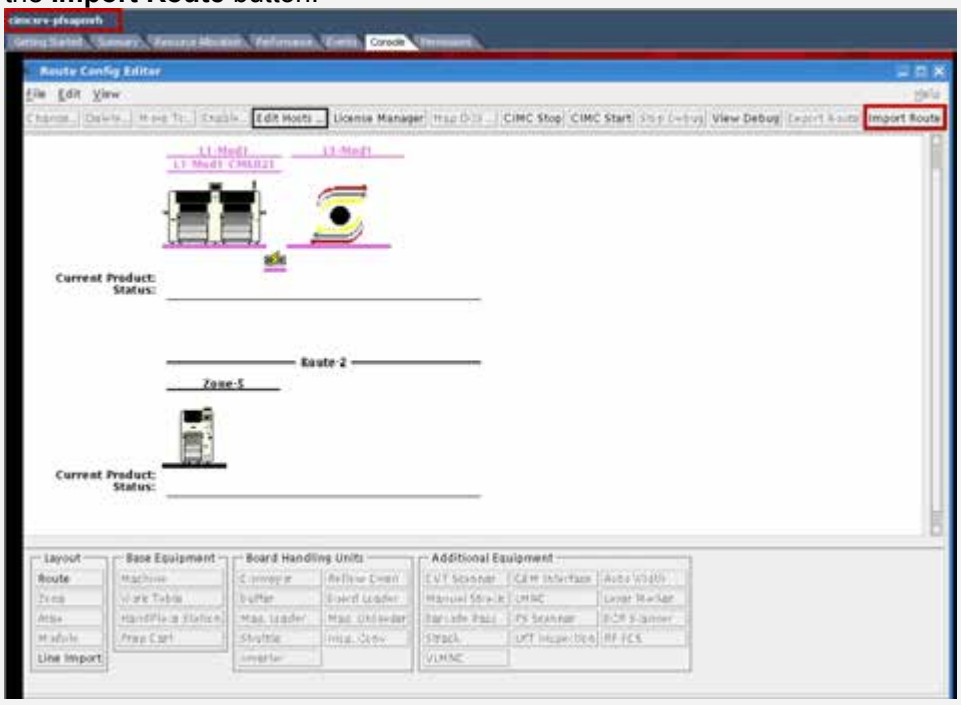
Step	Procedure
1	Select the route whose configuration needs to be exported and click the **Export Route** button. 
2	After clicking the **Export Route** button, following screen appears: 
3	As shown above, in the **DB Path** textbox, default path to which data is exported is /backups/route_exports. With this folder, db file will be created whose name will be of the format <Route Name>_Export_YYYYMMDDHHMMSS.db. Export File name location and name can be changed if the customer wants to.

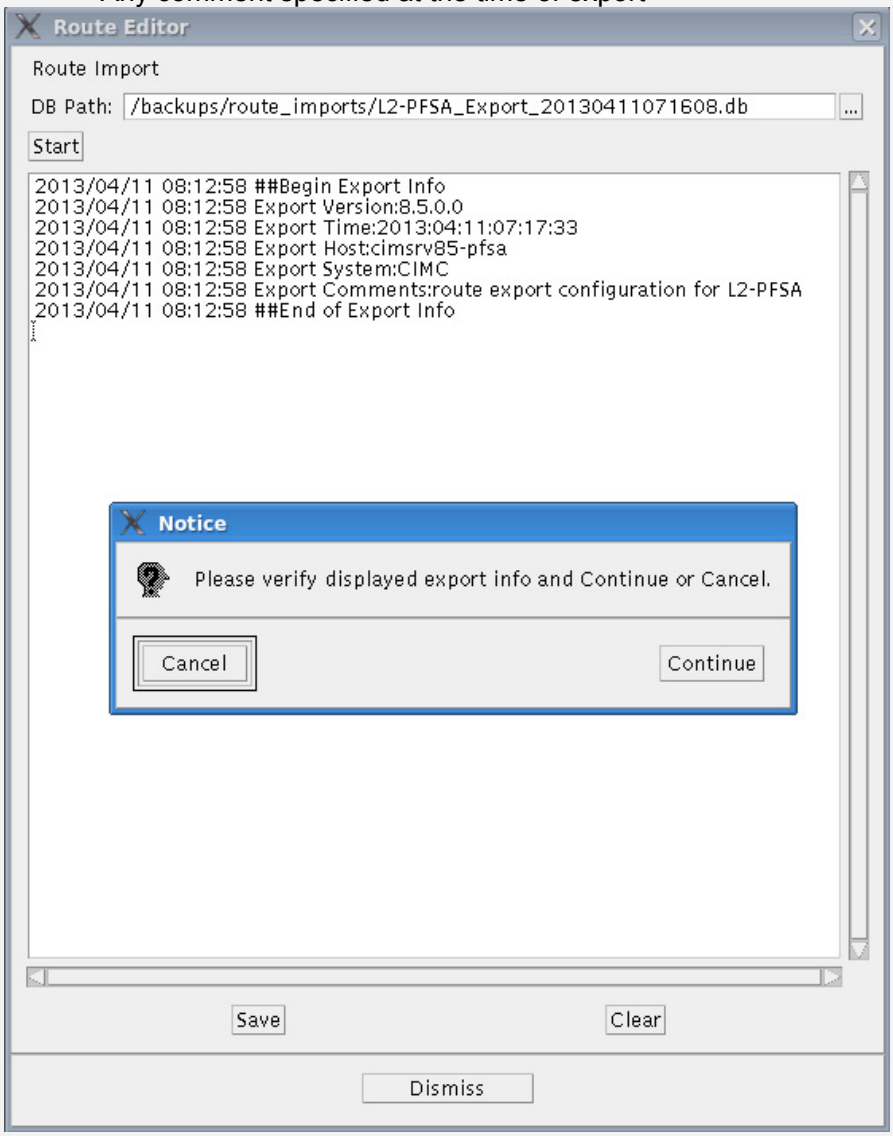
4	Click the **Start** button. Upon doing so, following dialog pops up: 
5	Click the **Continue** button. Upon doing so, you are prompted to enter comments for the route export. 
6	Click the **Apply** button. Messages appear in the listbox indicating the data pertaining to selected route that is exported and messages for the export process. 

7	Click the **Save** button to save the log messages in the text file. Upon clicking this button, following dialog appears indicating the file name to which log messages are saved. 
8	Upon navigating to the path specified at the time of export, name of file containing exported configuration data of a route is displayed. 

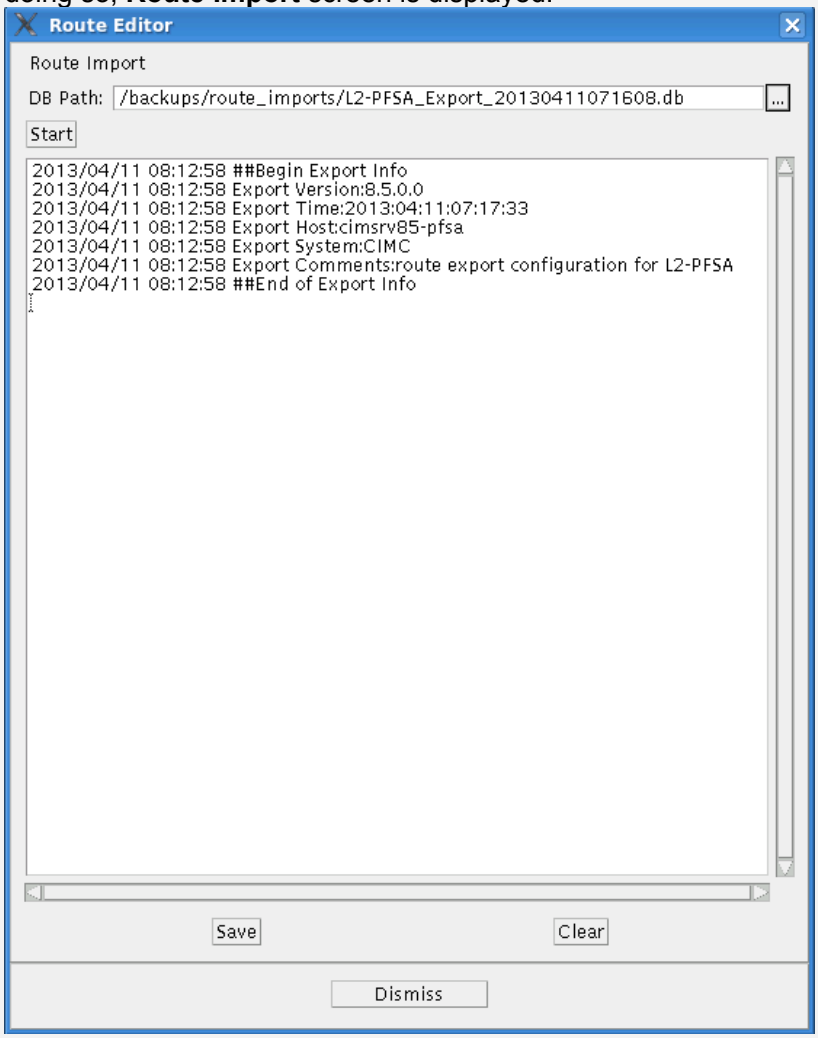
4.5.2 Importing Route Configuration to a Different CIMC

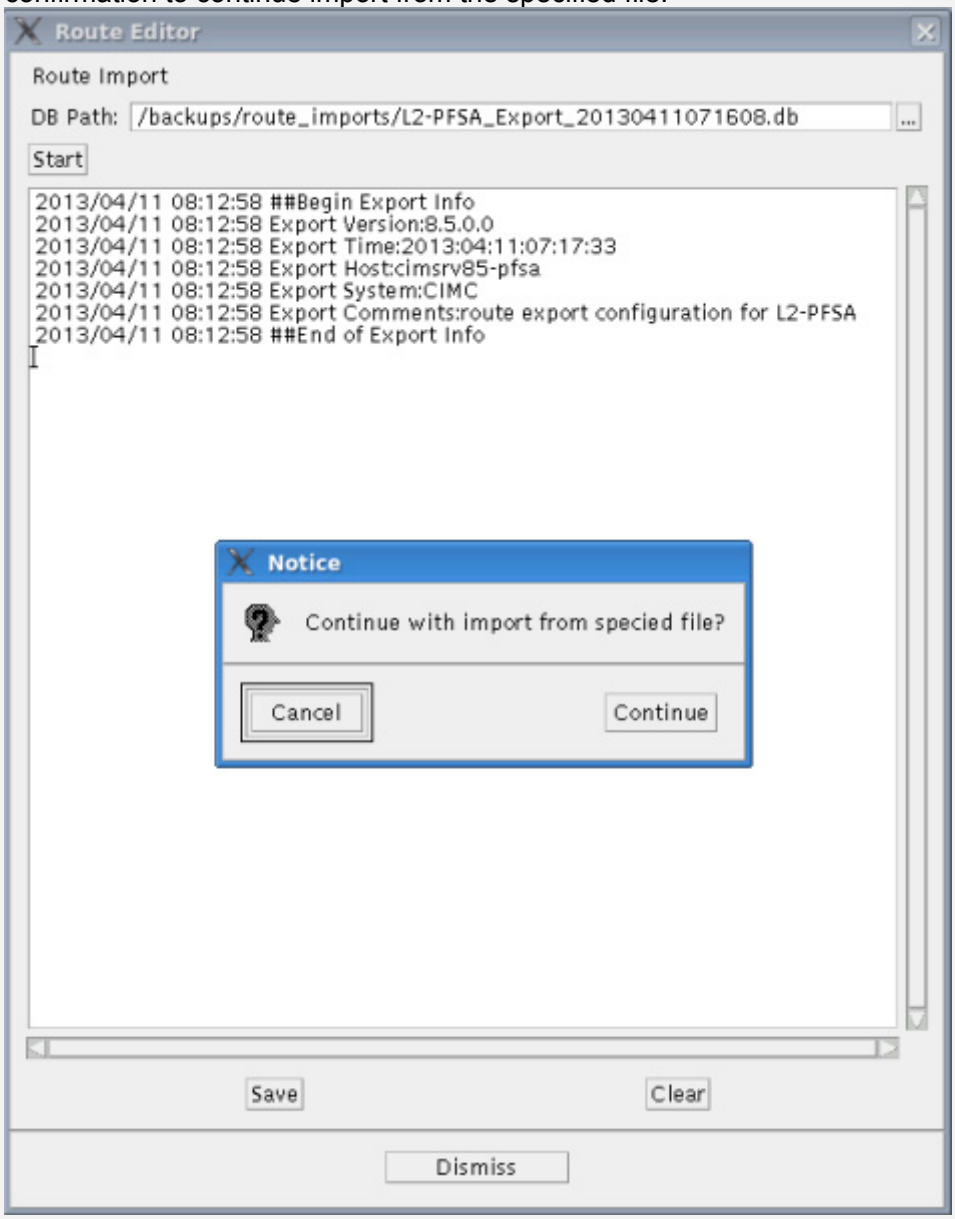
Take the following steps to import route configuration.

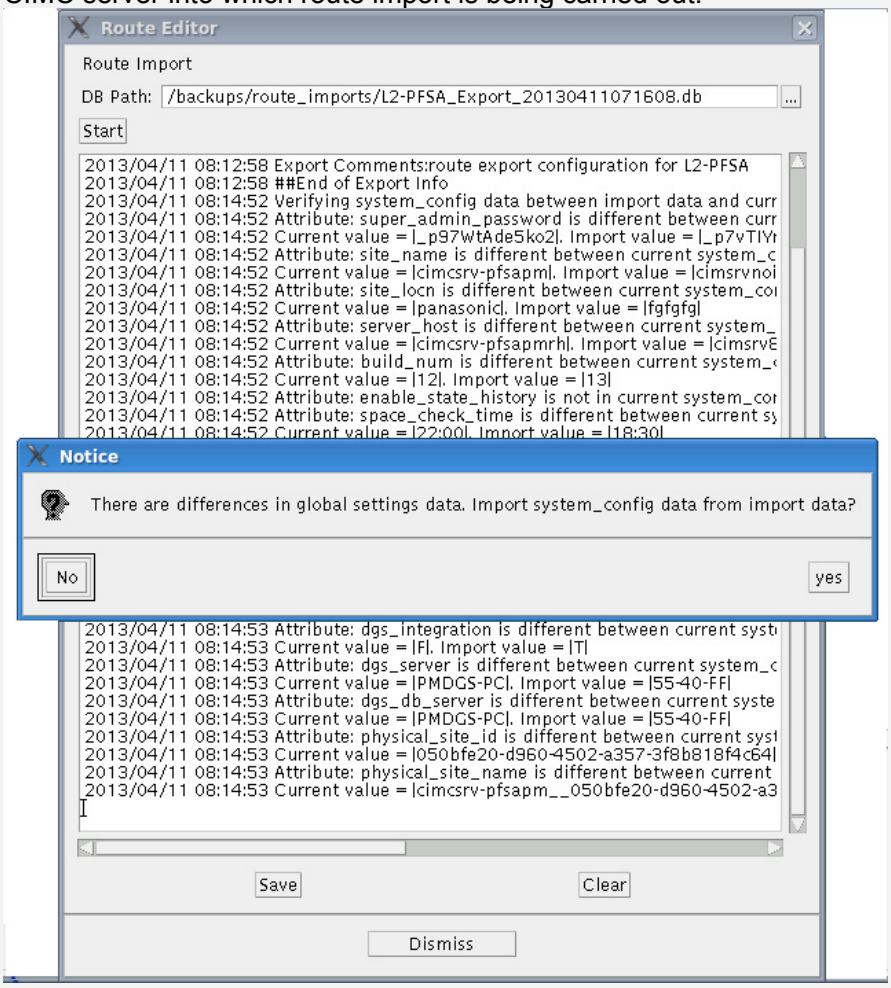
Step	Procedure
1	Copy the database file created after export to a different CIMC server in which route data needs to be imported. As shown below, database file (extension db) is copied to different CIMC server named **cimcsrv-pfsapmrh**. 
2	Go to another CIMC server in which route data needs to be imported and click the **Import Route** button. 

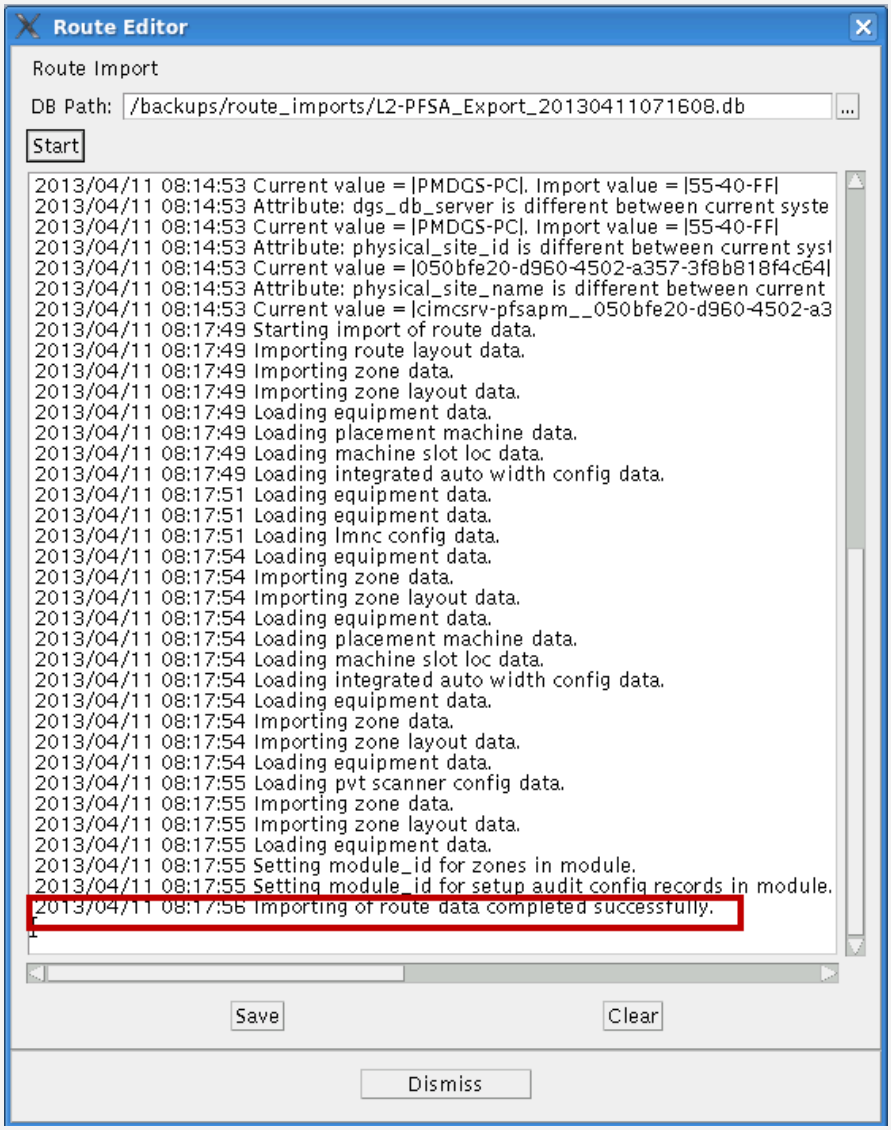
3	After clicking the **Import Route** button, following screen appears to choose the directory path containing route configuration data that needs to be imported. As shown below, file containing route configuration data is present in the path **/backups/route_imports**. Also, when a database file is selected from this path, exported info is displayed in the box below. Exported info contains following important information: • PanaCIM version from which route data was exported • Date and time at which export was carried out • Name of the CIMC server from which route data was exported • Any comment specified at the time of export 

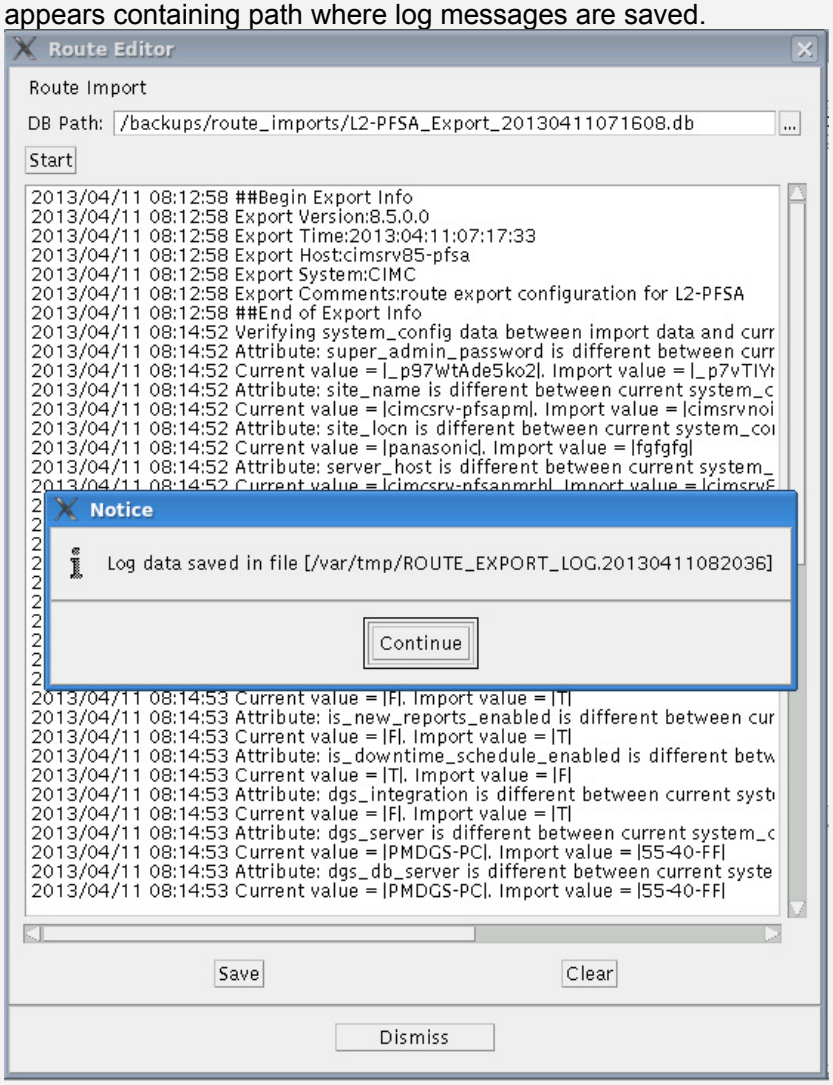
4	After verifying that exported info is correct, click the **Continue** button. Upon doing so, **Route Import** screen is displayed:



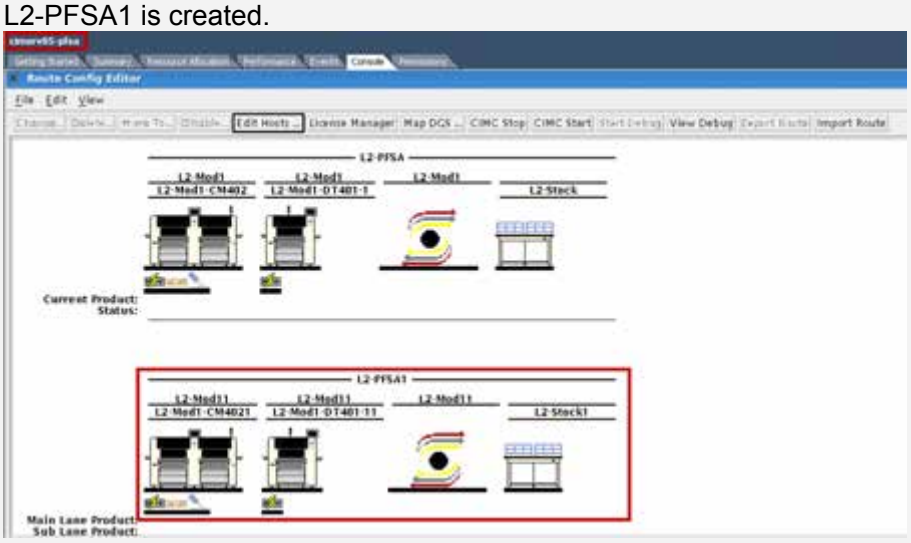
5	Click the **Start** button. Upon doing so, a message appears seeking confirmation to continue import from the specified file: 

6	Click the **Continue** button in the above shown message. **Note:** Upon doing so, a message appears in case global settings data present in **system_config** dialog of imported data is different from the one in current CIMC server into which route import is being carried out. 
7	Do one of the following depending upon your choice: • Click the **Yes** button to import system_config data from the database file being imported. This will overwrite the system_config data in the CIMC server into which import is carried out when same system config settings also exist in the file being imported. • Click the **No** button to preserve system config data in the CIMC server into which import is carried out.

8	After clicking a button in the message shown in step 5, import of route configuration data is started and messages pertaining to import are shown in the box. 
9	In case of successful import, a message appears at last confirming this.

10	List of log messages as part of route import can be saved to a file by clicking the **Save** button. Upon clicking the **Save** button, a notification message appears containing path where log messages are saved. 
11	Click the **Dismiss** button to close the **Route Import** screen. **Route Config Editor** will now contain the newly imported route along with all its configuration that existed in the imported database file. Note that route name created after import has 1 appended to the route name whose data exists in the database file. 

Note:
Route data from an existing route can also be imported to same CIMC server. As shown below after importing configuration of route L2-PFSA, a new route L2-PFSA1 is created.



4.5.3 Setting the Route - LNB

4.5.4 Setting the Route - PT200

4.5.5 Setting a Zone

4.5.6 Setting the Machine - NPM

4.5.7 Setting the Machine - SP

4.5.8 Setting the Machine - CM

4.5.9 Setting the Module - LNB

4.5.10 Setting the Module - PT200

4.6 Configuration License

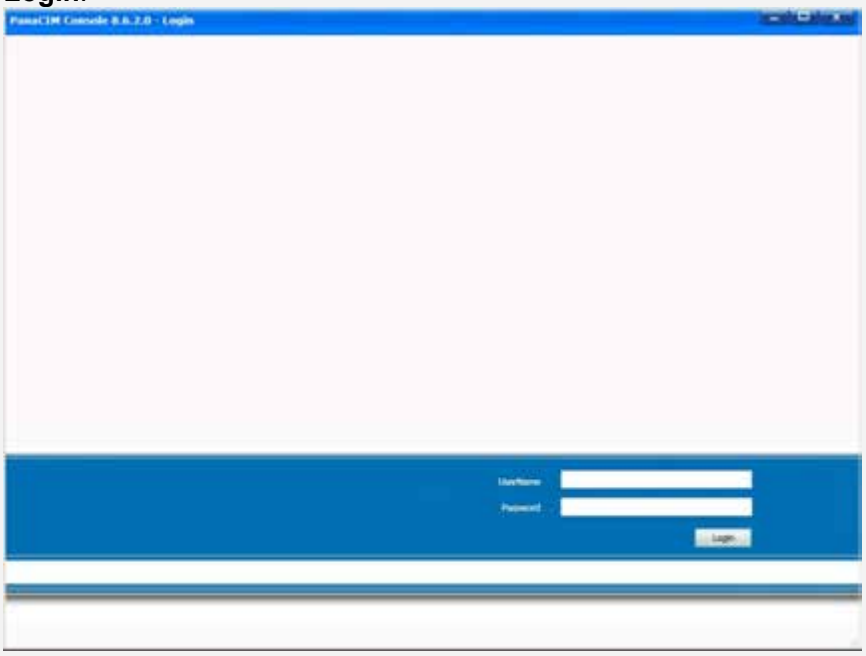
Before using the PanaCIM system, enter the license information via the PanaCIM Console Application.

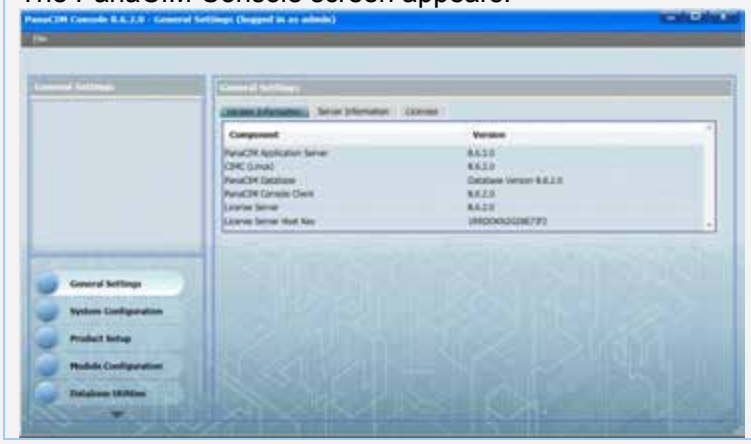
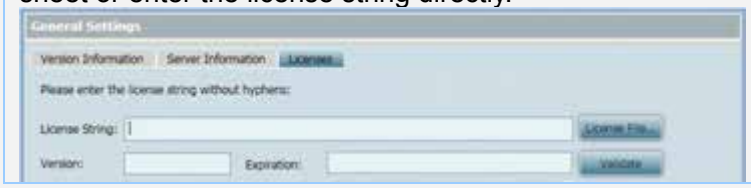
Prepare the following in advance:

1. License string sheet
   (Electronic file contained in the included license CD.)
2. USB dongle
   (Required for license authentication. Supplied with the PanaCIM system.)

Note: If you have lost the USB dongle, order it with the product code: N610120980AA.

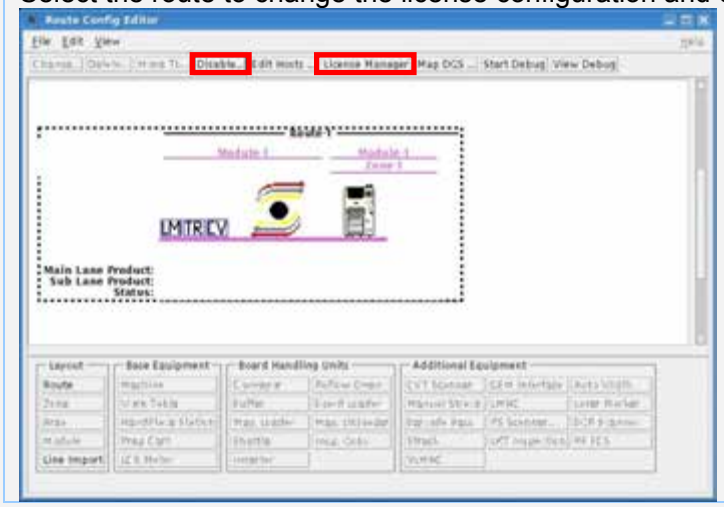
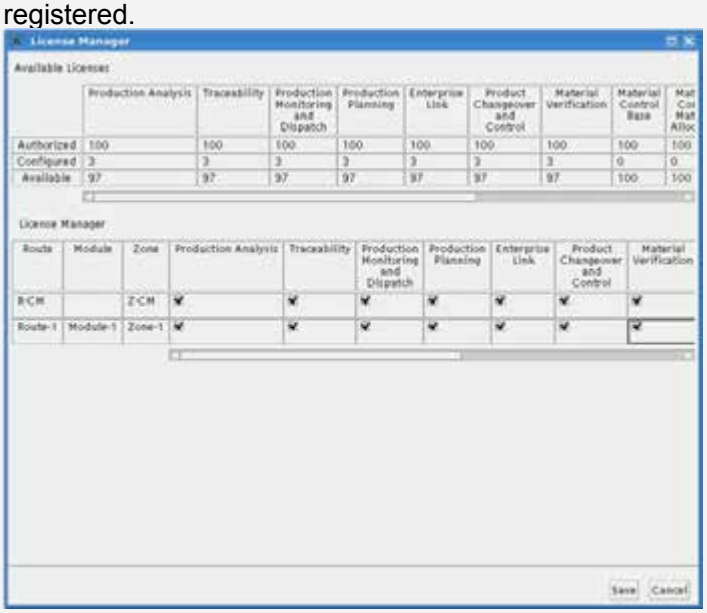
■ Registering license string

Step	Procedure
1	Insert the USB dongle into a USB port of the PanaCIM Application Server computer. Then, start the Windows Server, followed by the PanaCIM Application Server.
2	Launch the Red Hat Linux and run the PanaCIM Controller Server.
3	From the Programs menu, select **Panasonic** > **PanaCIM** > **PanaCIM Services** > **PanaCIM Console**.
4	The Login screen appears. Enter your **UserName** and **Password,** and then click **Login**.  Default account User Name: **admin** Password: **pass** **Note:** To change the default account, select **Security Manager** > **User** > **admin**. Then click **Change User** on the toolbar and enter a new password.

5	The PanaCIM Console screen appears. 
6	Click the **General Settings** tab. Three tabs: **Version Information**, **Server Information** and **License** are shown under **General Settings**. 
7	Click the **License** tab, followed by the **License File** button to open the License String sheet or enter the license string directly. 
8	Make sure the USB dongle has been recognized by Windows, and click the **Validate** button. **Version number**, **Expiration date**, and **Available licenses** are shown.  **Note:** If license aucentication fails due to an error, check the following again, and restart the License Service from the Service Controller (Described in 6.3). • PanaCIM Controller is running. • USB dongle has been recognized. • Correct license string has been entered. • SQL Server is running.

■ Allocating Licenses

Next, allocate licenses to the registered machines.

Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Application Launcher toolbar. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.
4	Refer to the Available Licenses section of this page to determine how many licenses you have available for use in the factory for each Module. *For iLNB, existing license of PMD or PA for the third party's machine can be registered.  Notes: 1) Currently enabled zones are shown in green on the Route Editor. To modify the license setting of a zone, disable the zone. 2) Authorized licenses are not displayed if PanaCIM Application Service (MES, License Service, etc.) is not running. Check the Service Controller (described in 6.3) and make sure the PanaCIM Service is running.
5	Select the disabled route and enable it.
6	Click **Save** to exit the License Manager.

4.7 Enabling the Route

4.7.1 How to Enable and Disable Route/Zone/Module/Area

Follow the steps below to change setting on Route Setting Editor.

- Disable the target Route/Zone/Module/Area from tool bar.
- You can add or delete Zone/Module/Area only when Route is disabled.
- You can add or delete equipment only when zone is disabled.
- If equipment is added in a module, you cannot delete equipment/module.
  To delete equipment, you need to remove the equipment from module.
- You can edit equipment only when Zone/Area is disabled.
- Data will not be collected when Route/Zone/Module/Area is disabled.
- After you delete Route/Zone/Module/Area/Equipment once, you cannot access to data in database including history data.

Step	Procedure
1	Select Route/Zone/Module/Area to disable.
2	Click **Disable**.
3	Click **Next** and continue. Or click **Cancel** and end.
4	To enable again, select Route/Zone/Module/Area from toolbar and click **Enable**.

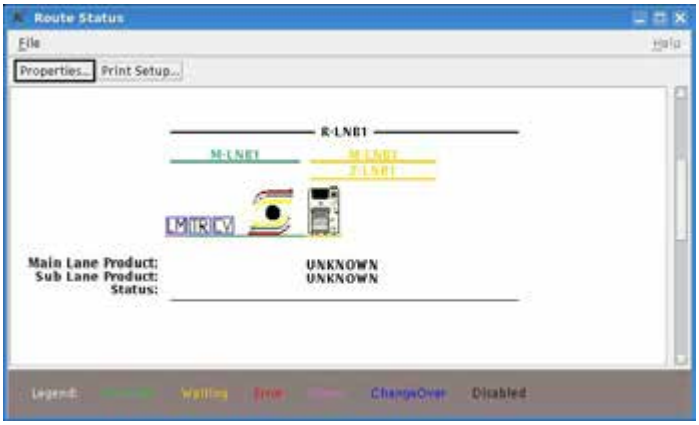
4.8 Confirming the Route Status

The Route Status Application graphically displays the status of items contained within a Route being monitored by PanaCIM.
Select **Route Status** from the PanaCIM toolbar.



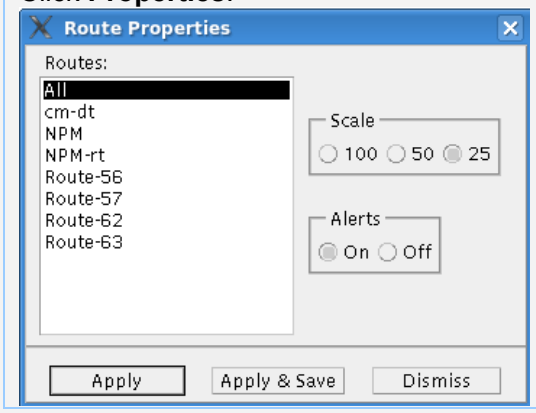
Features of the Route Status Application include:

- Route names are displayed above the route.
- Icons represent the Panasonic NM, Panasonic KX/Universal machines, Conveyors, and Ancillary equipment.
- The status of the line communicating with PanaCIM is updated in real time.
- A color coded key is displayed at the top of the Zones that depict the status of each Zone. The legend beneath the Equipment represents the status of that piece of equipment.
- Click within a Zone in Route Status to display information about each Zone, Ancillary equipment, and Placement machine. The Zone Popup is divided into various tabs, depending on the equipment and the licensing for the zone.



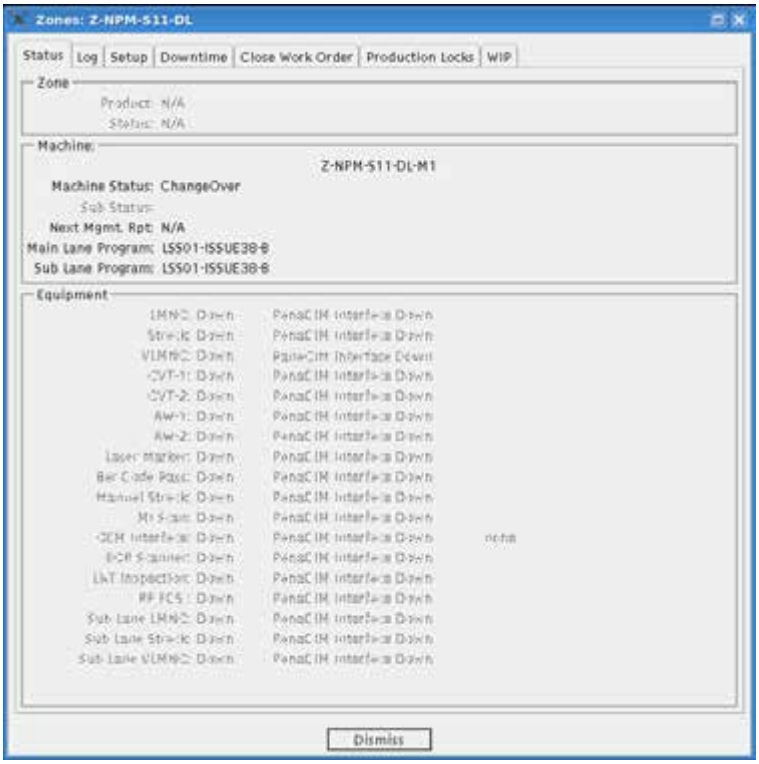
4.8.1 Route Property

Refer to the action table below to configure the way route properties are displayed.

Step	Procedure
1	Select **Route Status** from the PanaCIM Main Menu.
2	Click **Properties**. 
3	Select which routes are displayed on the Route Status screen. Select the **route name**, or select **All** in the **Routes** scrolling list.
4	The **Scale** setting allows for changing the size of the icons on the Route Status screen. Select one of the following from the **Scale** choice list: • **100%** to display the icons at full size • **50%** to display the icons at half size • **25%** to display the icons at one quarter size
	The **Alerts** setting allows PanaCIM to signal when it detects an error in a route, zone or piece of equipment. The Route Status window will be raised to the front and the system bell will beep. Select one of the following from the **Alert** choice list: • **On**. Sets the Alert feature. • **Off**. The system will not signal, but it will still log any errors in the Log Section of the Zone Popup.
5	Click **Apply** to reinitialize the Route Status window, and display the route(s) selected, or click **Apply and Save** to save the desired routes, size and alert settings. These will be the default settings used when the Application is run on that workstation.

4.8.2 Zone Popup Window

If you click an area in zone on <Route Status> screen, the information of the zone appears on <Zone> screen.



Tabs appear on <Zone> screen vary depending on the license assigned to equipment/zone.

- Status of zones (Zone/devices/equipment) and received messages are shown.
- [Status] and [Log] tag appear for all zones. [Status] tab is divided into three sections: Zone, Machine, and Equipment (peripheral). It shows producing board, production order, zone status, machine status, production program and next management report information of the zone.

4.8.3 Display and Print of Log Message

The Log Tab displays a list of the last 20 events that occurred within the zone.

Refer to the action table below to display and print log messages.

Step	Procedure
1	Select **Route Status** from the PanaCIM Main Menu.
2	Click within a Zone to invoke the **Zone Popup**.
3	Select the **Log** tab. The Log Section shows any messages in the window.
4	To Print a hard copy of the Log messages, click **Print** and a copy of the log will be sent to the default printer.

4.8.4 Display Communication Status on Route Status / Route Editor

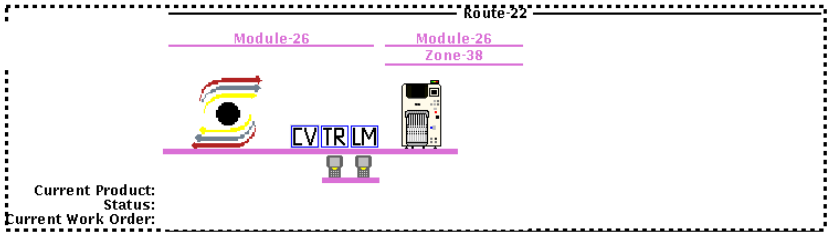
The LNB Communication status of the CV, TR, and LM on NPM Route is shown in green when in normal and in pink in case of failure.

Causes of communication failure (shown in pink) are as follows:

The CV will show down for the following reasons - Interface is not running and Communication is broken between lnb_interface and LNB.

The TR will show down for the following reasons -Interface is not running and Interface is not able to mount LNB file systems.

The LM will show down for the following reasons- Interface is not running and Communication is broken between lnb_interface and LNB.



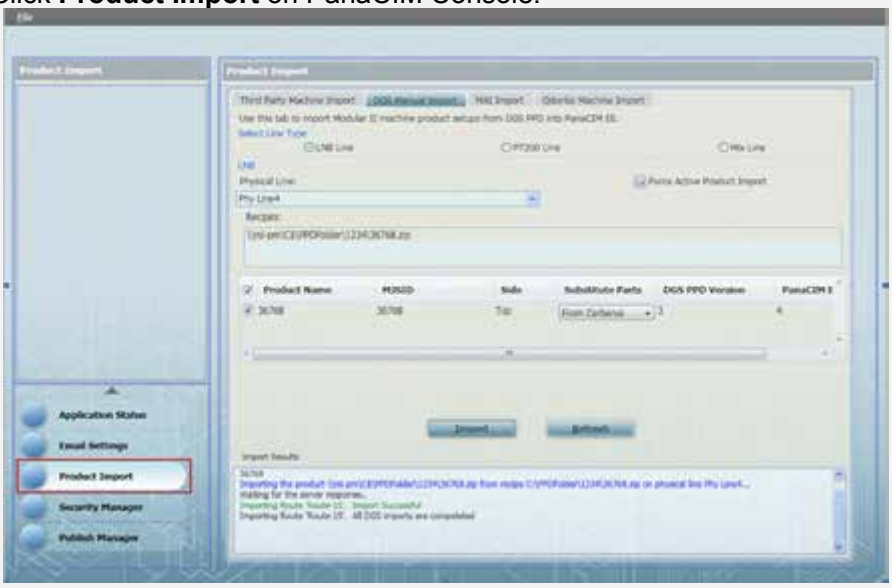
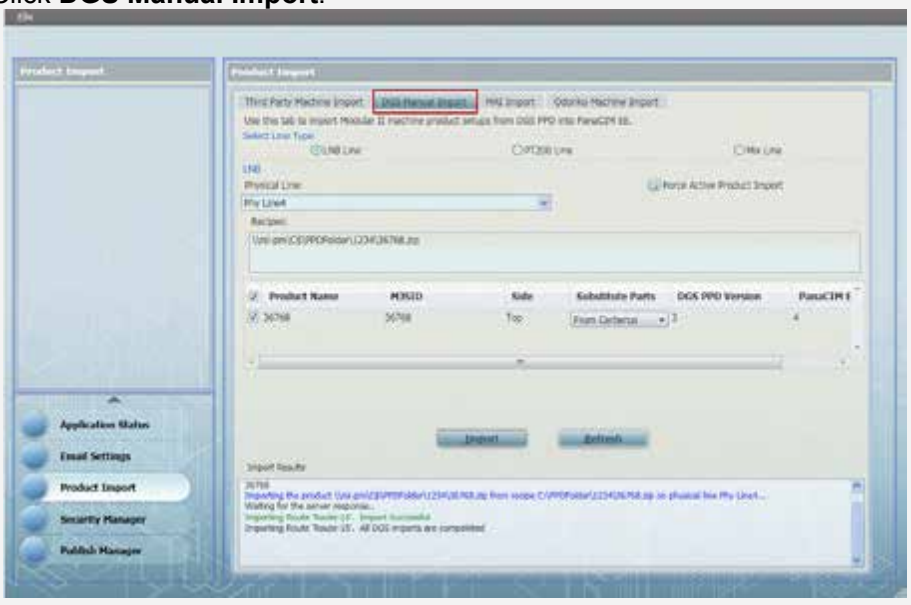
5. OPERATING SETTING

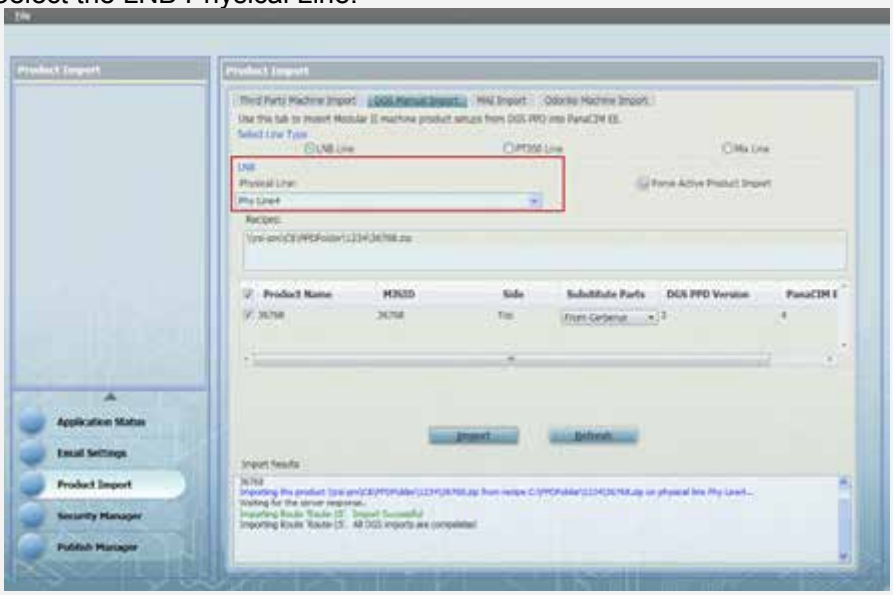
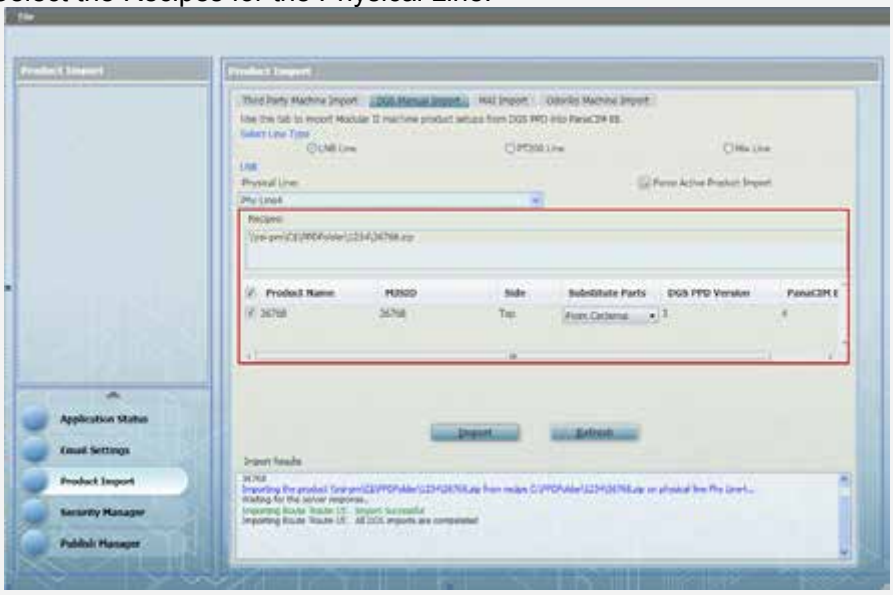
This chapter explains Operating Setting.

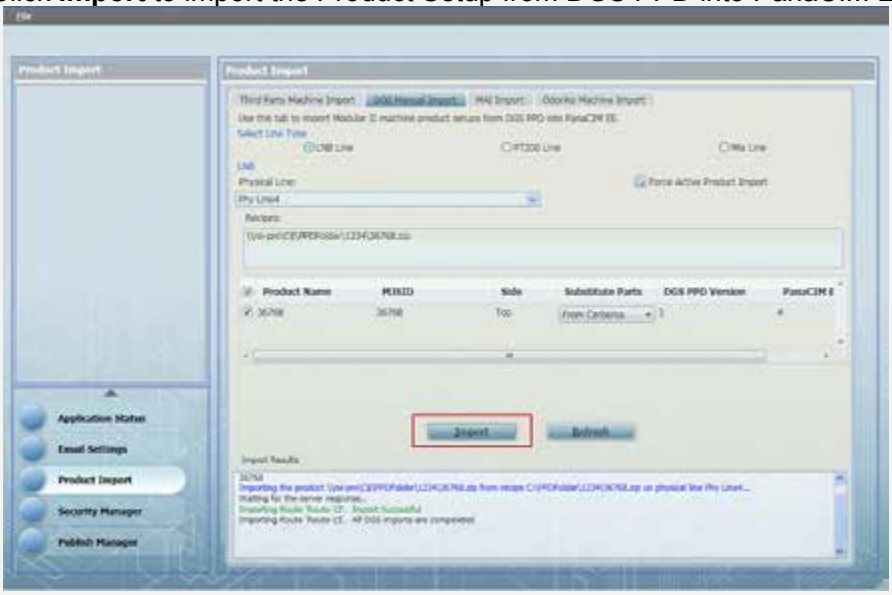
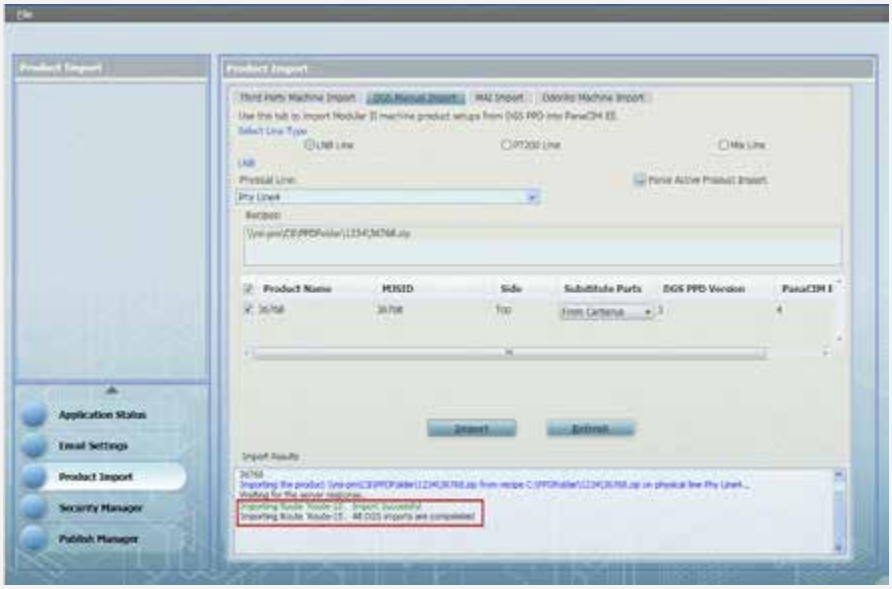
5.1 Product Setup

5.1.1 Importing from DGS

The DGS Manual Import on PanaCIM Console is used to import NPM machine Product Setup from DGS PPD into PanaCIM EE.

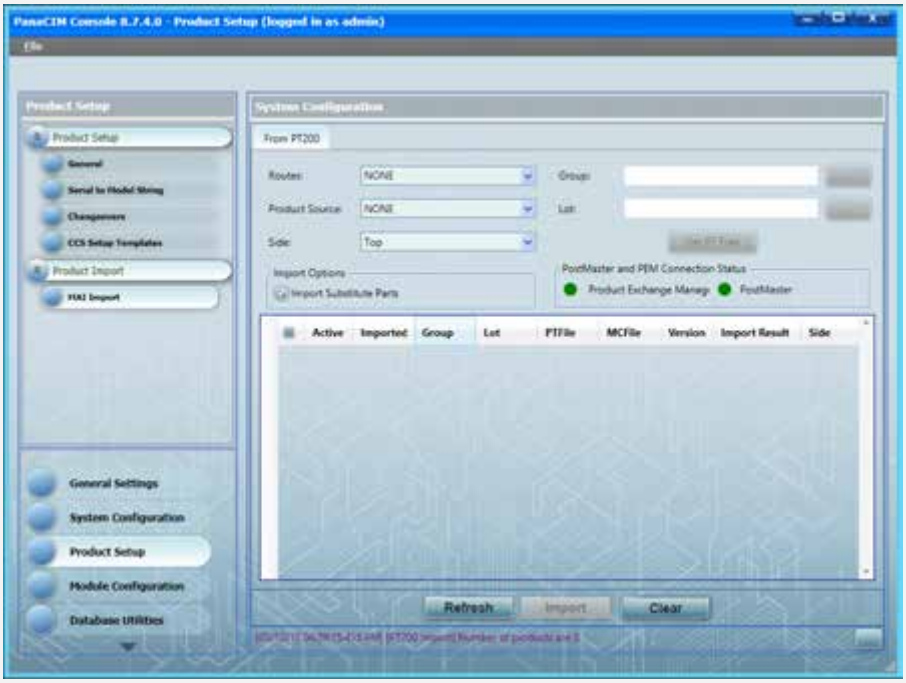
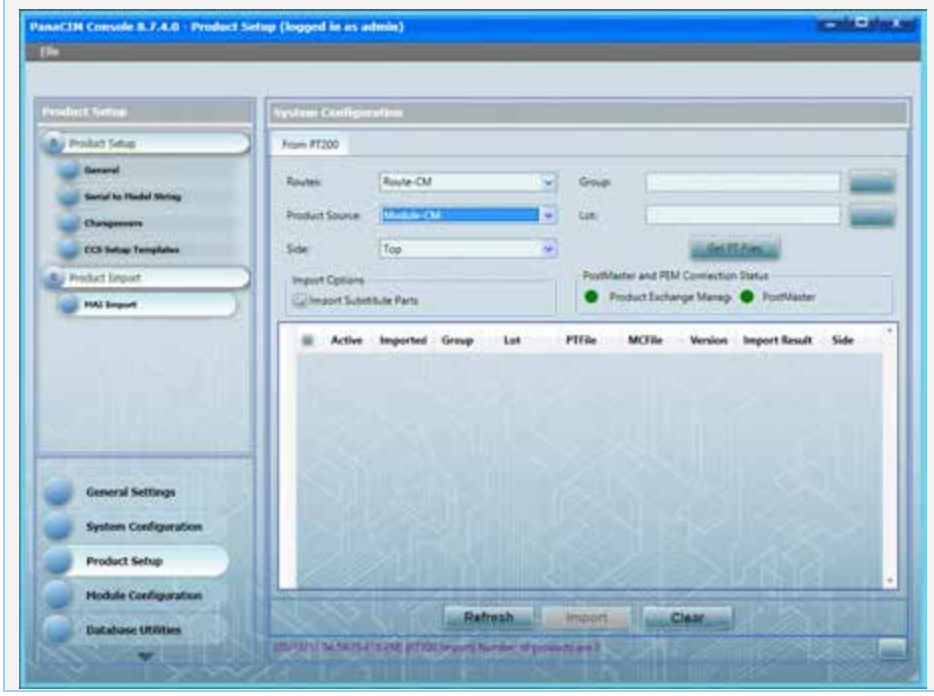
Step	Procedure
1	Click **Product Import** on PanaCIM Console. 
2	Click **DGS Manual Import**. 

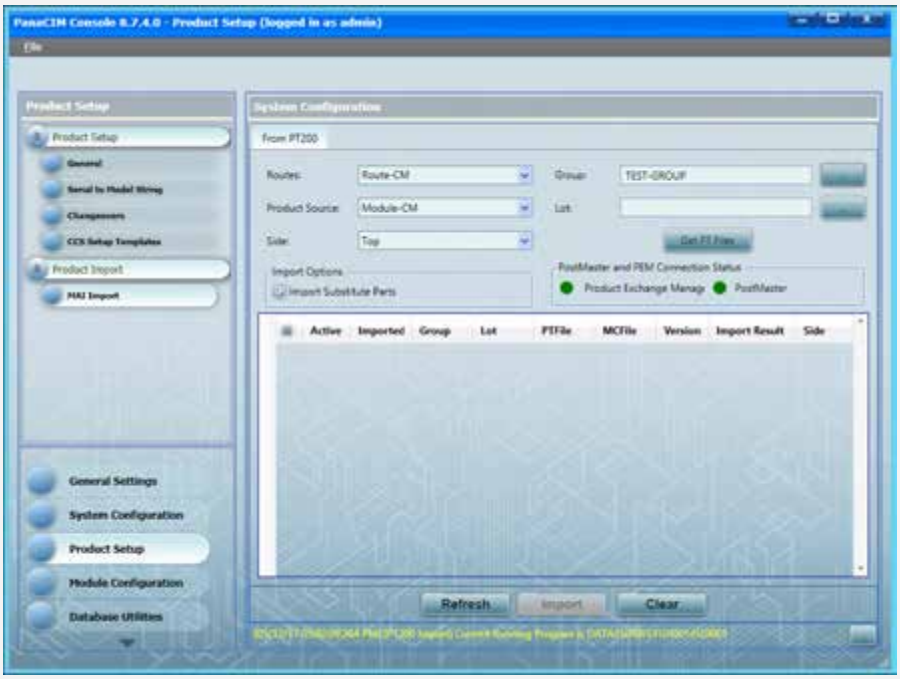
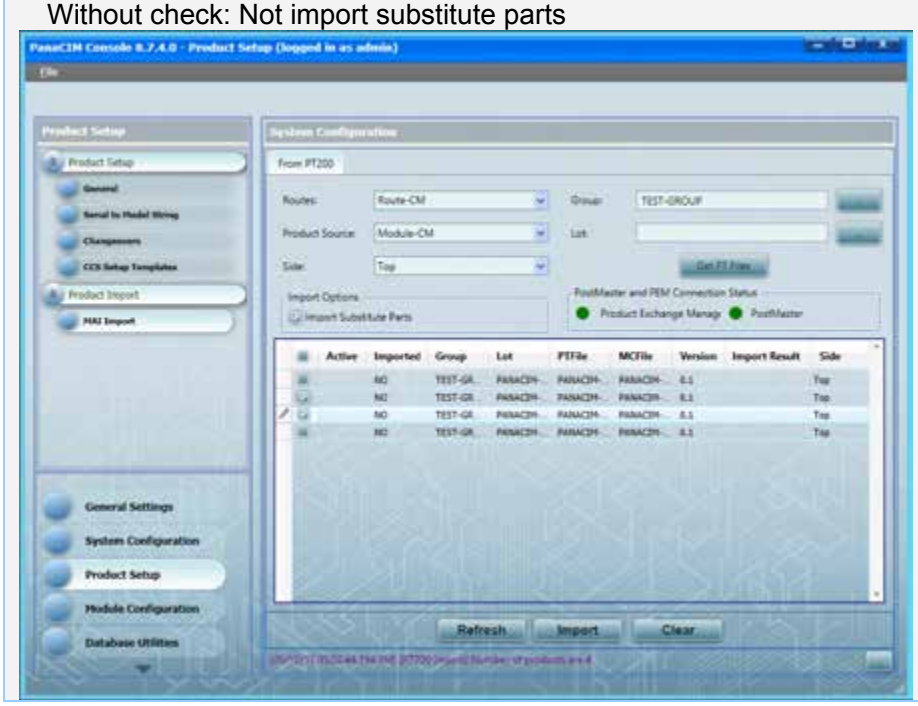
3	Select the LNB Physical Line. 
4	Select the Recipes for the Physical Line. 

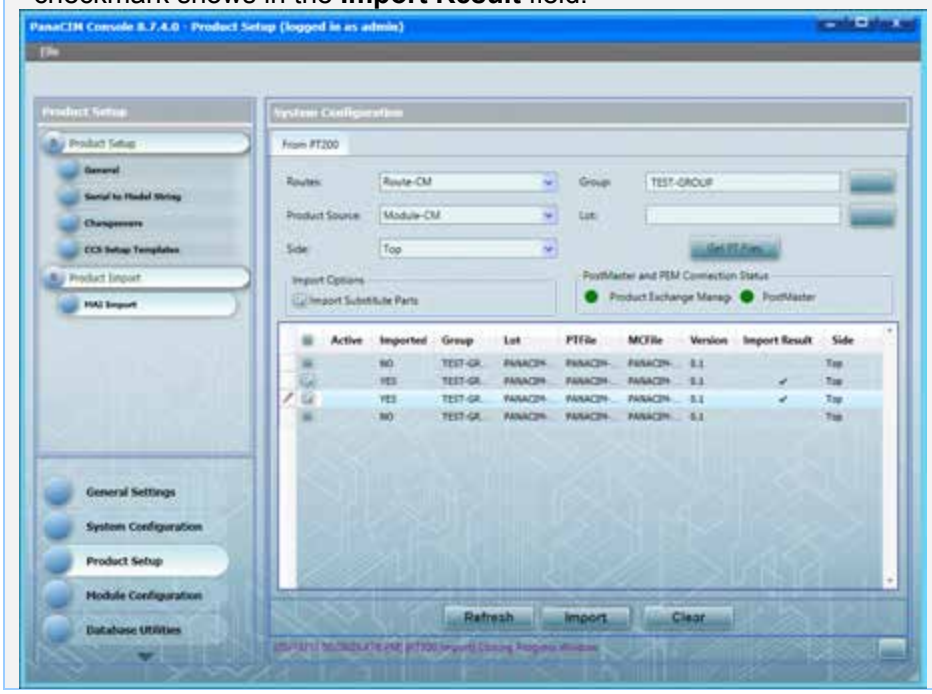
5	Click **Import** to import the Product Setup from DGS PPD into PanaCIM EE. 
6	Check the Import Result to verify the successful import of the Product Setup from DGS PPD into PanaCIM EE. 

5.1.2 Importing from PT200

The product setup for CM series can be imported to PanaCIM EE from the PT200 file system using MAI import on the PanaCIM console.

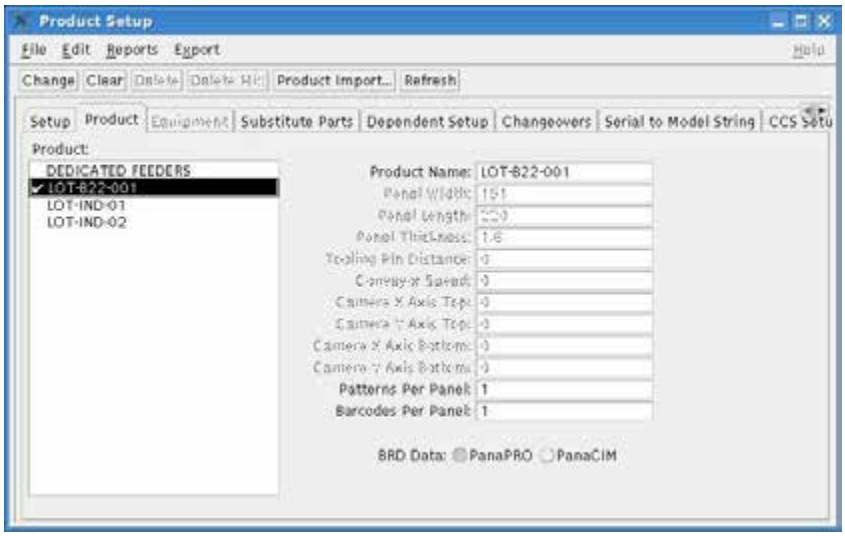
Step	Procedure
1	On the PanaCIM Console, select **Product Setup**->**Product Import**->**MAI Import**. 
2	Select **Routes** and **Product Source** from the drop-down lists. 

3	Specify '**Group**' or '**Group and Lot**' as necessary, and click **Get PT Files** to get a PT file list. 
4	Select whether substitute parts are imported or not by marking the checkbox (either single or multiple checkboxes) of program(s). Then click **Import**. Import Options [Import Substitute Parts] With check: Import substitute parts Without check: Not import substitute parts 

5	After substitute parts are successfully imported, **YES** shows in the **Imported** field. Also, checkmark shows in the **Import Result** field. 

5.1.3 Configure Panel Parameters

Select the **Product** tab in Product Setup Application to configure panel parameters.



Product Name
After creating a product in PanaPRO, the Product Name in PanaCIM will be the same as the **PCB file** name from PanaPRO. For easy identification, provide a descriptive name for the product to identify the product for PanaCIM.
Note: This optional change will only need to be made the first time a product is processed by PanaPRO. Subsequent runs of Automode will not modify this data.

Panel Parameters
Panel Parameters describe the physical characteristics of panels used to build the product. PanaCIM Product Changeover and Control uses these parameters to control aspects of the line, including conveyors and Autowidth Controllers. **Note**: A value of zero for Panel Width, Panel Length, Panel Thickness, or Tooling Pin Distance disables any automatic adjustments (for the selected field) for that product.

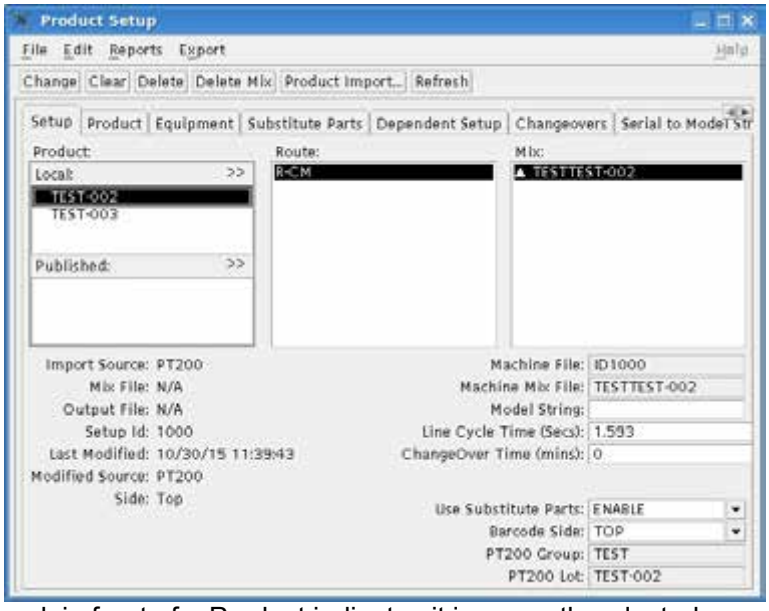
Refer to the action table below to configure panel parameters.

Step	Procedure
1	Click the product corresponding to the file name from the **Product** scrolling list. The same name appears in the **Product Name** text entry.
2	Optionally click the **Product Name** text entry and edit the name of the product.
3	Next, complete the remaining fields to configure the panel parameters according to the following: **Panel Width; Panel Length; Panel Thickness:** Value in millimeters. **Tooling Pin Distance:** The distance between tooling holes. **Patterns Per Panel:** Show the number of patterns to be produced on a panel. **Barcodes Per Panel:** Shows the number of barcodes to be scanned on a panel. Set 0 if the panel has no barcode. **BRD Data:** To use the PanaPRO imported BRD files or PanaCIM generated BRD files for BM series machines only. • **PanaPRO**. Editing of the panel dimensions will not be allowed since they will be updated upon subsequent imports. • **PanaCIM**. Values will be made available for editing and used to generate the BRD file.

	Note: The Conveyor Speed field is used by PanaCIM to create a PCB file for download to certain machines if the “BRD Data” radio button is set to PanaCIM, and the machine supports download of PCB files. The Panel Width, Panel Length, Panel Thickness, and Tooling Pin Distance are all used for this purpose as well. The Camera Axis fields are used during product changeover to automatically adjust an Autowidth Controller for a new PCB, if the controller has a barcode camera. The Panel Width is used for this purpose as well.
4	Click **Change**.

5.1.4 Delete a Product and the Product Setup

Select the **Product** tab in Product Setup Application to configure product name and various parameters.



Note: A checkmark in front of a Product indicates it is currently selected on one or more machines, and cannot be edited.
To build a product, it needs to be set up on a specific route with a specific mix (feeder arrangement). Before transferring a product, the name and the data file required for the product, route, and mix need to be configured.

Refer to the action table below to define the Product Setup.

Step	Procedure
1	Select a **Product**.
2	Select a **Route**.
3	Select **Mix** name. Product setup information
4	Mix File, Output File, Setup Id, Last Modified, Side, and DGS Version are read only attributes. **Note: DGS Version** will only be displayed if the route is a DGS Route. Model String data is used to identify the correct panel for the product; PanaCIM uses the Machine File name to uniquely identify the product setup running in each workcell. Model Strings and Machine File Names must be unique to the route. **Note:** PT Group String and PT Lot String **must** match those on the PT200 machine. Edit the following: • **Model String**. Guidelines: ○ Must match the Barcode parser. • **PT200 Group String**. Guidelines: ○ This field displays only if the route has CM/DT series machines. ○ This field displays blank by default when the product is imported from PanaPRO and must be set before the product can be downloaded to the PT200 machine. ○ Once a setup has been downloaded to the PT200 machine, the value cannot be changed for the setup from PanaCIM. ○ If the route is set for PT200 Import, this field will be displayed only.

	• **PT200 Lot String**. Guidelines: ○ This field displays only if the route has CM/DT series machines. ○ This field displays blank by default when the product is imported from PanaPRO and must be set before the product can be downloaded to the PT200 machine. ○ Once a setup has been downloaded to the PT200 machine, the value cannot be changed for the setup from PanaCIM. ○ If the route is set for PT200 Import, this field will be displayed only.
5	Click Change to save changes. The **Last Modified time** will be updated.

■ **Tips for searching product setup**

On the areas of Panel - Local, Panel - Published, Route, and Mix, you can search for the entered string (forward match) to find matches in the product, route, and mix file names.

<Product - Local area>

Step	Procedure
1	Select the first panel name shown under **Panel** - **Local**.
2	With the product name highlighted, enter the string to be searched.
3	The entered string is searched for (to find forward match) and the cursor moves to the applicable product name containing a matched string.

<Product - Published area>

Step	Procedure
1	Select the first panel name shown under **Panel** - **Published**.
2	With the product name highlighted, enter the string to be searched.
3	The entered string is searched for (to find forward match) and the cursor moves to the applicable product name containing a matched string.

<Route area>

Step	Procedure
1	Select the first route name shown under **Route**.
2	With the route name highlighted, enter the string be searched.
3	The entered string is searched for (to find forward match) and the cursor moves to the applicable route name containing a matched string.

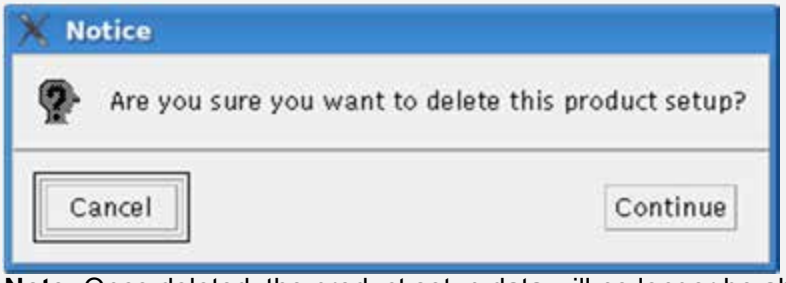
<Mix area>

Step	Procedure
1	Select the first route name shown under **Mix**.
2	With the mix file name highlighted, enter the string to be searched.
3	The entered string is searched for (to find forward match) and the cursor moves to the applicable mix file name containing a matched string.

5.1.5 Deleting Panel/Product Setup

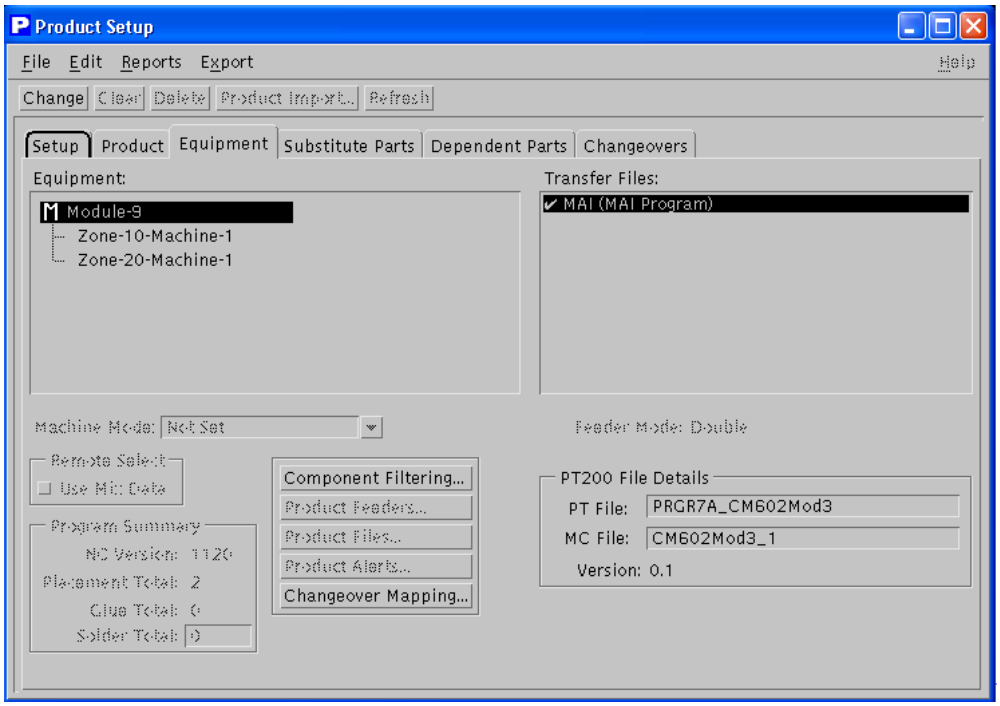
Existing data can be deleted from the product setup according to the following guidelines:

- Once panel data has been deleted from the setup information it is necessary to perform optimization via PanaPRO.
- Once a product setup has been deleted, it is necessary to perform optimization again via PanaPRO.
- Once the product setup in the route or mix has been deleted, the panel data specified in the route will no longer be available.
- Before deleting panel data, delete all setup information including the panel.

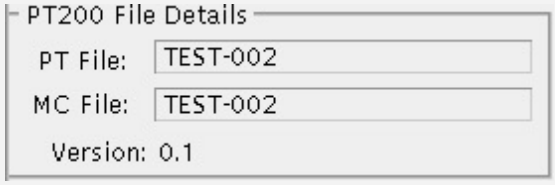
Step	Procedure
1	Under the **Setup** tab, select the product setup to be deleted.
2	Click **Delete**.
3	The dialog shown below appears. Click **Continue** to delete the setup.  **Note:** Once deleted, the product setup data will no longer be able to be obtained from the Product Transfer application.
4	To delete programs, click the **Panel** tab.
5	Click the desired panel name in the **Panel** list.
6	Click **Delete**.
7	A confirmation dialog appears. Click **Continue** to delete the panel.
8	The corresponding program name will be deleted from the program list.

5.1.6 Configuring Equipment

Selecting a product setup under the **Setup** tab enables the **Equipment** tab.



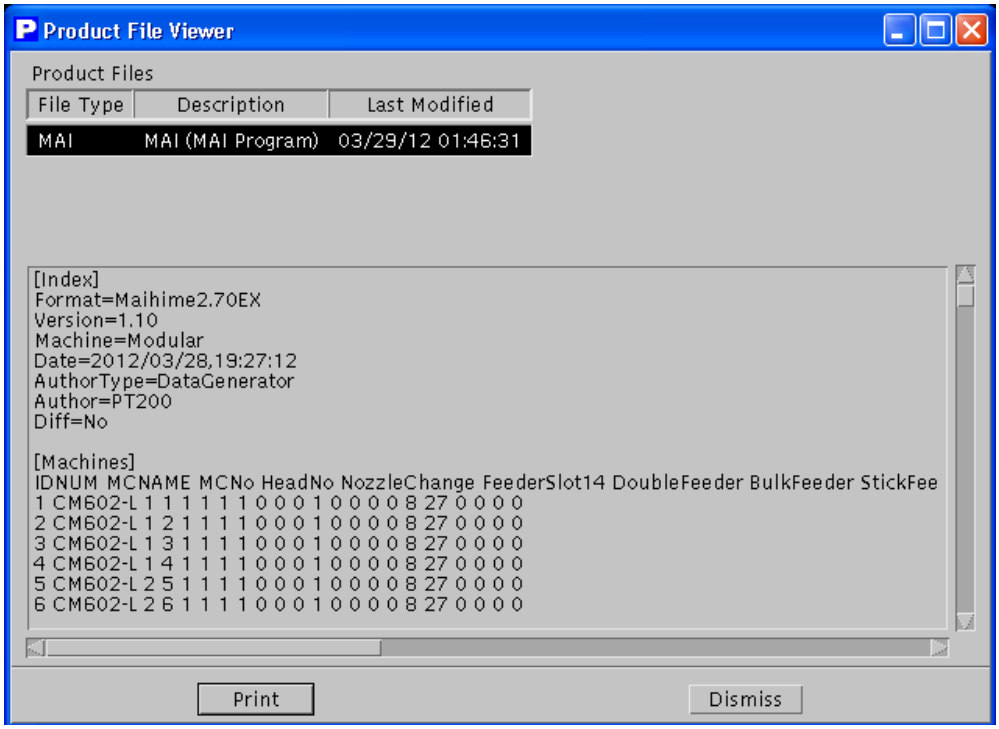
You can configure the files to transmit to particular workcells on the route. This configuration keeps PanaCIM from sending unnecessary files to the workcells.

Step	Procedure
1	Select the necessary product setup from the **Setup** tab, and select the **Equipment** tab
2	Select a workcell from the **Equipment** scrolling list.
3	Select the files to be transferred from the **Transfer Files** scrolling list.
4	Edit the **PT200 File Details** field. Guidelines:  • This field appears, and the value can be changed only if a CM/DT machine is selected. • Once a setup has been downloaded to the PT200 machine, the value cannot be changed for the setup from PanaCIM. • If the route is set for PT200 Import, this field will be displayed only.
5	Click the **Change** button.
6	Repeat this process for any other workcells that need a different transfer file setup.

Product File Viewer

Viewing the product files for a workcell can be useful when you would like to check an XY coordinate in a POS file, or Part Number in an Array file. All supported types of files imported to PanaCIM will be displayed.

After selecting the appropriate equipment, click the **Product Files** button to invoke the Product File Viewer.



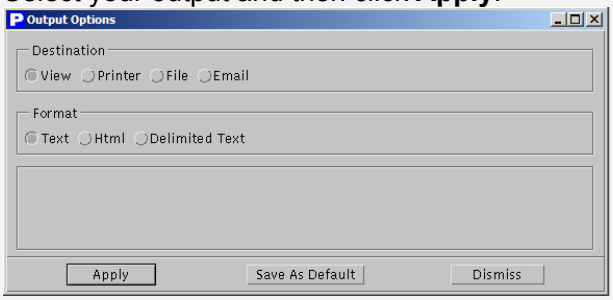
Step	Procedure
1	Select the file type you wish to view (POS, SET, PRT, etc).
2	Data of the selected file will be displayed as shown above.
3	Click **Print** to print the information.
4	Click **Dismiss** to close the Viewer.

5.1.7 Reports

The following reports can be run from the Product Setup Application. Click the Reports button to view the several reports available.

- **Route-Mix-Product Relation Report**. This report allows the user to specify a sort order of Routes, Mixes, and Products to generate a report of which products and mixes are available on which routes.
- **Active Files Report**. This report will generate a summary list of all the machine files configured for a product.
- **Product Report.** This report provided a summary of the data on the Product Tab.

Refer to the action table below to run reports directly from the Application.

Step	Procedure
1	Select the report from the **Reports** menu.
2	Select your output and then click **Apply**.  **Note:** Reports by **Email** are currently not supported.

5.2 Configuring Machine Locking

The user will use the System Configuration application to indicate which lock types are tracked by the system. When the system configuration changes, each affected interface refreshes its lock configuration and begins or ends lock condition tracking accordingly.

Lock types could be enabled or disabled using **System Config** dialog. Only enabled lock types would be considered for locking the machine. For example, if 'missing traceability data' is enabled in System Config dialog and PanaCIM could not acquire the traceability data for any reason, then this would constitute a lock condition and machine would be locked. But if 'missing traceability data' is disabled, even though PanaCIM fails to acquire the traceability data, the machine will not be locked.

The following lock types are fixed and enabled in the [Machine Locking] tab.

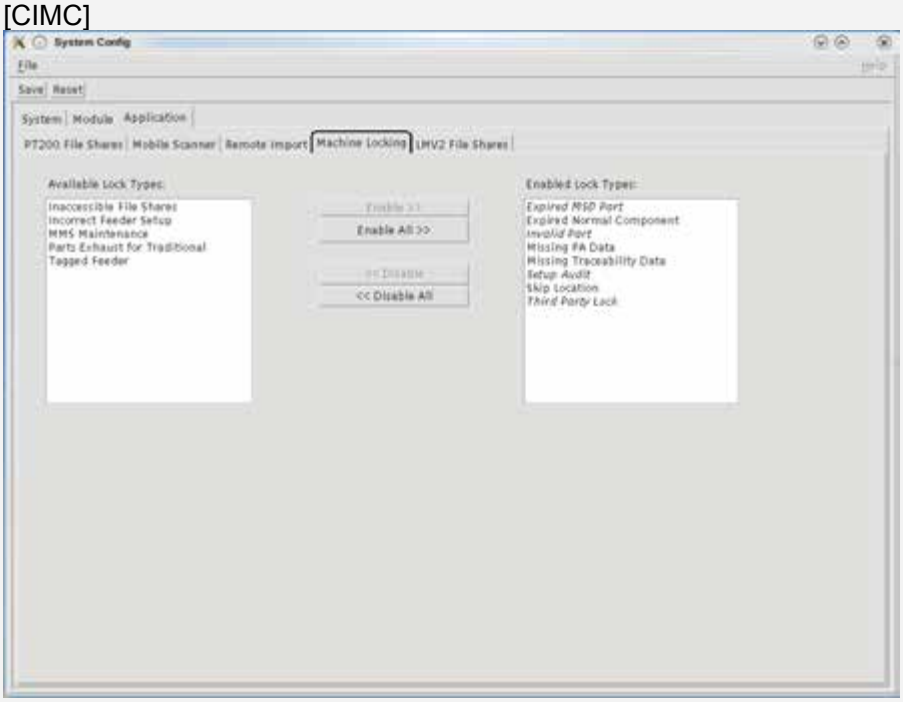
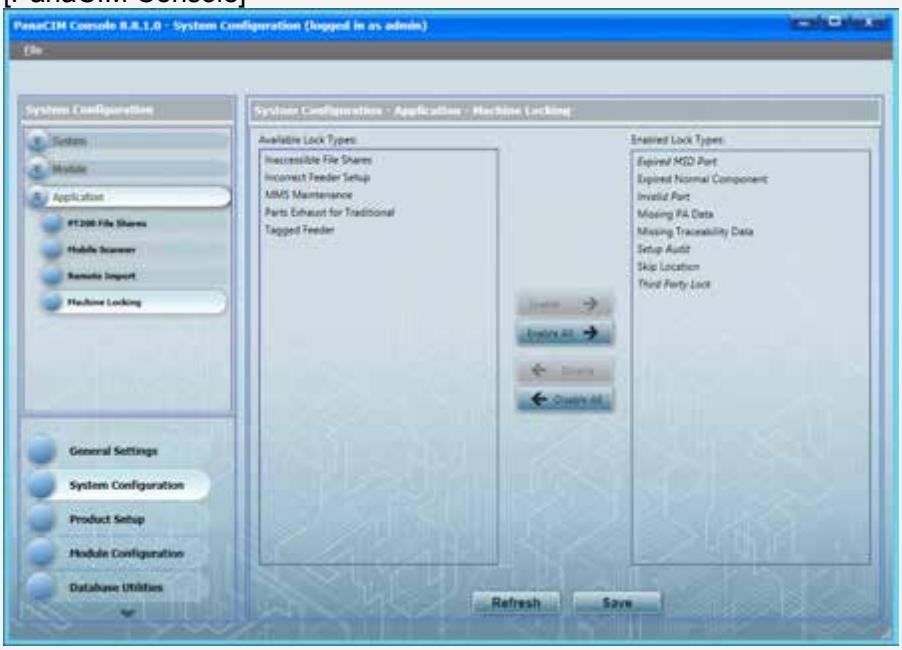
- **Third Party Lock:** This will constitute a lock condition of the third party's machine.
- **Invalid Part:** This will constitute a lock condition if a part locates at the incorrect feeder position, that is, the part exists in the position where it is not specified by the downloaded program.
- **Setup Audit:** This will constitute a lock condition if the setup audit has not been completed within the predetermined period of time, or an abnormality occurs twice when a certain feeder position is confirmed.
- **Expired MSD Part:** This will constitute a lock condition if the expired MSD part reel is mounted.
- **Expired Normal Part:** If a reel with expired parts is installed, the machine is locked according to the setting.

It is to be noted that following lock types can be enabled / disabled using Machine locking tab.

- **Tagged Feeder:** This will constitute a lock condition, when feeder tagged for maintenance, is mounted on a machine.
- **Missing Traceability Data:** This will constitute a lock condition if traceability data is not generated for a machine.
- **Incorrect Feeder Setup:** This will constitute a lock condition if one or more incorrect feeders are mounted for a product.
- **Missing PA Data:** Machine will be locked if Production Analytics data is not generated for a machine.
- **Inaccessible File Shares:** This will constitute a lock condition if access to the shared folder in LNB or PT200 is denied.
- **Skip Location:** This will constitute a lock condition if a part that the setup is skipped due to a stock component exists.
- **Parts Exhaust for Traditional:** *Out of target

Refer to the action table below to enable/disable lock types.

Step	Procedure
1	Open **System Config** dialog by clicking **System Cfg** in the PanaCIM Launcher bar or PanaCIM Console.
2	Activate **Machine Locking** tab within **Application** tab of **system config** dialog. Following screen appears:

	[CIMC]  [PanaCIM Console] 
3	Under **Available Lock types** list, select one of the lock types which needs to be enabled and click the **Enable >>** button. Selected lock type is populated in the **Enabled Lock Types** list. To enable all the lock types, click the **Enable All** button.
4	If all the lock types are enabled, they can be disabled by selecting **Disable All** button. To disable one of them, select that lock type from the **Enabled Lock Types** list and click the **Disable** button.
5	Click the **Save** button to save changes or **Refresh** button to undo the changes made and return to default settings. After changing click the **Save** button.
6	The setting will be applied after restarting the PanaCIM(CIMC).

5.3 Production

■ Adding modules (CM/DT/NPM series (including AM100) machines only)

Modules can be added to the CM/DT/NPM series (including AM100) machines or setup carts.
Select a route where you want to ad add a module, and click the Module button to launch the Module Editor.

- **General Setup**
  This page is used to make basic setting of the module.



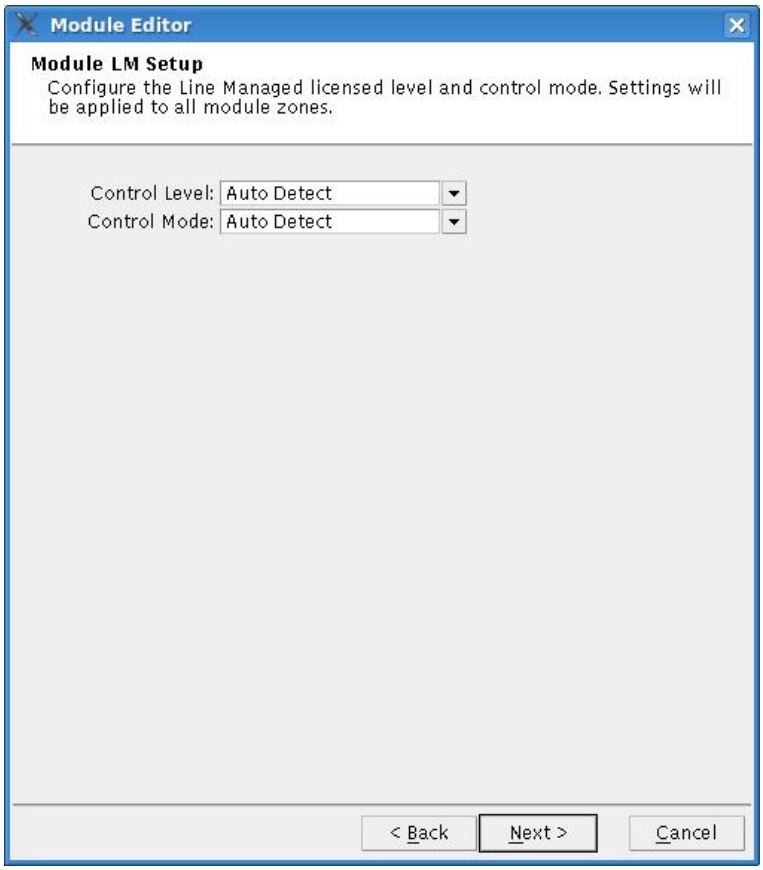


Step	Procedure
1	Type a unique **Module Name**.
2	Select the desired **Module Type** from the drop-down list. • Select PT200 for the CM/DT series machines. • Select LNB for the NPM series (including AM100) machines.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and exit the editor.
4	In order to use the integrated version*1 of verification system, [Enable LNB Integrated Verification] needs to be checked off.

*1: **For information on the integrated version of verification system, refer to 3.1.1 Overview in the instruction manual [Material Verification].**

- **LM Module Setting**

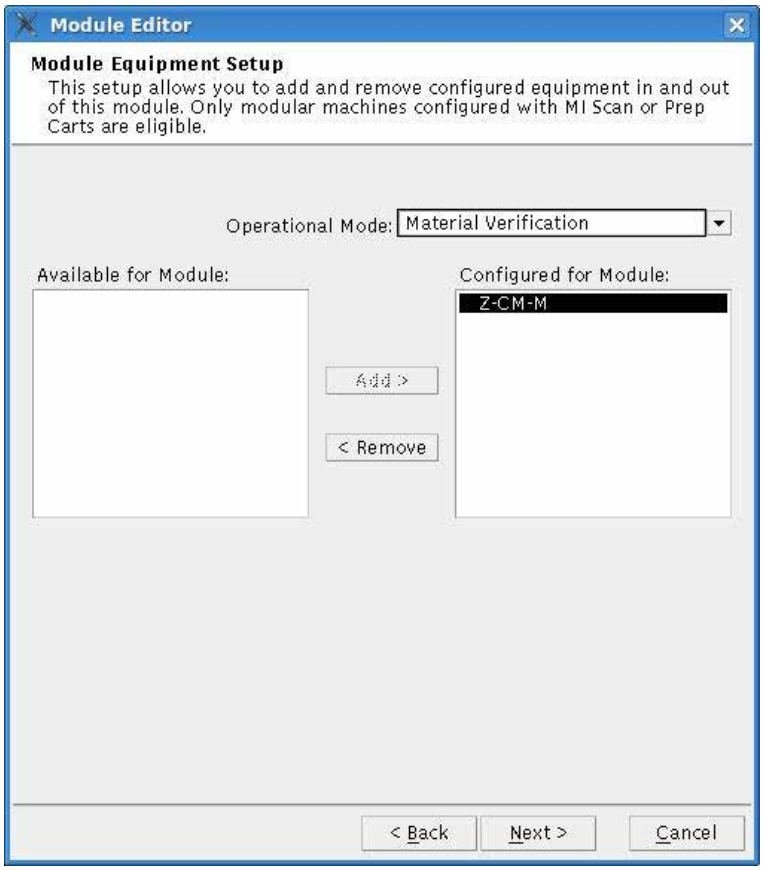
This page is for setting control level and mode for Product Changeover and Control.



Step	Procedure
1	Set **Control Level** and **Control Mode** from the drop-down lists: • **Auto Detect:** Select this option to detect the changeover of product (panel). • **Auto Download**: Select this option to download the applicable product program after the product changeover has been detected. • **Line Managed**: Select this option to start production automatically after product program has been downloaded. **Note:** To execute setup automatically, PanaCIM Product Changeover and Control license is required.
2	Click **Next**.

Module Equipment Setting

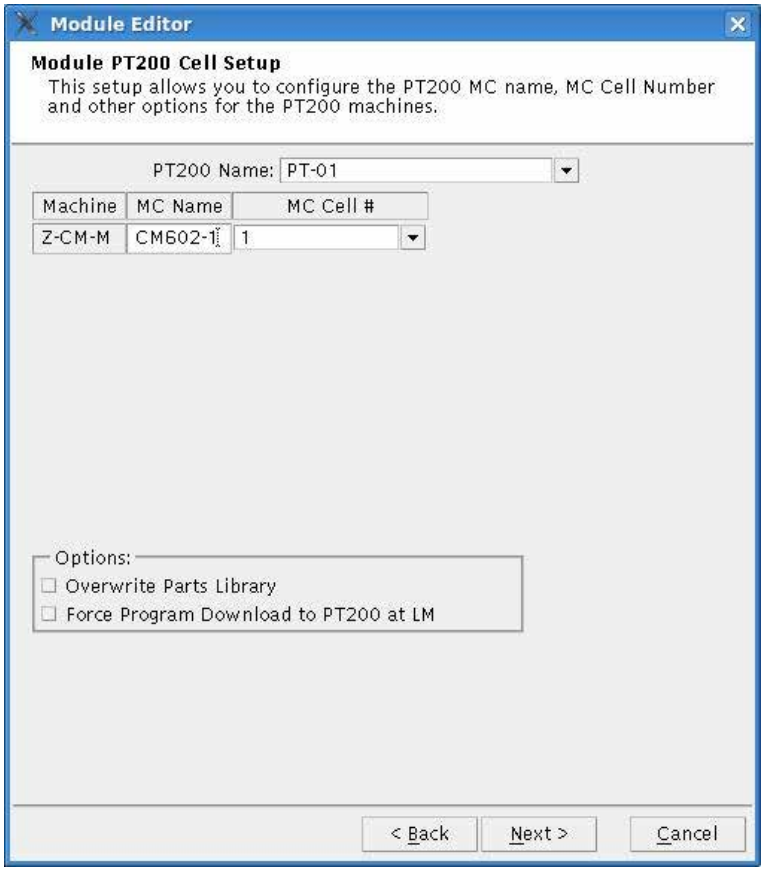
Use this page to add/remove equipment to/from the module.



Step	Procedure
1	Select either of the following **Operation Modes** from the drop-down list. • **Material Verification:** Select this mode when using Material Verification. • **Material Control Base**：Select this mode when using Material Control.
2	Select the machines or carts to be added to the module from the **Available Equipment** list.
3	Click **Add** to add the selected machine or cart to the **Module Configuration** list.
4	To delete machines from the module, select the desired machine from the **Module Configuration** list and click **Remove**.
5	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and exit the editor.

PT200 Machine Setting

For the CM series machines, use this page to set the configuration information on PT200.
Note: The PT200 set in the PanaCIM System Configuration in advance can be selected. Refer to “4.1.2 PT200 File Shares” for details.



Step	Procedure
1	Select the PT200 to be used for controlling the selected CM series machines from the **PT200 Name** drop-down list.
2	Set the **MC Cell #** (machine number) for each machine from the drop-down list according to the setting of PT200.
3	Click **Next** to continue the setting. Click **Cancel** to discard any changes made and exit the editor.

* Option to forcibly download programs during product changeover:
Enables the PanaCIM to forcibly download programs to PT200 and machines at product changeover regardless of whether there are programs on the PT200. This option requires longer time for product changeover, but it is helpful for the users wishing to use the latest product programs stored in the PanaCIM system whenever product is changed over.

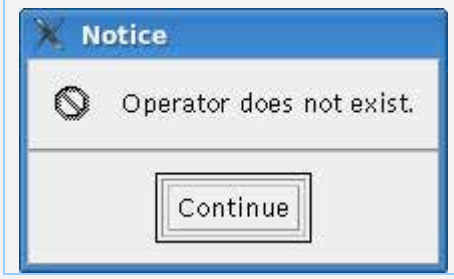
Use the Module Editor to select the PT200 cell. Machine name, MC cell name, MC cell number, and Forced program download from LM to PT200 can be set as well.

■ Password protection against deletion of routes, zones and machines

Machine and equipment in the route and individual or all zones in the route can be deleted. Before route, zone or machine is deleted, a dialog appears, prompting you to enter the password that was entered when installing the system.

* Note that once the routes, zones and/or machines have been deleted, access to the corresponding data including those remaining in the database will be disabled.

Take the following steps to delete routes, zones and machines using the Route Editor.

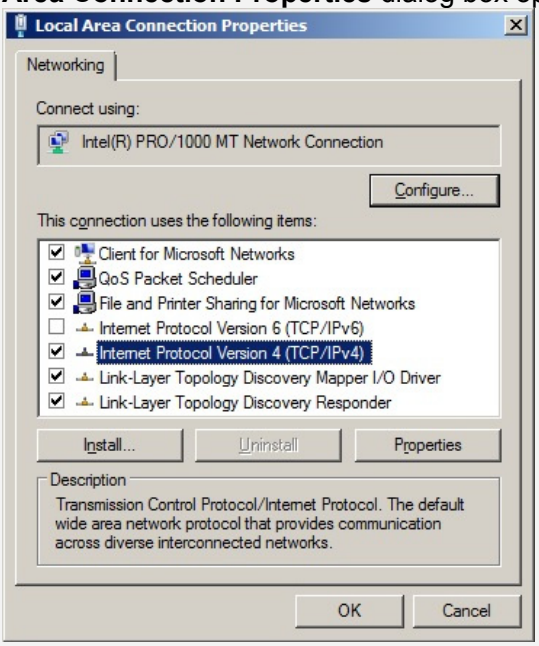
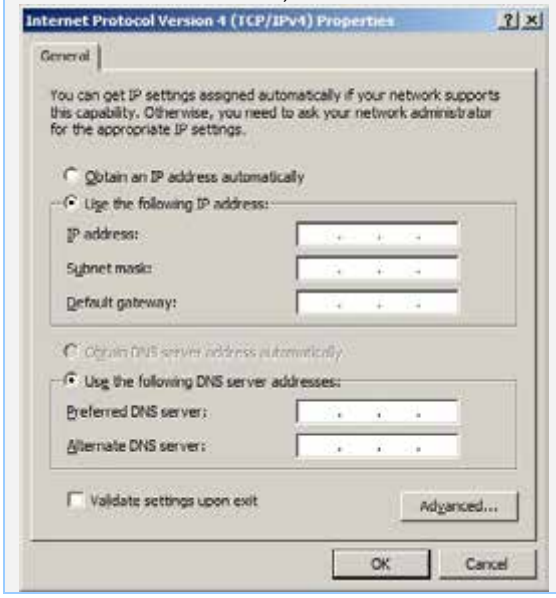
Step	Procedure
1	Open the Route Editor from the PanaCIM main menu.
2	Select the items to be deleted.
3	Click **Delete**.
4	The following dialog pops up to prompt you to enter the Operator ID and Password. 
5	Enter the Operator ID and Password, and click the **Apply** button. The Password required here is the one entered when installing the PanaCIM system. Operator ID refers to the operator barcode shown in the **Barcode** column under [System] → [Operator] tab on the System Configuration dialog. If the entered Operator ID is incorrect, or an operator name is entered instead of barcode, the following error dialog will be displayed. 
6	A confirmation message appears asking you whether or not to delete the selected items. Click **Continue** to delete your selection. Click **No** to cancel it.

a) If the zone is not a part of module, the modules, zones and machine in the zone can be deleted.
b) Clicking the CIMC Start or CIMC Stop button will display the dialog prompting you to enter the password for administrator privilege.

■ Communication between PanaCIM-EE and machines other than CM/DT/NPM series (including AM100)

● Machines using TCP/IP

Set the machine to be added to or removed from the module.
The Internet Protocol Suite (TCP/IP) is the set of communication protocols used on the Internet or similar computer networks.
Make sure the TCP/IP protocol has been installed.

Step	Procedure
1	Make sure the local area network (LAN) has been connected.
2	From the Windows taskbar, open **Network Connection** and make sure the TCP/IP protocol has been installed. Select **Local Area Connection** and right click it. The **Local Area Connection Properties** dialog box opens as shown below.  Select **Internet Protocol Version 4 (TCP/IPv4)** and click the **Properties** button.
3	Enter the **IP address**, **Subnet mask** and **Default gateway** of the machine. 
4	Click **OK**.
5	Restart the machine.

Next, change the machine settings to switch from RS232 to LAN communication.
Setting procedure varies with the machines used, as summarized below:

NM series (MSR, MSF, BM or later) machines

MSF:
1. Login as a level 4 user.
2. Machine Setup > Administrator (Password Input) > Added Options > Communication.
3. Enable Panaprotocol LAN.
4. Restart the machine.
5. Using LAN communication, acquire IP address and host name from the /etc/hosts file in the PanaCIM NIS Server, and input them to each machine.

MSR:
1. Login as a level 4 user.
2. Select Systems > System Parameters > RS232C Parameters.
3. Select **LAN** under the **Port Selection** field.
4. Restart the machine.
5. Using LAN communication, acquire IP address and host name from the /etc/hosts file in the PanaCIM NIS Server, and input them to each machine.

NM series (IPAC series) machines

Step	Procedure
1	Login as a level 4 user.
2	Select Menu > Systems > Administrator (Password Input). Select **Added Options**.
3	When using CVT, enter 5 to **Parts Count Control** under the **General** tab.
4	Under the **Communication** tab, set the following values: Host communication: 0 Mount log: 1 (Set to 0 only if ULT data is collected from the shared memory) LAN communication: 0 (Latest) Library addition mode: 0, 1, 2 (User-defined) Enable C2 commands: 1 (Yes) Enable C3 commands: 1 (Yes) Enable C5 commands: 1 (Yes) Enable R commands: 1 (Yes) Disable R command remote mode: 1 (Yes) SPC: 1 (Yes) C command port number: 49152 R command port number: 49153 Send value-added information with R command: 1 (Yes) Request host for upload when changing product program: 1 (DA only) Machine state setting: 1 (Yes)
5	Click **OK**.
6	Select Menu > Systems > Communication. Click **Network**.
7	Enter IP address, Subnet mask, and Default gateway of the machine.
8	Save the changes made and restart the machine.
9	Using LAN communication, acquire IP address and host name from the /etc/hosts file in the PanaCIM NIS Server, and input them to each machine.

YB series (UV Scriber) machines

Step	Procedure
1	Click **Maintenance** to open the **System Parameters** screen.
2	Login as a level 3 user.
3	Under the System tab, set the following values: 104 Image sensor communication timeout: 120 313 Host port 1 port number: 49152 314 Host port 2 port number: 49153 315 Communication mode start system: 1 (Online)

Universal GSM1 and GSM2 machines

Step	Procedure
1	System Configuration > UPS > Host Interface > Communication status. Set the following parameters: Communication status: Enabled Establish communication 20-second: Enabled
2	System Configuration > UPS > Host Interface > Control state. Set the following parameters: Control state: Online Online sub-state: Remote Online trial failure substate: Equipment offline Offline sub-state: Equipment offline
3	System Configuration > UPS > Host Interface > Default. Set the following parameters: Host interface active: Check the box. IP address: PanaCIM server or client host address Protocol: HSMS94 Entity type: Passive
4	Restart the machine. When restarted, a phone icon appears first on the screen. Icon color represents the status of communication between the machine and host: Red - Communication disabled Yellow - Communication enabled, but host not connected. Green - Communication enabled. Host connected.
5	Selecting the phone icon enables additional parameters to be set. **a.** Select Enterprise Link > Host > Communication status to enable the current communication. **b.** Select Enterprise Link > Host > Control State. Set as follows: Current control state: Remote online Control state: Not selected Online substate: Local

Machine using serial communication(RS232)

To perform communication between BM series machines and PanaCIM, configure the machines.

Pre-Windows NT machine setting

Step	Procedure
1	LAN: No Check sum: Yes R commands: Yes (Older Panasert machines may require this option to be set to No.) Momentary: No Timeout: Yes End command: ETX Baud rate: 9600

Windows NT machine setting

Step	Procedure
1	Service > MMII > PanaProtocol. Set the parameters as follows: C3 commands: ON C5 commands: ON R commands: ON SPC: ON
2	Service > MMII > RS232-Parameters > Host Port. Set the parameters as follows: Check sum: ON R commands: ON Timeout: ON End command: ETX Baud rate: 9600 *9 Momentary: OFF
3	Service > System Parameters > Host Port. Set the parameters as follows: Check sum: ON R commands: ON Timeout: ON End command: ETX Baud rate: 9600* Momentary: OFF

*9: Baud rates lower than 9600 may be set as long as it agrees with the work cell setting. When the cable is longer, data communication speed may not increase even if a higher baud rate (19200) is set, and trouble may occur.
For older Panasert machines such as MV150 and MPA40/80, it may be better to avoid using real-time messages to ensure stable communication for a long time.

5.4 Product Transfer

Product Transfer application provides the capability to transfer product programs set by product setup to the machines in the specified zone, or from the specified zone to the PT200/BM series machines. The destination route/zone and target panel mix and machine type can be set by the user.



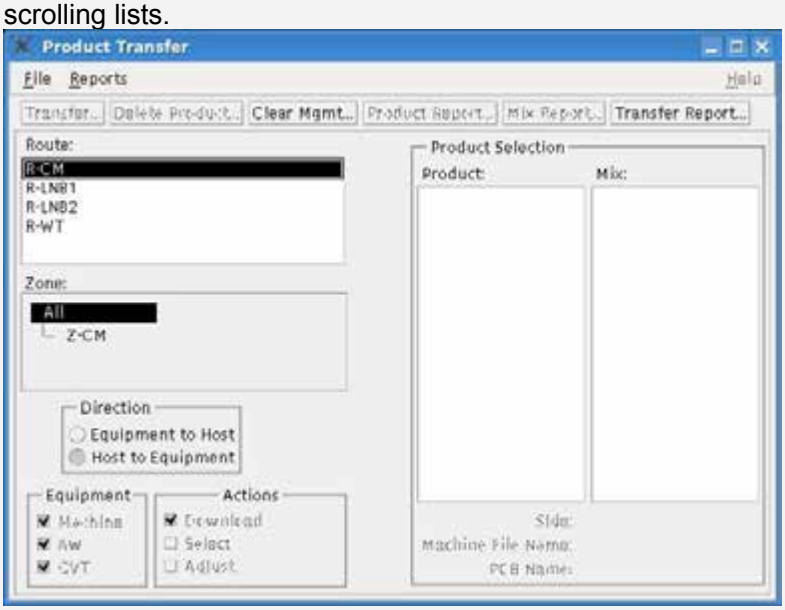
Clicking **Selected** when downloading the NC data on the BM series machine permits you to remotely manipulate the active files on the BM series machine.

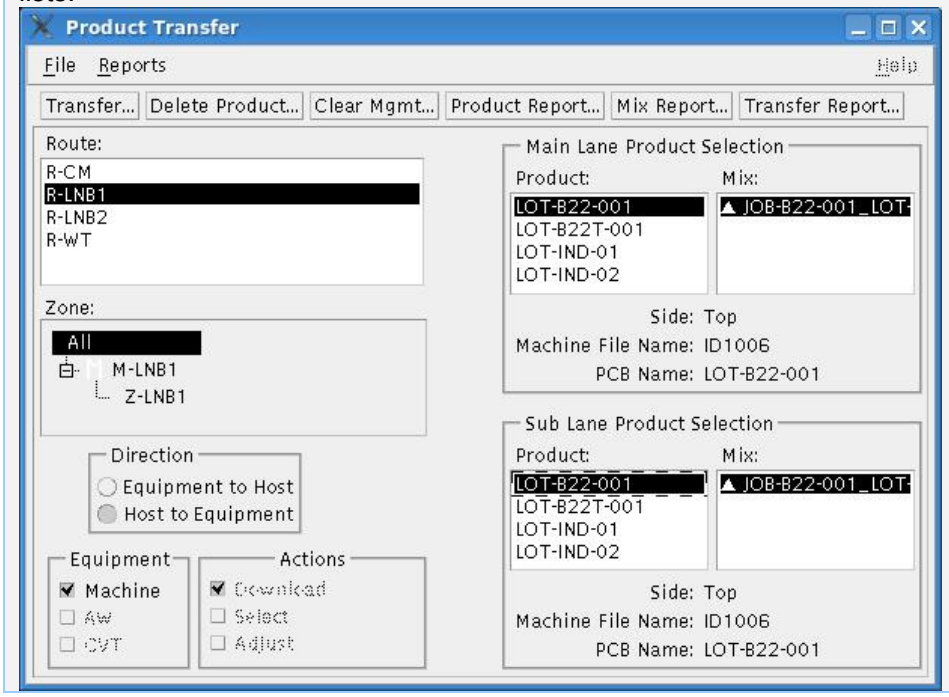
* Some BM series machines do not support this function.

Select Program Transfer from the PanaCIM Application toolbar.

Once a route, a product and a mix have been selected, the zone(s) in the destination route can be selected.

5.4.1 Program Transfer Setting

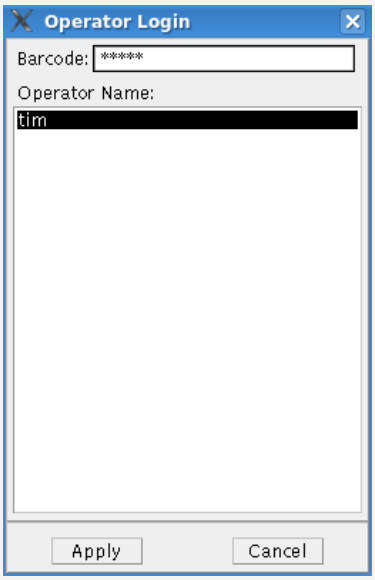
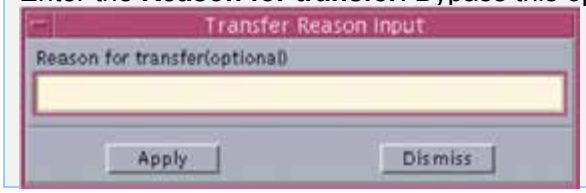
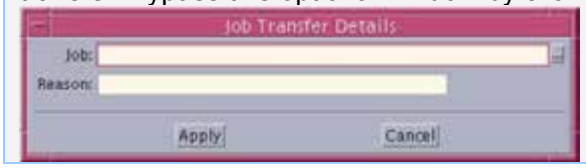
Step	Procedure
1	Select a **Route**. Selections vary dependent upon whether you are using a single or dual program route.
2.1	**Single Route** Select the **Route** from scrolling list containing the product being transferred. The Zones and Products associated with this Route will then be displayed in their respective scrolling lists. 
2.2	Select the product to be transferred from the **Product** scrolling list in the **Product Selection** section. Its respective mixes will then be displayed in the Mix scrolling list.
2.3	Select the mix of the product being transferred from the **Mix** scrolling list in the **Product Selection** section.
2.4	Select which Zone(s) to transfer from the **Zones** list.

3.1	**Dual Route** Select the **Route** from scrolling list containing the product being transferred. The Zones and Products associated with this route will then be displayed in their respective scrolling lists. 
3.2	Select the first product to be transferred from the **Product** scrolling list in the **Main Lane Product Selection** section. Its mixes will then be displayed in the Mix scrolling list.
3.3	Select the mix of the product to be transferred from the **Mix** scrolling list in the **Main Lane Product Selection** section.
3.4	Select the second product to be transferred from the **Product** scrolling list in the **Sub Lane Product Selection** section. Its mixes will then be displayed in the Mix scrolling list.
3.5	Select the mix of the product being transferred from the **Mix** scrolling list in the **Sub Lane Product Selection** section.
3.6	Select which Zone(s) to transfer from the **Zones** list.
4	**Direction:** Select the transfer direction to determine which way to transfer the files. Select either: • **Equipment to Host:** This selection transfers a product from the workcell to the Host PC (PT200/PanaCIM). • **Host to Equipment:** Transfers a product from the Host PC (PT200/PanaCIM) to the workcell. **Note:** Operator level logins (i.e. panaoper) can be restricted from transferring from Equipment to Host; in which case only the Host to Equipment choice is made available.
5	**Equipment:** Use these checkboxes to select what equipment in the selected zones should receive the download or supply the upload. • **Machine:** Select whether the Machine should transfer its product to the host, or receive a new product from the host. Checking the Machine checkbox selects the machine in the transfer. • **AW:** Select whether the AW conveyor should receive a new product from the host. Checking the AW checkbox selects the Autowidth Controller in the transfer. • **CVT:** Select whether the scanner should receive a new product from the host. Select the CVT checkbox to select the CVT in the transfer.

6	**Actions:** Choose any actions. The Selected option is used to determine whether the downloaded programs for the product should be automatically set to active in the workcells once downloaded. For those workcells that do not support this command, the programs must be set manually. • **Download:** Select whether the program should be transferred from the host to the machine by selecting the Download checkbox. Selecting this option will prompt an attempt to transfer the program on the workcell. • **Select:** Select whether the program should be set on the machine from the **Select** checkbox. Checking this will prompt an attempt to set the program on the workcell. If the job is enabled, then this will be replaced with Initiate Job. • **Adjust:** Select whether the Autowidth Controllers should automatically Adjust after receiving a program from the host. Checking this prompts an attempt to adjust. **Note:** BM series machines and Autowidth Controllers must be in the online mode to respond to these selections. Autowidth Controller interlocks must be set to allow movement (i.e. no panels within equipment). For CM/DT series machines, programs must be downloaded to the PT200 and then data check must be done on the PT200 machine. Then it can be selected to go to the machine.

5.4.2 Transfer

Press the **Transfer** button to start transfer.

Step	Procedure
1	After making your configurations, click the **Transfer** button to initiate the transfer.
2	**Operator Login** dialog appears.  **Note:** This step is dependent upon whether the security option was set in System Configuration. For further information, see the System Configuration section above under the subheading Base 1.
3	Select your **name** from the list of known operators and then enter your operator **barcode** as confirmation. Then, click **Apply**.
4	Enter the **Reason for transfer**. Bypass this optional window by clicking **Apply**. 
5	Enter the **Job** name for the Selected Product Setup. Next, enter a **Reason** for the job transfer. Bypass this optional window by clicking **Apply**. 
6	A popup will display the progress of data for each piece of equipment being tracked. Click **Continue** when it has finished. The window will disappear after successful transfer. If the transfer was not successful: • Click **Abort** to leave any errors as failures. • Click **Retry** to reattempt a successful transfer.

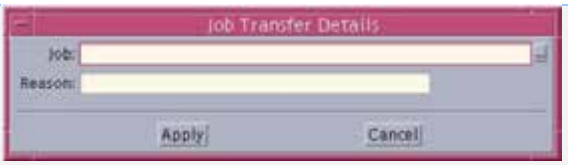
Reference the below table for descriptions of possible return errors.

Reason	Description
Failed	A generic communication error has occurred. This usually means the machine is not responding to PanaCIM communication attempts.
Aborted	A request to abort the transfer is completed.
Comm Fault	The communication link to the equipment is down. This could mean the equipment is powered off, or there is a physical problem in the serial cabling to the equipment.
Comm Param Err	The PanaCIM communication parameters (baud rate, etc.) do not match the parameters at the equipment.
Off-line	The equipment is in the off-line state and will not allow the requested operation to occur. Equipment must be switched to the On-line state for communication to succeed. This error code can also mean that there is no more free space in the machine. Delete one or more programs to create free space.
Protocol Error	Most commonly, the Group and Lot in Product Setup were not added after transferring a new Program to PanaCIM.
No Panasert File	The requested program file does not exist on the BM series machine. (Upload only).
Machine Full	There is no room on the equipment to store the program file. (Download only).
Not Supported	The equipment does not support the requested operation. This usually means the protocol parameters for the machine setup in **Route Configuration Editor** have been improperly configured.
Parse Failed	PanaCIM was unable to interpret a file uploaded from the BM series machine. Make sure the BM series machine type is correctly configured in PanaCIM.
Host File Error	This indicates a file system level problem in PanaCIM. For downloads it can mean the program file is corrupt or has had its Linux permission modified so there is no access to the file. For uploads it indicates a permission problem or it is possible that the Linux file system is full.
No PanaCIM File	The requested program file does not exist in PanaCIM. This will only occur during downloads and means that a file type has been selected for transfer that does not have the triangular glyph next to it. The file may not exist because it was not created by Panatools, or it has never been uploaded from the machine.
Wrong Scanner	For sites with CVT, a download to a CVT scanner has been attempted, and the CVT scanner that corresponds to the zone being transferred is not in the pod for the zone. This means you might have placed the CVT scanner for another zone into this zone's pod.
AW Interlock Err	The Autowidth Controller cannot move due to a master interlock. Make sure the Zone is clear of all panels and retry the operation.
Axis Interlock Err	One or more of the Auto-width controller axis could not move due to an interlock. Make sure the Zone is clear of all panels and retry the operation.
Current Prog Err	PanaCIM attempted to download a file to the BM series machine with the same name as the currently selected NC program on the BM series machines. Select a different NC program and retry the operation.

Interface Down	The PanaCIM daemon Application that performs the communication to the equipment is down. Normally, this Application is automatically re-started. This usually indicates there is a problem with the software.
No Files Marked	No files have been marked for transfer in the product setup (equipment setup popup for the specified equipment). This is informational only, and indicates that no actual transfer has occurred with the equipment.
No Equipment	A zone has been selected for transfer that contains no equipment that PanaCIM is configured to communicate with. This is for informational purposes.
In Progress	Indicates that a “select” or “adjust” operation has been initiated.

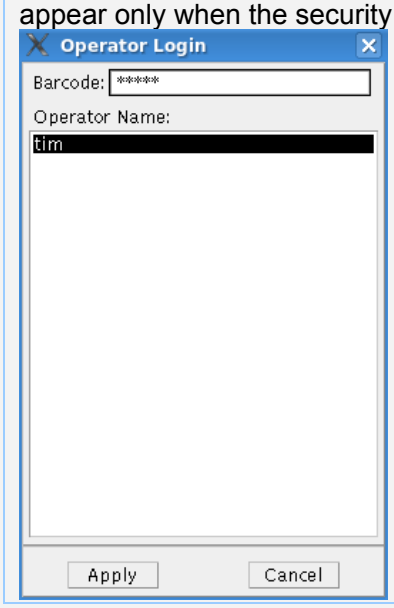
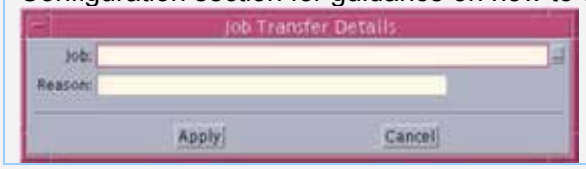
5.4.3 Clearing Management Data

The uploaded production management data for the current program can be cleared on the **Clear Mgmt** page. Refer to the action table below to clear management data for the current product.

Step	Procedure
1	Select the desired route from the **Route** scrolling list. The Zones and Products associated with this route will then be displayed in their respective scrolling lists.
2	Select the desired zone(s) from the **Zones** scrolling list. Selection of a product and mix are not required.
3	Click  to perform an upload and clear of the current product.
4	Enter the **Operator Login**. The operator login window requires the operator to select their name from the list of known operators, and then enter their operator barcode as confirmation. Click **Apply**. **Note:** The Operator Login dialog and the Job Transfer Details dialog in step 5 appear only when the security option has been set in the System Configuration.
5	 Enter the **Job Name** for the Selected Product Setup. Next, enter a **Reason** for the job transfer. Bypass this optional window by clicking **Apply**.

5.4.4 Deleting the Program

Refer to the action table below to remove the selected product from the machine.

Step	Procedure
1	Select the desired product and click **Delete Product** to remove the selected product from the selected zone(s)
2	Enter the **Operator Login**. The operator login window requires the operator to select their name from the list of known operators, and then enter their operator barcode as confirmation. Click **Apply**. **Note:** The Operator Login dialog and the Job Transfer Details dialog in step 5 appear only when the security option has been set in the System Configuration. 
3	Enter the **Job** name for the selected Product Setup. Next, enter a **Reason** for the job transfer. Bypass this optional window by clicking Apply. See the System Configuration section for guidance on how to enable/disable this feature. 

5.4.5 Output Reporting

The following reports can be run from the Product Transfer Application:

- **Product Changeover Report/Mix Report:** This report details all the feeders that need to be set up on each Machine to produce a new product. Types of Changeover reports include the Product Report and Mix Report.
- **Running a Transfer Report:** This report shows a historical log of the files that have been accessed through Product Transfer. This type of information is only available if transfer logging has been activated.

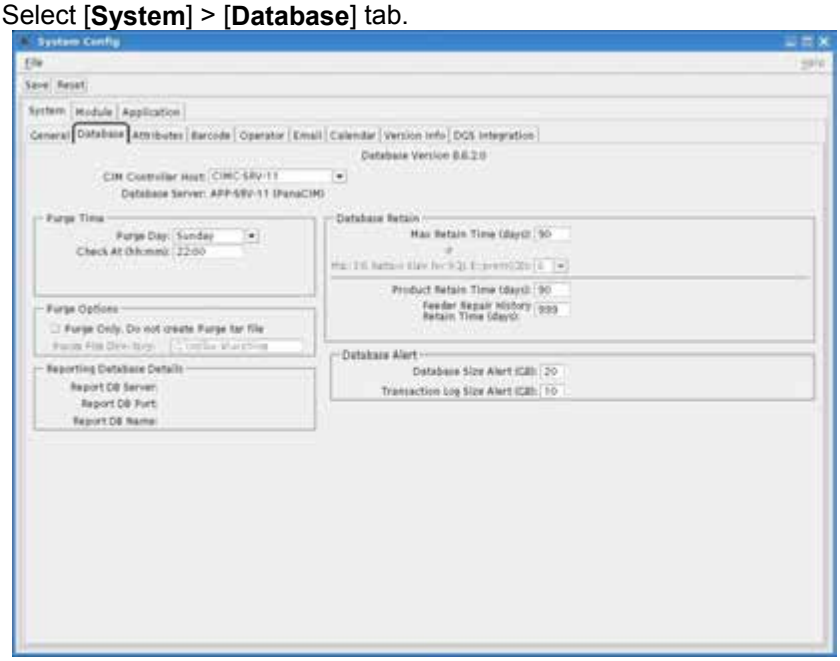
Refer to the action table below to run a Product Transfer report.

Step	Procedure
1	Select **Product Transfer** from the PanaCIM toolbar.
2	Select the **Product** and **Transfer Direction**.
3	Choose whether you would like a Product Report, Mix Report, or Transfer Report by clicking the appropriate button.
4	Choose your Output Options and click **Apply**.

5.5 Management PanaCIM DB

5.5.1 Setting the DB

Select [**System**] > [**Database**] tab.



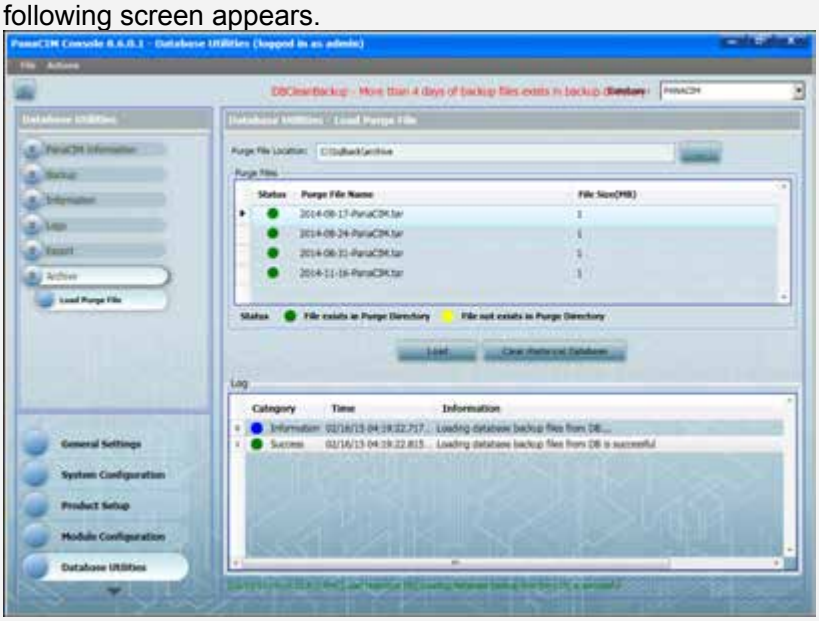
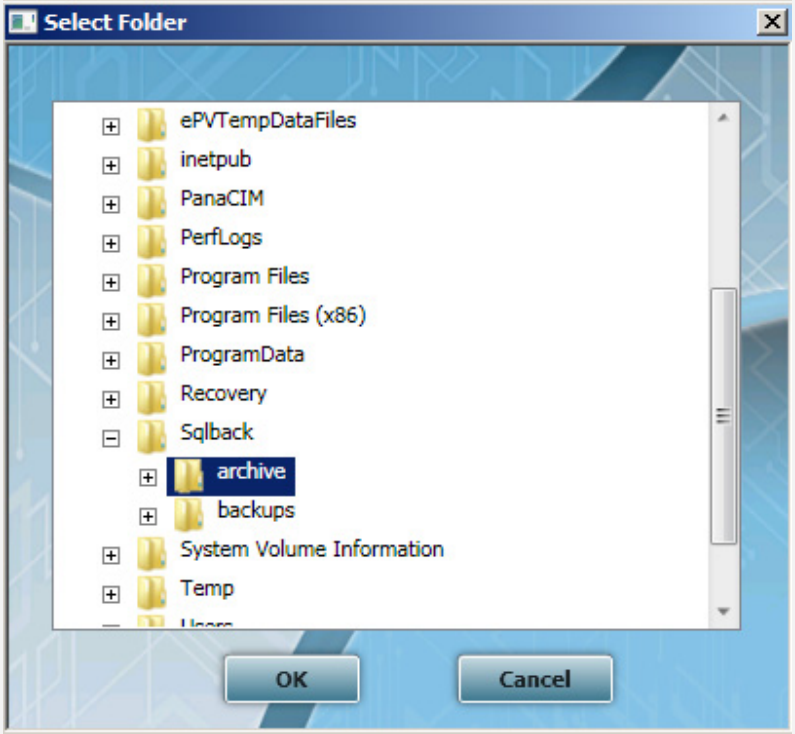
This page is used to set the following:

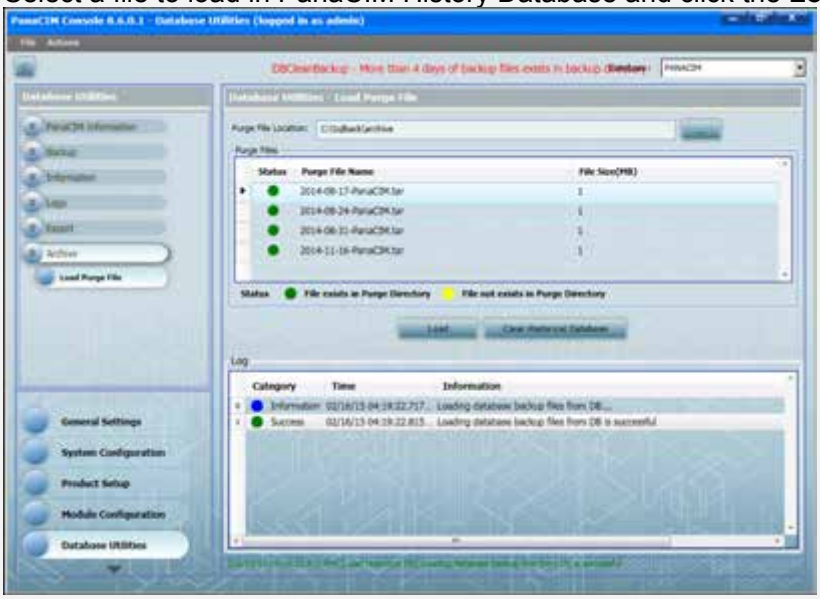
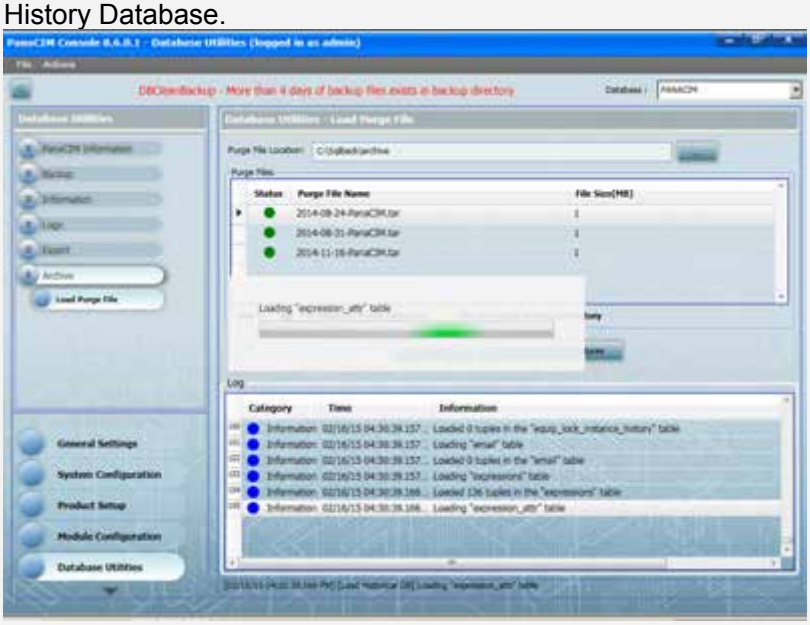
- PanaCIM controller server
- When to purge the information in the database
- Period of time to save data
- Other related options to purge the information in the database

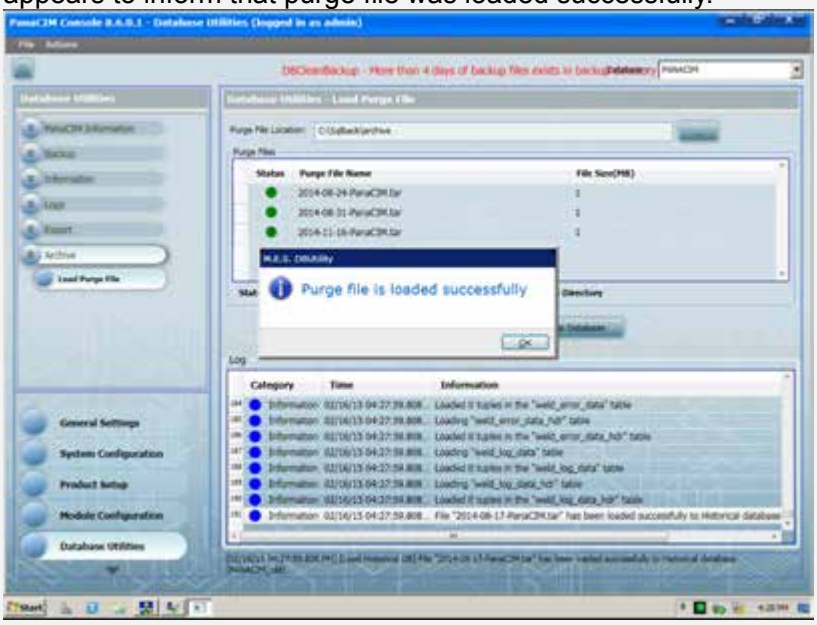
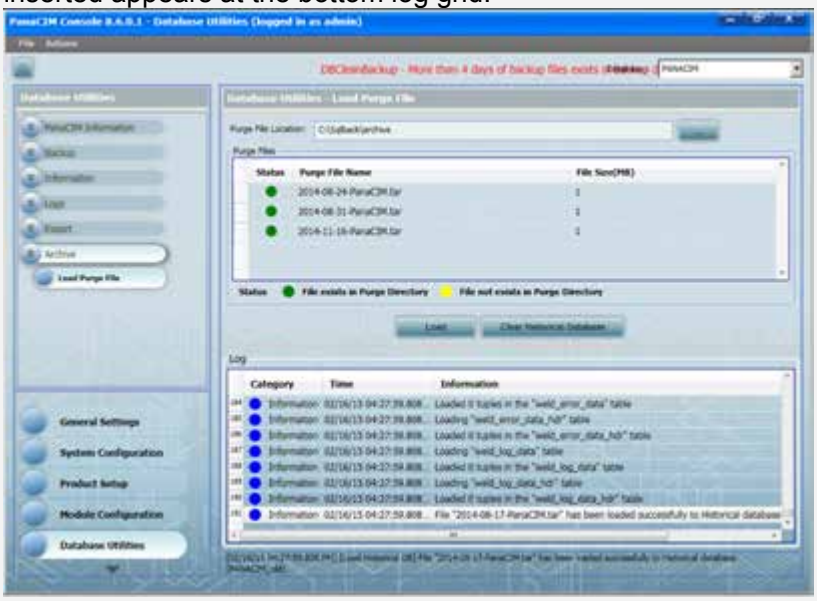
Step	Procedure
1	Select the applicable host name of the PanaCIM controller server from the drop-down list. **Note:** The database server information has been stored during installation, and therefore need not be added at this step.
2	**When to Purge Data:** This option sets when to purge the information in the database. • **When to Purge:** Sets the day of the week to purge the database. • **Specify by Day (day of the week):** Specify the day of the week on which the free space on the database is to be checked. • **Specify by Time:** Sets the time to check the free space in the database.
3	**Purge Option:** This option sets whether or not to compress the purged data (as TAR files). • **Purge Only, Not to Compress Files:** Selecting this option does not create compressed files. If not selected, the purged data will be saved in the directory specified in 'Path to save compressed files'. **Note:** The purged data files can be restored from the backup logs.
4	**Days to Save Data in Database:** This option sets the period of time to save the data in database until they are purged and stored as compressed files. • **Maximum period to save (days)**：Set the maximum period of time (in days) to save data in the database. • **Save Product Programs for (days):** Set the period of time (in days) to save the product program data in the database. • **Save Feeder Repair History for (days):** Set the period of time (in days) to save the feeder maintenance history in the database.
5	Click **Save** to save the changes made. Click **Reset** to discard any changes made.

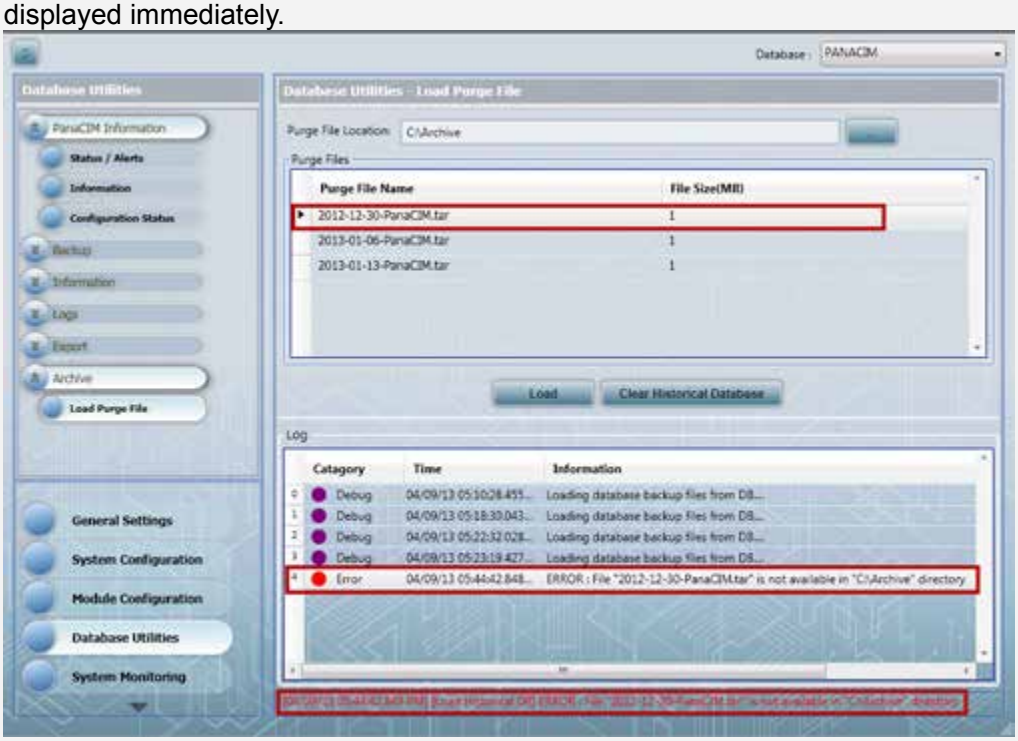
5.5.2 Restore Compressed Data to Historical DB

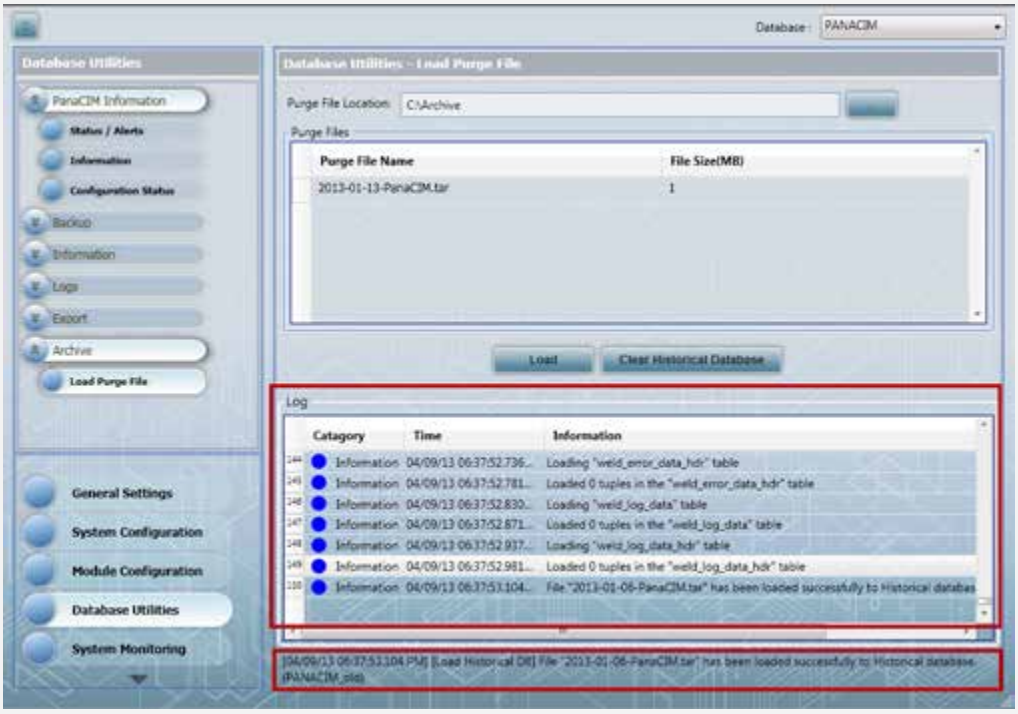
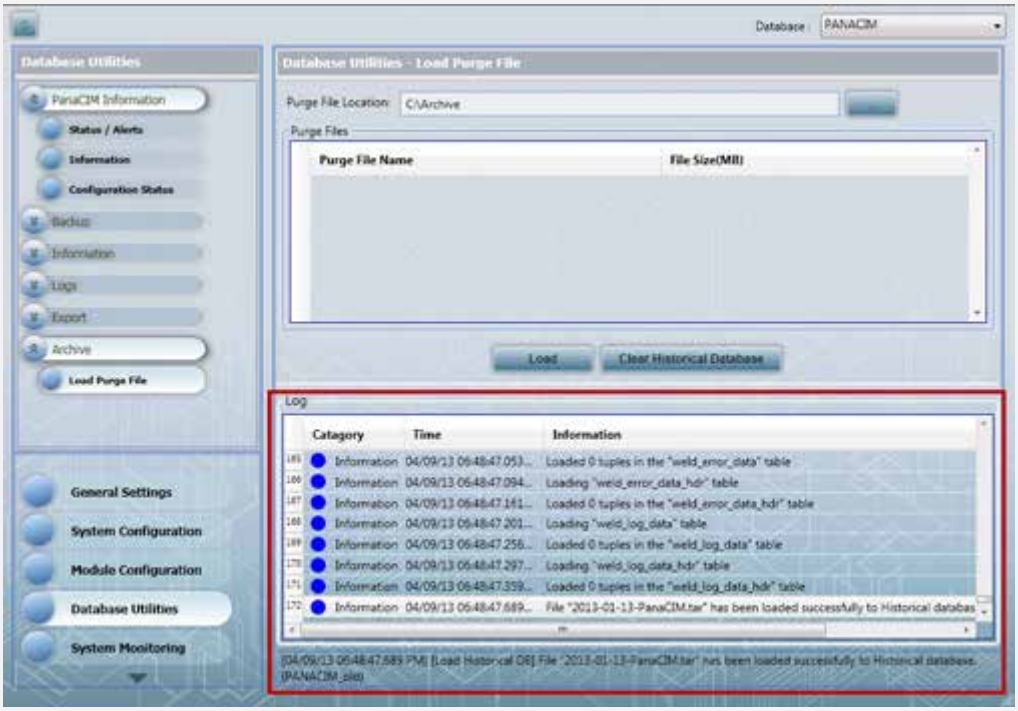
Add compressed database information to history database.

Step	Procedure
1	On application server, select **Database Utilities** at the lower left of PanaCIM Console. Then click **Archive plug-in** and select purge file reading below. The following screen appears. 
2	As above, the file is saved in a folder on the application serve. Saved in folder: [C:¥Archive] Click  right of text box and select the folder where tar file was copied. 

3	Select a file to load in PanaCIM History Database and click the **Load** button. 
4	After that, the screen shows that the data is being loaded (uploaded) to PanaCIM History Database. 

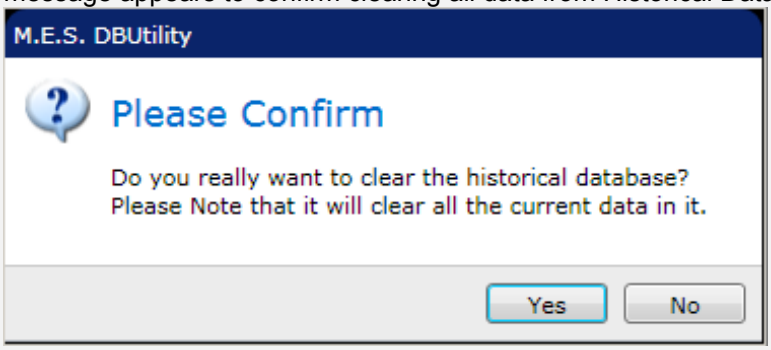
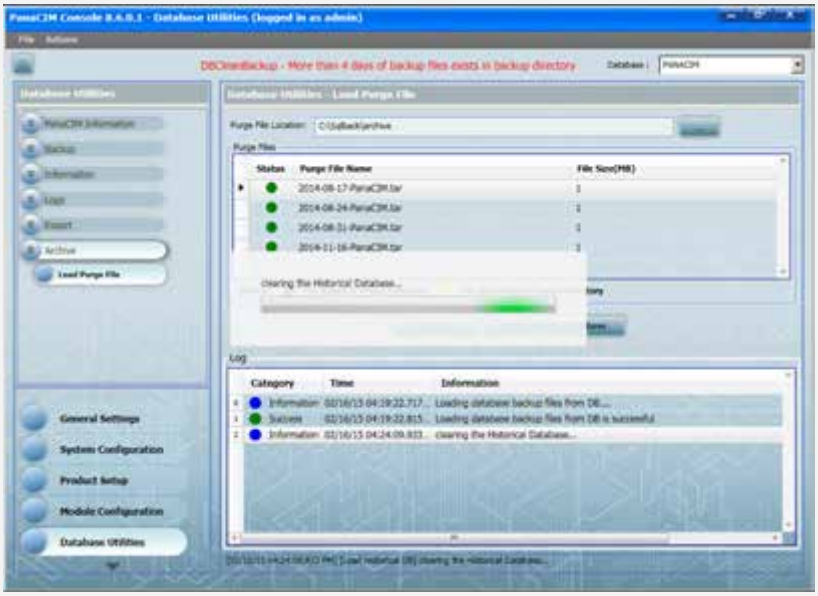
5	After all data was loaded from the file to PanaCIM History Database, a message appears to inform that purge file was loaded successfully. 
6	Purge files loaded successfully into PanaCIM History Database will be deleted from the file list. Data are added regarding table and the log where a row will be inserted appears at the bottom log grid. 

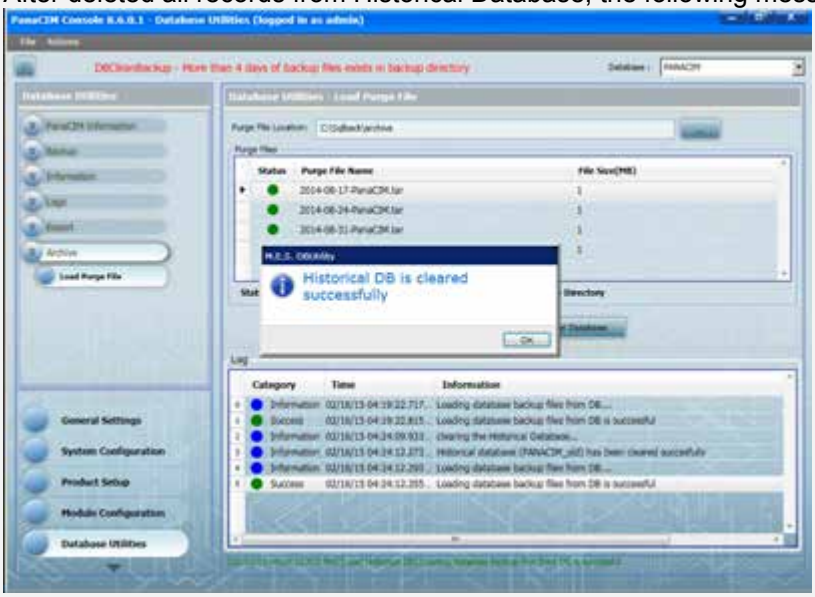
7	If invalid file or other error was found after clicked the **Load** button, they are displayed immediately. 

8	Select another purge file and click the **Load** button to continue loading another one.  
9	All purge file data loaded was saved in History Database.

5.5.3 Deleting inside of Historical DB

Follow the steps below to delete all information from History Database.

Step	Procedure
1	Click the **Clear Historical Database** button next to the **Load** button. The following message appears to confirm clearing all data from Historical Database. 
2	To delete all data from History Database, click **Yes**.
3	After clicked **Yes**, the following screen appears and a process starts to delete records from Historical Database. 

4	After deleted all records from Historical Database, the following message appears. 

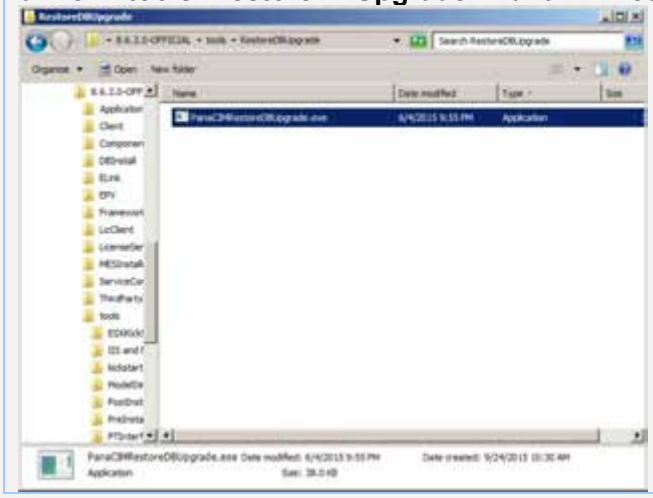
5.5.4 DataBase Restore

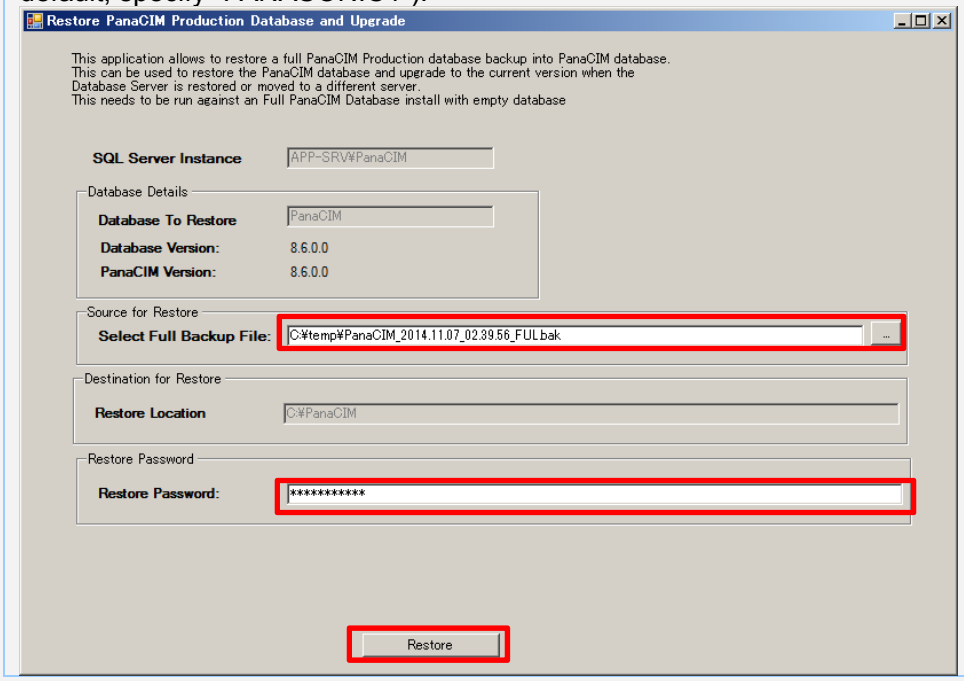
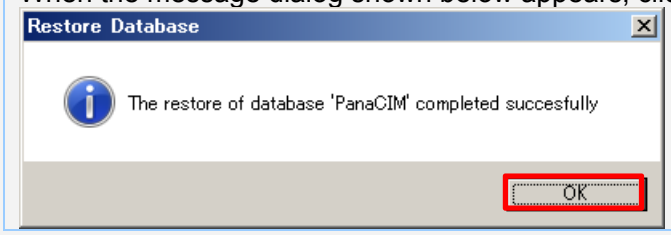
Take the following steps to restore a PanaCIM database full backup file to the PanaCIM database.
If a past different version* of a PanaCIM database full backup file is restored, it is upgraded to the currently installed PanaCIM database version.

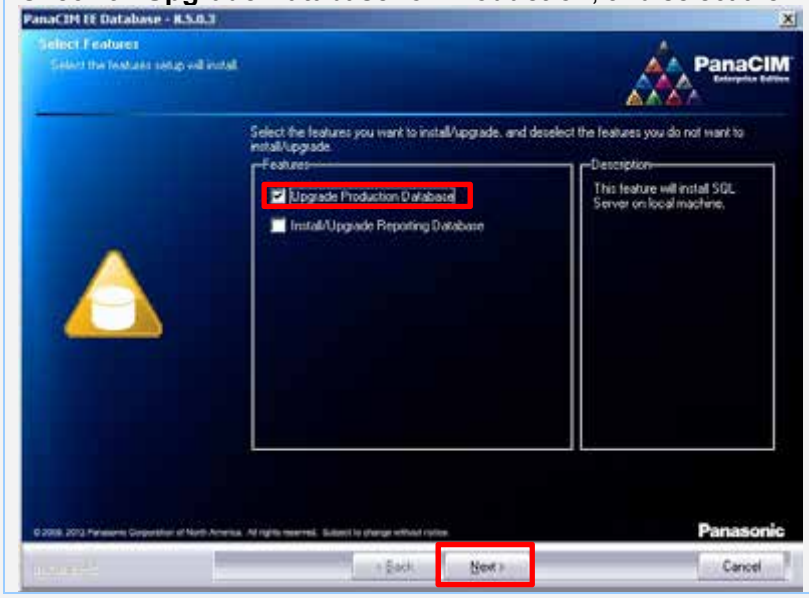
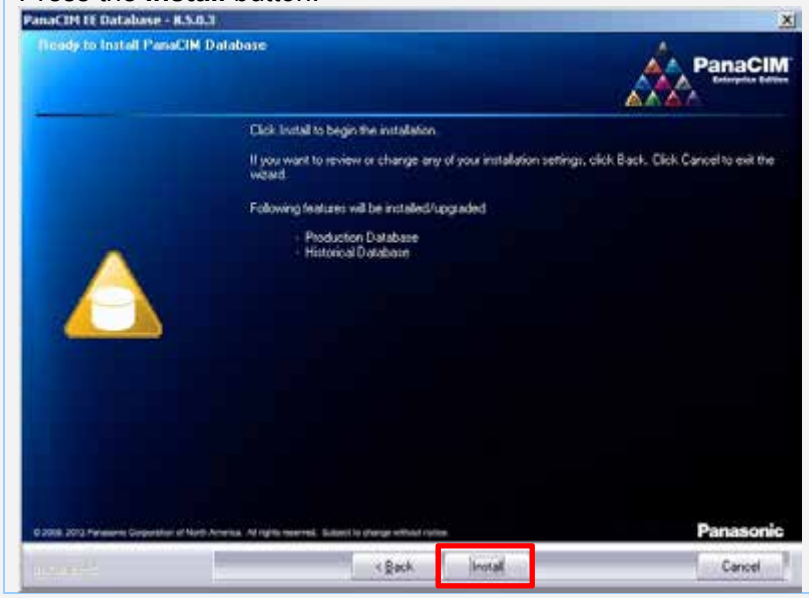
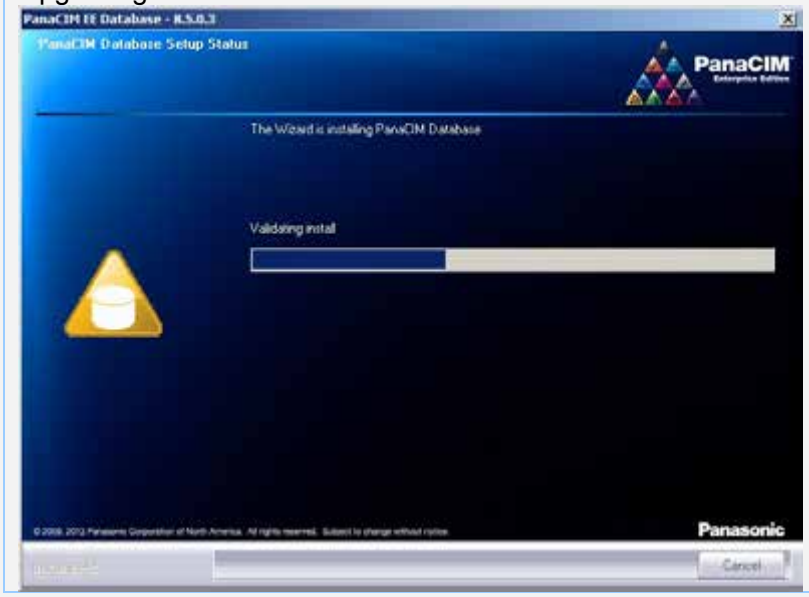
* The PanaCIM database full backup file created before PanaCIM 8.3.0.0 is not supported.

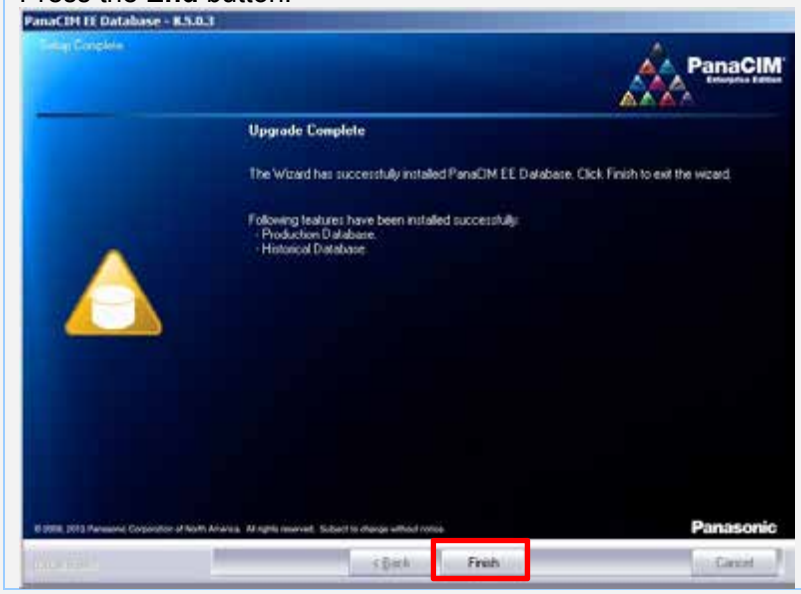
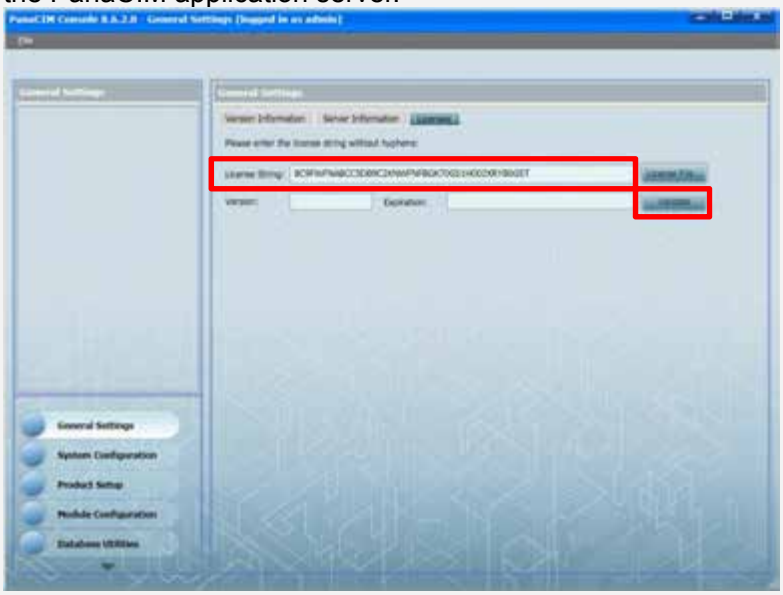
■What to be prepared before restoring data

- Full backup files of PanaCIM database
- PanaCIM Application Serve Windows DVD of the same version of the currently installed PanaCIM software
- License string for currently installed PanaCIM software version (required only if a different PanaCIM version of the database is restored)

Step	Procedure
1	Temporarily disconnect the PanaCIM control server from the network. (Remove the PanaCIM database reference from PanaCIM control server.)
2	From the PanaCIM Application Server, PanaCIM Service Controller and stop **All Services**. (Remove PanaCIM database reference from the PanaCIM Application Server.) 
3	Insert the PanaCIM Application Server Windows DVD of the same version as the currently installed PanaCIM software into the optical drive of the PanaCIM Application Server computer.
4	On the PanaCIM application server, launch the Explorer and double-click **Optical drive>:¥tools¥RestoreDBUpgrade ¥PanaCIMRestoreDBUpgrade.exe.** 

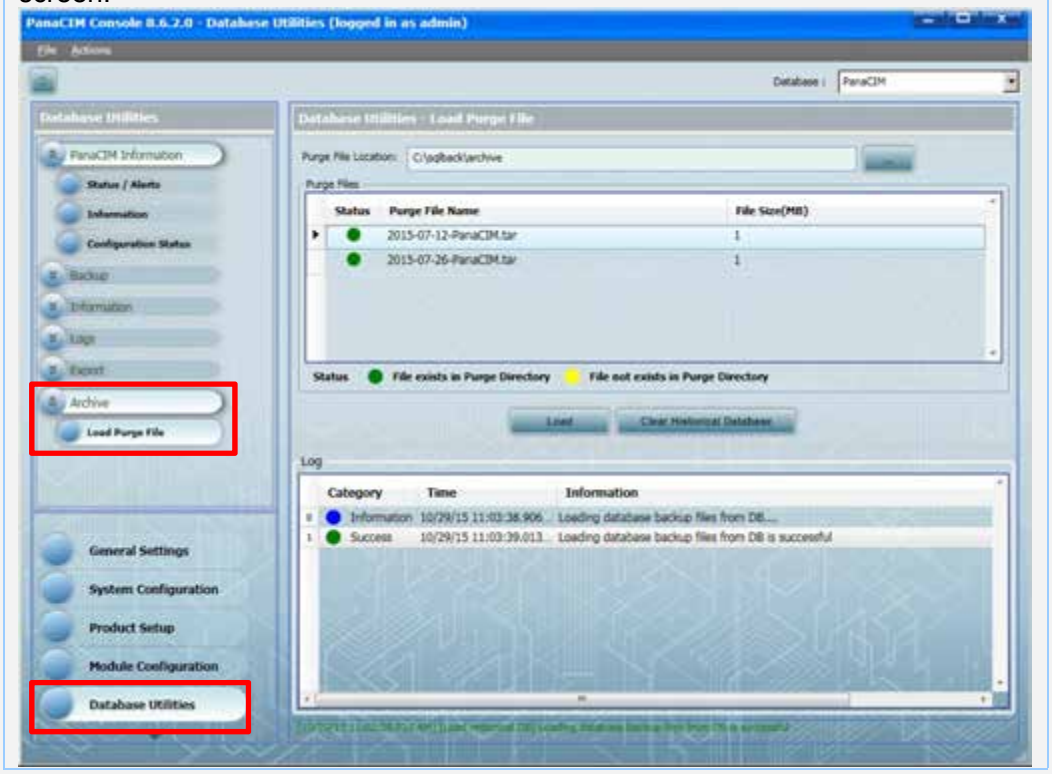
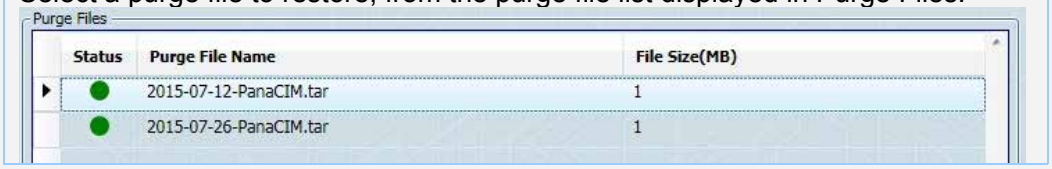
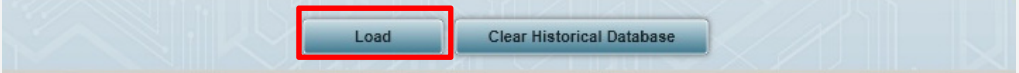
5	On the Restore PanaCIM Production Database and Upgrade screen, specify the following and select the **Restore** button. - **Select Full Backup File:** Specify the file path of the “PanaCIM database full backup file” to be used for restoring the database. - **Restore Password:** Specify the “sa user” password for the SQL Server (by default, specify “PANASONIC1”). 
6	When the message dialog shown below appears, click the **OK** button. 

7	Check off **Upgrade Database for Production**, and select the **Next** button. 
8	Press the **Install** button. 
9	Upgrading the database will start. 

10	Press the **End** button. 
11	Connect the PanaCIM control server to the network.
12	Restart the PanaCIM control server.
13	Restart the PanaCIM application server.
14	If a different PanaCIM version of the database is restored, perform re-authentication with the license string for the currently installed PanaCIM software version, on the PanaCIM Console - General Settings - License screen of the PanaCIM application server. 

5.5.5 Reference to Past Production Information

In order to prevent database bloat, the PanaCIM system outputs the past production information from the PanaCIM database to a purge file after a certain period of time has elapsed, and then deletes it from the database.
You can no longer check the deleted data on the report of the PanaCIM Web application; however, you will be able to check it by expanding that file on the PanaCIM history database.
The following describes the procedure for expanding the purge file on the PanaCIM history database and then referring to that data.

Step	Procedure
1	On the PanaCIM application server, start and log in to the PanaCIM Console.
2	On the PanaCIM Console, open the Database Utilities - Archive - Load Purge File screen. 
3	Select a purge file to restore, from the purge file list displayed in Purge Files. 
4	After selecting a purge file, press the **Load** button. 

5	When the restoration is complete, the completion dialog message will appear.  * It takes about 100 minutes to restore a 40 MB purge file.
6	Start the PanaCIM Web application, and select **History** for Database. 
7	On **New Chart and Report**, open the chart or the report to refer to.

■ Difference in destination location of purge data among versions
After PanaCIM 8.5.0.0, the destination location of purge files is changed from the PanaCIM control server to the PanaCIM application server.

Therefore, if upgrading the PanaCIM 8.5.0.0 or earlier version, you need to move purge files from the PanaCIM control server to the PanaCIM application server.

[Destination location of purge files before PanaCIM 8.4]
“/home/panadmin/backup” folder of PanaCIM control server

[Destination location of purge files after PanaCIM 8.5]
“archive” folder in database backup folder of PanaCIM application server
* The database backup folder can be checked at [Database Backup Location] on the [Database Utilities]-[Backup]-[Backup Settings] screen of the PanaCIM Console.

5.6 Best Mix

The best mix means a combination of two production programs (one is for surface mounting and the other is for back mounting), and the two programs (for surface and back) can be set for the same MJSID at a time. The user can execute either of the two programs, and the machine selects it inside. Since the DGS has configuration settings, you need not to make specific settings on the NPM or the LNB. However, it is affected by many settings on the NPM/LNB. When the operator selects a program, the NPM will issue a request to the DGS via the LNB. If this program has two best-mix programs, both the programs are sent to the machine.

Since both the best-mix programs are sent to the machine, correct programs are displayed on the active machine (mounting that part currently). However, only the **first** program is displayed on the PanaCIM. For example, if the operator selects a program A, the DGS sends both the best-mix programs A and B to the machine; however, even if the machine uses the programs A and B for production, only the program A (the first selected one) is displayed on the PanaCIM.

When production starts, pickup and placement of the programs for surface/back are selected inside the NPM machine. The production programs for surface/back are not downloaded, but selected regardless of the LNB.

At present, it is only two production programs (surface/back) that can be executed on one machine. In dual lane mode, it is possible to execute a combination of programs for surface/back on the lane 1 (example: programs C and D) and the lane 2 (programs A and B) separately.

5.6.1 Restrictions

- Only the NPM-series mounter is a target for execution. The other mounters are not supported.
- Neither of the best-mix production programs supports trace data.
- On the Route Status screen, only the production program first selected on equipment is displayed.
- Chart and report data are created only for the production program selected on equipment. (**SPC and MV charts are created**.)
- Substitute parts are not supported completely.
  Since the substitute parts created in Cerberus file format (crb file) on the DGS side are supported because they are common among all production programs. However, the list of substitute parts created on the PPD cannot be supported because it is managed for each production program. The substitute parts received modifications are not supported, either. **For the Nutshell, only the substitute parts in Cerberus file format are supported. The substitute parts in "DAT file" or CSV file format are not supported.**
- The PanaCIM uses the **"first program"** for restricted parts, and either of the two best-mix production programs is available. Therefore, the user will achieve correct function by setting the definition of restricted parts to both the production programs.
- The best mix becomes invalid if, during setup for production, a certain edit is made to a production program included in the best mix.

At the present time, the following modules and functions are not supported for the best mix.

- Production analysis
- Traceability
- Product changeover and control
- Material control (allocation, moisture control, maintenance):
- Free arrangement of feeders
- Guided presetup

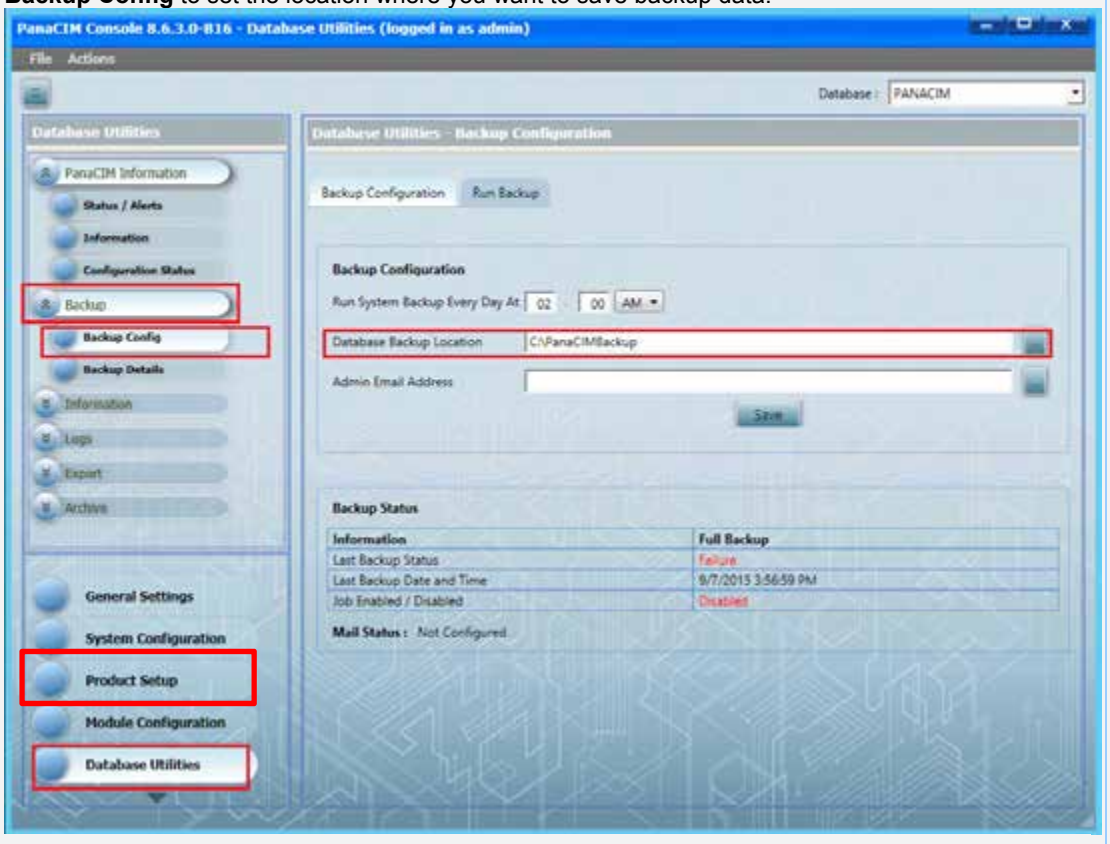
5.7 System Backup

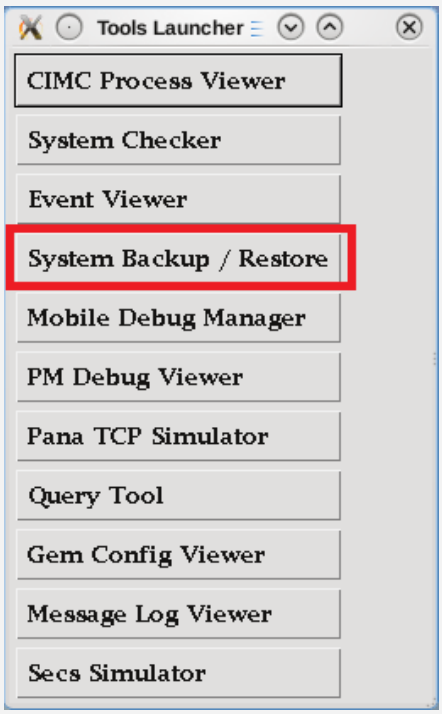
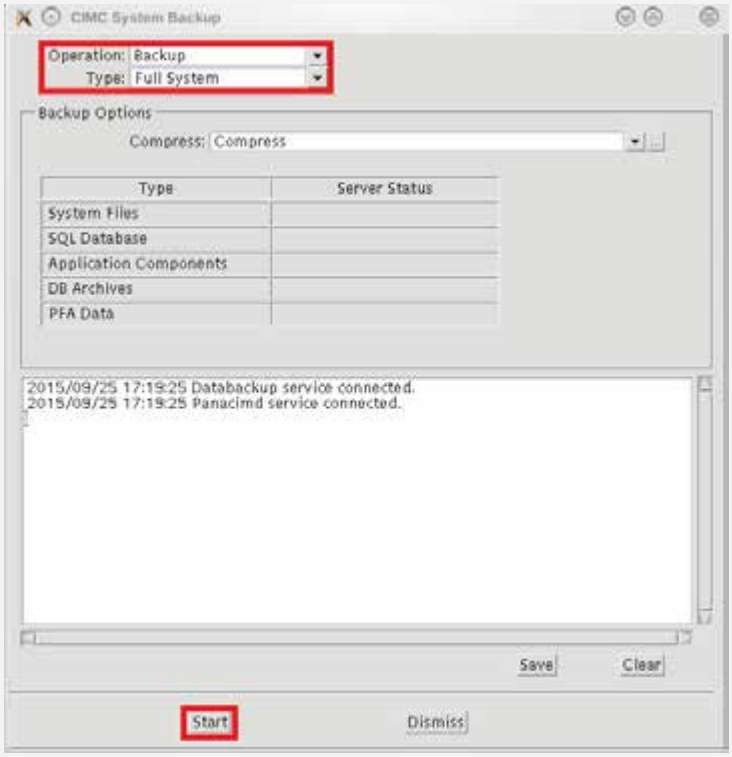
5.7.1 Backup

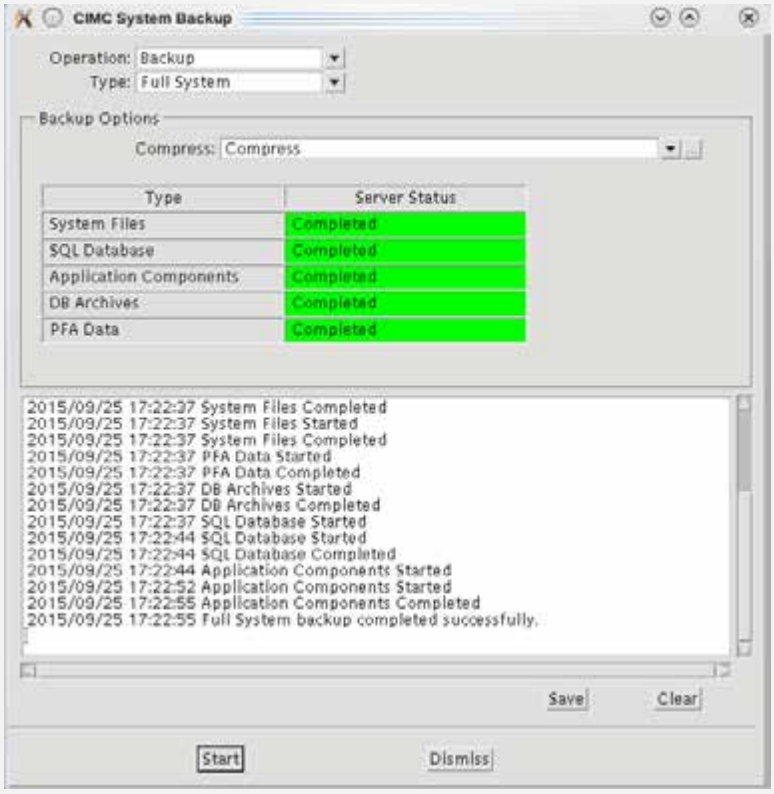
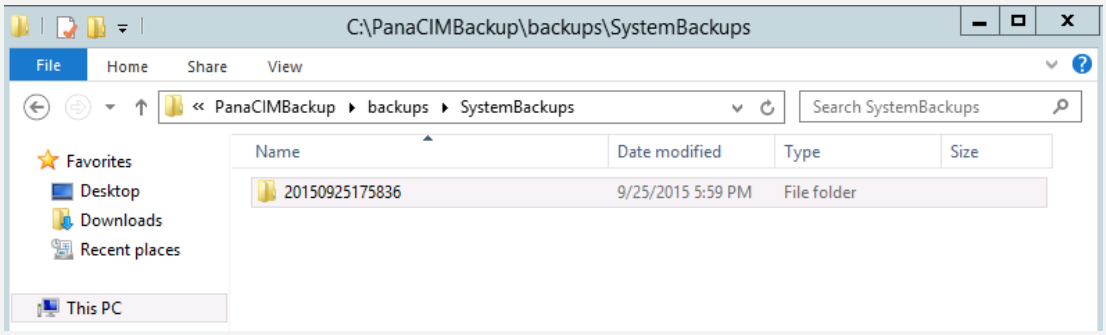
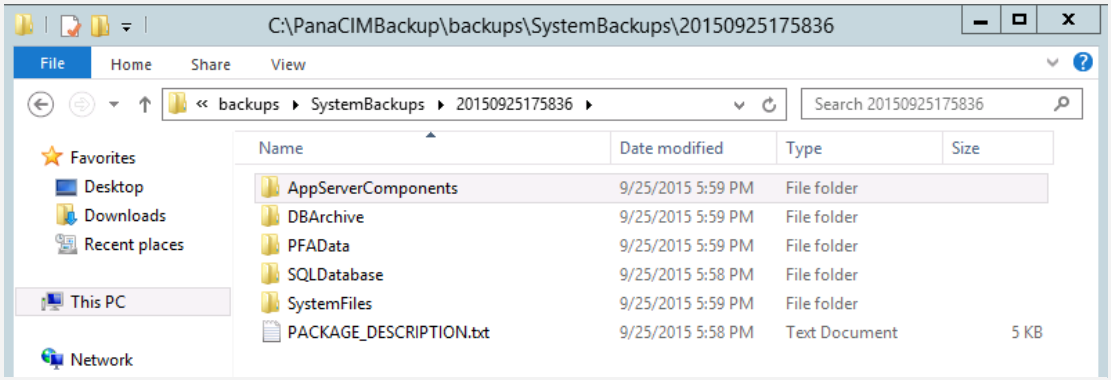
Files and data required for reconfiguration of PanaCIM environment are as follows.

1. System file (hosts files ("/etc/hosts"),
   product.conf file ("/home/panadmin/config/product. conf"))
2. SQL database
3. Application component
4. Database archive
5. PFA data

NOTE) For data loss prevention, we recommend that you move backup data to external storage devices such as an external hard drive or a USB flash drive. We also recommend that you set up the auto saving settings of backup data to the external storage device having enough storage capacity, and regularly back up data to magnetic tape or optical disks.

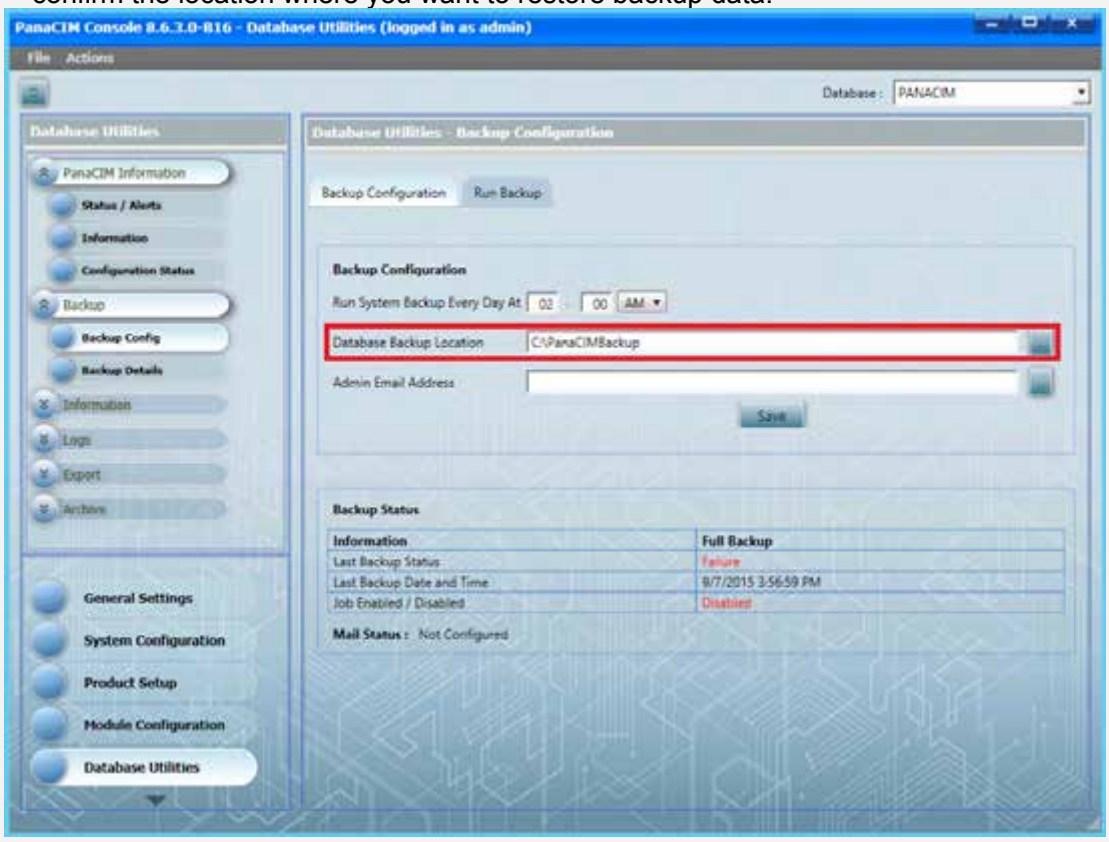
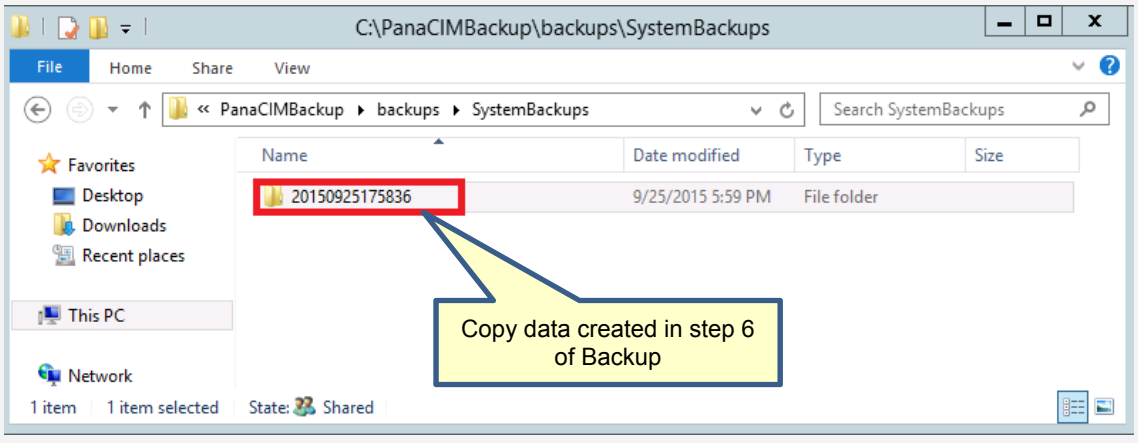
Step	Procedure
1	On the PanaCIM console screen in the controller server, select **Database Utilities** → **Backup** → **Backup Config** to set the location where you want to save backup data. 
2	Select **Diag Tools** from the PanaCIM controller server launcher. 

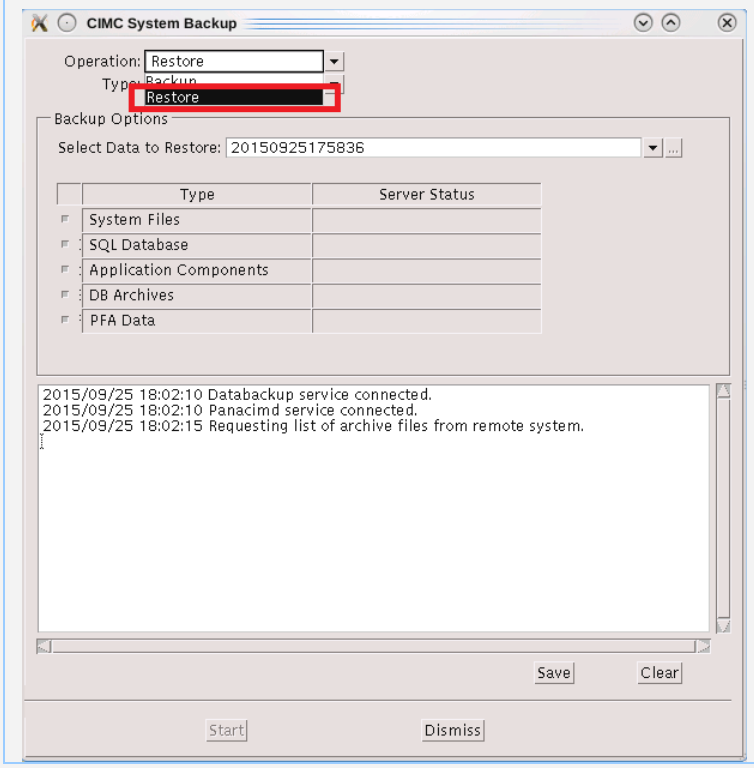
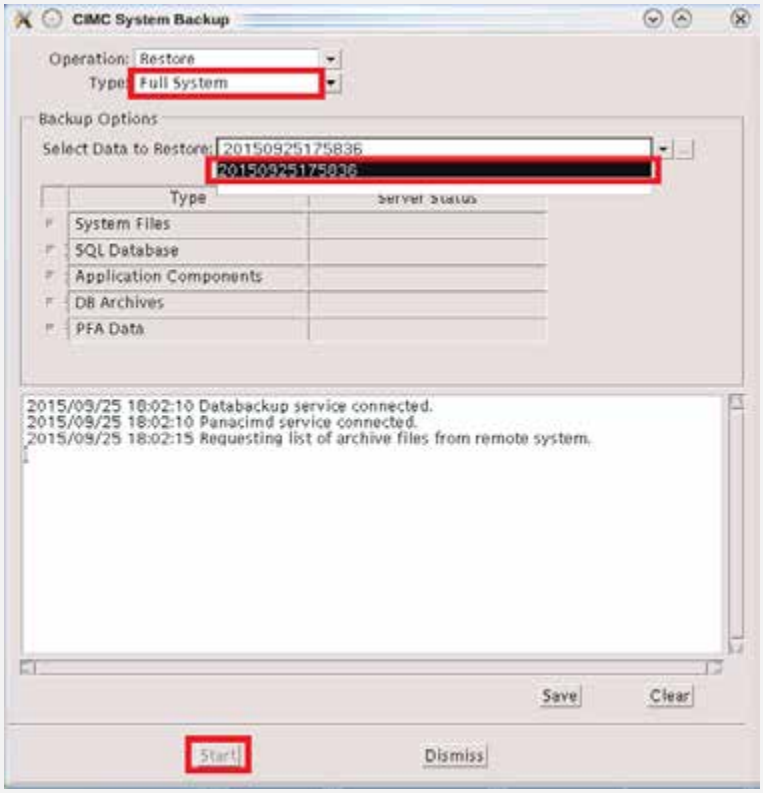
3	From the menu, select **System Backup / Restore**. 
4	On the CIMC System Backup screen, confirm the followings; Operation: Backup, Type: Full System. Then press the **Start** button. 

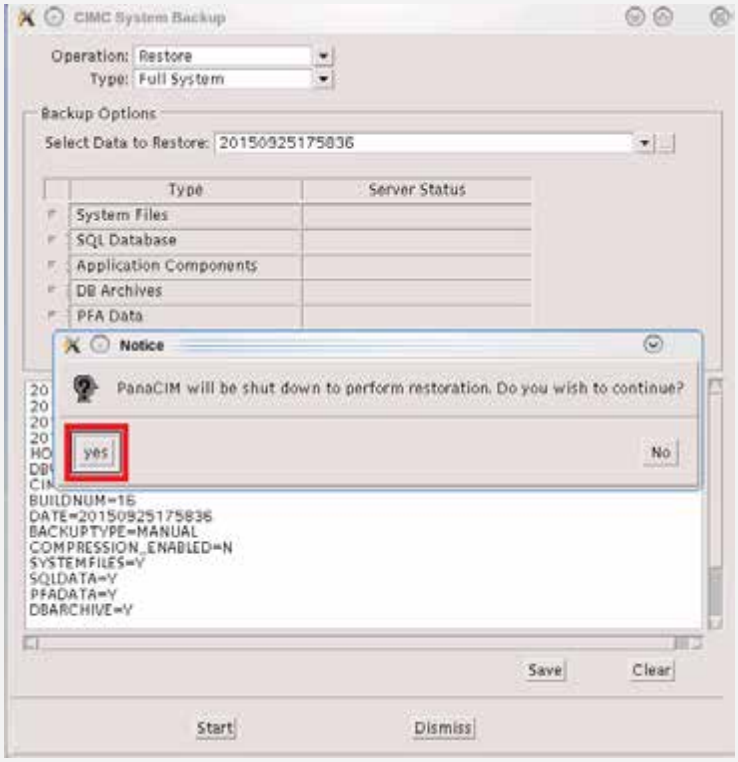
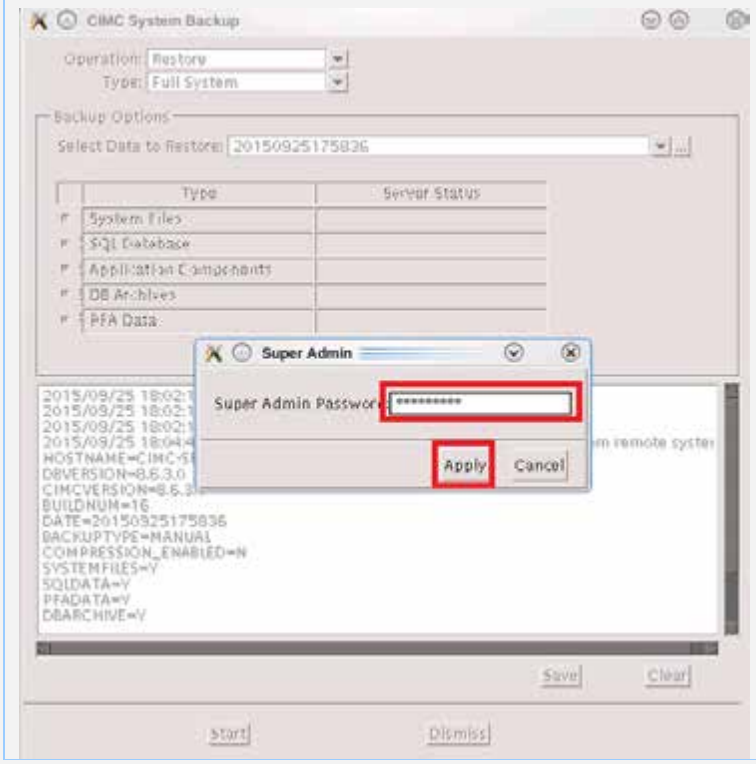
5	After status of all servers changes **Completed**, backup is successfully ended. 
6	A folder named for the date and time of the backup is created under “backups¥SystemBackups” which is located under the backup save folder set in the step 1. 
7	Five types of backup data are saved in the folder. 

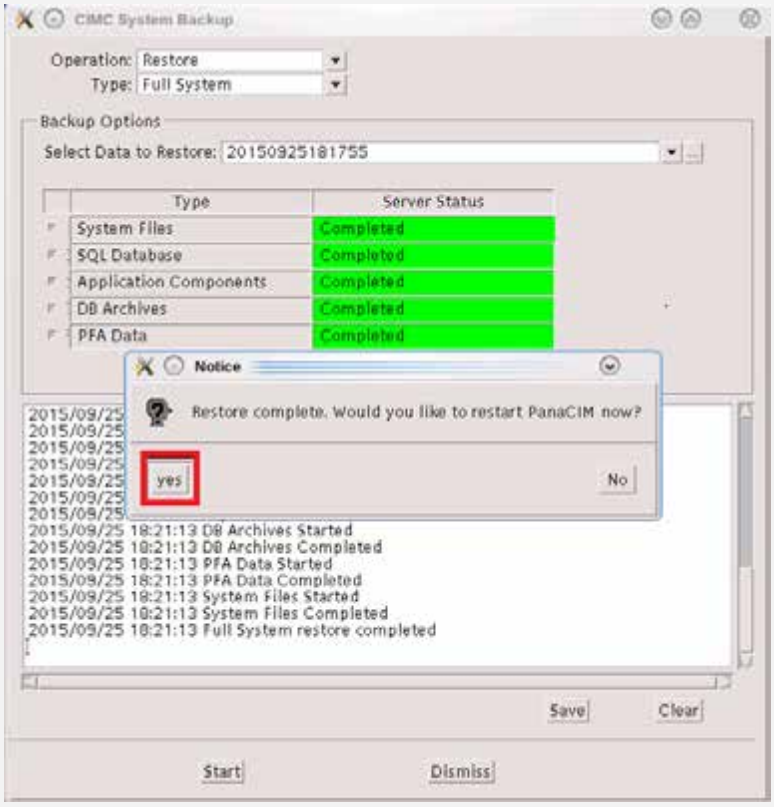
5.7.2 Restore

NOTE) When you restore data, make sure that PanaCIM is properly operated, and production is ended before starting restore.

Step	Procedure
1	On the Database Utilities – Backup Configuration screen in the PanaCIM controller server, confirm the location where you want to restore backup data. 
2	Create the “backups¥SystemBackups” folder in the backup data save folder that you are confirmed in Step1, and copy the folder named for the date and time of the backup data to be saved. 

3	On the CIMC System Backup screen of the PanaCIM controller server, change the Operation field to **Restore**. 
4	Select the folder name copied in step 2 as a target data, and after confirming that **Full System** is selected in the Type filed, press the **Start** button. 

5	After the following message appears, press **yes**. 
6	When you are asked for the password, type **panasonic** and then press the **Apply** button. 

7	After status of all servers changes **Completed**, restore is successfully ended. Press the **yes** button on the following screen and close the CIMC System Backup screen. 

5.8 [Serial – Model Conversion] Function

5.8.1 Overview

[Serial – Model Conversion] function is used to identify production data by its serial number when the PCB barcode attached to a PCB being produced carries no information available to identify the production data.

[Serial - Model String Mapping Utility], previously offered as a Windows execution program, was incorporated into the PanaCIM system as a function, and can now be operated without requiring a dedicated execution program [Serial - Model String Mapping Utility] separately.

By presetting the place where to store a CSV file to be imported to the PanaCIM side, the CSV file will be automatically imported at the timing of its storage. Also, the imported CSV file will be automatically deleted.

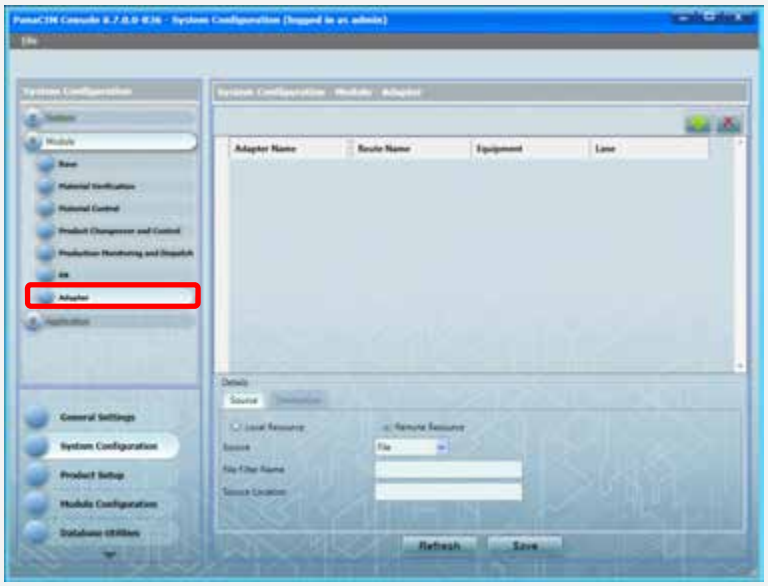
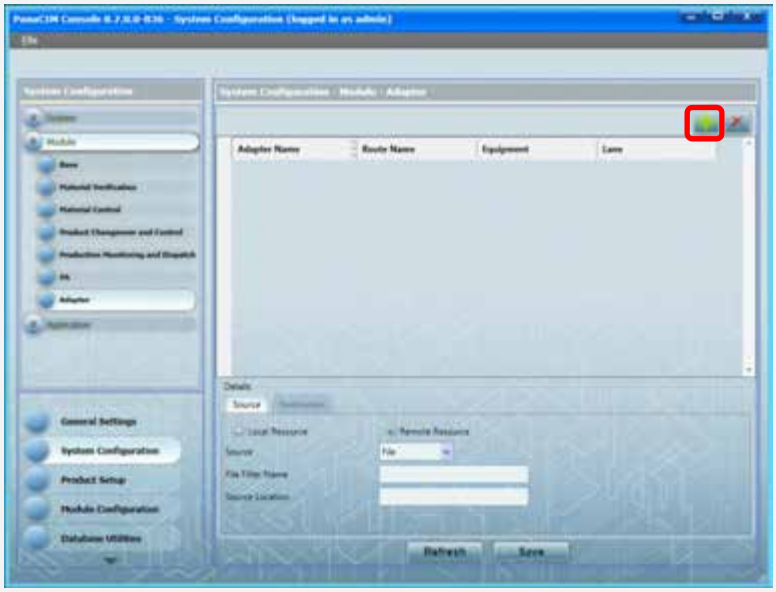
5.8.2 Restrictions

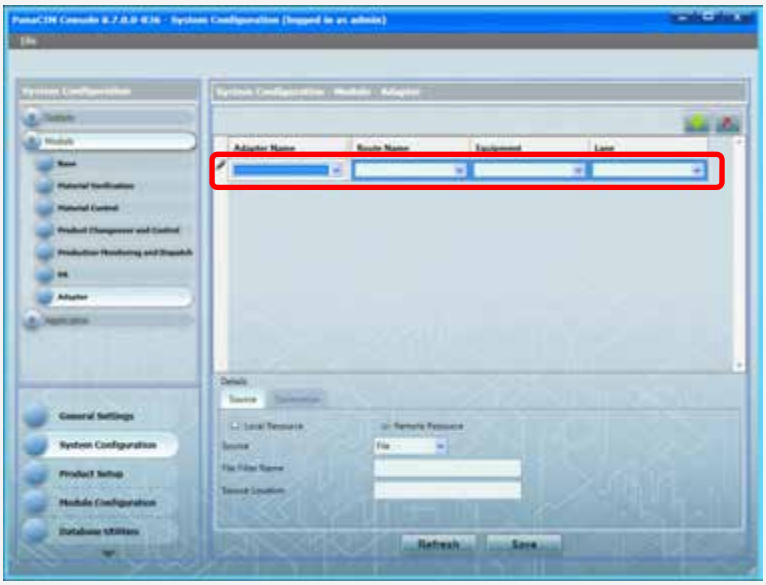
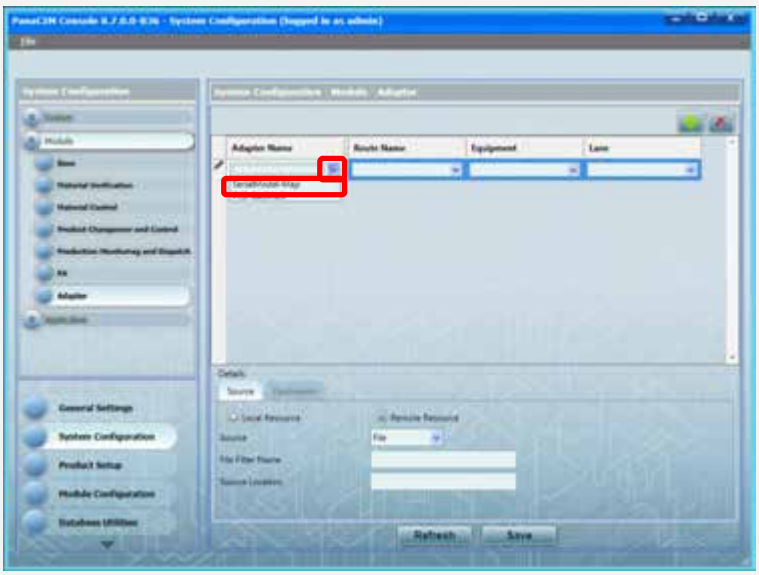
- All the routes registered with the PanaCIM system are to be executed.
  *Note that specifications cannot be made on a per-route basis.
- The model ID and serial number specified in CSV file will be up to 128 characters long each.
- The pieces of data specified in one line in CSV file will be a model ID and serial number and will be separated with a comma(,).
- A sharp (#) character, used to identify comment lines, cannot be used in CSV file.
- The serial number range will be specified as the start serial number－the end serial number.
  (Example: When specified as L2602S1MA, M01S012211001- M01S012211010, ten line of serial numbers from M01S012211001 to M01S012211010 can be registered.)
- When a CSV file describing the same serial number as an already-imported serial number is used, the CSV file cannot be imported.
- Already-imported serial numbers cannot be reregistered with different production data.
- Already-imported serial numbers cannot be reregistered with production data on another route.
- When you have changed or registered the [Adapter] settings with the PanaCIM console, be sure to restart the CIMC controller.

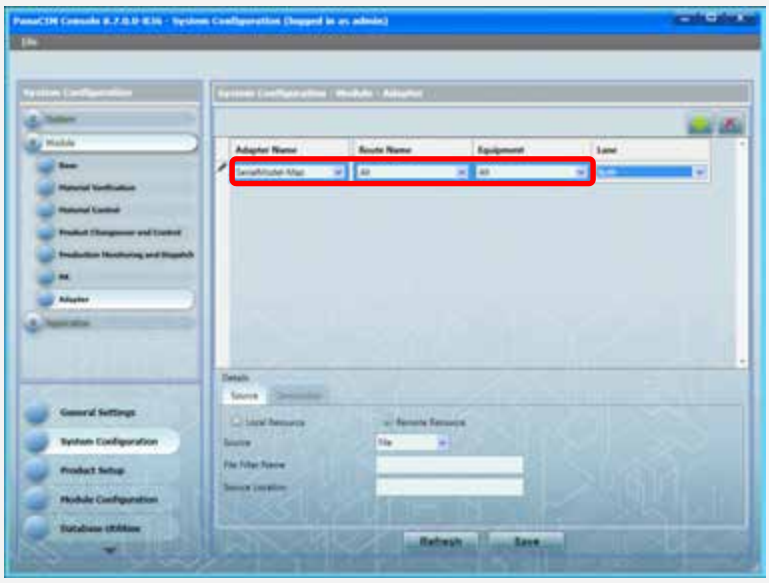
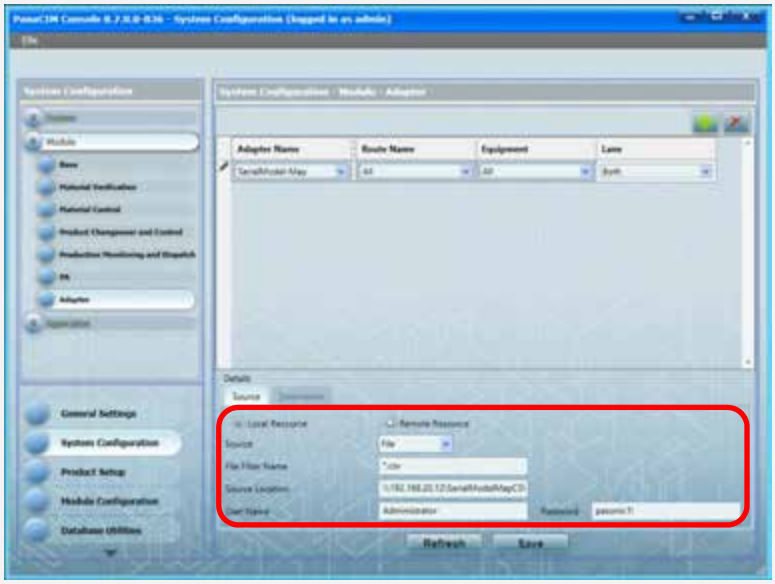
5.8.3 Settings

This section describes the basic settings to use the [Serial – Model Conversion] function with PanaCIM Console.

■ [Adapter] settings
Go to [Adapter] in the [System Configuration] – [Module] menu of PanaCIM Console to configure the settings for Serial – Model Conversion.

Step	Procedure
1	Start PanaCIM Console and select [System Configuration] - [Module] - [Adapter]. 
2	On the [Adapter] screen, click [+] icon and add line. 

	*Clicking the [+] icon can add line as shown in the screen below. 
3	On the [Adapter] screen, click the list box icon for [AdapterName] in the line added to the [Adapter] list, and select [SerialModelMap] from the displayed list. 

4	Configure [SerialModelMap] operation settings as follows. *It is possible to specify a particular route, equipment or lane; however, the settings will be configured here in terms of the system as a whole.  <Settings> ・Route Name: **Be sure to select All** ・Equipment: **Be sure to select All** ・Lane: **Be sure to select Both**
5	Set the following information for [Source] in [Detail] shown below the [Adapter] list. 

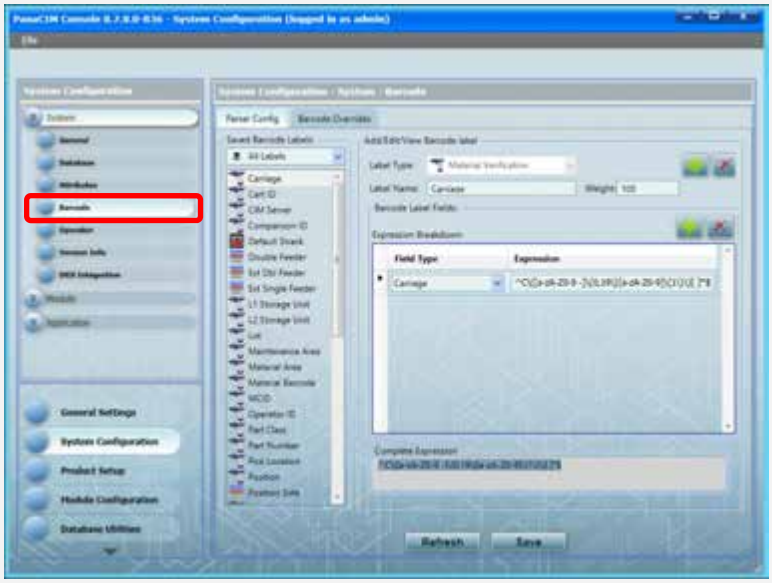
	<Settings for the Detail area> Assume that the shared folder (SerialModelMapCSV) for the application server (APP-SRV) is used to exchange a CSV file with a different server. ・Local Resource / Remote Resource: **Select Remote Resource** ・Source: File (Fixed) ・File Filter Name: **Specify *.csv** ・Source Location: **Specify the shared folder to store the CSV file** (e.g.,: ¥¥APP-SRV(192,168.20.13)¥SerialModelMapCSV) ・User Name: **Specify the authorized user name.** (Value authorized to access shared folder for APP-SRV) ・Password: **Specify the password for the authorized user name.** (Value authorized to access shared folder for APP-SRV)
6	Click the [Save] button to save the settings.

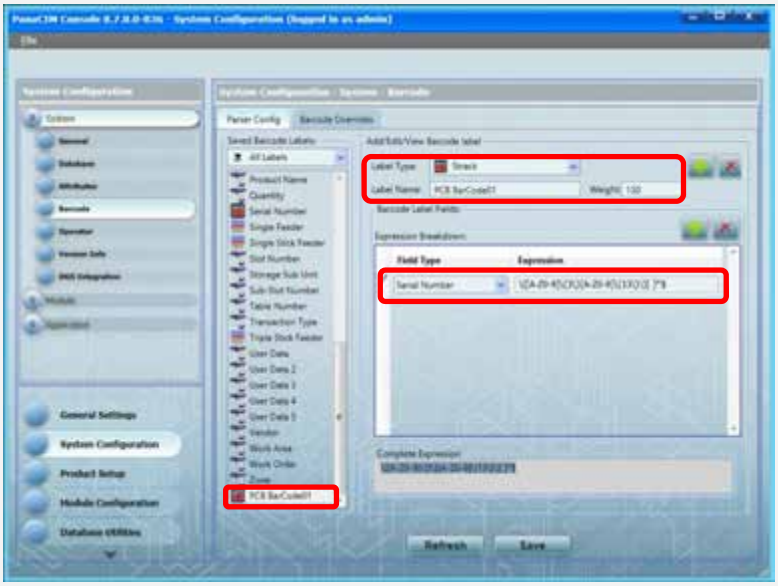
■ [Parser] configuration

◆ On the [Parser] configuration screen, set the following information that is used for the [Serial – Model Conversion] function.

- ・PCB barcode parser: Sets acceptable characters and digit number for PCB barcodes.
- ・Serial number parser: Sets acceptable characters and digit number for serial numbers.
- ・Model ID (Model string) parser: Sets acceptable characters and digit number for model IDs

Go to [Barcode] in the [System Configuration] - [System] menu of PanaCIM Console to configure the parser settings for Serial – Model Conversion.

Step	Procedure
1	Start PanaCIM Console and select [System Configuration] - [System] - [Barcode]. 
2	Select the PCB barcode parser settings [PCB BarCode01] from [Saved Barcode Labels], and change the settings as follows. *[PCB BarCode01] is an example. *If there is not any PCB barcode parser settings found in [Saved Barcode Labels], add PCB barcode parser settings.



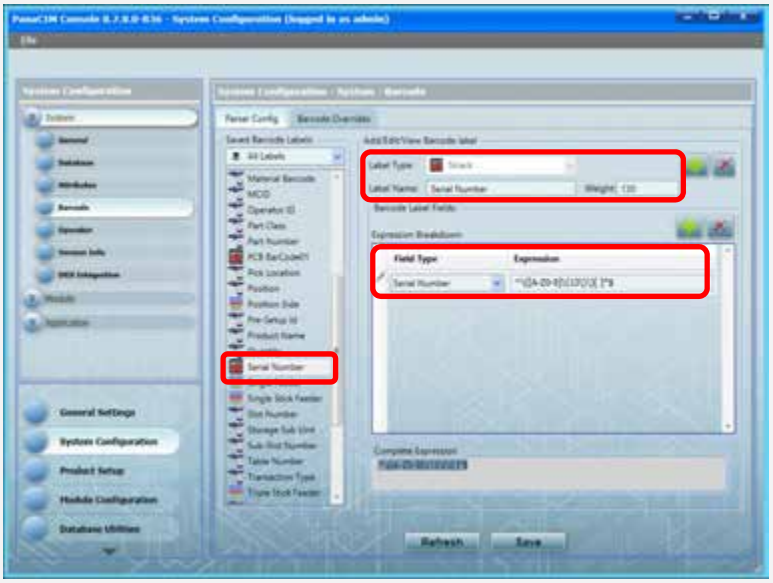
<Changes to the settings>

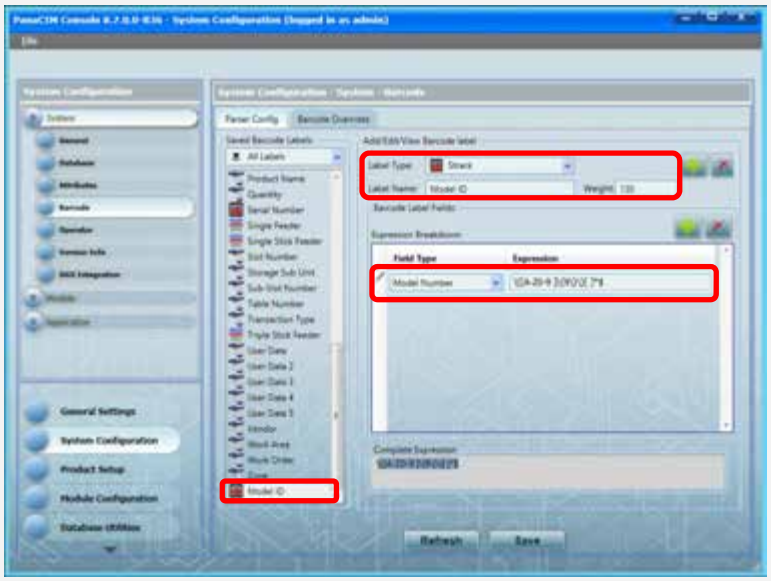
Label Type: **Strack**
Label Name: **Serial Number**
Weight: **130**

Barcode Label Fields
Field Type: **Serial Number**
Expression: **^[A-Z0-9]¥{3¥}¥([A-Z0-9]¥{9¥}¥)[ ]*$**

*The above parser settings are just an example, so configure the settings in a manner appropriate for the format of the PCB barcode that you actually use.

3 Select the serial number parser settings [Serial Number] from [Saved Barcode Labels], and change the settings as follows.



	<Changes to the settings> Label Type: **Strack** Label Name: **Serial Number** Weight: **130** Barcode Label Fields Field Type: **Serial Number** Expression: **^¥([A-Z0-9]¥{10¥}¥)[ ]*$** *The above parser settings are just an example, so configure the settings in a manner appropriate for the serial number format of the PCB barcode that you actually use.
4	Select the model ID parser settings [Model ID] from [Saved Barcode labels], and change the settings as follows. *[Model ID] is an example. *If there are not any model ID parser settings found in [Saved Barcode Labels], add model ID parser settings.  <Changes to the settings> Label Type: **Strack** Label Name: **Model ID** Weight: **130** Barcode Label Fields Field Type: **Model Number** Expression: **^¥([A-Z0-9]¥{9¥}¥)[ ]*$** *The above parser settings are just an example, so configure the settings in a manner appropriate for the model ID (model string) format of the production data that you actually use.

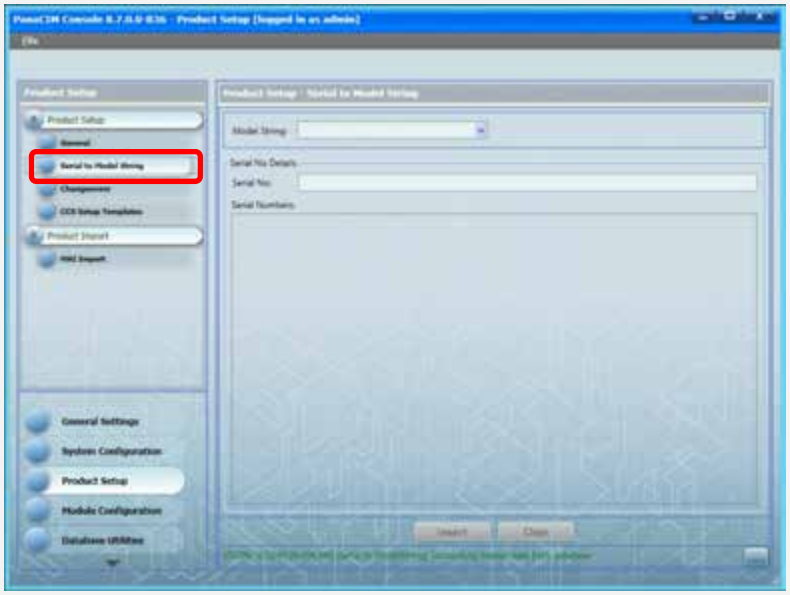
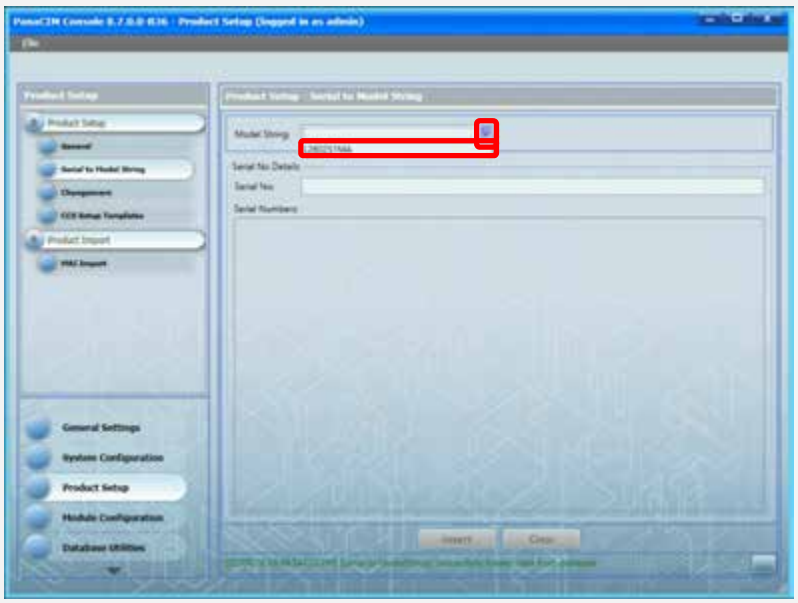
5.8.4 Operation

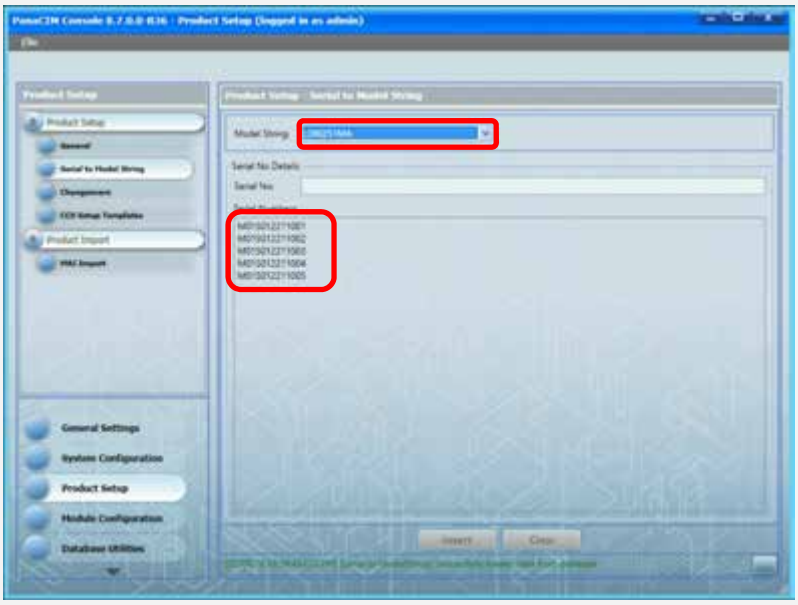
This section describes the procedure for operating the [Serial – Model Conversion] function using PanaCIM Console in regard to the two points shown below.

■ Importing [CSV file]
The following shows the procedure for placing the desired CSV file in a shared folder on the application server (APP-SRV) of PanaCIM.

Step	Procedure
1	Prepare a CSV file to be imported to the PanaCIM side. *The following CSV files are set up and used as examples. L2602S1MA,M01S012211001 L2602S1MA,M01S012211002 L2602S1MA,M01S012211003 L2602S1MA,M01S012211004 L2602S1MA,M01S012211005
2	Copy the CSV file PanaCIM set up in a shared folder on the application server (APP-SRV). *As an example, a CSV file will be copied onto the shared folder specified in "5.8.3Settings - [Adapter] settings." Copy to: [SerialModelMapCSV] folder set to be shared
3	After a while the copied folder will be deleted. *At this point, the data in the copied CSV file will be imported to the PanaCIM side.

■ Confirming [Serial – Model Map] registration information
The following shows the procedure for confirming the serial number actually imported on the [Product Setup] screen of PanaCIM Console.

Step	Procedure
1	Start PanaCIM Console, and select [Product Setup] - [Serial → Model ID]. 
2	On the [Product Setup – Serial → Model ID] screen, display the list box for [Model ID], and select the target model ID.  *In the example above, the model ID (Model String) below is selected. ・ Model ID: **L2602S1MA**

3	The serial number imported from the CSV file is displayed in the [Serial No] field on the [Product Setup – Serial → Model ID] screen.  *The above example shows the result of the import of the CSV file, for which any of the following serial numbers is specified, into the model ID (**L2602S1MA**). ・ Serial No.: **M01S012211001** **M01S012211002** **M01S012211003** **M01S012211004** **M01S012211005**

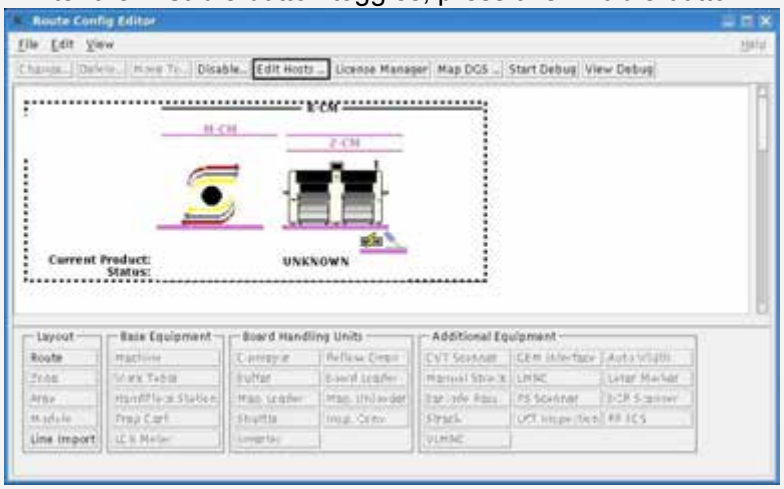
= MEMO =

6. TROUBLESHOOTING

This chapter explains troubleshooting.

6.1 Troubleshooting

Examples	Solutions
Cannot log into PanaCIM controller with panadmin/panasupr/panoper account and returns to Login screens.	May not be able to communicate with PanaCIM Database. Check starting/setting of SQL server and try again.
PanaCIM toolbar does not start and message appears to confirm if PanaCIM Postmaster has started.	PanaCIM system service application may have stopped. Open Linux application <Terminal> and run the following command. panadmin> **pana_start** Wait 1~2 minutes and restart PanaCIM tool bar.
PanaCIM controller server application is locked and does not work.	The application does not respond for some reason. Go to start menu [system] - [KSysGuard] and start KDE System Guard. Select executable files (see below) to be terminated from running process list and press [Kill] button to terminate application(s). Executable files: • Route Status: route_status • Route Editor: route_editor • Product Transfer: product_transfer • Product Setup: product_setup • System Configuration: system_config
PanaCIM does not communicate with equipment. (Equipment icon is not turned green on Route Status/Route Editor screen.)	The following factors may fail communication. Revise PanaCIM/ equipment settings. • Equipment not started: Start equipment. • PT interface not started: Start PT interface • Not connected to the network: Make sure all systems/equipment are connected with LAN. • PanaCIM application server not started: Start PanaCIM application server. • Network setting modified: Check and set network setting of all systems/equipment.

Examples	Solutions
Machine icons displayed on the Route Status/Route Editor of the PanaCIM controller remain pink, thereby disabling connection. => In this state, verification is disabled and operation information cannot be collected.	● It may take some time for the PC to start up. Wait for a while and check the display again. ● The PT interface may not have started up. When the PT interface PC has started up, open “PT-01 Cells Configuration” in the taskbar, and check that “Status” is ONLINE. ● If, in spite of the above solution, the icons still remain pink, open the Route Setting Editor at the PanaCIM controller PC, select the whole line, and then press the **Disable** button. After the **Disable** button toggles, press this **Enable** button.  In spite of the above solution, connection and verification are still disabled. => A certain mismatch may have occurred on the PanaCIM. ● Choose **Konsole** from the right-click menu on the Linux desktop, and execute the following commands in order. panadmin> pana_stop panadmin> pana_start Since it takes several minutes for the PanaCIM service to restart, wait for several minutes, and then click the **Application** icon in the upper left to start the toolbar.
You cannot log in to the PanaCIM controller PC when it starts up. Or you can log in, but the following message appears. "Unable to open mailbox. verify that the PanaCIM Postmaster is running."	Connection with the SQL server may not be established. ● Check that the PanaCIM application (Windows Server) PC has started up. ● It may take some time for the PC to start up. Wait for a while and check that again.
The following message appears when the PanaCIM controller (Linux) PC starts up. "PanaCIM is running base mode."	This message appears when the license check on starting is not complete. ● You can ignore it because the license check finishes in the background.

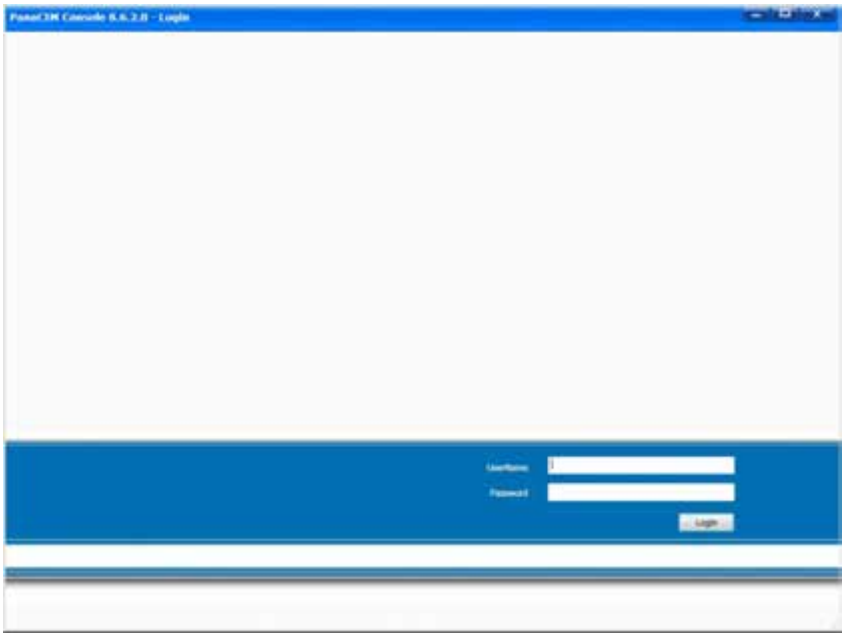
6.2 Analysis/Utilities

This section describes the PanaCIM Console that provides information about the database and a monitoring log of the system, and the Service Controller that starts, stops, and restarts various PanaCIM-related services.
These utilities are also used for troubleshooting.

Each application of the PanaCIM-EE Console supports various troubleshooting in the PanaCIM. They show the database, the history data, and the application states, and are equipped with the following applications.

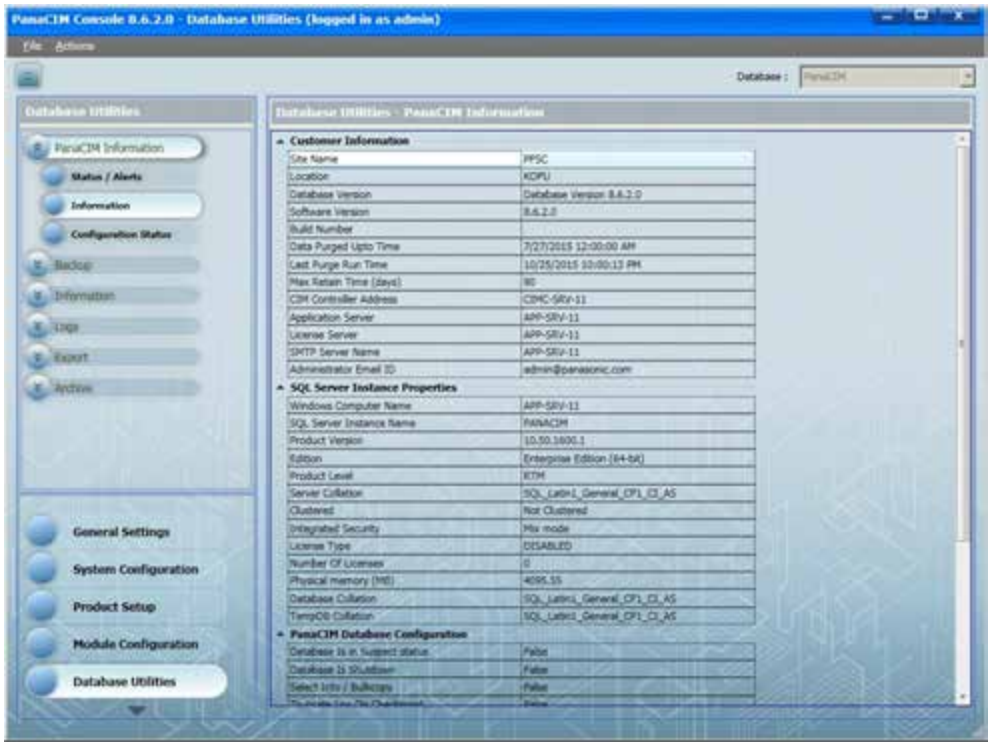
- Database utility
- System monitoring
- Application status

From the program menu, choose **Panasonic** > **PanaCIM** > **MES Services** > **PanaCIM Console**, and log in to it. Enter your user name and password, and click **Login**.



6.2.1 DataBase Utility

Database utility shows PanaCIM Database information. Select Database option from the drop-down menu on the left side of the page.



PanaCIM information

This section is divided into the following groups.

- **Situation/Warning:** Shows utility status. If Utility has a problem, it is shown on this page.
- **Information:** Shows database name, License character strings and database information.
- **Setting status:** Shows the setting details of overall factory (machine, route, database etc). If you click **Save** on this page, copy of setting information can be saved for troubleshooting.

Backup

This section is divided into the following groups.

- **Backup setting**: Set database backup during night.
  Note: Specify physical drive of database machine for database. Enter location details on **Run Backup** tab and make sure the location is correct.
- **Backup details**: Save HistoryLog of backups.

Information

This section is divided into the following groups.

- **File group details**
- **Table**
- **Index**
- **Job schedule**
- **Disk space**
- **Database connection**
- **Database management**
- **SQL server lock**
- **File archive details**

The information is used for the purpose of confirmation, but can also be used for troubleshooting. Factory information that are included in Factory tabs can be saved. In case of troubles, the information can be sent to us.

Log

This section is divided into the following groups.

- **SQL agent**: SQL agent detail logs
- **SQL server**: .SQL error details
- **Database mail**: Error logs in case of failures

Export

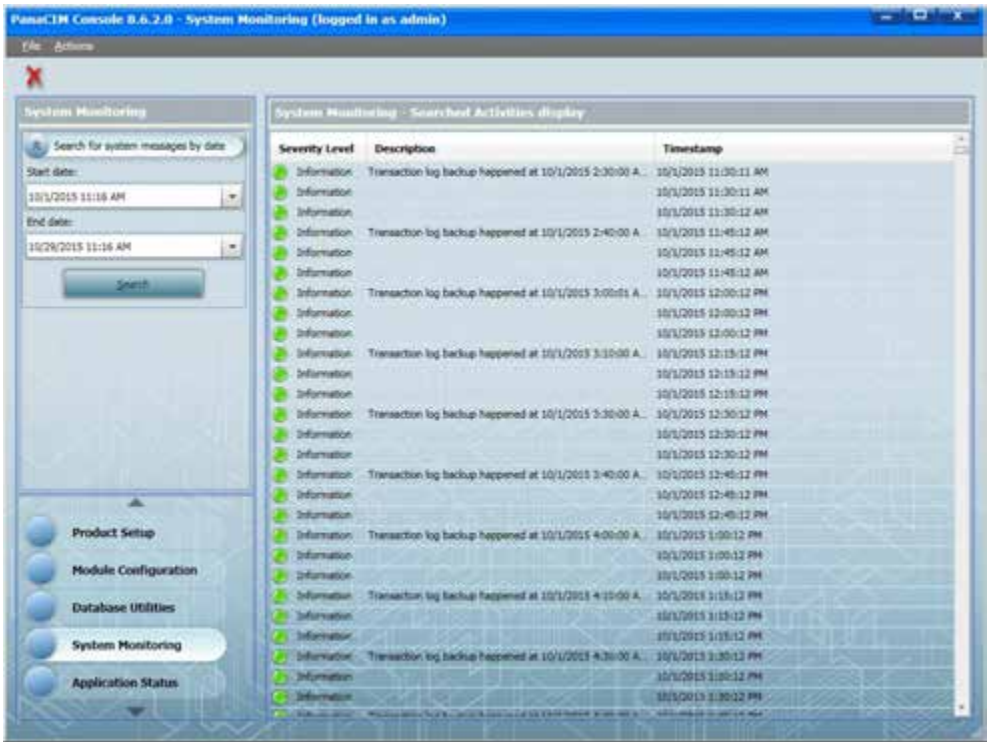
This section is divided into the following groups.

- **Download report**
- **E mail report**

You can collect database configuration and other information by another method. Save or send information directly by email from this page.

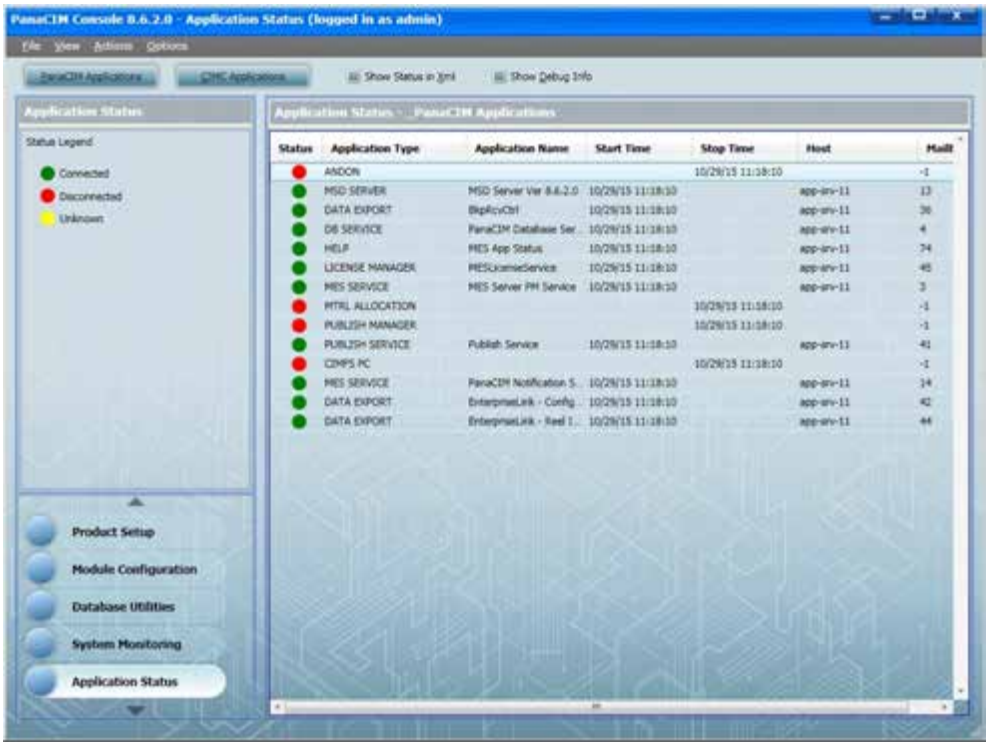
6.2.2 System Monitoring

History data from the specified start to end date is shown here.



6.2.3 Application Status

Check application status from this page. See the legends on the left side and check application status.



Click **PanaCIM application** or **Controller server application** to see the running status of respective applications.

6.3 Service Controller

The Service Controller consolidates all of the various services on the PanaCIM Windows Application server. Use the Service Controller to Stop, Start, or Restart programs.

Step	Procedure
1	Click **Start** > **Programs** > **Panasonic** > **PanaCIM** > **PanaCIM Services** > **ServiceController**.
2	Click the **PanaCIM Service Controller** icon  located in your system tray.
3	• Start, Stop, or Restart all Applications by selecting the action from the top left of the screen. • Start, Stop, or Restart Applications individually by selecting the corresponding action on the top right hand of the screen. You may also choose to Configure, Monitor, or start the Simulator from this section as well. 

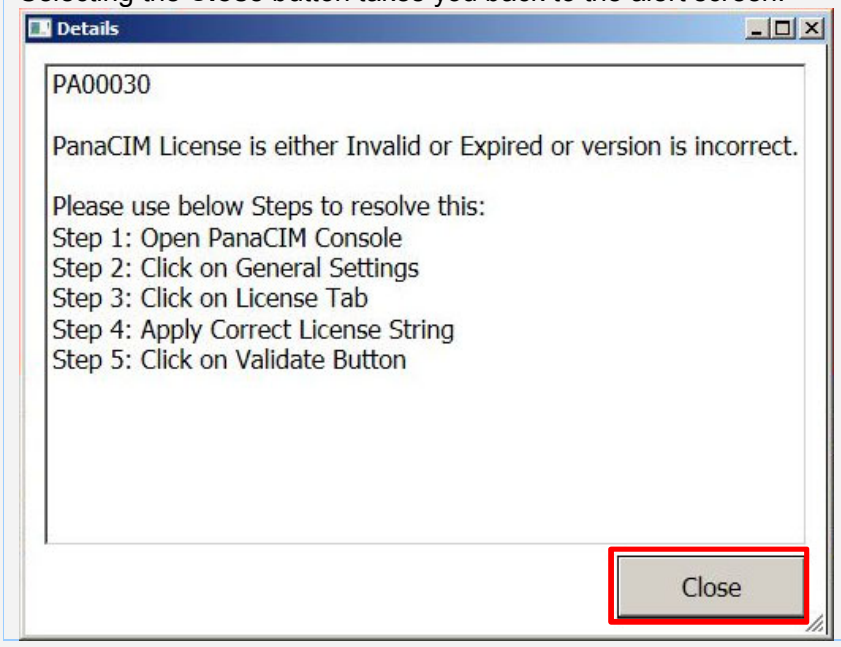
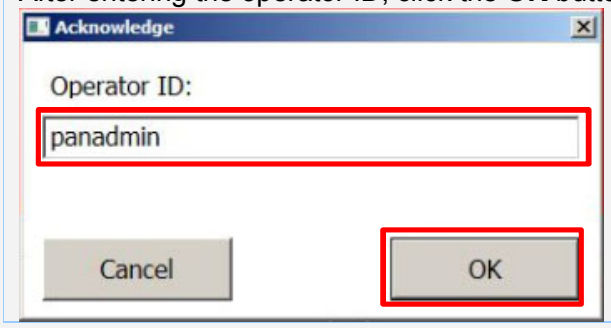
6.4 Production Alerts

The production alert function is executed in the background as a service of the PanaCIM application server. It displays the alert screen when a problem occurs in the PanaCIM system.
The alert screen remains displayed until the problem is solved, all the alerts have received user's "Check," or the **Abort** button is selected.



6.4.1 Alert Check Procedure

Step	Procedure
1	The alert screen is displayed when the service detects a problem.  Description: Description of the alert Date and Time of Occurrence: Date and time of the occurrence of the alert Period: Time elapsed since the alert occurred Level: Alert level (error, alert, information)

2	Selecting the **Detail** button of each alert displays the details of the alert and the procedure for solution. Selecting the **Close** button takes you back to the alert screen. 
3	To accept the description of the alert, place a checkmark in its checkbox, and select the **Check** button (if you place a checkmark in **Select All**, all the alerts are checked off). 
4	After selecting the **Check** button, you will be prompted to enter the operator ID. After entering the operator ID, click the **OK** button. 

5	The alerts that have received “Check” will be deleted from the alert screen. 
6	Selecting the **Abort** button hides the alert screen temporarily. The alert screen will be hidden for 5 minutes. If the problem is not solved after the time has passed, the alert screen will be displayed again. 

6.4.2 Displayed Items of Production Alert

Problems detected on PanaCIM control server:

No.	Displayed Problem and Operation
1	Pana_stop command executed
2	cimc_stop command executed
3	Route disabled
4	PT200 system disconnected from network
5	Abnormality in access to shared folders below of PT200 system ・Strace folder ・Product Report folder ・Navi folder ・External output data folder
6	File outputs below are not set by Proviewer of PT200 system. ・Production management information file output ・Machine event file output ・PT event file output
7	LNB system disconnected from network
8	Abnormality in access to shared folders below of LNB system ・/lnbroot/othersystem/MngData folder ・/lnbroot/lwseditor folder
9	Memory occupancy or CPU occupancy has exceeded 90% for 30 seconds or more.
10	PanaCIM database disconnected from network
11	PanaCIM application server disconnected from network
12	PanaCIM license server disconnected from network
13	PanaCIM control client and PMD client disconnected from network
14	Zone or interface duplicated
15	Abnormality has occurred in PanaCIM service (dbpurge/cimd/postmaster).
16	Difference in system time against other systems (PanaCIM application server, PanaCIM license server, PanaCIM database server, LNB system, PT200 system) has occurred for 30 seconds or more.
17	Email notification to PanaCIM application server failed.
18	In a certain partition, hard disk occupancy has exceeded 75%.
19	Firewall is enabled.
20	Linux security is enabled
21	State of program under execution is unknown for 60 seconds or more.

Problems detected on PanaCIM application server:

No.	Displayed Problem and Operation
1	Abnormality has occurred in a certain service.
2	PanaCIM license server disconnected from network
3	Abnormality in external output export
4	Abnormality in post master
5	User of PanaCIM Web application is invalid.
6	IIS service is stopped, or settings of IIS service are abnormal.
7	Memory occupancy or CPU occupancy has exceeded 80% for 30 seconds or more.
8	In a certain partition, hard disk occupancy has exceeded 75%.
9	Time synchronization with PanaCIM database server failed.

10	Security strengthening of Internet Explorer is enabled.
11	Message queue service is disabled.

Problems detected on PanaCIM database server:

No.	Displayed Problem and Operation
1	Backup failed.
2	Hard disk occupancy has exceeded 75%.
3	Transaction log output is disabled, or transaction log failed to be saved.
4	Memory occupancy or CPU occupancy has exceeded 80% for 30 seconds or more.
5	Database size has exceeded database size alert.

Problems detected on PanaCIM license server:

No.	Displayed Problem and Operation
1	Abnormality in key tag connection, expiration of license, incorrect license string or version
2	Time synchronization with PanaCIM database server failed.

Problems detected on PT interface:

No.	Displayed Problem and Operation
1	Memory occupancy or CPU occupancy has exceeded 80% for 30 seconds or more.
2	Time synchronization with PanaCIM database server failed.

Problems detected on DGS:

No.	Displayed Problem and Operation
1	Memory occupancy or CPU occupancy has exceeded 80% for 30 seconds or more.
2	Backup failed.
3	Hard disk occupancy has exceeded 75%.
4	Time synchronization with PanaCIM database server failed.

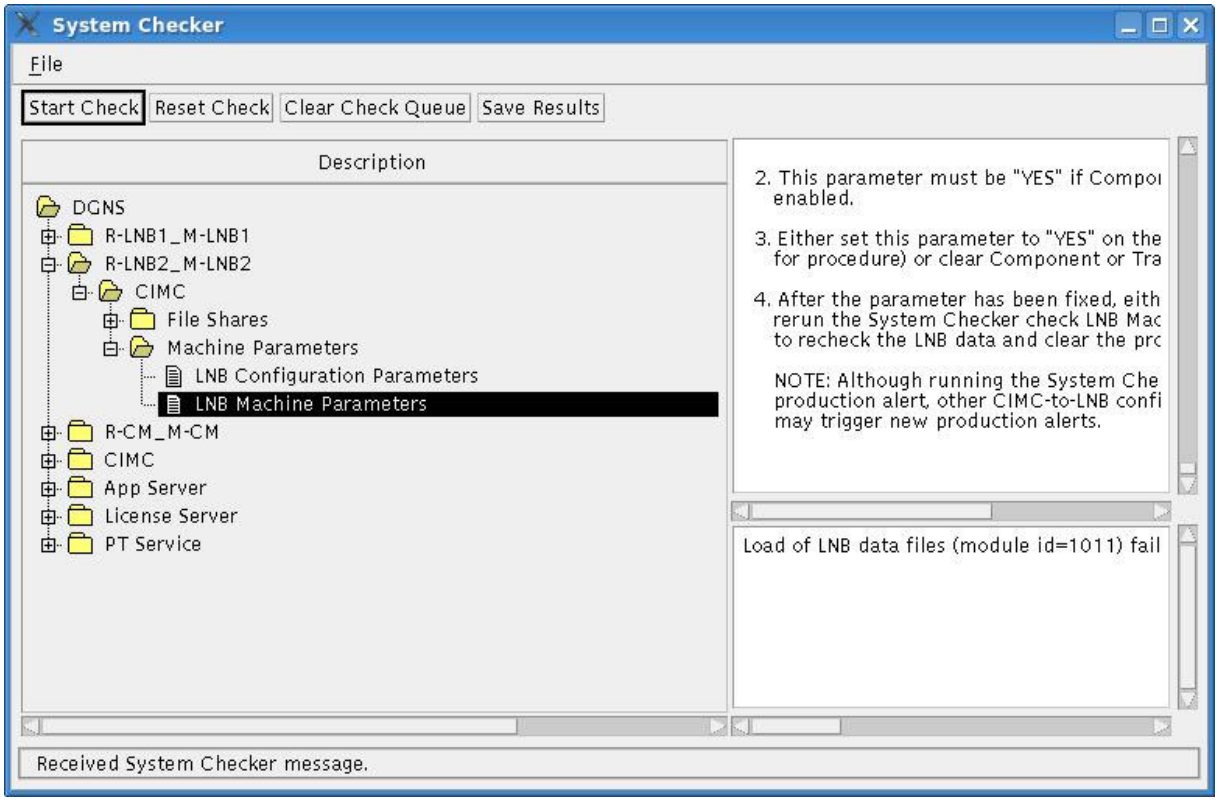
Problems detected on control client and PMD client:

No.	Displayed Problem and Operation
1	PanaCIM control server disconnected from network
2	PanaCIM database disconnected from network
3	PanaCIM application server disconnected from network
4	Hard disk occupancy has exceeded 75%.
5	Memory occupancy or CPU occupancy has exceeded 80% for 30 seconds or more.
6	Time synchronization with PanaCIM database server failed.

6.5 LNB/NPM/SP Parameter check

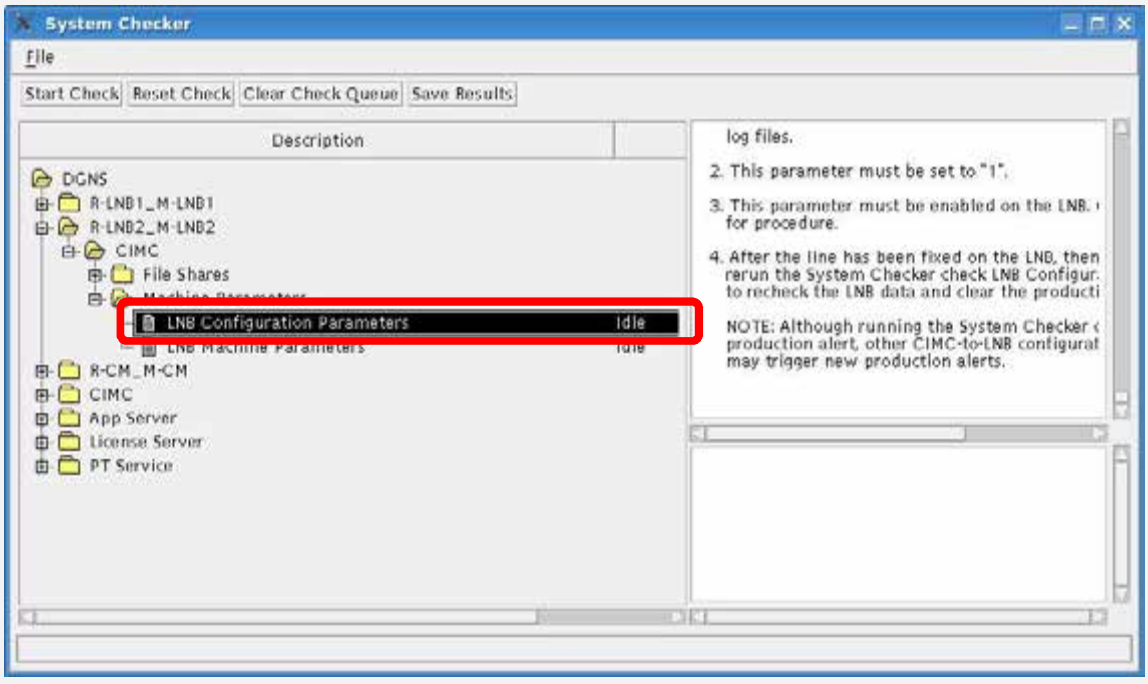
6.5.1 Overview

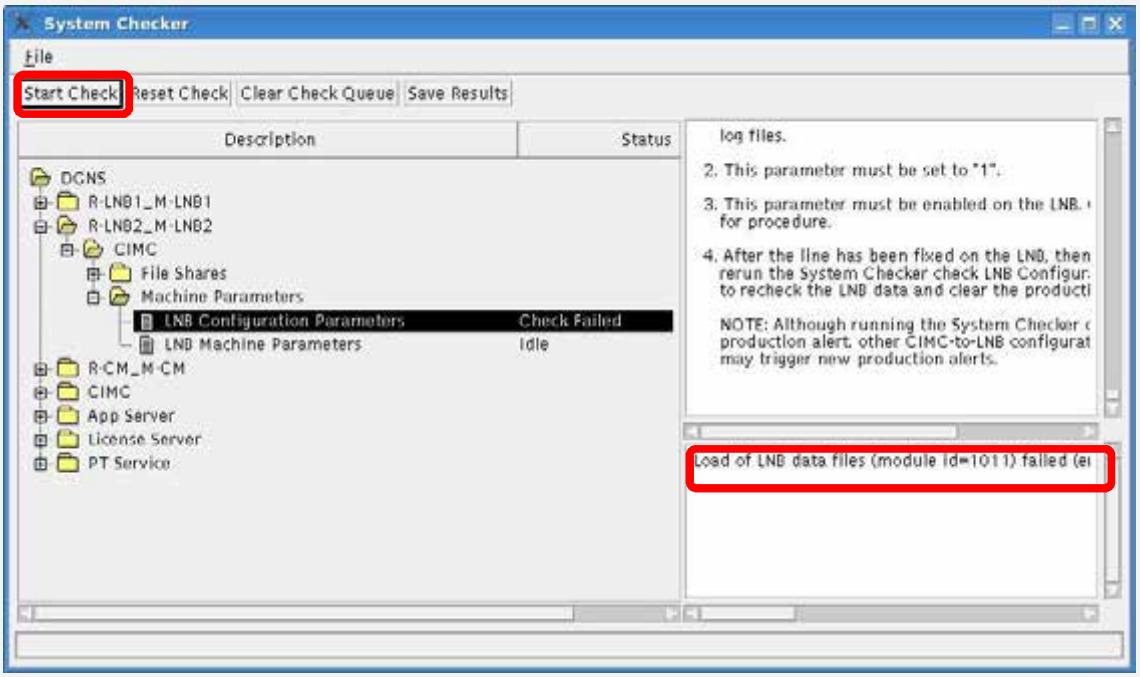
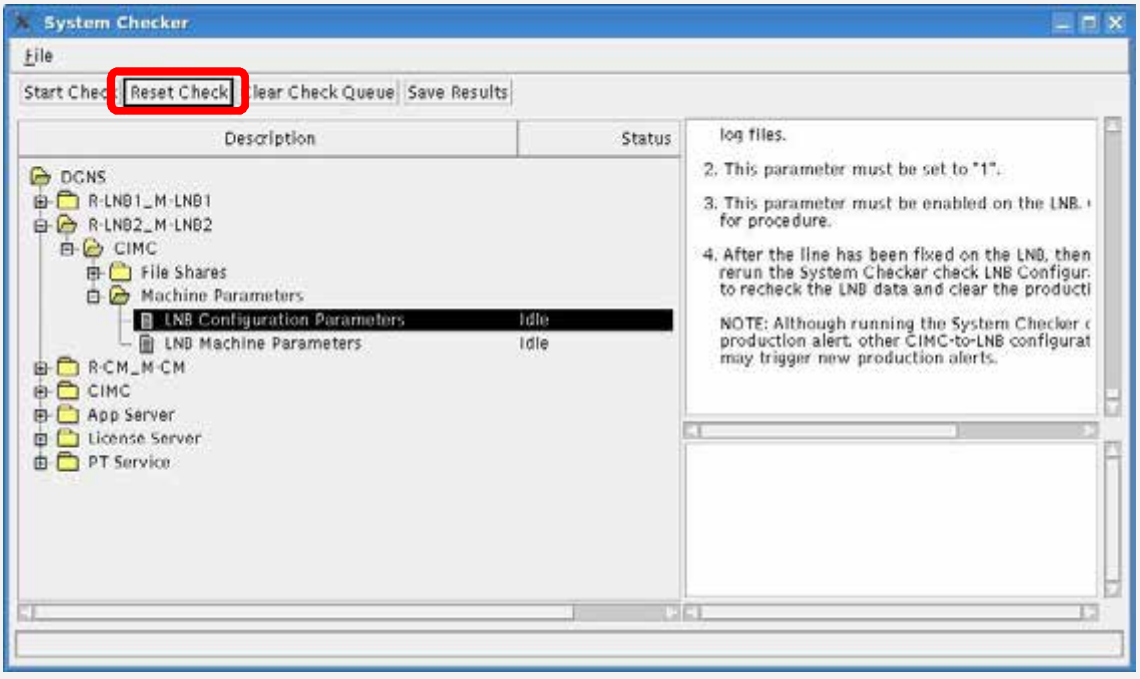
The [System Checker] tool of the CIMC controller can check the SQL server, the application server, the license server, the CIMC server, the LNB, and the PT200/PT interface computer for any problem with settings, or whether they can communicate correctly.
This section describes how to use this [System Checker] tool for checking the connection between PanaCIM and LNB/NPM/SP, and the various configuration parameters of machines and LNB.



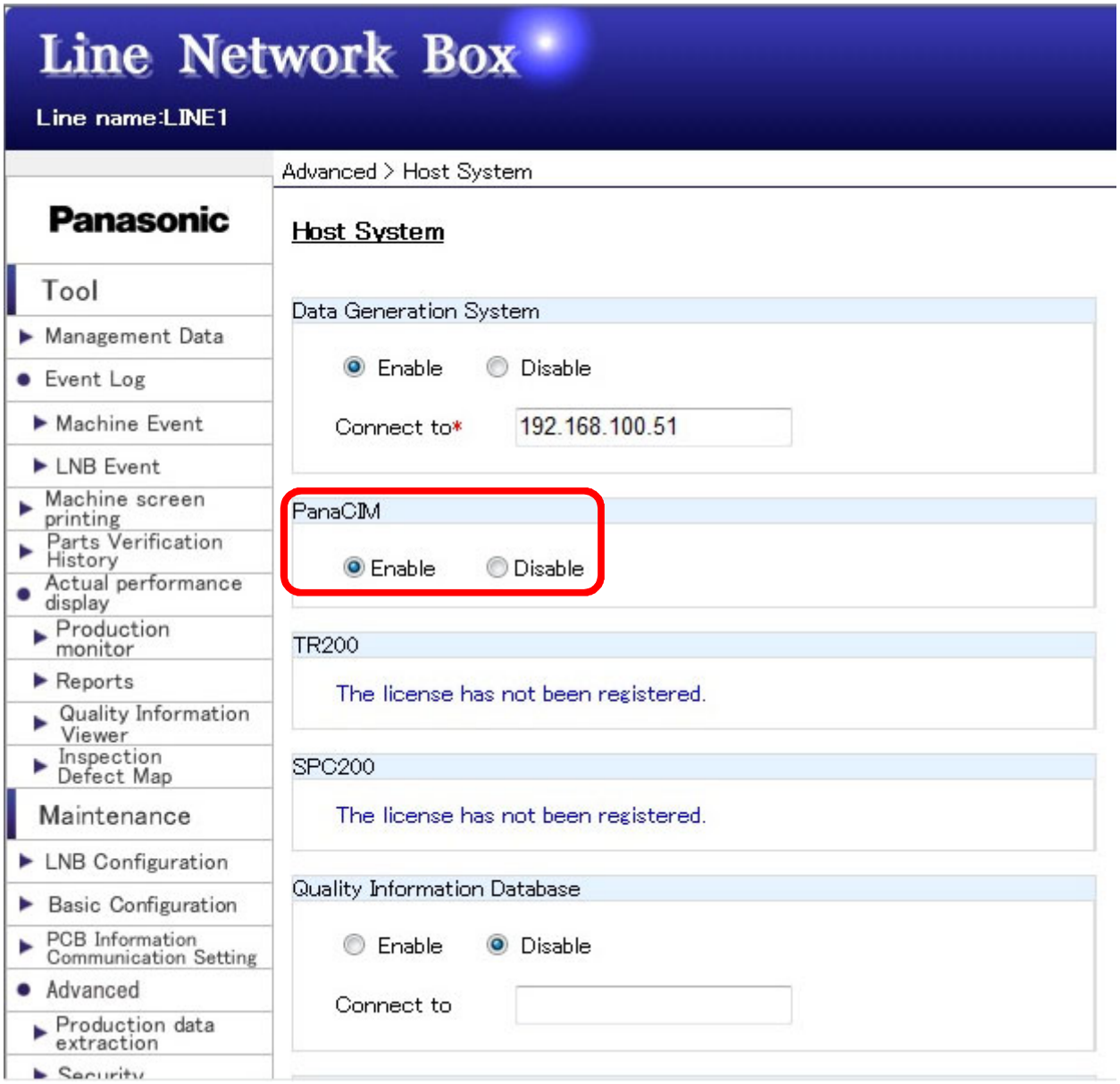
6.5.2 LNB Configuration Parameter Check

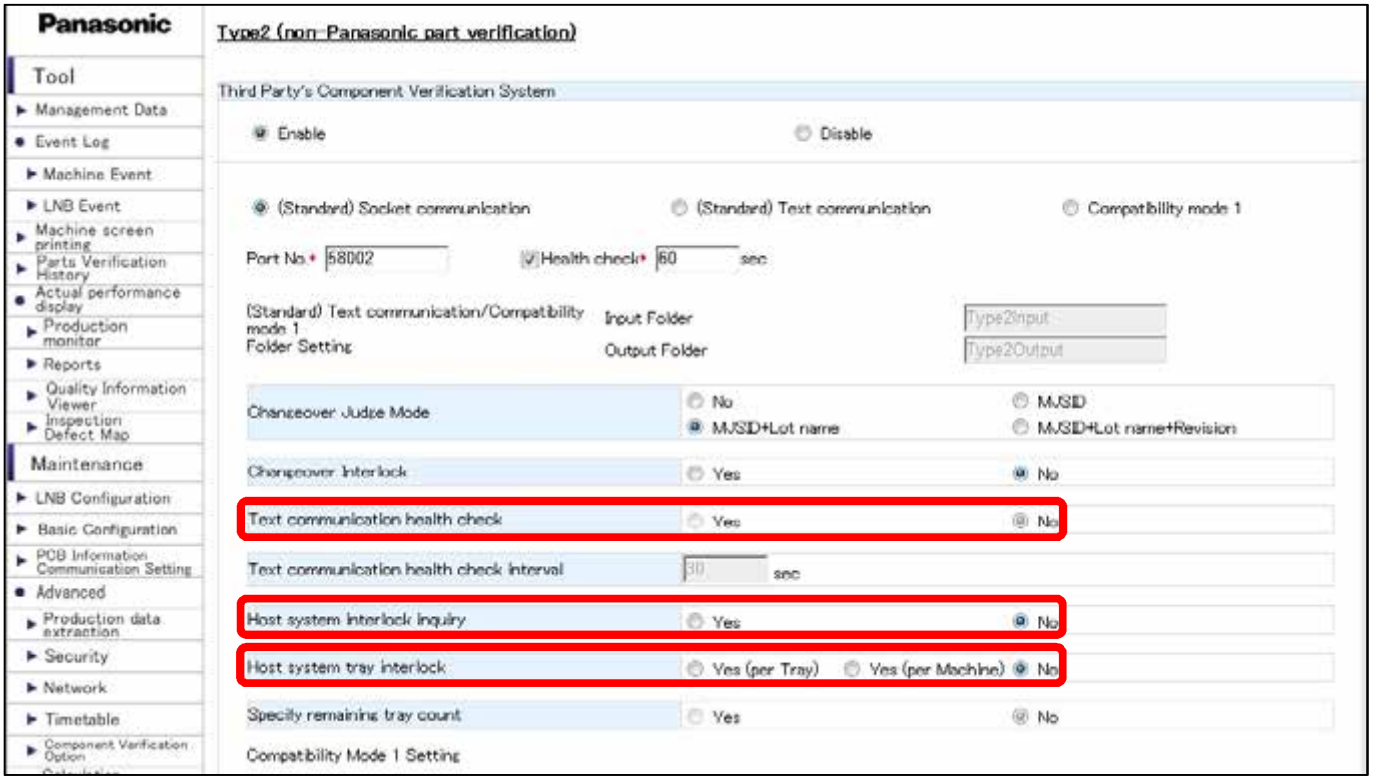
The following describes how to check the configuration parameters set on the LNB.

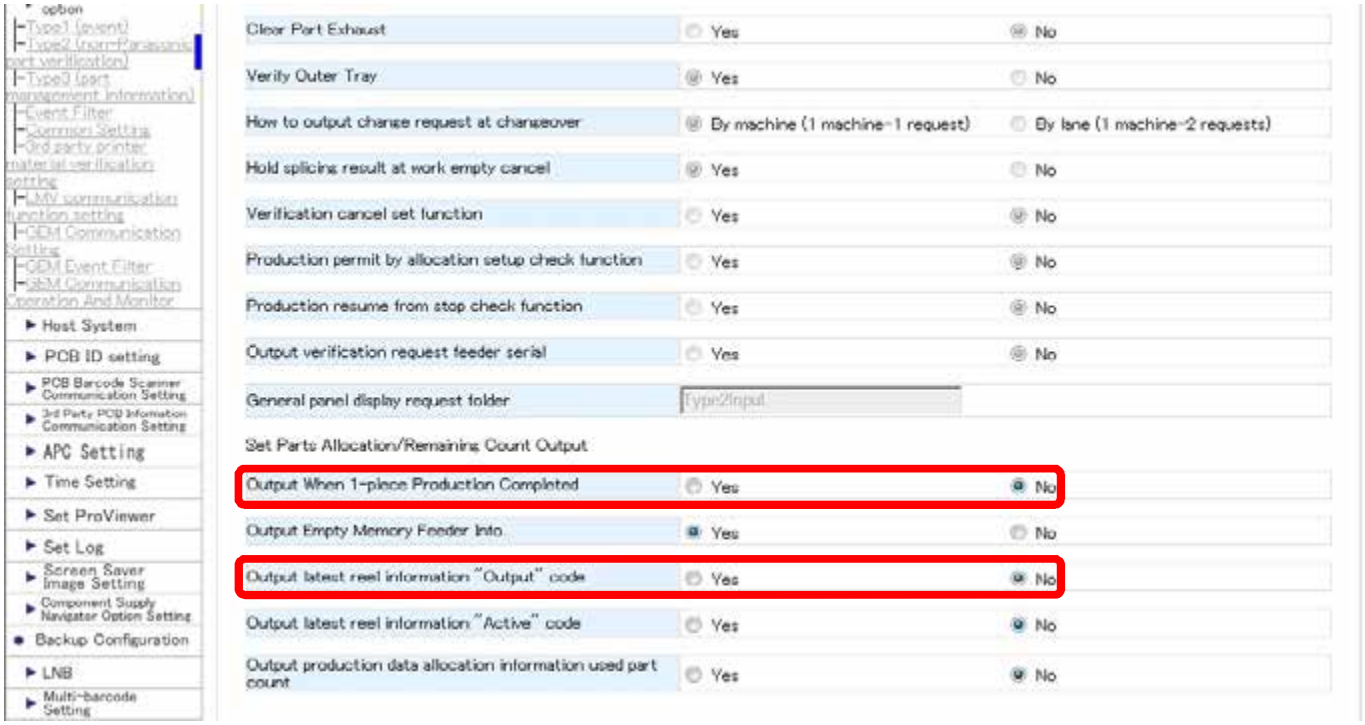
Step	Procedure
1	Modify the following file, and then restart the LNB. File to modify: **/Inbroot/configuration/CurrentResourceProprety.ini** Location to modify: **[Option]** **PanaCIMLnbConfiguration= 1 (0:Disable/1:Enable)**
2	Click **Diag Tools** in the PanaCIM launcher bar. 
3	Open the Tools Launcher screen. Choose **System Checker**. 
4	Choose **LNB Configuration Parameters** in **DGNS** - **LNB Route** - **CIMC** - **Machine Parameters**. * The user can execute the desired level of check by expanding the folder. 

5	Press the **Start Check** button. The result appears in the lower right window. OK: Check Passed NG: Check Failed In case of NG, the cause appears in the lower right window. Apply an appropriate solution to that. 
6	To execute a recheck, press the **Reset Check** button. When the status changes to **Idle**, perform step 4 again. 

• Items to be checked in LNB configuration

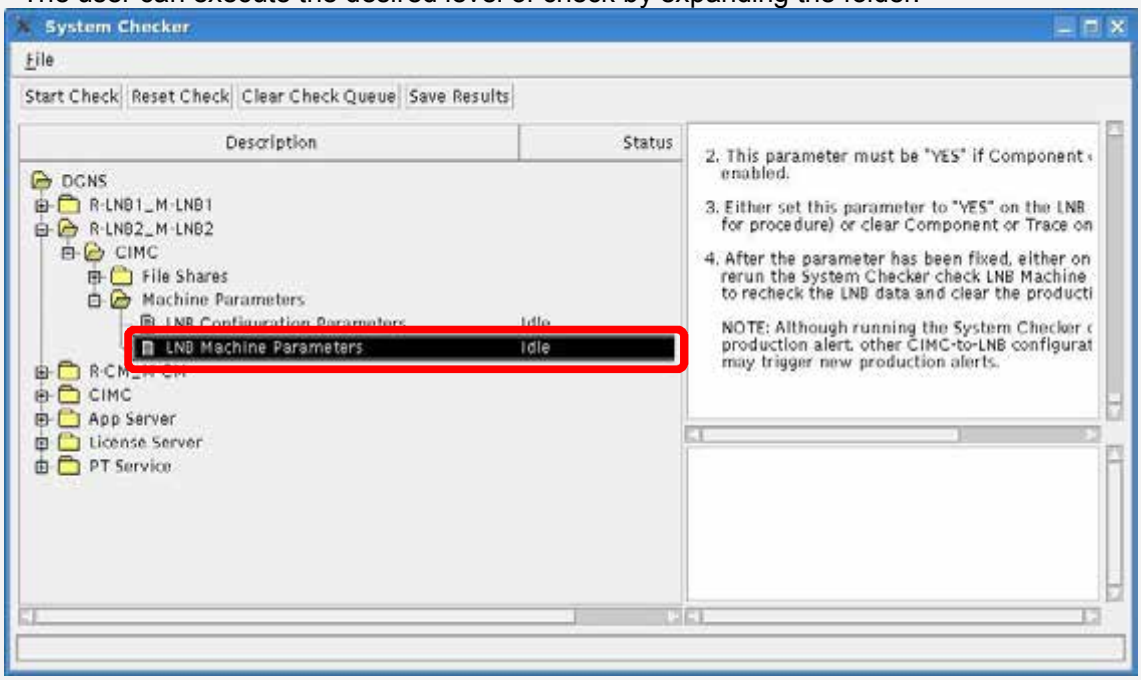


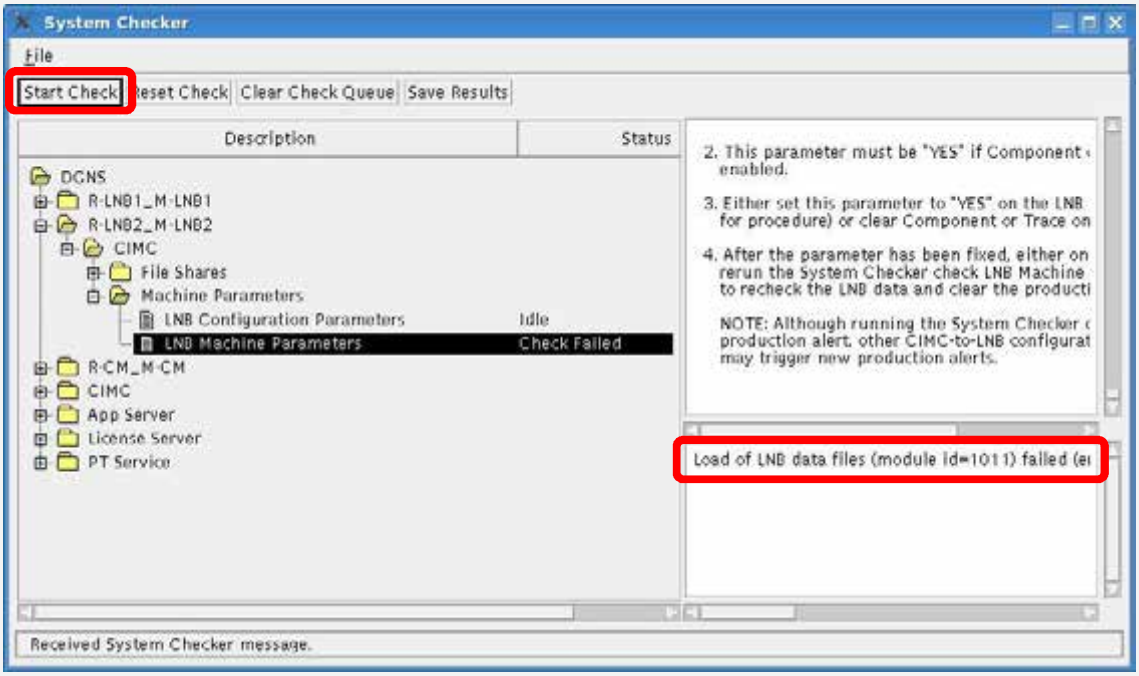
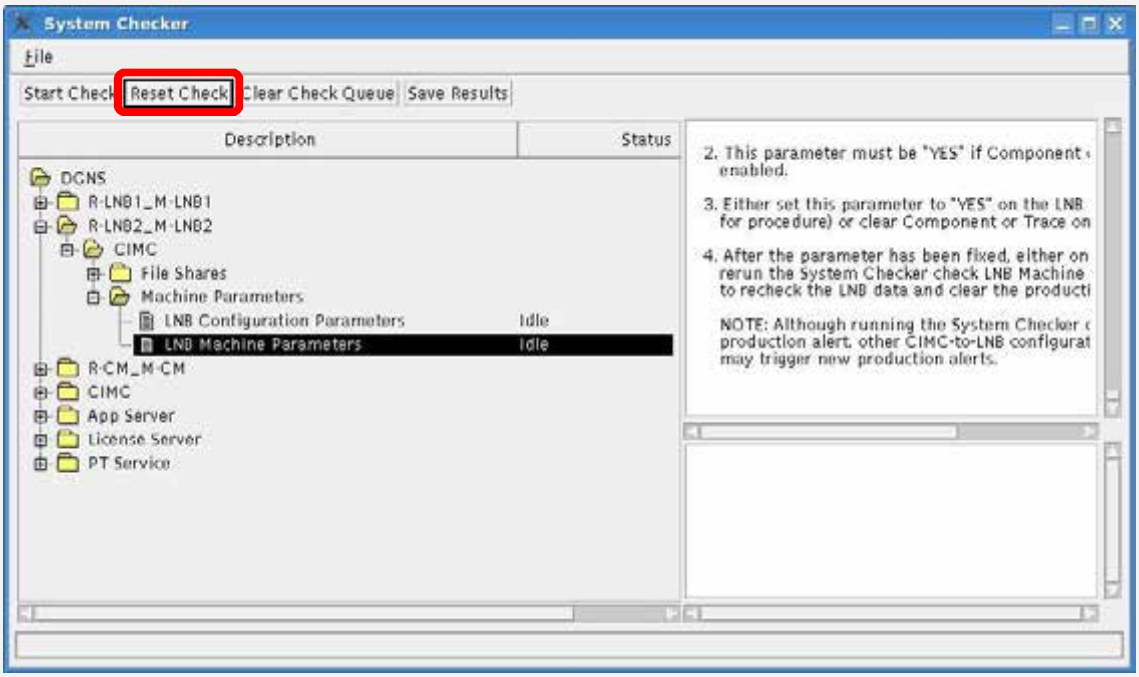

Panasonic
Tool
Management Data
Event Log
Machine Event
LNB Event
Machine screen printing
Parts Verification History
Actual performance display
Production monitor
Reports
Quality Information Viewer
Inspection Defect Map
Maintenance
LNB Configuration
Basic Configuration
PCB Information Communication Setting
Advanced
Production data extraction
Security
Network
Timetable
Component Verification Option
Type2 (non-Panasonic part verification)
Third Party's Component Verification System
Enable
Disable
(Standard) Socket communication
(Standard) Text communication
Compatibility mode 1
Port No.* 58002
Health check* 60 sec
(Standard) Text communication/Compatibility mode 1 Folder Setting
Input Folder Type2Input
Output Folder Type2Output
Changeover Judge Mode
No
M/SID
M/SID+Lot name
M/SID+Lot name+Revision
Changeover Interlock
Yes
No
Text communication health check
Yes
No
Text communication health check interval 30 sec
Host system interlock inquiry
Yes
No
Host system tray interlock
Yes (per Tray)
Yes (per Machine)
No
Specify remaining tray count
Yes
No
Compatibility Mode 1 Setting


option
Type1 (event)
Type2 (non-Panasonic part verification)
Type3 (part management information)
Event Filter
Common Setting
3rd party printer material verification setting
LMV communication function setting
GEM Communication Setting
GEM Event Filter
GEM Communication Operation And Monitor
Host System
PCB ID setting
PCB Barcode Scanner Communication Setting
3rd Party PCB Information Communication Setting
APC Setting
Time Setting
Set ProViewer
Set Log
Screen Saver Image Setting
Component Supply Navigator Option Setting
Backup Configuration
LNB
Multi-barcode Setting
Clear Port Exhaust
Yes
No
Verify Outer Tray
Yes
No
How to output change request at changeover
By machine (1 machine=1 request)
By lane (1 machine=2 requests)
Hold splicing result at work empty cancel
Yes
No
Verification cancel set function
Yes
No
Production permit by allocation setup check function
Yes
No
Production resume from stop check function
Yes
No
Output verification request feeder serial
Yes
No
General panel display request folder Type2Input
Set Parts Allocation/Remaining Count Output
Output When 1-piece Production Completed
Yes
No
Output Empty Memory Feeder Info
Yes
No
Output latest reel information "Output" code
Yes
No
Output latest reel information "Active" code
Yes
No
Output production data allocation information used part count
Yes
No

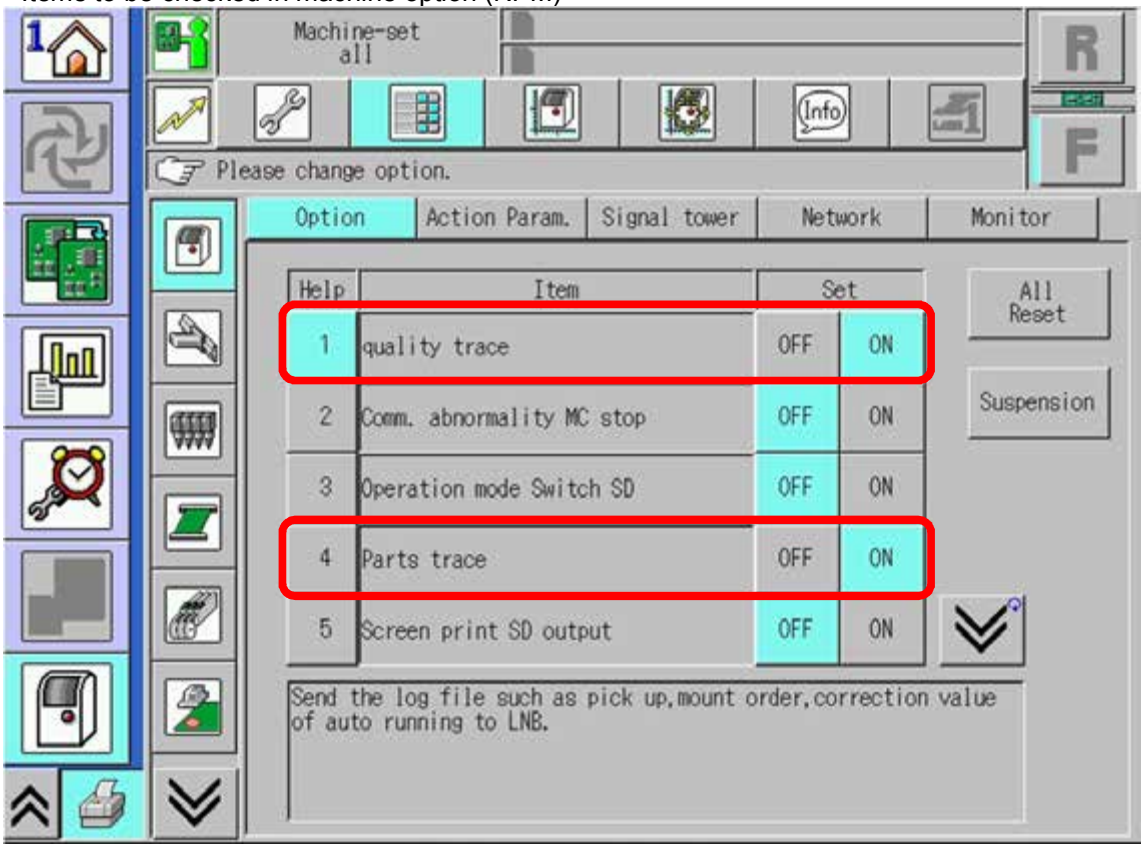
6.5.3 NPM/SP Machine Parameter Check

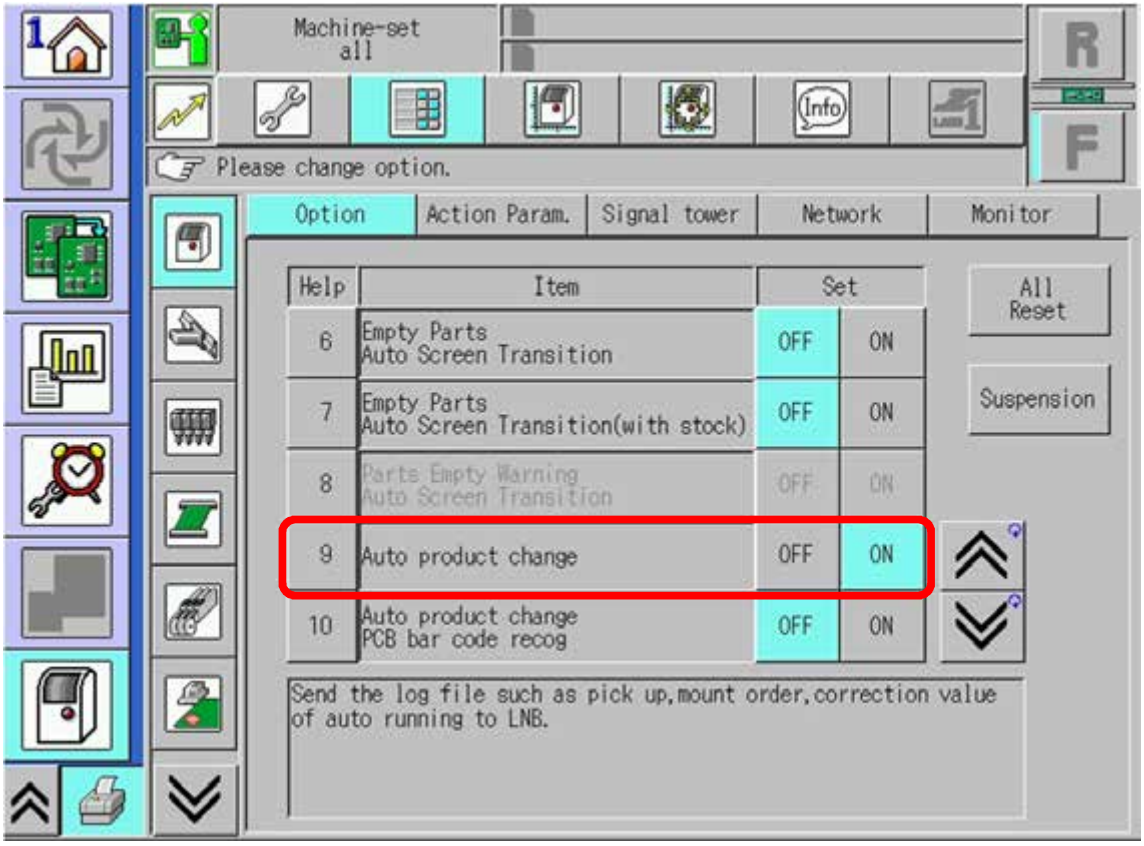
The following describes how to check the machine parameters set on the NPM/SP.

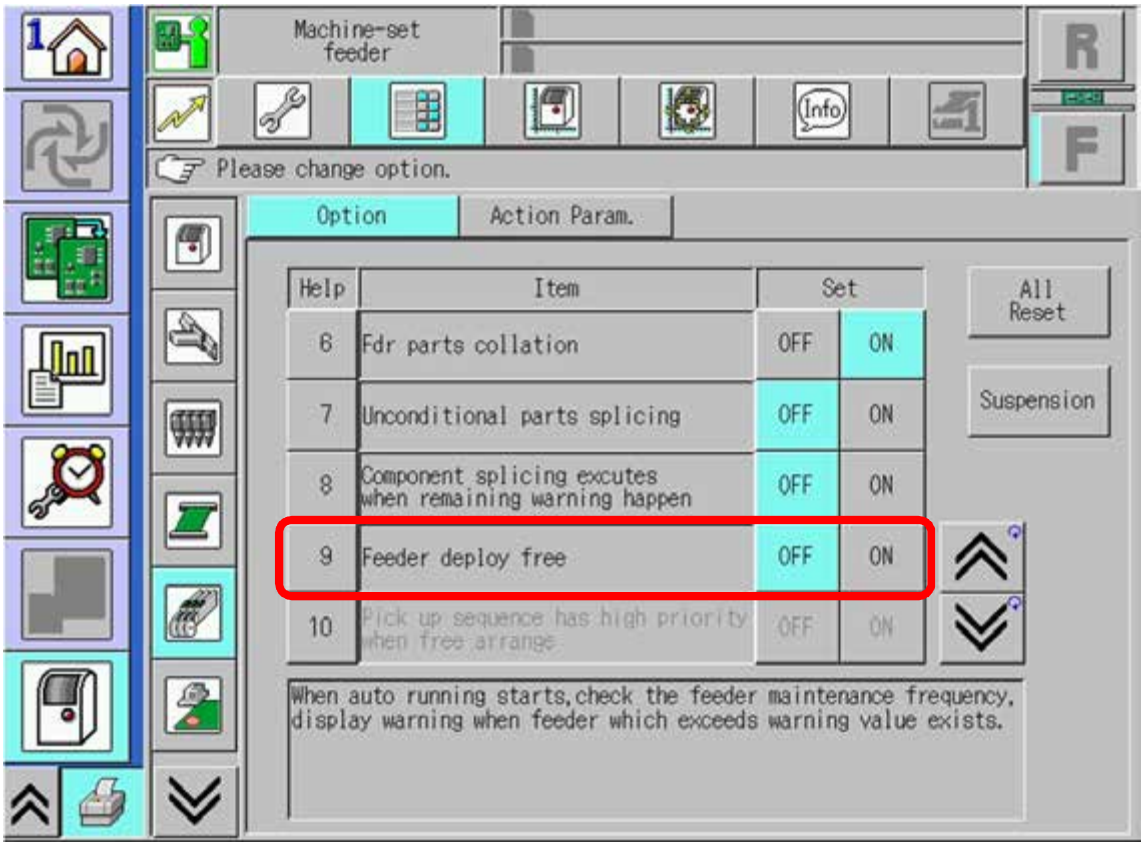
Step	Procedure
1	Modify the following file, and then restart the LNB. File to modify: **/lnbroot/configuration/CurrentResourceProprety.ini** Location to modify: **[Option]** **PanaCIMMachineParameter= 1 (0:Disable/1:Enable)**
2	Click **Diag Tools** in the PanaCIM launcher bar. 
3	Open the Tools Launcher screen. Choose **System Checker**. 
4	Choose **LNB Machine Parameters** in **DGNS** - **LNB Route** - **CIMC** - **Machine Parameters**. * The user can execute the desired level of check by expanding the folder. 

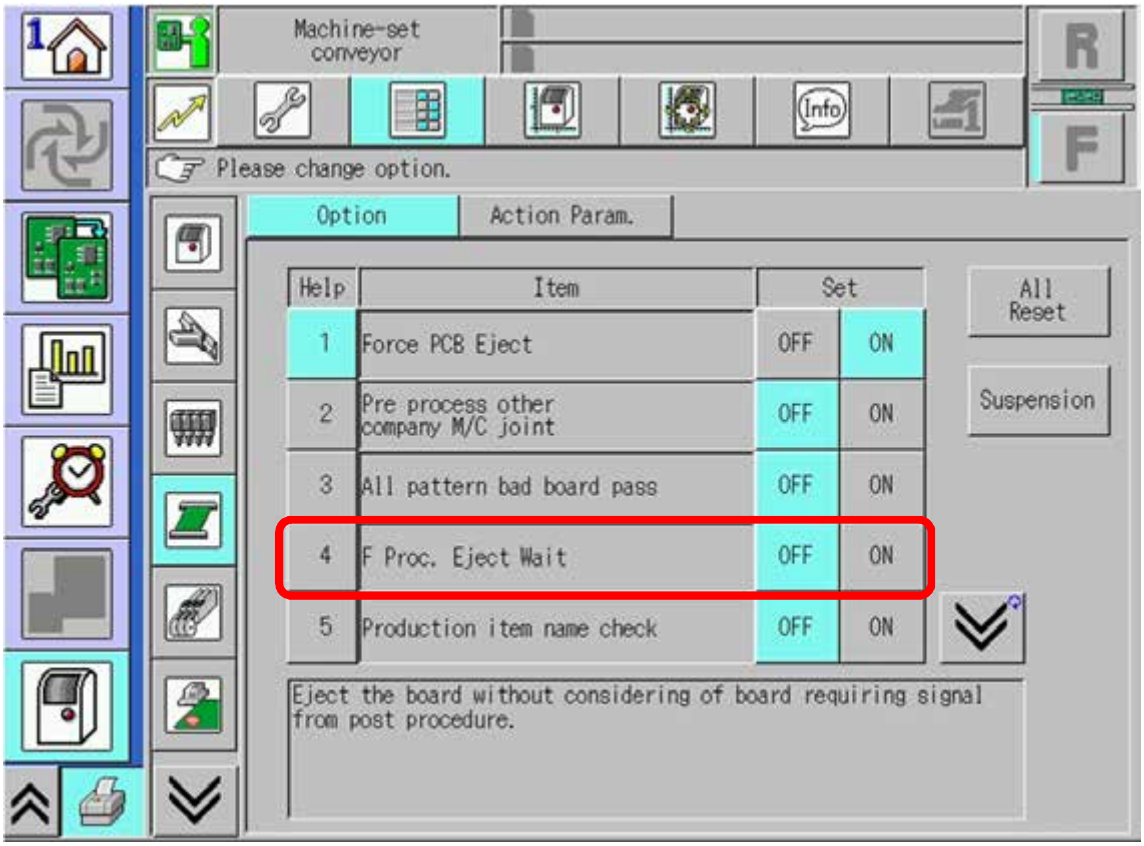
5	Press the **Start Check** button. The result appears in the lower right window. OK: Check Passed NG: Check Failed In case of NG, the cause appears in the lower right window. Apply an appropriate solution to that. 
6	To execute a recheck, press the **Reset Check** button. When the status changes to Idle, perform step 4 again. 

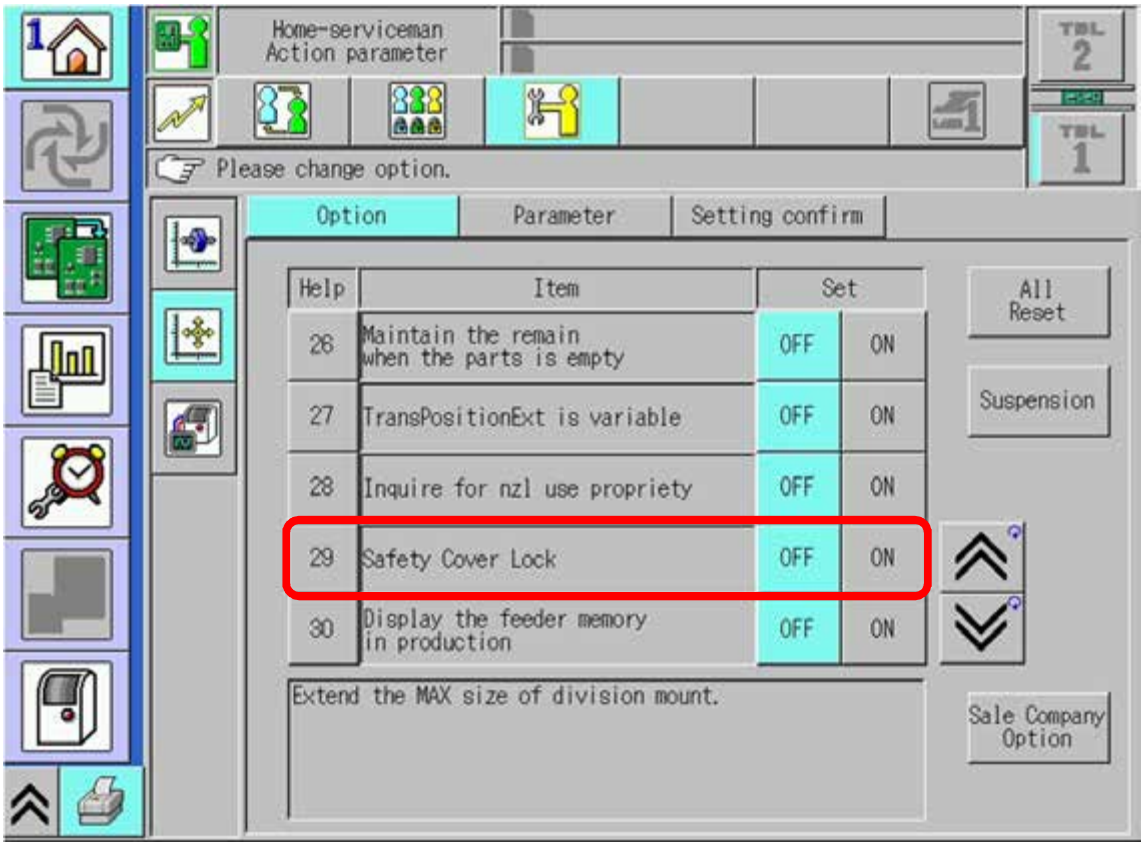
• Items to be checked in machine option (NPM)






Machine-set feeder
R
F
Info
Please change option.
Option
Action Param.
Help
Item
Set
All Reset
Suspension
6
Fdr parts collation
OFF
ON
7
Unconditional parts splicing
OFF
ON
8
Component splicing excutes when remaining warning happen
OFF
ON
9
Feeder deploy free
OFF
ON
10
Pick up sequence has high priority when free arrange
OFF
ON
When auto running starts,check the feeder maintenance frequency, display warning when feeder which exceeds warning value exists.


Machine-set conveyor
R
F
Info
Please change option.
Option
Action Param.
Help
Item
Set
All Reset
Suspension
1
Force PCB Eject
OFF
ON
2
Pre process other company M/C joint
OFF
ON
3
All pattern bad board pass
OFF
ON
4
F Proc. Eject Wait
OFF
ON
5
Production item name check
OFF
ON
Eject the board without considering of board requiring signal from post procedure.


Home-serviceman
Action parameter
TBL 2
TBL 1
Please change option.
Option
Parameter
Setting confirm
Help
Item
Set
All Reset
26
Maintain the remain when the parts is empty
OFF
ON
27
TransPositionExt is variable
OFF
ON
Suspension
28
Inquire for nzl use propriety
OFF
ON
29
Safety Cover Lock
OFF
ON
30
Display the feeder memory in production
OFF
ON
Extend the MAX size of division mount.
Sale Company Option

7. APPENDIX

7.1 Glossary

A

Additional equipment
Button at the bottom of route setting form. Used to add PanaCIM equipment to a route.

Allow feeder remove
Add feeder delete option on RF CVT scanner main menu.

Apply
Apply items edited on open dialog. The changes are then applied to the database.

Area Editor
Edit and save area settings. There are two area types.

- Materials Management: used for material such as reel or solder or feeder trace, or feeder pre setup
- Maintenance: location for feeder maintenance

Auto changeover
Manage changeover locations automatically after changeover was detected.

- Download PCB data to equipment
- Select PCB data to be used on equipment
- Adjust conveyor width in zone. Adjust support pins and guide adjusted by auto adjustment controller.

After all changeover was completed, machine starts in route and zone and a PCB of the new model enters into the zone.

Auto Download（Auto Download）
Download recipe automatically to equipment when changeover was detected. Detect changeover when PCB scanner or handy PCB scanner scans the next PCB or user enters the next PCB on LMNC screen. You will need PanaCIM/Changeover license to use this function.

Auto width adjustment
Set the type of auto width adjustment controller used on equipment and controller axis that controls the equipment.

Auto width adjustment controller
Used to adjust conveyor width for various PCB widths. Controller adjusts machine's conveyor width. Each zone accepts up to two auto width adjustment controller.

Available directory
Select remote directly supported by machine. Files being selected are highlighted. Machine default file is shown with check mark.

Axis
It shows the number of axis that corresponds with the number of auto width adjustment controller or the number of machines.
Example）Controller with six ports can adjust six PCB width.

Axis number
Specify the number of width controller axis. Controller can be used in multiple machines.

B

bit
The number of bits send at once between equipment and host.

Baud
Set this to set links from equipment to PanaCIM. Baud shows connection speed from S-232 to equipment.

C

Cable name
Set a unique name to be used in report. You can also omit this.

Carriage lock
Lock machine after duplication or part exhaust occurred until properly handled.

Cart
Cart set to machine.

Change
Change settings of the selected item.

Controller server stop/start
This appears only when Route Editor appears was started on controller server. If you stop controller server, all client interfaces stop. Postmaster (one of PanaCIM services) does not stop as PanaCIM Windows application is using. After you start controller server, all invalid interfaces turns valid on server and client machine.

D

Digital output mask
Set LMNC digital output response regarding logic status.

Digital input mask
Physical data from machine are controlled and converted to digital signal by LMNC. If multiple signals are set for monitoring, LMNC adjust voltage according to the influence that impulse gives on machine.

Delete
Delete route or zone or machine or other selected item. Deleting an item will deletes all information related to it.

Deleteable
Select a file that can be deleted remotely from machine. Selected item is highlighted.

Deleteable by operator
Delete pre-setup information from scanner pre-setup menu.

Destination
Start Position Editor after selected zone or machine or other equipment.

Device name
Shows serial port device name (/dev/ttyb) that is used for communication between machines and work stations.

DGS map
Maps DGS virtual line and machine with PanaCIM physical route and zone. This button is enabled only when server is set from <System Configuration> dialog and DGS is set to a route on Route Editor.

E

Enable tape splicing
Use this to joint tapes on reel before part exhaust occurs. Supported only by CM/DT/NPM series (includes AM100) and specific BM series machine.

F

Feeder remove method
Set the following changeover operation.

- Prompt: Select on dialog.
- True: Remove all feeders from machine slots

- False: Replace minimum number of feeders (that do not match with the next setup)

H

Host edit
Add and edit IP address in /etc/hosts file.

L

License control level
Determine control level for the selected zone. For each control level setting, license needs to be purchased accordingly.

Location Editor
Move change position of zone, machine, equipment within the same route or to another route.

- Move equipment within the zone or to another zone.
- Move zone within the same zone or to another zone.

Lock by ZNo.
Inform operator to supply parts to exhausted ones and release carriage lock.

Lot
Enter lot number of part reels.

M

Machine interface host
Select host name that machine interface communicates.

Machine Editor
Generate and edit data related to machine. The screen varies depending on type of machine.

Maintenance password
Password to change LMNC settings.

Mix setup data
Set setup operation.

- Prompt: Select on dialog
- True: Always setup for entire mix
- False: Setup only for target board

Module name
Specify module name.

Mount verification option
Verify slot, reel data attributes, feeder barcode, cart position, presetup ID and others when presetup cart was set in the machine. If those settings are not on, presetup cart is set to machine without verification.

N

Name
Set unique name for each route, zone and machine equipment.

No management (None)
If zone's management label is set to 'None', run changeover manually on machine. PanaCIM only checks if the selected program is correct before releasing PCB. This data is also used in PanaCIM/Auto changeover.

O

Offset
Use auto width adjustment controller scale to specify distance to be added or detected from the value

measured automatically. This is normally used to correct axis-origin offset error.

Omit slot verification
Select from the following changeover operation.
- True: Scan after feeders are on cart and inserted.
- False: No scan

Only difference valid
- Prompt: Appears when the dialog is at changeover operation when moving from current changeover to another one
- True: Always scan only differences
- False: Scan all feeders

Operation mode
Set scanner's operation mode. Operation mode depends on the set license.

Operator/Worker
Save operator ID in database and specify data input method to trace setting parts on reel.

P

Password
Overwrite carriage lock (of BM series machine) to release lock.

PCB transfer unit
Use the button below route setting form to add PCB transfer unit to zone.

Port type
Set port type that is used when machine or equipment communicate with host server. Not all equipment can use the same type of port. Only available ports are displayed as active in the menu.

Pre-setting ID
ID with 20 characters that expresses a group of feeders used in pre-changeover mix or program.

Production Management Data
Select the type of production management data to be uploaded from machine. Current setting is highlighted. This is checked at default.

Protocol
Set communication protocol used for communication between equipment and host server. BM series machine uses PanaProtocol.

S

Save format
- Use this option to save mix for which presetup was done on a removed feeder or all that can be used for PCB mix.
- Use this option on slot to save changeover information of entire slots. All existing feeder information will be removed and moved to presetup.

Save pre-setting
Set whether to save pre-changeover information when removing feeders. Setting True saves the information when removing feeders.

Scan & Key
Data can be scanned or manually input.

Scan Only
Use scanner to input data. Manual input not allowed.

Send sub-slot
Machine sends sub-slots upon IPC authentication request from BM series machine.

Server name
Host server name for interface application that communicates with equipment.

Settable
Specify data that allows remote operation. Selected item will be highlighted. Check mark shows default item.

Setup connection
Make sure the selected setup has been started or set it.

- Prompt: Select to continue changeover or start all over
- True: Pick up parts from feeders that have not been verified
- False: Finish changeover that is only partly finished and start from the first feeder.

Setup lock
Apply carriage lock when some parts necessary for production are not set. This function is enabled only when carriage lock function is set to ON.

Stick/Tape
Specify feeder on selected cart is stick or tape and if the cart is accepted in changeover area.

Supported command
Select remote commands supported by machine. Selected commands are highlighted. Checkmark shows default command.

T

Title bar
At the top of route or zone to show their name.

Toolbar
Click these buttons to create, set and delete route, zone and machine.

Tray
Shows feeders on the selected cart are tray feeders.

Type
Select type of feeder loaded on carts.

U

Uploadable
Select a type of file to be uploaded from machine. Selected item is highlighted. Check mark shows default item.

Use empty signal
Check this box when there is a digital cable to detect if PCB exist on equipment in the zone. This data is also used in PanaCIM/Auto changeover.

Use master/slave data
Prohibit carriage lock on a slot until the feeder in master/slave combination is empty. Once locked feeder carriage and scan only master feeder, the lock can be released.

User data
Set whether to collect user defined data by CVT

- Disable: Information not collected
- Key Only: Allows only key input
- Scan & Key: Allow scan and key input
- Scan Only : Allow input by scan only

User data name
Show name on RF CVT scanner user data prompt.

V

Vendor
Set vendor information to parts.

Verification
Set whether to perform carriage lock.

- No Locking: No carriage lock
- Part Exhaust: Carriage lock when part exhaust occurs. The lock is not released unless parts are replenished.
- PE & Setup Locking: Carriage lock when part exhaust occurs and feeder parts does not match with the current product setup.
- FCS: Select this to manage carriage lock.

W

Work cell
Show production equipment in general. (Mounter, Printer, Reflow and others)

Work table
Used by Materials Management and Maintenance Area. Feeder cart, maintenance area, storage container and flat table are included.

Z

Zone
Indicate equipment group with max one work cell. Work cell must be the equipment at the head of a zone. Extended conveyor is also set to a zone.

Zone Editor
Editor to add/correct/save zone.

Zone interface host
Set host PC as communication interface of a zone.

7.2 Bar Code Setting

7.2.1 The Input Method of Bar Code Type

PanaCIM scans barcodes and collects data. Configuration or contents of barcodes on parts or boards vary between their manufacturers. Therefore you need to set barcode to purse scanned data properly. You can set patterns for kind of data displayed.

Create pattern
There are many methods to define a pattern. For example, all of those below are the same.

^1PN¥([A-Z0-9]¥{3¥}¥)Q¥([0-9]¥{5¥}¥)$
^1PN¥([A-Z0-9][0-9A-Z]¥{2¥}¥)Q¥([0-9]¥{5¥}¥)$
^1PN¥([A-Z0-9][A-Z0-9][A-Z0-9]¥)Q¥([0-9][0-9][0-9][0-9][0-9]¥)$

The table below shows example steps to create barcode patterns.

Steps	Guideline	Example
Write barcode example using characters and numbers for variable number		1PNAAAQ99999 (AAA expresses part number and 99999 quantity)
Specify data to be extracted	*¥(* = Start extracting data *¥)* = End extracting data	1PN¥(AAA¥)Q¥(99999¥)
Specify start and end	^ = Start $ = End	^1PN¥(AAA¥)Q¥(99999¥)$
Replace characters or numbers of extracting data with possible value range	[A-Z0-9] = All upper cases and all numbers [0-9] = All numbers	^1PN¥([A-Z0-9][A-Z0-9][A-Z0-9]¥)Q¥([0-9][0-9][0-9][0-9][0-9]¥)$
Simplify repeating number	¥{#¥} = Repeat this value	^1PN¥([A-Z0-9]¥{3¥}¥)Q¥([0-9]¥{5¥}¥)$
Specify data with variable length Specify end space to ignore	¥{#,#¥} = Min. and Max value of the value	^1PN¥([A-Z0-9]¥{3,18¥}¥)Q¥([0-9]¥{5¥}¥)$

7.2.2 The Bar Code Standard to Support

This section explains barcode format and label supported by PanaCIM. PanaCIM supports 1D and 2D barcode format.

The following 1D barcode formats are supported.

- Code 39
- Code 128
- Code 93
- Code 2of5 Interleaved
- UPC
- EAN
- CODEBAR
- MSI

The following 2D barcode formats are supported.

- Data Matrix
- QR Code
- PDF417

Use formats that are supported by the scanner you are using.
To read barcodes properly, use matt type for labels. High gloss may cause reading problems.

7.2.3 Bar Code Type

PanaCIM allows the following barcode labels. These barcode labels are to be read at PanaCIM/Material Verification, Traceability, and changeover/part replenishment/PCB loading.

■ Part number

This is a label with variable length defining part number to be distinguished different part kinds. Part number must be unique for each part kind and be the same as the part number defined by Program Generation software. Please prepare labels.

*Any space cannot be used for part names.
Correct: ABCDEFGHIJ
Wrong: ABCDEFGH__ (_ is a space.)

Barcode specifications are as follows.

Length: 1-20 characters variable *For the NPM system, up to 30 characters are acceptable
Default value: P
Allowed characters: Numbers, single-byte upper cases, - . $ / + %
Label length: Px - Pxxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^P¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Quantity

This is a label with variable length defining quantity of parts.

Barcode specifications are as follows.

Length: 1-5 characters variable *For the NPM system, up to 6 digits are acceptable (Required to check the machine specs for the maximum value)
Default value: Q
Allowed characters: Number
Label length: Q1 - Q99999
Standard barcode parser: ^Q¥([0-9]¥{1,5}¥)[ ]*$

■ Vendor

This is a label with variable length defining part supply vendor.
Barcode specifications are as follows.
Length: 1-20 characters variable
Default value: V
Allowed characters: Numbers, single-byte upper case, - . $ / + %
Label length: Vx - Vxxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^V¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Lot

This is a label with variable length defining manufacture lot by part supply vendor.
Barcode specifications are as follows.
Length: 1-20 characters variable
Default value: L
Allowed characters: Numbers, single-byte upper cases, - . $ / + %
Label length: Lx - Lxxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^L¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Operator ID

This is a label with variable length showing operator who performs material verification.
Barcode specifications are as follows.
Length: 1-20 characters variable
Default value: O
Allowed characters: Numbers, single-byte upper cases, - . $ / + % space
Label length: Ox - Oxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^O¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ User defined data

This is a label with variable length defining a field that customers can use as desired.
Up to five pieces of data can be set. If it is defined by the same barcode, the definition with a higher priority is applied. If it has the same priority, the first definition is applied.
* For priority, please refer to “PanaCIM-EE Material Verification, 2.2.1 Setting of CIMC.
Barcode specifications are as follows.
Length: 1-20 characters variable
Default value: U
Allowed characters: Numbers, single-byte upper cases, - . $ / + % space
Label length: Ux - Uxxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^U¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Production area (CM/DT/NPM series (including AM100) machine only)

This is a label with variable length defining area for material verification.
Barcode specifications are as follows.
Length: 1-20 characters variable
Default value: W
Allowed characters: Numbers, single-byte upper cases, - . $ / + % space
Label length: Wx - Wxxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^W¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Model ID

This is a label with variable length defining to identify PCB.
The label is pasted directly on PCB. Paste the same model ID label on boards of the same recipe.
Barcode specifications are as follows.

Length: 1-20 characters variable
Default value: ML
Allowed characters: Numbers, single-byte upper cases, , - . $ / + % space
* ". (dot)" and "+" are not available.
The reason is that ". (dot)" is used for an extension and "+" is used for a delimiter when U01 and U03 files are imported into the PanaCIM.
Label length: MLx - MLxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^ML¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

■ Serial No.

This label is pasted to manage serial numbers. You cannot use same serial number in the same recipe.
Barcode specifications are as follows.

Length: 1-20 characters variable
Default value: SN
Allowed characters: Numbers, single-byte upper cases, , - . $ / + % space
Allowed characters: Numbers, single-byte upper cases, , - . $ / + % space
* ". (dot)" and "+" are not available.
The reason is that ". (dot)" is used for an extension and "+" is used for a delimiter when U01 and U03 files are imported into the PanaCIM.
Label length: SNx - SNxxxxxxxxxxxxxxxxxxx
Standard barcode parser: ^SN¥([A-Z0-9]¥{0,19¥}[A-Z0-9]¥{1¥}¥)[ ]*$

7.2.4 The Definition Library of Bar Code (Regular Expression)

■ Basic sign

Definition (Regular Expression)	Meaning
^	Definition start
$	Definition end
¥(	Valid item start
¥)	Valid item end

■ Definition of reading characters

Definition (Regular Expression)	Meaning
[]	Sign to conclude characters and signs in definition
[0-9]	Number 0 to 9,
[a-z]	a~z alphabet (lower cases only)
[A-Z]	A~Z alphabet (upper cases only)
[A-Za-z]	a~z,A~Z alphabet
[A-Za-z0-9]	a~z, A~Z alphabet, numbers

■ Reading characters (applied)

Definition (Regular Expression)	Meaning
[A-Z][0-9]	Combination of one character from A~Z and one number from 0 to 9, , such as A0,A1,B9
[x|yyy|zzz]	Means “or”; x or yy or zzz
[¥^]	“^” itself
[ ]	“(Space) ”
[.]	Any character
[¥.]	“.” (Period)

■ Repeating count (character count) definition

Definition (Regular Expression)	Meaning	Example
¥{m¥}	Repeat the character immediately before (left) m times.	
¥{n,m¥}	Repeat the character immediately before (left) n-m times.	
*	Repeat the character immediately before (left) 0 or more times.	ab*c: matches ac, abc, abbc, abbbc
+	Repeat the character immediately before (left) 1 or more times.	aback: matches abc, abbc, abbbc
?	Repeat the character immediately before (left) 0 or one time.	ab?c: matches ac, abc

7.3 LMNC

7.3.1 Appendix

LMNC (Line Manager Node Controller) is a controller to connect our software system on the host computer with equipment on site.
LMNC provides host software to detect and control digital signals send from equipment on site. It also has a function as serial communication concentrator that allows communication between host software and multiple equipment via LAN connection. LMNC operating software is to be downloaded from host computer. You can customize its control functions easily according to specific applications.

PanaCIM/Traceability or Auto changeover needs at least one LMNC installed on the conveyor of the head machine in a line.

LMNC supports the following device/functions.

- PCB scanner installed on the head conveyor
- Hold PCB until all setup requirements are fulfilled
- Special handy PCB barcode scanner called CBS scanner

PanaCIM/Material Verification on BM series machine allows interlock by LMNC to hold PCB on the conveyor until new setup has been confirmed.

Note: If you perform material verification without LMNC on BM series machine, you cannot use stop (interlock) even when setup or part exhaust verification has not been finished.

7.3.2 User Interface

You can use display mode key (yellow) to switch three of LMNC operator panel display modes.

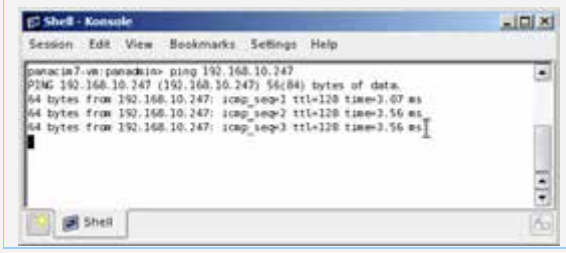
- **Normal Screen**: Shows PanaCIM message or options. Operators use function key or keyboard to input barcode or other information.
- **I/O Screen**: Shows interface information.
- **Config Screen**: Set LMNC unit network information. (IP address, subnet mask, gateway)



7.3.3 Configuration Procedure

To operate LMNC on a network, you need to configure IP address, subnet mask and other information. Configure them from <Config Screen> on LNMC display. See the following steps to configure network parameters.

Step	Procedure
1	Press the **Config Screen** button on upper left of keyboard.
2	
3	Select ‘Enter Password’ from keyboard screen. Default password: **lmnc**

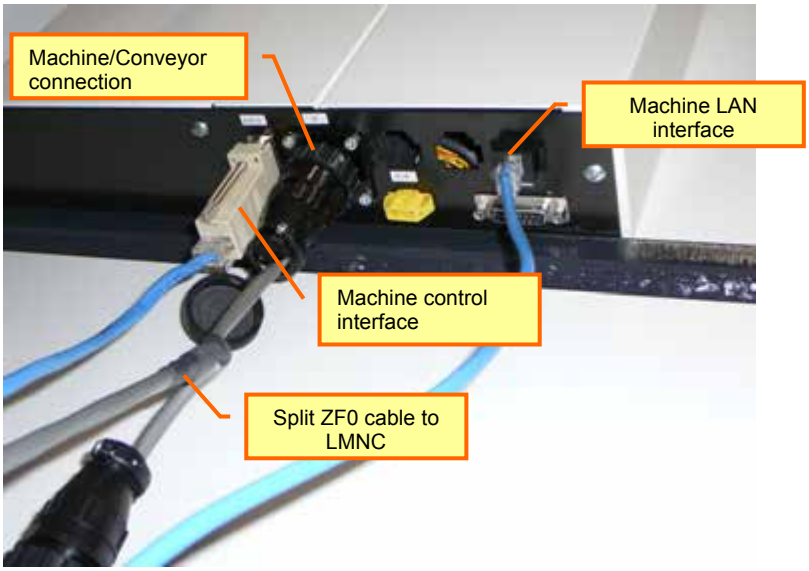
4	Select <IP Address>, <NET MASK>, <GATEWAY> on the screen and enter respective address. Select F1 to save changes. To discard changes and go back, select F10. 
5	Follow the steps below to check communication between LMNC-PanaCIM. • Open Console window from PanaCIM controller (Linux). • Use the command below to check communication with IP address set at 4 > ping␣(LMNC IP Address) • Make sure the result starts with “64 bytes from …” Otherwise communication failed, so check network configuration. 

7.3.4 Input/Output of LMNC



Connector	Description
JP2 - JP4	Connect JP2-INPUT and JP3-OUTPUT to the cables connected with the SMEMA cables between conveyor and machine. They prevent a PCB from being loaded before a setup is completed.
SERIAL PORT 3	Connect this to the PCB scanner.
SERIAL PORT 4	Connect this to the machine control interface.
SERIAL PORT 7	Connect this to the other display (Burr Brown etc.).
SERIAL PORT 8	Use this to set the IP address and subnet mask for LAN connection.
LAN	Connect this to the PanaCIM via a TCP/IP network.

Be sure to use a NULL cable or a crossover cable for the cable connecting from SERIAL PORT 4 to the machine. (Example: pin1 connecting to pin8, pin2 connecting to pin7, etc.)



CM/DT-series machine connection (equipment side as viewed from below)

Connector	Description
Machine control interface	Connect this to SERIAL PORT 4 of LMNC. Be sure to use a NULL cable with RJ45 plug for the LMNC side, and a NULL cable with 9-pin male DIN plug for the machine side.
Machine/Conveyor connection	This sends and receives signals with the conveyor before the machine.
ZF0 cable	This is a special cable that disables the connection from machine or conveyor until a setup requirement of the PanaCIM machine is satisfied.
Machine LAN access	This provides a network interface to the PT200 or the PanaCIM. The cable can also be connected directly to a hub inside the machine.

7.3.5 LMNC Message

This section shows messages appear on LMNC display and their explanations.

■ **Normal series**

Waiting for connection from PanaCIM

```
Waiting For Configuration Download
From Host
```

Downloading control program from PanaCIM to LMNC

```
PanaCIM Configuration Download
Currently In Progress
```

LMNC and PanaCIM are connected

```
Communications to Host Restored
** LMNC Now Re-initializing
```

LMNC main menu after control program was successfully downloaded from PanaCIM

```
<F1> Operator          <F3> Status
<F2> Maintenance     <F4> Config
```

Operator menu appears by pressing F1 on main menu

```
<F1> CHANGEOVER DONE    <F3> RETRY TRANSFER
<F2> BARCODE ENTRY            <F4> ZONE EMPTY
```

PCB menu after manually input barcode value

```
Enter the Panel Side
Press <T> for Top or <B> for Bottom:
```

Manual work required

```
Changeover to do: Feeders, Width,
Select NC: (nc_name), Manual Changes
```

Manual work required after program selected

```
(product_name) Changeover To Do:
Width, Manual Changes <F1> When Done
```

Auto changeover status (1)

Automatic Changeover in Progress Selecting Program: (Program name)

Auto changeover status (2)

Automatic Changeover in Progress Waiting for all Panels to Leave Zone

Auto changeover status (3)

Automatic Changeover in Progress Adjusting Width: (Product Name)

Auto changeover status (4)

Automatic Changeover in Progress Starting NC: (Program Name)

Changeover completed

CHANGEOVER DONE message sent to Host Changeover model: (Model Code)

Changeover completed

CHANGEOVER DONE message sent to Host

Waiting for next PCB (Show program name and previous PCB barcode)

Waiting For Panel NC: (Program Name) BC: (Panel barcode)

Barcode does not have model characters and LMNC waits for PanaCIM to send model character string

Requesting Model Info From Host NC: (Program Name) BC: (barcode)

Barcode does not exist in PanaCIM serial number

Barcode NOT In Database: (barcode) Remove Current Panel To Continue

After barcode was read properly from loaded PCB and post-process machine sends PCB request

Waiting For Downstream Panel Request NC: (Program Name) BC: (barcode)

LMNC receives PCB loading start signal from downstream machine

```
Releasing Panel
NC: (Program Name) BC: (barcode)
```

PCB does not have barcode and Zero barcode is set at Product Setup

```
Releasing Panel With Zero Bar Codes
NC(Program Name) me MOD: (Current Model)
```

Maintenance menu appears by pressing F2 on main menu

```
<F1> START DOWNTIME    <F3> CARRIAGE UNLOCK
<F2> LOG DOWNTIME          <F4> DISABLE STRACK
```

Downtime start screen appears by pressing F1 on Maintenance menu and start s equipment downtime

```
Downtime START message sent to Host
```

Select downtime kind

```
Choose the Type of Downtime Attribute
<F1> SCHEDULED    <F2> UNSCHEDULED
```

Select downtime display

```
Scroll Codes with Arrows, then <ENTER>
01 > Tooling Setup
```

Finish downtime period

```
Downtime END message sent to Host
```

Finish downtime log output

```
Downtime LOG message sent to Host
```

Changeover disable menu appears by pressing F4 on Maintenance menu. Password needed to disable changeover system,

```
Press <2nd> <CLEAR> To Abort
Enter Password: _
```

Changeover system disabled.

```
*** STRACK Disabled ****
NC: NC_program BC: ** Disabled **
```

Status menu appears by entering F3 at Maintenance menu

```
<F1> SERIAL PORTS 1-4
<F2> SERIAL PORTS 5-8     <F4> INPUT/OUTPUT
```

SERIAL PORT1-4 menu displayed by entering F1 at status menu. It shows current status of the first four SERIAL PORTs.

```
1) PanaCIM: OK       3) CVT: Unused
2) Display: OK       4) STRACK: Fault
```

SERIAL PORT 5-8 menu displayed by entering F1 at status menu. It shows current status of the last four SERIAL PORTs.

```
5) A Wdth 1: OK      7) Chngover: Unused
6) A Wdth 2: OK      8) BC Pass: Unused
```

I/O menu appears by entering F4 at status menu.
It shows binary status of input/ output port

```
IM: 1111 0000 0000 0000    OM:0000 0000
I: 0000 0000 0000 0011    O:0000 0001
```

Configuration menu appears from main menu F4

```
<F1> CURRENT PRODUCT    <F3> ZONE INFO
<F2> EQMT STATE
```

Current program menu displayed from configuration menu F1

```
PRODUCT: (Product Name)
NC NAME: (Program Name)
```

Current status menu displayed from configuration menu F2 (when machine is not down)

```
EQ STATE: (State)
EQ COMM STATE: (Comm State)
```

Current status menu displayed from configuration menu F2 (when machine is down)

```
The Machine in this Zone is DOWN
Reason: (Down Reason)
```

Zone information displayed by entering F3 at status menu

```
ZONE NAME: (Zone Name)
MACHINE: (Machine Name)
```

■ Error series

Communication establishment failed between LMNC and PanaCIM (when PanaCIM is stopped)

```
Check communications link to Host!
```

Explanation: PanaCIM controller does not communicate with LMNC
Action: Check network connection between PanaCIM controller host and LMNC. Make sure correct host name is set at LMNC on Route Editor.

Invalid program selected

```
Select NC: (Program Name)
Wrong/Unknown Machine Program Selected
```

Explanation: For PCB at PCB scanner has a program different from the Product Setup selected on machine.
Action: Select the program specified in Product Setup or change Product Setup program to machine.

Invalid program selected

```
Unknown NC Program: (Program Name)
Select NC: (Correct Program Name)
```

Explanation: The PCB linked with the current program selected from machine does not exist in Product Setup.
Action: Edit Product Setup or set it as program for the selected program or select another program from machine.

PCB scanner failed to read barcode of loaded PCB

```
No Barcode Received For Current Panel
Check Barcode Reader and Re-Scan Panel
```

Explanation: Failed to read **b**arcode of PCB loaded into the pre-process conveyor before machine.
Action: Enter barcode on LMNC display or remove PCB and make sure scanner reads barcode and sends it to LMNC.

PCB scanner sends error message after reading barcode of loaded PCB

```
** Bad Barcode Read **
<F2> to Enter Barcode Or Re-Scan Panel
```

Explanation: PCB scanner sends barcode reading failed message to LMNC.
Action: Enter barcode again on LMNC display or scan it again.

Operator tried to enter barcode while machine was processing PCB.

Cannot Accept Operator Input While A Panel Is Released to the Zone

Explanation: Operator tried to enter barcode on LMNC display while loading PCB into machine.
Action: You cannot enter barcode while loading PCB.

Barcode of loaded PCB does not match with PanaCIM barcode format

Unable to Parse Barcode: *(barcode)* <F2> to Enter Barcode Or Re-Scan Panel

Explanation: Barcode parser set at System Configuration does not support read barcode.
Action: Correct parser from PanaCIM controller - System Configuration application

Model changed after PCB loading started

Product Changed While Releasing Panel Verify Zone is Configured For Last Panel

Explanation: Program was changed on machine while conveying PCB to machine
Action: Remove PCB from machine and rescan PCB.

Wrong password entered

Invalid Password ! Retry Password (Y or N)? : _

Explanation: Wrong password entered on LMNC display
Action: Select ‘Y’ and enter correct password

7.3.6 Troubleshooting

Troubles that may occur in LMNC and their solutions are shown below.

LMNC Dispaly	Solution
Waiting for Configuration Download from PanaCIM	Communication with host has been established and waiting for LMNC configuration. Wait for communication to complete.
Check Communications Link to PanaCIM	Serial port1 RS-232 connection is down. Make sure PanaCIM controller is running. Make sure on PanaCIM controller that zone_interface Service is running. Check actual RS-232C, or LAN connection.
No Barcode Received For Current Panel	Current PCB does not have barcode. LMNC did not receive data within 10 seconds after changeover barcode reader detected PCB on the edge of conveyor. Make sure the scanner is turned on and its safety shield is up (not interfering laser). Check RS-232 cable connecting with scanner.
Bad Barcode Read	Remove PCB from the edge of conveyor and scan it again. Or if there is only one barcode, input barcode manually.
Unable to Parse Barcode	Barcode format is not defined by PanaCIM. If barcode is wrongly input, scan it again or input. If barcode has been input properly, check barcode expression from System Configuration.
Barcode NOT in Database	Barcode only with serial is not defined in PanaCIMDatabase. See the list of serial number you can use.
Product Changed While Releasing Panel	Changeover started while the last PCB was loading into the zone. 1. Make sure machine NC program is correct. 2. Remove the last PCB loaded into the zone and rescan.
Manually Change to Product	Auto download is not set for the zone. Operator selects and downloads a program as needed. Run feeder setup and adjust zone equipment and press LMNC operator sub-menu [CHANGEOVER DONE].
Auto-Download in Progress	PCB was rescanned or barcode was manually input during auto changeover. If a wrong PCB was downloaded, wait download to finish and scan correct PCB.
Downloading Error	One or more file(s) was/were not transferred properly to equipment in the zone.
Wrong/Unknown Product Selected	Program selected by operator does not match with the recipe expected from reading the barcode of PC. Select program again or run changeover for new PCB.
Invalid Model Supplied	Scanned or input barcode’s model section is not defined in PanaCIM. Set model ID from Product Setup.

LMNC Action	Solution
PCB transfer does not start. Display shows “Releasing Panel”	Timer to control transferring (Pre-reflow) PCB starts transferring PCB after its time is over. Make sure input status 0 and 1 are active.
PCB stays at the end of zone after reestablished PanaCIM link	Model of PCB on the edge of conveyor is now known to LMNC. Remove and rescan the PCB. Run changeover manually if needed.
Feeder replacement completed but LMNC display changeover operation list shows that feeder needs to be replaced	Check if CVT communication unit is communicating properly with PanaCIM. Press Product Transfer application print button to print feeder setup of a program for which changeover was completed. This changeover report shows the setup before replacing feeder required by PanaCIM. Use Material Verification scanner to change specified feeder.
LMNC display key does not work	Make sure LMNC power is on.

7.4 Allowable Characters (Prohibited Characters) in PanaCIM

There are limitations on allowable characters (characters that can be used) in PanaCIM. In addition, note that not only the data type and length but allowable characters are specified according to the attribute type as listed below.

***If you use the characters other than described below, an operation cannot be assured.**

- Product (Product-related information)
- Route (Route-related information)
- Material (Material-related information)
- Strack (PCB scanner-related information)
- Operator (Operator-related information)

Product (Product-related information)

Attribute name	Data type	Length	Allowable characters (characters which can be used)
Product Name	String	64	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Machine File	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Mix Name	String	150	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Model String (Mode ID)	String	40	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Group	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Lot	String	40	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

Route (Route-related information)

Attribute name	Data type	Length	Allowable characters (characters which can be used)
Route Name	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Zone Name	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Area Name	String	64	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

Module Name	String	18	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Equipment Name	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Work Area Barcode	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Storage Unit	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

Material (Material-related information)

Attribute name	Data type	Length	Allowable characters (characters which can be used)
Part Number	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Vendor	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Lot	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Quantity	Integer	18	Half-width digit: 0 to 9,
Material Barcode	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Part Class	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Cart ID	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
MCID	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Comparison ID	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
User Data(1-5)	String	30	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

Strack (PCB scanner-related information)

Attribute name	Data type	Length	Allowable characters (characters which can be used)
Model Number	String	40	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Serial Number	String	40	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

Operator (Operator-related information)

Attribute name	Data type	Length	Allowable characters (characters which can be used)
Name	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)
Barcode	String	20	Half-width digit: 0 to 9, Half-width uppercase characters: A to Z, Half-width symbols (”+”,”-“,”_”)

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